

Haar measure and quotient integration

4 course lessons, 7 prerequisite chapters and 1 supplementary proof · Published four-lesson companion; source authors self-checked; no independent AI or human review claimed

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About this edition

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Measure and Hilbert space tools for Haar integration

Written by GPT-6.1 Sol (OpenAI), Ultra reasoning effort, October 2026. Self-checked by the writing AI, GPT-6.1 Sol. Independent AI review remains separate. Original exposition is public domain (CC0).

Haar integration needs convergence theorems even when the group has uncountably many open components. It also needs a precise meaning for a tensor product of Hilbert spaces. This lesson proves the measure results used in the next lesson and explains which parts require countability. The finite-measure arguments are deliberately separated from the arguments valid on every measure space.

The earlier programme lesson **Hilbert spaces and compact operators**, Theorems 2.1–2.3, 4.1 and 8.1, proves orthogonal projection, representation of a bounded Hilbert-space functional, orthonormal bases and the Hilbert tensor product. Those complete proofs are the Hilbert-space prerequisites below. Elementary set theory, including the axiom of choice, is assumed. No topology is needed until the following Haar lesson. The core course **Measure and Integration** uses Fremlin’s human text. The core **Real Analysis I** and **Real Analysis II** use J. Lebl’s open *Basic Analysis*. Our Lemma 6.1 supplies the one-variable integral identities used here. Basic references for measure theory are [Fremlin] and [Lebesgue].

1. From an outer measure to a measure

A sigma-algebra on a set X is a family containing X and closed under complements and countable unions. A measure on it is a function into $[0, \infty]$ taking the empty set to zero and countably additive on disjoint sets. An outer measure m^* is defined on every subset of X , is zero at the empty set, is monotone, and satisfies

$$m^* \left(\bigcup_n A_n \right) \leq \sum_n m^*(A_n).$$

Theorem 1.1 (Carathéodory). The sets $E \subseteq X$ satisfying

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E) \quad (A \subseteq X) \quad (1.1)$$

form a sigma-algebra. The restriction of m^* to it is a complete measure. To verify (1.1) it suffices to prove its greater-than-or-equal inequality for A with finite outer measure.

Proof. The other inequality is subadditivity; when $m^*(A) = \infty$, the greater-than-or-equal inequality is automatic. Complements preserve the condition. If E, F satisfy it, split A first by E , and then split $A \setminus E$ by F . The sum of the first two resulting terms is at least $m^*(A \cap (E \cup F))$, by subadditivity. This proves the condition for $E \cup F$; differences and finite intersections follow.

Let E_n be disjoint sets satisfying the condition, and put $E = \bigcup_n E_n$. Repeated finite splitting gives

$$m^*(A) = \sum_{n=1}^N m^*(A \cap E_n) + m^* \left(A \setminus \bigcup_{n=1}^N E_n \right) \geq \sum_{n=1}^N m^*(A \cap E_n) + m^*(A \setminus E).$$

Let N increase. Subadditivity gives $\sum_n m^*(A \cap E_n) \geq m^*(A \cap E)$, so E satisfies the condition. An arbitrary countable union is reduced to this case by replacing E_n with $E_n \setminus \bigcup_{j < n} E_j$. Thus the family is a sigma-algebra. Taking $A = E$ in the finite-splitting inequality proves countable additivity on the disjoint E_n , together with subadditivity. If N has outer measure zero, every subset of N satisfies (1.1): monotonicity makes the first term zero, while monotonicity and subadditivity force the second term to equal $m^*(A)$. This also proves completeness. $\square$

Example. Assign to a subset of the real line the infimum of the sums of lengths of countable open interval covers. This is an outer measure: combine covers with errors $\varepsilon 2^{-n}$ to prove subadditivity. Splitting every covering interval at a fixed point splits its total length between the two half-lines; enlarge the split intervals by errors whose sum is arbitrarily small. This gives the reverse Carathéodory inequality for each half-line. Half-lines generate the Borel sigma-algebra, so Theorem 1.1 makes every Borel set measurable.

The measure of $[a, b]$ is $b - a$. An enlarged interval gives the upper bound. For any countable open cover of $[a, b]$, compactness selects a finite subcover. The sum of its interval lengths is at least $b - a$: arrange its endpoints in order, and each successive subinterval of $[a, b]$ is covered by at least one member. Summing lengths proves the lower bound. Singletons have measure zero, so the same length formula holds for open and half-open bounded intervals. Translating or dilating interval covers shows directly that $m^*(E + t) = m^*(E)$ and $m^*(aE) = |a|m^*(E)$ for $a \neq 0$, using the inverse operation for the reverse inequalities. The resulting Borel Lebesgue measure, or its completion, is therefore translation invariant. The next lesson instead constructs an outer measure using continuous functions on an arbitrary LCH space.

2. Integration and convergence without countability assumptions

Fix a measure space (X, Σ, μ) . A measurable real function is one for which $\{f > t\} \in \Sigma$ for every real t . Countable suprema and infima are measurable: use $\{\sup_n f_n > t\} = \bigcup_n \{f_n > t\}$, and negate for infima. Limits are measurable because $\liminf f_n = \sup_N \inf_{n \geq N} f_n$. Complex functions are measurable when their real and imaginary parts are.

For a nonnegative simple function $s = \sum_{j=1}^r a_j 1_{E_j}$, with disjoint measurable E_j and $a_j \geq 0$, define $\int s = \sum_j a_j \mu(E_j)$, with $0 \cdot \infty = 0$. Refining two partitions proves independence of the presentation, monotonicity, and additivity for simple functions. For measurable $f \geq 0$, define

$$\int f = \sup \left\{ \int s : 0 \leq s \leq f, s \text{ simple} \right\}.$$

There are increasing simple $s_n \rightarrow f$: truncate f at n and round down to multiples of 2^{-n} . The truncation bounds and the refining dyadic grids ensure monotonicity.

Countable additivity implies continuity from below: if $E_n \uparrow E$, write E as the disjoint union of $E_1, E_2 \setminus E_1, \dots$. Then $\mu(E_n) \uparrow \mu(E)$. If $\mu(E_1) < \infty$ and $E_n \downarrow E$, apply this to $E_1 \setminus E_n$ to obtain continuity from above.

Theorem 2.1 (Monotone convergence). If $0 \leq f_n \uparrow f$ almost everywhere, then $\int f_n \uparrow \int f$.

Proof. Discard one measurable null set to obtain pointwise inequalities; changing values there changes none of the integrals. Monotonicity gives $\lim \int f_n \leq \int f$. For simple $s \leq f$ and $0 < t < 1$, the sets $A_n = \{f_n \geq ts\}$ increase and cover $\{s > 0\}$. Consequently continuity from below on the finitely many level sets of s gives $\int s 1_{A_n} \uparrow \int s$. Since $f_n \geq ts 1_{A_n}$, the limit of their integrals is at least $t \int s$. Let $t \uparrow 1$ and then take the supremum over s . The same argument works when $\int s = \infty$. $\square$

Applying this to increasing simple approximations of $f, g \geq 0$ proves $\int(f + g) = \int f + \int g$. Applying it to partial sums proves

$$\int \sum_n f_n = \sum_n \int f_n \quad (f_n \geq 0). \quad (2.1)$$

These identities justify defining the integral of an integrable real function as $\int f^+ - \int f^-$, and then treating complex functions by real and imaginary parts. Here integrable means $\int |f| < \infty$. Linearity follows from nonnegative additivity by moving all negative parts to the opposite side of the desired equality. Choosing a phase so that $e^{i\theta} \int f$ is nonnegative real gives $|\int f| \leq \int |f|$.

Theorem 2.2 (Fatou and dominated convergence). For measurable $f_n \geq 0$,

$$\int \liminf_n f_n \leq \liminf_n \int f_n.$$

If complex measurable $u_n \rightarrow u$ almost everywhere and $|u_n| \leq g$ almost everywhere for one integrable $g \geq 0$, then u is integrable and $\int |u_n - u| \rightarrow 0$. In particular $\int u_n \rightarrow \int u$.

Proof. The functions $v_N = \inf_{n \geq N} f_n$ increase to the $\liminf$, and $\int v_N \leq \inf_{n \geq N} \int f_n$. Monotone convergence proves Fatou. For the second assertion, $|u| \leq g$ almost everywhere. Fatou applied to the nonnegative functions $2g - |u_n - u|$, whose limit is $2g$, gives

$$2 \int g \leq 2 \int g - \limsup_n \int |u_n - u|.$$

All integrals here are finite, so the $\limsup$ is zero. The integral inequality proved above gives the last assertion. $\square$

Example. On an uncountable set with counting measure, an integrable function has countable support: for each positive integer n , only finitely many points can have $|f| > 1/n$. Nevertheless the whole space is not sigma-finite. Theorems 2.1 and 2.2 remain valid. A later interchange of integrals will need a separate theorem; convergence alone does not supply it.

3. The complete spaces of integrable functions

For $1 \leq p < \infty$, let $L^p(\mu)$ be measurable functions modulo almost-everywhere equality with norm $\|f\|_p = (\int |f|^p)^{1/p}$. Let $L^\infty(\mu)$ consist of essentially bounded functions, with essential supremum norm. These definitions use the given sigma-algebra, with no assumption that the measure is complete.

Theorem 3.1 (Hölder and Minkowski). If $1/p + 1/q = 1$, including the endpoints, then $\int |fg| \leq \|f\|_p \|g\|_q$. For $1 \leq p \leq \infty$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

Proof. For $1 < p < \infty$, minimization of $a^p/p - ab + b^q/q$ in $a \geq 0$ gives Young's inequality $ab \leq a^p/p + b^q/q$. Apply it to $a = |f|/\|f\|_p$, $b = |g|/\|g\|_q$ and integrate. A zero norm means the corresponding function vanishes almost everywhere. The endpoints follow by bounding the essentially bounded factor outside its exceptional null set.

For Minkowski with $1 < p < \infty$, the elementary convexity inequality $(a + b)^p \leq 2^{p-1}(a^p + b^p)$ first proves $f + g \in L^p$. Put $h = |f + g|$. Hölder gives

$$\int h^p \leq \int (|f| + |g|)h^{p-1} \leq (\|f\|_p + \|g\|_p) \left(\int h^p \right)^{1/q}.$$

Divide when $\|h\|_p > 0$. The cases $p = 1, \infty$ are the pointwise triangle inequality. $\square$

Theorem 3.2 (Completeness and simple approximation). Every $L^p(\mu)$, $1 \leq p \leq \infty$, is complete. For finite p , simple functions with support of finite measure are dense, and every $f \in L^p$ vanishes outside a sigma-finite measurable set.

Proof. Let f_n be Cauchy in L^p , $p < \infty$. Choose a subsequence f_{n_j} with $\|f_{n_{j+1}} - f_{n_j}\|_p \leq 2^{-j}$, and measurable representatives. The partial sums of

$$v = \sum_{j \geq 1} |f_{n_{j+1}} - f_{n_j}|$$

have L^p norm at most one by Minkowski. Monotone convergence applied to their p -th powers gives $\int v^p \leq 1$. Thus $v < \infty$ outside a measurable null set, and the subsequence converges there to a measurable f , defined as zero on that null set. Its tail difference satisfies

$$|f - f_{n_j}| \leq \sum_{k \geq j} |f_{n_{k+1}} - f_{n_k}|, \quad \|f - f_{n_j}\|_p \leq \sum_{k \geq j} 2^{-k}.$$

The latter follows by Minkowski for finite sums and monotone convergence. Hence $f \in L^p$ and the subsequence converges in norm; the original Cauchy sequence does too.

For $p = \infty$, choose a subsequence with the same summable bounds. Remove a measurable null set on which its first term fails its essential bound, as well as the countably many null sets on which successive differences fail their bounds. On the complement the first term is bounded and the series of differences converges uniformly. Define the limit as zero on the removed measurable set. It is bounded and measurable, and the same tail estimate proves norm convergence.

For finite p , the sets $E_n = \{1/n < |f| \leq n\}$ increase to $\{f \neq 0\}$, and $\mu(E_n) \leq n^p \|f\|_p^p < \infty$. Dominated convergence shows $f 1_{E_n} \rightarrow f$ in L^p . Approximate the bounded complex function $f 1_{E_n}$ uniformly by simple functions supported on E_n , with error at most $\varepsilon/(1 + \mu(E_n))^{1/p}$. This proves density and the support assertion. $\square$

In particular L^2 is a Hilbert space for $\langle f, g \rangle = \int f \bar{g}$. Hölder proves that the inner product exists, and the theorem proves completeness. This is the point at which the earlier Hilbert-space results may be applied to measure spaces.

4. Densities and bounded functionals

Absolute continuity $\nu \ll \mu$ means that every μ -null measurable set is ν -null. We use the classical Hilbert-space method of J. von Neumann, also presented in S. Axler's *Measure, Integration & Real Analysis*. The

following proof first produces a bounded density relative to $\mu + \nu$; division then gives the density relative to μ . It avoids assuming a Radon–Nikodym theorem inside a representation theorem.

Theorem 4.1 (Radon–Nikodym). If μ, ν are sigma-finite measures on the same sigma-algebra and $\nu \ll \mu$, there is a measurable $h \geq 0$, unique μ -almost everywhere, such that $\nu(E) = \int_E h d\mu$ for every measurable E .

Proof. First assume both measures finite and put $\lambda = \mu + \nu$. The functional $u \mapsto \int u d\nu$ on $L^2(\lambda)$ is bounded: Cauchy–Schwarz gives a bound $\nu(X)^{1/2} \|u\|_{L^2(\lambda)}$. The Riesz–Fréchet theorem in the earlier Hilbert lesson, Theorem 2.3, supplies $g \in L^2(\lambda)$ with $\int u d\nu = \int ug d\lambda$; conjugate the representing vector because our inner product is linear in its first variable.

Testing measurable indicators shows that g is real with $0 \leq g \leq 1$ almost everywhere. Indeed $\int_E g$ is real and between zero and $\lambda(E)$ for every E ; the level sets where the imaginary part has one sign, or where $g < 0$ or $g > 1$, would violate these inequalities. Redefine g on the exceptional measurable null set. Then $\nu = g\lambda$ and $\mu = (1 - g)\lambda$ on all sets. On $A = \{g = 1\}$, $\mu(A) = 0$, so $\nu(A) = 0$, hence $\lambda(A) = 0$. Set $h = g/(1 - g)$ outside A , and $h = 0$ on A . The indicator identities extend to nonnegative functions by simple approximation and monotone convergence. Consequently

$$\int_E h d\mu = \int_E h(1 - g) d\lambda = \int_E g d\lambda = \nu(E).$$

For sigma-finite measures choose a disjoint countable measurable partition into pieces finite for both measures: intersect their two finite-measure covers and disjointify. Apply the finite argument on each piece and join the densities. Countable additivity and monotone convergence give the formula on X . For uniqueness, a candidate density is finite almost everywhere on each piece where the target measure is finite: its infinite level set must be null for the reference measure. On a finite-measure piece where both candidate densities are bounded, integrate over $\{h \geq k + 1/n\}$. Equality of the integrals forces its measure to vanish. Countably many such pieces cover the set where $h > k$, up to null sets; interchange them for the reverse inequality. $\square$

Theorem 4.2 (The dual of L^1). On a sigma-finite measure space, every bounded complex linear functional Φ on $L^1(\mu)$ has the form $\Phi(u) = \int bu d\mu$ for a unique $b \in L^\infty(\mu)$, with $\|\Phi\| = \|b\|_\infty$.

Proof. If $\mu(X) < \infty$, restrict Φ to L^2 , using $\|u\|_1 \leq \mu(X)^{1/2} \|u\|_2$. Hilbert representation gives a measurable $b \in L^2$ with $\Phi(u) = \int bu$ there. For $c > \|\Phi\|$, take $E = \{|b| > c\}$ and $u = 1_E \bar{b}/|b|$, with zero value when $b = 0$. This bounded function belongs to L^2 , and

$$c\mu(E) \leq \int_E |b| = |\Phi(u)| \leq \|\Phi\|\mu(E).$$

Thus $\mu(E) = 0$, and $\|b\|_\infty \leq \|\Phi\|$. Simple functions are in L^2 and dense in L^1 , by Theorem 3.2, so the identity extends to L^1 .

In the sigma-finite case partition X into finite-measure pieces E_n , apply the argument to $u \mapsto \Phi(u1_{E_n})$, and join the bounded densities. They have the uniform bound $\|\Phi\|$. The series $u = \sum_n u1_{E_n}$ converges in L^1 , so the joined density represents Φ . Its uniqueness follows by testing the phase of the difference on finite-measure pieces. Hölder gives $\|\Phi\| \leq \|b\|_\infty$, proving norm equality. $\square$

The sigma-finite hypothesis in this last theorem is explicit. The Haar lesson later decomposes a general group into open sigma-compact cosets and applies the theorem on each coset; it will not assume the hypothesis globally.

Why sigma-finiteness matters for a density

Example 4.3. Let $X = [0, 1]$ with its Borel sigma-algebra, let μ be counting measure, and let ν be Lebesgue measure. Then $\nu \ll \mu$, but no nonnegative measurable function h satisfies $\nu(E) = \int_E h d\mu$ for every Borel E .

Proof. Counting measure has no nonempty null set, so absolute continuity is automatic. If a density existed, then for every $x \in [0, 1]$,

$$0 = \nu(\{x\}) = \int_{\{x\}} h d\mu = h(x).$$

Thus h would vanish everywhere, contradicting $\nu([0, 1]) = 1$. A finite-counting-measure set is finite, and a countable union of finite sets is countable, whereas $[0, 1]$ is uncountable. Consequently μ is not sigma-finite. This example pinpoints the failure of the reference-measure hypothesis in Theorem 4.1, even though the target measure is finite. $\square$

5. Tensor products as Hilbert–Schmidt operators

For Hilbert spaces H, K , a tensor is built in **Hilbert spaces and compact operators**, Theorem 8.1: choose orthonormal bases indexed by I, J , take $\ell^2(I \times J)$, and send (ξ, η) to the array $\langle \xi, e_i \rangle \langle \eta, f_j \rangle$. The theorem proves basis independence through its isometric universal property. No separability is assumed.

There is a second useful interpretation. Write $\overline{H}$ for the conjugate Hilbert space: $\bar{\xi}$ denotes a vector with scalar rule $a\bar{\xi} = \overline{a\xi}$ and inner product $\langle \bar{\xi}, \bar{\eta} \rangle = \langle \eta, \xi \rangle$. A bounded operator $T : H \rightarrow K$ is Hilbert–Schmidt if

$$\|T\|_{\text{HS}}^2 = \sum_{i \in I} \|Te_i\|^2 < \infty,$$

where an arbitrary nonnegative sum means the supremum of its finite subsums.

Theorem 5.1. This definition and norm do not depend on the orthonormal basis. The Hilbert–Schmidt operators form a Hilbert space canonically isometric to $K \otimes \overline{H}$, through

$$k \otimes \bar{h} \mapsto [\xi \mapsto \langle \xi, h \rangle k]. \quad (5.1)$$

Every Hilbert–Schmidt operator is compact, and $\|T\| \leq \|T\|_{\text{HS}}$.

Proof. Fix bases $(e_i), (f_j)$. Parseval gives

$$\sum_i \|Te_i\|^2 = \sum_{i,j} |\langle Te_i, f_j \rangle|^2 = \sum_j \|T^* f_j\|^2. \quad (5.2)$$

The interchange is valid for arbitrary sums of nonnegative numbers: both iterated sums equal the supremum over finite rectangles, and each finite subset of $I \times J$ lies in a finite rectangle. Fixing the f_j in (5.2) proves independence of (e_i) ; the left expression then proves independence of (f_j) .

A square-summable array (a_{ji}) determines an operator. On finite linear combinations set $Te_i = \sum_j a_{ji} f_j$. For finite-support (c_i) , Cauchy–Schwarz gives

$$\left\| \sum_i c_i Te_i \right\|^2 \leq \left(\sum_i |c_i|^2 \right) \left(\sum_{i,j} |a_{ji}|^2 \right).$$

It extends uniquely to a bounded operator on H , with matrix (a_{ji}) and Hilbert–Schmidt norm equal to its ℓ^2 norm. Conversely every Hilbert–Schmidt operator has this matrix, so its space is complete and has the inner product of these arrays.

For the rank-one operators in (5.1), summing their matrix inner products gives $\langle k, k' \rangle \langle h', h \rangle$. This is exactly the elementary-tensor inner product on $K \otimes \overline{H}$. Finite-support matrices are finite sums of these rank-one operators and are dense among the square-summable arrays. The universal property of the tensor product therefore extends (5.1) to a unitary. Finally finite-support matrices approximate every such array in Hilbert–Schmidt norm, and hence in operator norm by the displayed estimate. They give finite-rank operators. A norm limit of finite-rank operators is compact: approximate its image of the unit ball by a finite net for the image under one finite-rank approximant. $\square$

Example. A diagonal operator on $\ell^2(I)$ is Hilbert–Schmidt exactly when its diagonal (a_i) is square summable. For uncountable I , at most countably many a_i can be nonzero. The identity is Hilbert–Schmidt exactly when I is finite. This explains why an uncountable basis causes no uncountable convergence problem in (5.2).

6. One-variable integral identities

The explicit Haar and arithmetic examples use elementary calculus. Here are the integral identities needed, with their proofs, so no general multidimensional change-of-variables theorem is implicit.

Lemma 6.1. If h is continuous on a compact interval, its Lebesgue integral equals its Riemann integral. For $H \in C^1([a, b])$, $\int_a^b H'(t) dt = H(b) - H(a)$. If $\phi \in C^1([a, b])$ and h is continuous on an interval containing its image, then

$$\int_{\phi(a)}^{\phi(b)} h(u) du = \int_a^b h(\phi(t))\phi'(t) dt. \quad (6.1)$$

In particular, the integral of a compactly supported continuous derivative on the real line is zero. Differentiating a parameter integral is valid when its difference quotients converge pointwise and have a common integrable bound.

Proof. Recall the elementary differential facts involved. A continuous function on a compact interval attains its extrema. If its endpoints agree, either it is constant or one extremum occurs in the interior, where its derivative is zero: the left and right difference quotients have opposite weak signs. This is Rolle's theorem. Subtract the line joining the endpoint values to obtain the mean value theorem. The chain rule follows by substituting $\phi(t+h) - \phi(t) = \phi'(t)h + o(h)$ into the corresponding first-order expansion of the outer function; differentiability makes the inner increment $O(h)$.

Uniform continuity makes the difference between the upper and lower step functions on a sufficiently fine partition arbitrarily small in integral; both bound h . Their Lebesgue integrals are the same interval-length sums that define the Riemann integral. For H , the mean value theorem on each partition interval expresses its increment as $H'(\xi)$ times the interval length. Summing telescopes to $H(b) - H(a)$; uniform continuity of H' makes the sums converge to its integral.

Define $A(v) = \int_{v_0}^v h(u) du$, with oriented integrals. The difference quotient of A is an interval average of h , tending to $h(v)$ by continuity. Thus $(A \circ \phi)' = (h \circ \phi)\phi'$. Apply the preceding identity to $A \circ \phi$ to obtain (6.1). The zero-integral assertion follows by choosing endpoints outside the support. Finally apply dominated convergence, Theorem 2.2, to the difference quotients to justify the parameter derivative. ■

7. Exercises with complete solutions

1 (easy): finite-measure support. Prove that $f \in L^p$, $p < \infty$, has sigma-finite support without assuming X sigma-finite. Explain why the same assertion fails for $p = \infty$.

Solution. The support in the measure-theoretic sense $\{f \neq 0\}$ is $\bigcup_n \{|f| > 1/n\}$. On each set, $n^{-p}\mu(\{|f| > 1/n\}) \leq \int |f|^p$, so its measure is finite. On an uncountable counting space, $1 \in L^\infty$ has all of X as support, which cannot be covered by countably many finite sets.

2 (medium): a summable series in L^p . If $\sum_n \|u_n\|_p < \infty$, $p < \infty$, prove that $\sum_n u_n$ converges almost everywhere and in L^p , and estimate every tail.

Solution. Minkowski bounds the norm of $\sum_{n=1}^N |u_n|$ by $\sum_n \|u_n\|_p$. Monotone convergence of its p -th power shows that $\sum_n |u_n| < \infty$ almost everywhere. Thus the series converges there. Apply the same argument to a tail, obtaining $\|\sum_{n \geq N} u_n\|_p \leq \sum_{n \geq N} \|u_n\|_p \rightarrow 0$. Define the sum as zero on the measurable exceptional set.

3 (medium): failure of domination. On $(0, 1)$ with Lebesgue measure put $u_n = n1_{(0, 1/n)}$. Find its pointwise limit and its integral. Prove that no integrable common dominator exists.

Solution. Every $x > 0$ eventually lies outside $(0, 1/n)$, so $u_n(x) \rightarrow 0$. Yet $\int u_n = 1$. If $|u_n| \leq g$ for an integrable g , dominated convergence would force these integrals to tend to zero, a contradiction. The shrinking support does not replace domination.

4 (hard): weighted finite reduction. For sigma-finite μ , construct a strictly positive measurable w with $\int w d\mu < \infty$. Prove that $w\mu$ has the same null sets, and that multiplication by $w^{-1/2}$ gives a unitary from

$L^2(\mu)$ to $L^2(w\mu)$.

Solution. Partition X into finite-measure measurable E_n and take $w = \sum_n 2^{-n}(1 + \mu(E_n))^{-1}1_{E_n}$. Then $w > 0$ and $\int w \leq 1$. For any measurable E , $\int_E w = 0$ implies $\mu(E \cap \{w \geq 1/n\}) = 0$ for each n , hence $\mu(E) = 0$. The converse is immediate. The norm identity is $\int |w^{-1/2}f|^2 w d\mu = \int |f|^2 d\mu$, and the inverse is multiplication by $w^{1/2}$.

5 (hard): which tensor represents an operator? Explain why $K \otimes H$ is not the canonical tensor for complex-linear rank-one maps $H \rightarrow K$, and verify (5.1) on two rank-one maps.

Solution. The map $h \mapsto [\xi \mapsto \langle \xi, h \rangle k]$ is conjugate-linear in h , while elementary tensors are bilinear in their factors. Passing to $\overline{H}$ makes it linear. For $T\xi = \langle \xi, h \rangle k$ and $S\xi = \langle \xi, h' \rangle k'$, Parseval gives $\sum_i \langle Te_i, Se_i \rangle = \langle k, k' \rangle \sum_i \langle e_i, h \rangle \overline{\langle e_i, h' \rangle} = \langle k, k' \rangle \langle h', h \rangle$, the required tensor inner product.

What this enables

The next lesson uses Theorem 1.1 to construct Radon measures, Theorems 2.1–3.2 to pass to limits and complete L^p , and Theorems 4.1–4.2 on the sigma-finite pieces of a general group. Its Radon-product theorem proves the additional hypotheses required for iterated integration; none are implicit in this lesson. Hilbert tensors then identify L^2 of that product, and Theorem 5.1 interprets square-integrable kernels as Hilbert–Schmidt operators.

References

- [Erdman] J. M. Erdman, *Functional Analysis and Operator Algebras: An Introduction*, version 4 October 2015, [author's text and editable sources](#). The core Functional Analysis course uses this text. Its Hilbert–Schmidt exercises concern separable spaces; Theorem 5.1 above supplies the full matrix and conjugate-tensor proof for arbitrary Hilbert spaces.
- [Paschke] W. L. Paschke, *Lecture Notes in Functional Analysis*, edition 0.9, [author's Hilbert–Schmidt chapter](#). It gives a complementary completeness argument and a square-integrable kernel example on separable L^2 .
- [Hilbert tools] *Hilbert spaces and compact operators*, programme lesson by Claude Opus 5.5 (Anthropic), October 2026, CC0, Theorems 2.1–2.3, 4.1 and 8.1. These are preceding internal full-proof providers.
- [Fremlin] D. H. Fremlin, *Measure Theory*, Volumes 1 and 2, Torres Fremlin, 2000–2001. [Freely accessible text and supplied editable TeX](#). Its general localizable-measure duality complements the sigma-finite proof here; the non-sigma-finite Haar case is proved in the following lesson.
- [Lebesgue] H. Lebesgue, *Leçons sur l'intégration et la recherche des fonctions primitives*, Gauthier-Villars, 1904, [freely accessible historical edition](#). This is historical reading, rather than a prerequisite for the proofs above.
- [Lebl] J. Lebl, *Basic Analysis I* and *Basic Analysis II*, [author's open reader](#). The fundamental theorem of calculus is a complementary human proof for Lemma 6.1.
- [Axler] S. Axler, *Measure, Integration & Real Analysis*, author's freely accessible edition dated 12 June 2026. It presents the von Neumann Hilbert-space proof and explains the role of sigma-finiteness. The proof and counterexample above are independently written.

Haar measure on locally compact groups

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This lesson constructs Haar measure on an arbitrary locally compact group and develops the measure theory that goes with it. Sections 2–6 work on locally compact spaces. They prove the Riesz representation theorem for Radon measures, treat image measures and densities, build the product of two Radon measures with its Tonelli and Fubini theorems, and identify L^2 of a product with a Hilbert tensor product. Sections 7–11 turn to groups: topological groups, existence and uniqueness of Haar measure, the modular function, and the inversion formula. Sections 12–15 treat products of groups, groups that are not σ -compact, convolution, and approximate identities. Section 16 has exercises with solutions.

No countability assumption is made anywhere. The group need not be σ -compact, second countable or unimodular, and its Haar measure need not be σ -finite. This generality needs care: a Borel set of infinite measure can have only null compact subsets, and Fubini's theorem can fail. Section 13 shows what goes wrong, and the rest of the lesson is arranged so that it does no harm.

Haar measure makes a locally compact group G into a measure space on which G acts by translations. Then $L^1(G)$ is a Banach $*$ -algebra under convolution, and $L^2(G)$ carries the left and right regular representations. These objects are the starting point of abstract harmonic analysis, and of the group von Neumann algebras studied in the course on modular theory and weights.

The lesson builds on four sources, cited where they are used:

- **Measure and Hilbert space tools for Haar integration**, Theorems 1.1–4.2, for the full measure-theory proofs; the measure convention is compared with Fremlin below;
- **The Stone–Weierstrass theorem for functions vanishing at infinity**, for locally compact spaces and Urysohn's lemma;
- **Weak topologies: Tychonoff, Banach–Alaoglu, Mazur, bipolars, Krein–Milman and Eberlein–Šmulian**, for Tychonoff's theorem;
- **Hilbert spaces and compact operators**, for Hilbert tensor products and multiplication operators.

The exact statements are collected at the end, under "Results used from other lessons". Core preparation is [Point-Set Topology](#); [Measure and Integration](#) provides Fremlin's broader measure-theory setting. The arbitrary-product and Hilbert-space proofs used here are the advanced programme results named above.

Freely accessible complementary treatments are Fremlin's [Measure Theory](#) and F. van Doorn's [Formalized Haar Measure](#). Fischer and Ruzhansky's freely accessible [Quantization on Nilpotent Lie Groups](#) supplies further convolution and Heisenberg-group examples. The historical development from Haar and von Neumann to Weil and Cartan is discussed in F. van Doorn's paper; the original accessible von Neumann and Cartan papers are linked below.

1. Conventions

Spaces. An *LCH space* is a Hausdorff space that is locally compact. Its compact subsets are closed. $C_c(X)$ is the space of continuous complex functions with compact support, $C_c^+(X)$ is the set of $f \in C_c(X)$ with $f \geq 0$ and $f \neq 0$, and $\|f\|_{\sup} = \sup_x |f(x)|$. For an open set U we write $f \prec U$ when $f \in C_c(X)$, $0 \leq f \leq 1$ and $\text{supp } f \subseteq U$. For a compact set K we write $K \prec f$ when $f \in C_c(X)$, $0 \leq f \leq 1$ and $f = 1$ on K . A function $h: X \rightarrow [0, \infty]$ is *lower semicontinuous* (lsc) if every set $\{h > c\}$ is open; lsc functions are Borel.

Four tools from topology. The first two tools are proved in the lesson on the Stone–Weierstrass theorem; we derive the other two from them.

- **(T1) (Shrinking)** If $K \subseteq U$ with K compact and U open, there is an open V with compact closure such that $K \subseteq V \subseteq \overline{V} \subseteq U$. This is [Proposition 4.2\(2\) of the Stone–Weierstrass lesson](#).
- **(T2) (Urysohn's lemma)** If $K \subseteq U$ as in (T1), there is f with $K \prec f \prec U$. *Proof.* Take V from (T1). [Corollary 5.2 of the Stone–Weierstrass lesson](#), applied to the compact set K and the closed set $X \setminus V$, gives $f \in C_c(X)$ with $0 \leq f \leq 1$, $f = 1$ on K and $f = 0$ off V . Then $\text{supp } f \subseteq \overline{V} \subseteq U$. $\square$
- **(T3) (Partitions of unity)** If K is compact and $K \subseteq U_1 \cup \dots \cup U_n$ with each U_i open, there are $h_i \prec U_i$ with $\sum_i h_i = 1$ on K and $\sum_i h_i \leq 1$ everywhere. *Proof.* For each $x \in K$ pick an index i with $x \in U_i$, and by (T1) an open $V_x \ni x$ whose compact closure lies in U_i . Finitely many V_x cover K . Let F_i be the union of the closures of the chosen V_x that were assigned to i . Then F_i is compact, $F_i \subseteq U_i$, and

$K \subseteq \bigcup_i F_i$. Choose $F_i \prec g_i \prec U_i$ by (T2) and set $h_1 = g_1$ and $h_k = (1 - g_1) \cdots (1 - g_{k-1})g_k$. By induction, $\sum_{k \leq m} h_k = 1 - \prod_{k \leq m} (1 - g_k)$. This lies in $[0, 1]$, and it equals 1 on each F_i . $\square$

- **(T4) (Tube lemma)** If $K \subseteq X$ and $L \subseteq Y$ are compact and $K \times L \subseteq W$ with W open in $X \times Y$, there are open $U \supseteq K$ and $V \supseteq L$ with $U \times V \subseteq W$. *Proof.* Fix $x \in K$. Each (x, y) , $y \in L$, lies in an open box inside W ; finitely many boxes $P_j \times Q_j$ of this kind have Q_j covering L . Put $P^x = \bigcap_j P_j$ and $Q^x = \bigcup_j Q_j$; then $P^x \times Q^x \subseteq W$. Cover K by finitely many P^{x_i} , and let $U = \bigcup_i P^{x_i}$ and $V = \bigcap_i Q^{x_i}$. $\square$

Measures and functions. A measure is positive and countably additive. $\mathcal{B}(X)$ denotes the Borel sets of X , the σ -algebra generated by the open sets; functions on X are Borel unless said otherwise. A Borel set is σ -finite for a measure when countably many Borel sets of finite measure cover it. For $1 \leq p < \infty$, $L^p(\mu)$ consists of the Borel functions with $\int |f|^p d\mu < \infty$, modulo functions that vanish almost everywhere. Hilbert spaces are complex, and the inner product $\langle \xi, \eta \rangle = \int \xi \bar{\eta} d\mu$ is linear in ξ . For a Borel $f: X \rightarrow [0, \infty]$ we use the *layer functions*

$$s_n = 2^{-n} \sum_{i=1}^{n2^n} 1_{\{f > i2^{-n}\}}. \quad (1.1)$$

They are simple Borel functions, and they vanish where f does. They increase, $s_n \leq s_{n+1}$, because the i -th term of s_n is at most the sum of the terms of s_{n+1} with indices $2i - 1$ and $2i$. And $s_n(x) \rightarrow f(x)$ at every point: if $f(x) < n$, then $f(x) - 2^{-n} \leq s_n(x) \leq f(x)$, and if $f(x) = \infty$, then $s_n(x) = n$. The monotone and dominated convergence theorems, Hölder's inequality, and the completeness of L^p are proved in **Measure and Hilbert space tools for Haar integration**, Theorems 2.1, 2.2, 3.1 and 3.2. The exact uses are listed under "Results used from other lessons".

Groups. A group G is a *locally compact group* when it carries a locally compact Hausdorff topology for which $(x, y) \mapsto xy$ and $x \mapsto x^{-1}$ are continuous; e is its identity. For a function f on G and $y \in G$ put

$$L_y f(x) = f(y^{-1}x), \quad R_y f(x) = f(xy), \quad \check{f}(x) = f(x^{-1}).$$

Then $L_{yz} = L_y L_z$ and $R_{yz} = R_y R_z$. The *left and right regular representations* of G on $L^2(G)$ are $\lambda(g) = L_g$ and $\rho(g) = \Delta(g)^{1/2} R_g$, where Δ is the modular function of Section 10; Exercise 16.3 shows that they are unitary representations.

2. Radon measures and the Riesz representation theorem

Definition 2.1 (Radon measure). A *Radon measure* on an LCH space X is a measure μ on $\mathcal{B}(X)$ with the following three properties.

- **(R1)** $\mu(K) < \infty$ for every compact K .
- **(R2) (outer regularity)** $\mu(E) = \inf\{\mu(U) : U \supseteq E \text{ open}\}$ for every Borel set E .
- **(R3) (inner regularity on open sets)** $\mu(U) = \sup\{\mu(K) : K \subseteq U \text{ compact}\}$ for every open U .

This is the notion of regularity used throughout the lesson. Inner regularity is not required on all Borel sets, because it can fail (Example 13.4). Proposition 2.3 shows that it holds on every σ -finite set.

Theorem 2.2 (Riesz representation theorem). Take a positive linear functional I on $C_c(X)$. There is exactly one Radon measure μ on X with $I(f) = \int f d\mu$ for every $f \in C_c(X)$. Moreover, every Radon measure μ satisfies, with $I(f) = \int f d\mu$,

$$\mu(U) = \sup\{I(f) : f \prec U\} \quad (U \text{ open}), \quad \mu(K) = \inf\{I(f) : f \in C_c(X), 1_K \leq f\} \quad (K \text{ compact}). \quad (2.1)$$

Regularity warning: The stronger requirement that a measure be inner and outer regular on every Borel set is inappropriate here. In that form it fails for the Haar integral of $\mathbb{R} \times \mathbb{R}_d$ (Example 13.4).

Proof. *Step 1: formula (2.1) and uniqueness.* Let μ be Radon. If $f \prec U$, then $\int f d\mu \leq \mu(U)$. If $K \subseteq U$ is compact, Urysohn's lemma (T2) gives $K \prec f \prec U$, and then $\mu(K) \leq \int f d\mu$. With (R3) this proves the first formula. If $1_K \leq f$, then $\mu(K) \leq \int f d\mu$; if $U \supseteq K$ is open, (T2) gives $K \prec f \prec U$ with $\int f d\mu \leq \mu(U)$. With

(R2) this proves the second formula. By the first formula, two Radon measures that give every $f \in C_c(X)$ the same integral agree on open sets, and then on all Borel sets by (R2).

Step 2: an outer measure. For open U define $\mu(U) = \sup\{I(f) : f \prec U\}$, and for any $A \subseteq X$ put $\mu^*(A) = \inf\{\mu(U) : U \supseteq A \text{ open}\}$. Then $\mu^* = \mu$ on open sets, and μ^* is monotone. Let $U = \bigcup_j U_j$ with U_j open, and let $f \prec U$. The compact set $\text{supp } f$ lies in $U_1 \cup \dots \cup U_n$ for some n . The partition of unity (T3) gives $h_j \prec U_j$ with $\sum_{j \leq n} h_j = 1$ on $\text{supp } f$. So $f = \sum_{j \leq n} f h_j$ with $f h_j \prec U_j$, and $I(f) \leq \sum_j \mu(U_j)$. Hence $\mu(U) \leq \sum_j \mu(U_j)$. For arbitrary sets A_j , choose open $U_j \supseteq A_j$ with $\mu(U_j) \leq \mu^*(A_j) + \varepsilon 2^{-j}$; this gives $\mu^*(\bigcup_j A_j) \leq \sum_j \mu^*(A_j) + \varepsilon$. So μ^* is an outer measure.

Step 3: open sets are measurable. Let U be open. By Carathéodory's theorem [the preceding tools lesson](#), [Theorem 1.1](#) it is enough to show $\mu^*(A) \geq \mu^*(A \cap U) + \mu^*(A \setminus U)$ whenever $\mu^*(A) < \infty$. First let $A = V$ be open. Given $\varepsilon > 0$, choose $f \prec V \cap U$ with $I(f) > \mu(V \cap U) - \varepsilon$. The set $V \setminus \text{supp } f$ is open; choose $g \prec V \setminus \text{supp } f$ with $I(g) > \mu(V \setminus \text{supp } f) - \varepsilon$. The supports of f and g are disjoint, so $f + g \prec V$ and

$$\mu(V) \geq I(f + g) > \mu(V \cap U) + \mu(V \setminus \text{supp } f) - 2\varepsilon \geq \mu^*(V \cap U) + \mu^*(V \setminus U) - 2\varepsilon,$$

since $V \setminus U \subseteq V \setminus \text{supp } f$. For general A , take an open $V \supseteq A$ with $\mu(V) \leq \mu^*(A) + \varepsilon$, apply the open case, and use monotonicity. By Carathéodory's theorem the μ^* -measurable sets form a σ -algebra on which μ^* is countably additive. It contains the open sets, hence $\mathcal{B}(X)$. Let μ be the restriction of μ^* to $\mathcal{B}(X)$. It satisfies (R2) by construction.

Step 4: compact sets. Let K be compact and $1_K \leq f \in C_c(X)$; in particular $f \geq 0$. For $0 < c < 1$ the open set $U_c = \{f > c\}$ contains K , and every $g \prec U_c$ satisfies $g \leq f/c$. So $\mu(K) \leq \mu(U_c) \leq I(f)/c$, and letting $c \rightarrow 1$ gives $\mu(K) \leq I(f)$. Some f with $K \prec f$ exists by (T2) with $U = X$, so $\mu(K) < \infty$, which is (R1). Given $\varepsilon > 0$, choose an open $U \supseteq K$ with $\mu(U) \leq \mu(K) + \varepsilon$ and then f with $K \prec f \prec U$: $I(f) \leq \mu(U) \leq \mu(K) + \varepsilon$. This proves the second formula of (2.1).

Step 5: (R3). If $f \prec U$, then $f \prec V$ for every open $V \supseteq \text{supp } f$, so $I(f) \leq \mu(V)$, and $I(f) \leq \mu(\text{supp } f)$ by (R2). Taking the supremum over $f \prec U$ gives $\mu(U) \leq \sup\{\mu(K) : K \subseteq U \text{ compact}\}$. The reverse inequality is monotonicity.

Step 6: μ represents I . By linearity it suffices to treat real $f \in C_c(X)$ with $0 \leq f \leq 1$. Fix $N \geq 1$. Put $K_0 = \text{supp } f$, $K_j = \{f \geq j/N\}$ for $1 \leq j \leq N$ (compact sets), and $f_j = \min(\max(f - \frac{j-1}{N}, 0), \frac{1}{N})$. Then $f = \sum_{j=1}^N f_j$, each f_j is in $C_c(X)$ with $\text{supp } f_j \subseteq K_{j-1}$, and $\frac{1}{N} 1_{K_j} \leq f_j \leq \frac{1}{N} 1_{K_{j-1}}$. Integrating, $\frac{1}{N} \mu(K_j) \leq \int f_j d\mu \leq \frac{1}{N} \mu(K_{j-1})$. On the other side, $\mu(K_j) \leq I(Nf_j)$ by Step 4. Also $Nf_j \prec U$ for every open $U \supseteq K_{j-1}$, so $I(Nf_j) \leq \mu(U)$, and $I(Nf_j) \leq \mu(K_{j-1})$ by (R2). Summing over j , both $I(f)$ and $\int f d\mu$ lie between $\frac{1}{N} \sum_{j=1}^N \mu(K_j)$ and $\frac{1}{N} \sum_{j=0}^{N-1} \mu(K_j)$. These bounds differ by at most $\mu(\text{supp } f)/N$. Let $N \rightarrow \infty$. $\square$

Proposition 2.3 (Regularity on σ -finite sets). Every Radon measure μ on X has the following properties.

1. If E is Borel and $\mu(E) < \infty$, then $\mu(E) = \sup\{\mu(K) : K \subseteq E \text{ compact}\}$.
2. The same holds for every Borel set that is σ -finite for μ .
3. Every σ -finite Borel set E is the union of a σ -compact set and a Borel null set.

Proof. (1) Let $\varepsilon > 0$. By (R2) choose an open $U \supseteq E$ with $\mu(U) < \mu(E) + \varepsilon$, so $\mu(U \setminus E) < \varepsilon$. Again by (R2) choose an open $W \supseteq U \setminus E$ with $\mu(W) < \varepsilon$. By (R3) choose a compact $F \subseteq U$ with $\mu(F) > \mu(U) - \varepsilon$. The compact set $K = F \setminus W$ lies in E : a point of F outside E is in $U \setminus E \subseteq W$. And $\mu(K) \geq \mu(F) - \mu(W) > \mu(E) - 2\varepsilon$. (2) Write E as an increasing union of Borel sets E_n of finite measure; then $\mu(E) = \lim \mu(E_n)$, and (1) applies to each E_n . (3) With E_n as in (2), choose compact $K_{n,m} \subseteq E_n$ with $\mu(E_n \setminus K_{n,m}) < 1/m$. The σ -compact set $S = \bigcup_{n,m} K_{n,m}$ lies in E , and $\mu(E_n \setminus S) = 0$ for every n , so $\mu(E \setminus S) = 0$. $\square$

Where the hypotheses enter. Local compactness and the Hausdorff property are used only through (T1)–(T3): they supply enough functions $f \prec U$. No countability is used, and the measure μ need not be σ -finite. A measure that is inner and outer regular on every Borel set would be more convenient, but Example 13.4 shows that on the group $\mathbb{R} \times \mathbb{R}_d$ no measure of that kind represents Haar integration. So the lesson never uses inner regularity on arbitrary Borel sets.

Bounded functionals on $C_0(X)$

The operator-algebra lessons also use the bounded, not necessarily positive, linear functionals on $C_0(X)$, the space of continuous functions vanishing at infinity with the norm $\|\cdot\|_{\sup}$. Call a functional φ *hermitian* if $\varphi(f)$ is real for every real f . Two facts are used below.

- $C_c(X)$ is dense in $C_0(X)$. For $f \in C_0(X)$ and $\varepsilon > 0$, the set $K = \{|f| \geq \varepsilon\}$ is compact. Choose $K \prec u$ by (T2). Then $uf \in C_c(X)$ and $\|f - uf\|_{\sup} \leq \varepsilon$.
- *Norms of positive functionals.* A positive linear functional ψ on $C_0(X)$ is hermitian, since every real f is $f^+ - f^-$. If ψ is bounded, then $\|\psi\| = \sup\{\psi(f) : 0 \leq f \leq 1\}$, and by (2.1) this is $\nu(X)$ for the Radon measure ν that represents ψ on $C_c(X)$. Indeed, for $\|\psi\|_{\sup} \leq 1$ choose θ with $e^{i\theta}\psi(f) \geq 0$, and write $e^{i\theta}f = u + iv$ with u, v real. Then $|\psi(f)| = \psi(u) + i\psi(v)$ forces $\psi(v) = 0$, and $|\psi(f)| = \psi(u) \leq \psi(u^+)$ with $0 \leq u^+ \leq 1$. The functions $f \prec X$ suffice, by the density just proved.

Theorem 2.4 (Bounded functionals on $C_0(X)$). Let X be an LCH space and φ a bounded linear functional on $C_0(X)$.

1. (*Jordan decomposition.*) Let φ be hermitian. For $f \geq 0$ put $\varphi_+(f) = \sup\{\varphi(g) : g \in C_0(X), 0 \leq g \leq f\}$. Then φ_+ extends to a bounded positive linear functional, $\varphi_- = \varphi_+ - \varphi$ is positive, and $\|\varphi\| = \|\varphi_+\| + \|\varphi_-\|$. Every bounded φ is $\varphi_1 + i\varphi_2$ with φ_1, φ_2 bounded and hermitian.
2. (*Representation.*) There are finite Radon measures $\nu_0, \dots, \nu_3$ with $\varphi(f) = \sum_{k=0}^3 i^k \int f d\nu_k$ for every $f \in C_0(X)$. The set function $\mu = \sum_k i^k \nu_k$ on $\mathcal{B}(X)$ does not depend on the choice of the ν_k . It is the *complex Radon measure* of φ , and we write $\varphi(f) = \int f d\mu$. Conversely, every such μ defines a bounded functional in this way.
3. (*Norm.*) $\|\varphi\| = |\mu|(X)$, where $|\mu|(E) = \sup \sum_j |\mu(E_j)|$, the supremum over the finite partitions of E into Borel sets.
4. (*Positivity.*) φ is positive if and only if $\mu \geq 0$; then μ is the measure of Theorem 2.2. If φ is hermitian, then $\mu = \nu_+ - \nu_-$, where $\nu_{\pm}$ are the Radon measures of $\varphi_{\pm}$, and $|\mu| = \nu_+ + \nu_-$. In particular $|\mu|$ is a finite Radon measure.

Proof. (1) *Additivity.* For $f_1, f_2 \geq 0$, clearly $\varphi_+(f_1 + f_2) \geq \varphi_+(f_1) + \varphi_+(f_2)$. Conversely, let $0 \leq g \leq f_1 + f_2$. Put $g_1 = \min(g, f_1)$ and $g_2 = g - g_1$. Then $0 \leq g_1 \leq f_1$ and $0 \leq g_2 \leq f_2$, and $\varphi(g) = \varphi(g_1) + \varphi(g_2) \leq \varphi_+(f_1) + \varphi_+(f_2)$.

- φ_+ is positively homogeneous, and $0 \leq \varphi_+(f) \leq \|\varphi\| \|f\|_{\sup}$.
- So $\varphi_+(f^+) - \varphi_+(f^-)$ for real f , and then real and imaginary parts, extend φ_+ to a positive linear functional. The extension is well defined: if $f = a - b$ with $a, b \geq 0$, then $a + f^- = b + f^+$.
- $\varphi_+(f) \geq \varphi(f)$ for $f \geq 0$, so φ_- is positive.

The norm. $\|\varphi\| \leq \|\varphi_+\| + \|\varphi_-\|$. Conversely, let $\varepsilon > 0$.

- Choose $0 \leq f, h \leq 1$ with $\varphi_+(f) > \|\varphi_+\| - \varepsilon$ and $\varphi_-(h) > \|\varphi_-\| - \varepsilon$.
- Choose $0 \leq g \leq f$ with $\varphi(g) > \varphi_+(f) - \varepsilon$.
- Since $\varphi_-(h) = \sup\{-\varphi(g') : 0 \leq g' \leq h\}$ (substitute $g' = h - g''$ in the definition), choose $0 \leq g' \leq h$ with $-\varphi(g') > \varphi_-(h) - \varepsilon$.
- Then $g - g'$ takes values in $[-1, 1]$, and $\|\varphi\| \geq \varphi(g - g') > \|\varphi_+\| + \|\varphi_-\| - 4\varepsilon$.

General φ . Put $\varphi^\dagger(f) = \overline{\varphi(f)}$, $\varphi_1 = \frac{1}{2}(\varphi + \varphi^\dagger)$ and $\varphi_2 = \frac{1}{2i}(\varphi - \varphi^\dagger)$. For real f , $\varphi_1(f) = \operatorname{Re} \varphi(f)$ and $\varphi_2(f) = \operatorname{Im} \varphi(f)$.

(2) Apply (1) to φ_1 and φ_2 , and Theorem 2.2 to the four positive parts restricted to $C_c(X)$. This gives Radon measures ν_0, ν_2 for $\varphi_{1,+}, \varphi_{1,-}$ and ν_1, ν_3 for $\varphi_{2,+}, \varphi_{2,-}$. They are finite, with $\nu_k(X)$ the norm of the corresponding positive part. The identity $\varphi(f) = \sum_k i^k \int f d\nu_k$ holds on $C_c(X)$, and both sides are continuous for $\|\cdot\|_{\sup}$, so it holds on $C_0(X)$.

Independence. Let $\sum_k i^k \int f d\nu_k = \sum_k i^k \int f d\nu'_k$ on $C_0(X)$, with finite Radon measures.

- Taking real parts at real f , the finite measures $\nu_0 + \nu_2'$ and $\nu_0' + \nu_2$ give the same integrals on $C_c(X)$.

- A sum of two finite Radon measures is Radon: (R1) is clear; for (R2) intersect the two open sets; for (R3) take the union of the two compact sets.
- By the uniqueness in Theorem 2.2 the two sums are equal, so $\nu_0 - \nu_2 = \nu'_0 - \nu'_2$ as set functions. The imaginary parts are treated in the same way.

Integrals. For a bounded Borel h , put $\int h d\mu = \sum_k i^k \int h d\nu_k$. This depends only on μ : it does for simple functions, and every bounded Borel function is a uniform limit of simple ones. Also $|\int h d\mu| \leq \int |h| d\nu$ with $\nu = \sum_k \nu_k$. The converse statement of (2) is clear, with $\|\varphi\| \leq \nu(X)$.

(3) $\|\varphi\| \leq |\mu|(X)$. For a Borel simple function $s = \sum_j c_j 1_{E_j}$, with the E_j a partition of X , $|\int s d\mu| \leq \max_j |c_j| \sum_j |\mu(E_j)| \leq \|s\|_{\sup} |\mu|(X)$. Let $f \in C_0(X)$. Cutting the range of f into small Borel pieces, and taking on each piece a value of f , gives simple $s_n \rightarrow f$ uniformly with $\|s_n\|_{\sup} \leq \|f\|_{\sup}$. So $|\varphi(f)| \leq \|f\|_{\sup} |\mu|(X)$.

$|\mu|(X) \leq \|\varphi\|$. Let $E_1, \dots, E_n$ be a Borel partition of X and $\varepsilon > 0$. The measure $\nu = \sum_k \nu_k$ is a finite Radon measure.

- By Proposition 2.3(1), choose compact $K_j \subseteq E_j$ with $\nu(E_j \setminus K_j) < \varepsilon/n$.
- Disjoint compact sets in a Hausdorff space have disjoint open neighbourhoods: separate each point of one from the other set by compactness of the other, then use compactness of the first. Doing this for each pair and intersecting, we get disjoint open $W_j \supseteq K_j$. By (R2) choose open U_j with $K_j \subseteq U_j \subseteq W_j$ and $\nu(U_j \setminus K_j) < \varepsilon/n$.
- By (T2) choose $K_j \prec g_j \prec U_j$. Let $c_j = \overline{\mu(E_j)}/|\mu(E_j)|$, or $c_j = 0$ if $\mu(E_j) = 0$, and $f = \sum_j c_j g_j$. The supports are disjoint, so $\|f\|_{\sup} \leq 1$.
- $|g_j - 1_{E_j}| \leq 1_{U_j \setminus K_j} + 1_{E_j \setminus K_j}$, so $|\int g_j d\mu - \mu(E_j)| < 2\varepsilon/n$.
- Hence $\sum_j |\mu(E_j)| = \sum_j c_j \mu(E_j) \leq |\varphi(f)| + 2\varepsilon \leq \|\varphi\| + 2\varepsilon$.

The variation is itself a finite Radon measure, also for complex μ . Refining finite partitions proves finite additivity of $|\mu|$ on disjoint Borel sets, and $|\mu|(E) \leq \nu(E)$ with the finite Radon measure $\nu = \sum_k \nu_k$. For a disjoint sequence, the ν -measure of the union of the tail tends to zero. The same domination makes its variation tend to zero, turning finite additivity into countable additivity. To check regularity, approximate a Borel E from within by a compact K and from outside by an open U in ν -measure, using Proposition 2.3(1) and (R2). Then $|\mu|(E \setminus K) \leq \nu(E \setminus K)$ and $|\mu|(U \setminus E) \leq \nu(U \setminus E)$, proving inner and outer regularity.

(4) If φ is positive, then $\varphi_+ = \varphi$, $\varphi_- = 0$ and $\varphi_2 = 0$, so μ is the measure of Theorem 2.2. If $\mu \geq 0$, then $\int s d\mu \geq 0$ for simple $s \geq 0$, and every $f \geq 0$ in $C_0(X)$ is a uniform limit of such s ; so φ is positive.

Let φ be hermitian, so $\varphi_2 = 0$ and $\mu = \nu_+ - \nu_-$.

- $|\mu|(E) \leq \nu_+(E) + \nu_-(E)$ for every Borel E , since $|\mu(E_j)| \leq \nu_+(E_j) + \nu_-(E_j)$.
- $|\mu|$ is additive on disjoint Borel sets: partitions of E and F combine to a partition of $E \cup F$, and a partition of $E \cup F$ can be refined by E and F without decreasing the sum.
- By (3) and (1), $|\mu|(X) = \|\varphi\| = \|\varphi_+\| + \|\varphi_-\| = \nu_+(X) + \nu_-(X)$.
- So $|\mu|(E) + |\mu|(X \setminus E) = (\nu_+ + \nu_-)(E) + (\nu_+ + \nu_-)(X \setminus E)$, with $\leq$ in each term. Hence $|\mu|(E) = (\nu_+ + \nu_-)(E)$. $\square$

For compact X , $C_0(X) = C(X)$, and the finite Radon measures are regular: outer regular by (R2), and inner regular on all Borel sets by Proposition 2.3(1). So Theorem 2.4 identifies the dual of $C(X)$ with the complex regular Borel measures on X , normed by total variation.

3. Image measures, densities, and approximation by continuous functions

This section collects four operations on a Radon measure that the rest of the lesson uses constantly: moving it by a homeomorphism, multiplying it by a positive continuous density, integrating lower semicontinuous functions, and approximating L^p functions by continuous ones.

Proposition 3.1. Fix a Radon measure μ on an LCH space X .

1. (*Homeomorphisms*) Let $\theta: X \rightarrow X'$ be a homeomorphism onto an LCH space. Then $\theta_*\mu(E) = \mu(\theta^{-1}(E))$ defines a Radon measure on X' , and $\int f d\theta_*\mu = \int f \circ \theta d\mu$ for every Borel $f \geq 0$ and every $f \in L^1(\theta_*\mu)$.
2. (*Positive continuous densities*) Let $\varphi: X \rightarrow (0, \infty)$ be continuous. Then $\varphi\mu(E) = \int_E \varphi d\mu$ defines a Radon measure; $\int f d(\varphi\mu) = \int f\varphi d\mu$ for every Borel $f \geq 0$; and $\varphi\mu$ has the same null sets and the same σ -finite sets as μ . Consequently, if a Radon measure ν satisfies $\int f d\nu = \int f\varphi d\mu$ for all $f \in C_c(X)$, then $\nu = \varphi\mu$.
3. (*Lower semicontinuous functions*) For every lsc $h: X \rightarrow [0, \infty]$,

$$\int h d\mu = \sup \left\{ \int g d\mu : g \in C_c(X), 0 \leq g \leq h \right\}. \quad (3.1)$$

4. (*Density*) Let $1 \leq p < \infty$. Every $f \in L^p(\mu)$ vanishes outside a σ -finite Borel set, and hence, after a change on a null set, outside a σ -compact set. The space $C_c(X)$ is dense in $L^p(\mu)$.

Proof. (1) Since θ is a homeomorphism, θ^{-1} maps compact, open and Borel sets of X' onto sets of the same kind in X , and it preserves inclusions. So (R1)–(R3) transfer from μ to $\theta_*\mu$. The integral formula holds for indicators by definition, and by linearity for simple functions. The layer functions (1.1) and monotone convergence extend it to Borel $f \geq 0$, and splitting f into four nonnegative parts extends it to integrable f .

(2) The formula for integrals holds for indicators by definition, hence for Borel $f \geq 0$ by the layer functions (1.1) and monotone convergence; in particular $\varphi\mu$ is countably additive, so it is a measure. Since $\varphi > 0$, $\int_E \varphi d\mu = 0$ exactly when $\mu(E) = 0$. The sets $\{1/n \leq \varphi \leq n\}$ cover X , and on each of them the two measures are comparable, so the σ -finite sets agree. It remains to check (R1)–(R3) for $\varphi\mu$. (R1) holds because φ is bounded on compact sets. (R2): Let E be Borel with $\varphi\mu(E) < \infty$ (otherwise take $U = X$), and let $\varepsilon > 0$. For $j \in \mathbb{Z}$, the open sets $V_j = \{2^{j-1} < \varphi < 2^{j+1}\}$ cover X . Put $E_j = E \cap V_j$. Then $\mu(E_j) \leq 2^{1-j}\varphi\mu(E_j) < \infty$, so by (R2) for μ there is an open U_j with $E_j \subseteq U_j \subseteq V_j$ and $\mu(U_j \setminus E_j) < \varepsilon 2^{-|j|-j-3}$. Then $\varphi\mu(U_j \setminus E_j) \leq 2^{j+1}\mu(U_j \setminus E_j) < \varepsilon 2^{-|j|-2}$. The set $U = \bigcup_j U_j$ is open, contains E , and $U \setminus E \subseteq \bigcup_j (U_j \setminus E_j)$. So $\varphi\mu(U \setminus E) < \varepsilon \sum_j 2^{-|j|-2} < \varepsilon$. (R3): Let U be open and $c < \varphi\mu(U)$. If $\mu(U \cap \{\varphi > s\}) = \infty$ for some $s > 0$, then (R3) for μ gives a compact $K \subseteq U \cap \{\varphi > s\}$ with $\mu(K) > c/s$, and $\varphi\mu(K) \geq s\mu(K) > c$. Otherwise all these sets have finite μ -measure. Put $D_j = U \cap \{2^j < \varphi \leq 2^{j+1}\}$ for $j \in \mathbb{Z}$. These Borel sets are disjoint, cover U , and have finite μ -measure. Choose a finite set $J \subseteq \mathbb{Z}$ with $\sum_{j \in J} \varphi\mu(D_j) > c + \eta$ for some $\eta > 0$. By inner regularity on sets of finite measure (Proposition 2.3(1)), choose compact $K_j \subseteq D_j$ with $2^{j+1}\mu(D_j \setminus K_j) < \eta/|J|$. The compact set $K = \bigcup_{j \in J} K_j \subseteq U$ satisfies $\varphi\mu(K) \geq \sum_{j \in J} (\varphi\mu(D_j) - 2^{j+1}\mu(D_j \setminus K_j)) > c$. The last sentence of (2) follows from the uniqueness in the Riesz theorem 2.2.

(3) The inequality $\geq$ is clear. For $\leq$, let $c < \int h d\mu$. The layer functions (1.1) of h have the form $\sum_i (t_i - t_{i-1})1_{\{h > t_i\}}$ with $t_i = i2^{-n}$, and their integrals tend to $\int h d\mu$ by monotone convergence. So there is a partition $0 = t_0 < t_1 < \dots < t_k$ with $\sum_i (t_i - t_{i-1})\mu(\{h > t_i\}) > c$. Each set $\{h > t_i\}$ is open, so (R3) gives compact $K_i \subseteq \{h > t_i\}$ with $\sum_i (t_i - t_{i-1})\mu(K_i) > c$. Choose $K_i \prec \varphi_i \prec \{h > t_i\}$ by (T2) and put $g = \sum_i (t_i - t_{i-1})\varphi_i \in C_c(X)$. At a point x , $\varphi_i(x) \neq 0$ forces $h(x) > t_i$, so $g(x)$ is at most the sum of $t_i - t_{i-1}$ over the indices i with $t_i < h(x)$, which is less than $h(x)$ (or is 0). So $0 \leq g \leq h$ and $\int g d\mu \geq \sum_i (t_i - t_{i-1})\mu(K_i) > c$.

(4) The set $\{f \neq 0\}$ is the union of the sets $\{|f| > 1/n\}$, and by Chebyshev's inequality each has measure at most $n^p \|f\|_p^p$. So $\{f \neq 0\}$ is σ -finite, and Proposition 2.3(3) gives the σ -compact set. Simple functions are dense in $L^p(\mu)$, and a simple function in L^p is a combination of indicators of Borel sets of finite measure. So it suffices to approximate 1_E with $\mu(E) < \infty$. Given $\varepsilon > 0$, inner regularity (Proposition 2.3(1)) and outer regularity (R2) give a compact K and an open U with $K \subseteq E \subseteq U$ and $\mu(U \setminus K) < \varepsilon$. Take $K \prec f \prec U$. Then $|f - 1_E| \leq 1_{U \setminus K}$, so $\|f - 1_E\|_p \leq \varepsilon^{1/p}$. $\square$

Part (4) needs no σ -finiteness of μ : an L^p function with $p < \infty$ automatically lives on a σ -finite set, and there inner regularity holds. For a Haar measure (Section 8) and $p = 2$, it says that $C_c(G)$ is dense in $L^2(G)$. In (2) the density must be strictly positive; Exercise 16.1 shows what goes wrong with a density that vanishes on a closed set.

4. Products of Radon measures

Let X, Y be LCH spaces with Radon measures μ, ν . For $F \in C_c(X \times Y)$ write $F_x = F(x, \cdot)$, and let K_F and L_F be the projections of $\text{supp } F$ to X and Y ; they are compact. The product of μ and ν is built from iterated integrals of such F , and the first step is to approximate F by sums of products.

Lemma 4.1 (Tensor approximation). Let $F \in C_c(X \times Y)$.

1. The map $x \mapsto F_x$ is continuous from X to $C_c(Y)$ with the norm $\|\cdot\|_{\text{sup}}$, and F_x vanishes outside L_F .
2. Let $U \supseteq K_F$ be open with compact closure. For every $\varepsilon > 0$ there are $\varphi_1, \dots, \varphi_n \prec U$ and $g_1, \dots, g_n \in C_c(Y)$ vanishing outside L_F such that $|F(x, y) - \sum_i \varphi_i(x)g_i(y)| \leq \varepsilon$ for all (x, y) .

Proof. (1) Fix x_0 and $\varepsilon > 0$. For each $y \in L_F$, continuity of F at (x_0, y) gives open $P_y \ni x_0$ and $Q_y \ni y$ with $|F(x, y') - F(x_0, y)| < \varepsilon/4$ on $P_y \times Q_y$. Finitely many Q_{y_k} cover L_F ; let $P = \bigcap_k P_{y_k}$. For $x \in P$ and $y' \in L_F$, pick k with $y' \in Q_{y_k}$; comparing both $F(x, y')$ and $F(x_0, y')$ with $F(x_0, y_k)$ gives $|F(x, y') - F(x_0, y')| < \varepsilon/2$. Off L_F both values are 0. (2) By (1), each $x \in K_F$ has an open neighbourhood $P_x \subseteq U$ with $\|F_{x'} - F_x\|_{\text{sup}} < \varepsilon$ for $x' \in P_x$. Cover K_F by $P_{x_1}, \dots, P_{x_n}$, take a partition of unity $\varphi_i \prec P_{x_i}$ from (T3), and let $g_i = F_{x_i}$. Then

$$F(x, y) - \sum_i \varphi_i(x)F(x_i, y) = \left(1 - \sum_i \varphi_i(x)\right)F(x, y) + \sum_i \varphi_i(x)(F(x, y) - F(x_i, y)).$$

The first term vanishes: $\sum_i \varphi_i = 1$ on K_F , and $F_x = 0$ off K_F . In the second, $\varphi_i(x) \neq 0$ forces $x \in P_{x_i}$, so the term is at most $\varepsilon \sum_i \varphi_i(x) \leq \varepsilon$ in absolute value. $\square$

Proposition 4.2 (Iterated integrals). Let $F \in C_c(X \times Y)$.

1. The function $x \mapsto \int F(x, y) d\nu(y)$ is in $C_c(X)$ and vanishes off K_F ; similarly with the roles of the factors exchanged.
2. The two iterated integrals agree:

$$\int \left(\int F(x, y) d\nu(y) \right) d\mu(x) = \int \left(\int F(x, y) d\mu(x) \right) d\nu(y). \quad (4.1)$$

3. F is measurable for the product σ -algebra $\mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Proof. (1) $\left| \int F(x, y) d\nu(y) - \int F(x_0, y) d\nu(y) \right| \leq \|F_x - F_{x_0}\|_{\text{sup}} \nu(L_F)$, which tends to 0 as $x \rightarrow x_0$ by Lemma 4.1(1). (2) Choose an open $U \supseteq K_F$ with compact closure by (T1). For a sum $F' = \sum_i \varphi_i \otimes g_i$ as in Lemma 4.1(2), both sides equal $\sum_i \int \varphi_i d\mu \int g_i d\nu$. All these functions vanish outside the compact set $\overline{U} \times L_F$, and $|F - F'| \leq \varepsilon$, so each side of (4.1) for F differs from the same side for F' by at most $\varepsilon \mu(\overline{U}) \nu(L_F)$. Let $\varepsilon \rightarrow 0$. (3) The approximants are $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ -measurable, and F is their pointwise limit as $\varepsilon = 1/m \rightarrow 0$. $\square$

Definition 4.3 (Radon product). The *Radon product* $\mu \hat{\times} \nu$ is the Radon measure on $X \times Y$ that the Riesz theorem 2.2 gives for the positive linear functional $F \mapsto \int \left(\int F(x, y) d\nu(y) \right) d\mu(x)$ on $C_c(X \times Y)$. By (4.1) it also represents the other iterated integral.

Proposition 4.4 (Rectangles). For open $U \subseteq X$, $V \subseteq Y$ and compact $K \subseteq X$, $L \subseteq Y$, with the convention $0 \cdot \infty = 0$,

$$(\mu \hat{\times} \nu)(U \times V) = \mu(U) \nu(V), \quad (\mu \hat{\times} \nu)(K \times L) = \mu(K) \nu(L). \quad (4.2)$$

Proof. If $F \prec U \times V$, then $x \mapsto \int F(x, y) d\nu(y)$ vanishes off U and is at most $\nu(V)$, so $\int F d(\mu \hat{\times} \nu) \leq \mu(U) \nu(V)$; when $\mu(U) = 0$ the integral is 0. If $f \prec U$ and $g \prec V$, then $f \otimes g \prec U \times V$ and its integral is $\int f d\mu \int g d\nu$. Taking suprema and using (2.1) for μ, ν and $\mu \hat{\times} \nu$ gives the first formula. For the second, if $F \in C_c(X \times Y)$ and $1_{K \times L} \leq F$, then $\int F(x, y) d\nu(y) \geq \nu(L) 1_K(x)$, so $\int F d(\mu \hat{\times} \nu) \geq \mu(K) \nu(L)$. If $1_K \leq f$ and $1_L \leq g$, then $1_{K \times L} \leq f \otimes g$, whose integral is $\int f d\mu \int g d\nu$. Taking infima and using (2.1) gives the second formula. $\square$

5. Tonelli's and Fubini's theorems for Radon products

Let X, Y be LCH spaces with Radon measures μ, ν , and let $\pi = \mu \hat{\times} \nu$ be their Radon product (Definition 4.3). For $E \subseteq X \times Y$ put $E_x = \{y : (x, y) \in E\}$ and $E^y = \{x : (x, y) \in E\}$; if E is Borel, so are all sections, because $y \mapsto (x, y)$ is continuous. A function on X is μ -a.e. Borel if it agrees with a Borel function outside a Borel μ -null set; its integral is that of the Borel function.

Theorem 5.1. Everything below also holds with the roles of X and Y exchanged, since π represents both iterated integrals (4.1).

1. For every open $W \subseteq X \times Y$, the function $x \mapsto \nu(W_x)$ is lsc and $\pi(W) = \int \nu(W_x) d\mu(x)$.
2. For every compact $C \subseteq X \times Y$, the function $x \mapsto \nu(C_x)$ is Borel and $\pi(C) = \int \nu(C_x) d\mu(x)$.
3. For every Borel E with $\pi(E) < \infty$, the function $x \mapsto \nu(E_x)$ is μ -a.e. Borel and $\pi(E) = \int \nu(E_x) d\mu(x)$. If $\pi(E) = 0$, then $\nu(E_x) = 0$ for μ -almost every x .
4. (Tonelli) Let $F: X \times Y \rightarrow [0, \infty]$ be Borel and vanish outside a set that is σ -finite for π . Then $x \mapsto \int F(x, y) d\nu(y)$ is μ -a.e. Borel, $y \mapsto \int F(x, y) d\mu(x)$ is ν -a.e. Borel, and

$$\int F d\pi = \int \left(\int F(x, y) d\nu(y) \right) d\mu(x) = \int \left(\int F(x, y) d\mu(x) \right) d\nu(y). \quad (5.1)$$

5. (Fubini) Let $F \in L^1(\pi)$. Then $F(x, \cdot) \in L^1(\nu)$ for μ -almost every x , the μ -a.e. defined function $x \mapsto \int F(x, y) d\nu(y)$ is μ -integrable, and (5.1) holds.
6. (Rectangles) Let $A \subseteq X$ and $B \subseteq Y$ be Borel and σ -finite for μ and ν . Then $A \times B$ is σ -finite for π , and $\pi(A \times B) = \mu(A)\nu(B)$ with $0 \cdot \infty = 0$. In particular $A \times B$ is π -null when A is μ -null.
7. If X and Y are σ -compact, $\pi(E) = \int \nu(E_x) d\mu(x)$ for every $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, so π extends the usual product measure. If X and Y are second countable, then $\mathcal{B}(X \times Y) = \mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Scope warning: Functions vanishing outside a σ -compact set meet the support hypothesis, but (4) and (5) need only a σ -finite set. Omitting this hypothesis and requiring a product measure that is inner and outer regular on every Borel set would be incorrect; Example 13.5 shows that no such product measure exists on $\mathbb{R}_d \times \mathbb{R}$.

Proof. (1) *Lower semicontinuity.* Let $c < \nu(W_{x_0})$. The section W_{x_0} is open, so (R3) gives a compact $L \subseteq W_{x_0}$ with $\nu(L) > c$. By the tube lemma (T4) applied to $\{x_0\} \times L \subseteq W$, there is an open $P \ni x_0$ with $P \times L \subseteq W$. For $x \in P$, $L \subseteq W_x$, so $\nu(W_x) > c$. The inequality $\leq$. If $F \prec W$, then $\int F(x, y) d\nu(y) \leq \nu(W_x)$, and integrating gives $\int F d\pi \leq \int \nu(W_x) d\mu(x)$. Take the supremum over F and use (2.1). The inequality $\geq$. Put $h(x) = \nu(W_x)$ and let $c < \int h d\mu$. By (3.1) choose $g \in C_c(X)$ with $0 \leq g \leq h$ and $\int g d\mu > c$. Let $K = \text{supp } g$, and choose $\eta > 0$ with $\int g d\mu - \eta \mu(K) > c$. For each $x \in K$, (R3) gives a compact $L_x \subseteq W_x$ with $\nu(L_x) \geq g(x) - \eta/2$ (empty if $g(x) \leq \eta/2$). By (T4) there are open $P_x \ni x$ and $Q_x \supseteq L_x$ with $P_x \times Q_x \subseteq W$; shrinking P_x , we may also assume $g < g(x) + \eta/2$ on P_x . Cover K by $P_{x_1}, \dots, P_{x_n}$, take a partition of unity $\alpha_i \prec P_{x_i}$ with $\sum_i \alpha_i = 1$ on K (T3), and $L_{x_i} \prec \beta_i \prec Q_{x_i}$ (T2). The function $F = \sum_i \alpha_i \otimes \beta_i$ satisfies $0 \leq F \leq 1$ and $\text{supp } F \subseteq \bigcup_i P_{x_i} \times Q_{x_i} \subseteq W$, so $F \prec W$. For $x \in K$,

$$\int F(x, y) d\nu(y) = \sum_i \alpha_i(x) \int \beta_i d\nu \geq \sum_i \alpha_i(x) \nu(L_{x_i}) \geq g(x) - \eta,$$

because $\alpha_i(x) \neq 0$ forces $x \in P_{x_i}$, and then $\nu(L_{x_i}) \geq g(x_i) - \eta/2 > g(x) - \eta$. Hence $\pi(W) \geq \int F d\pi \geq \int_K (g - \eta) d\mu > c$. (2) Let K and L be the projections of C . By (T1) choose open $U \supseteq K$ and $V \supseteq L$ with compact closures, and put $W = U \times V$. Then $\nu(W_x) = 1_U(x)\nu(V) < \infty$, $\pi(W) = \mu(U)\nu(V) < \infty$ by (4.2), and $W \setminus C$ is open. So $\nu(C_x) = \nu(W_x) - \nu((W \setminus C)_x)$ is a difference of finite lsc functions, hence Borel, and by (1), $\pi(C) = \pi(W) - \pi(W \setminus C) = \int \nu(C_x) d\mu(x)$. (3) By outer regularity (R2) and inner regularity on sets of finite measure (Proposition 2.3(1)), choose open $W_n \supseteq E$ and compact $C_n \subseteq E$ with $\pi(W_n) < \pi(E) + 1/n$ and $\pi(C_n) > \pi(E) - 1/n$. Replacing W_n by $W_1 \cap \dots \cap W_n$ and C_n by $C_1 \cup \dots \cup C_n$, we may assume W_n decreases and C_n increases. Let $a(x) = \lim_n \nu((W_n)_x)$ and $b(x) = \lim_n \nu((C_n)_x)$; both are Borel, and $b(x) \leq \nu(E_x) \leq a(x)$. By (1), $\int \nu((W_1)_x) d\mu = \pi(W_1) < \infty$. Its infinite-value set is therefore Borel and null; define all section bounds as zero on that exceptional set before taking differences. Dominated convergence now gives $\int a d\mu = \lim \pi(W_n) = \pi(E)$. By (2) and monotone convergence, $\int b d\mu = \lim \pi(C_n) = \pi(E)$. The finite-a.e. nonnegative difference $a - b$ has integral zero, so $a = b$ outside a Borel μ -null set, and there

$\nu(E_x) = b(x)$. If $\pi(E) = 0$, then $\int b d\mu = 0$, so $b = 0$ almost everywhere. (4) By (3), (5.1) holds for $F = 1_E$ when $\pi(E) < \infty$. If E is σ -finite, write it as an increasing union of Borel sets of finite measure and use monotone convergence on all three terms; countably many null sets have a null union. By linearity (5.1) holds for nonnegative simple Borel functions vanishing outside a σ -finite set. For general F , the layer functions (1.1) of F are such simple functions and increase to F ; apply monotone convergence again. (5) Split F into the four nonnegative functions $(\operatorname{Re} F)^\pm$ and $(\operatorname{Im} F)^\pm$, and apply (4) to each. They vanish outside $\{F \neq 0\}$, which is σ -finite by Proposition 3.1(4), and their iterated integrals are finite, so their inner integrals are finite almost everywhere. Subtract. (6) By Proposition 2.3(3), write $A = S_A \cup N_A$ and $B = S_B \cup N_B$ with S_A, S_B σ -compact and N_A, N_B null. For compact $L \subseteq Y$ and $\varepsilon > 0$, choose an open $U \supseteq N_A$ with $\mu(U) < \varepsilon$ and an open $V \supseteq L$ with compact closure; by (4.2), $\pi(N_A \times L) \leq \mu(U)\nu(V) \leq \varepsilon\nu(\bar{V})$. So $N_A \times L$ is null, and so is $N_A \times S_B$. Choosing also an open $V' \supseteq N_B$ with $\nu(V') < \varepsilon$, we get $\pi(N_A \times N_B) \leq \mu(U)\nu(V') < \varepsilon^2$. In the same way $S_A \times N_B$ is null. So $A \times B$ differs from the σ -compact set $S_A \times S_B$ by a null set, and it is σ -finite by (R1). Now (4) with $F = 1_{A \times B}$ gives $\pi(A \times B) = \int 1_A(x)\nu(B) d\mu(x) = \mu(A)\nu(B)$. (7) If X and Y are σ -compact, so is $X \times Y$, and every Borel set is σ -finite by (R1); apply (4) to 1_E , noting $\mathcal{B}(X) \otimes \mathcal{B}(Y) \subseteq \mathcal{B}(X \times Y)$ because the projections are continuous. For E in the product σ -algebra, $\int \nu(E_x) d\mu(x)$ is the usual product measure of σ -finite factors. If X and Y have countable bases, every open $W \subseteq X \times Y$ is the union of the countably many products of basic sets contained in W , so $W \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. $\square$

Where the hypotheses enter, and why they are needed. Parts (1)–(3) use only the Radon properties. σ -finiteness enters in (4)–(6), and it cannot be dropped. Example 13.5 gives a closed set in $\mathbb{R}_d \times \mathbb{R}$ whose two iterated integrals are 0 and ∞ , and Example 13.6 gives a set $\{e\} \times G$ with $\pi(\{e\} \times G) = \infty$ although $\mu(\{e\}) = 0$. In particular, (4) applies to every Borel $F \geq 0$ that vanishes outside a σ -compact set, since σ -compact sets are σ -finite by (R1).

Example 5.2 (Borel sets of products). If X and Y are second countable, $\mathcal{B}(X \times Y) = \mathcal{B}(X) \otimes \mathcal{B}(Y)$ (Theorem 5.1(7)). Without second countability this fails even for discrete groups. Let G be a discrete group of cardinality greater than $2^{\aleph_0}$, for instance a free abelian group on such a set. The diagonal $\Delta_G \subseteq G \times G$ is open, but it is not in $\mathcal{B}(G) \otimes \mathcal{B}(G)$, which here is the σ -algebra generated by all rectangles. Proof: a set in the σ -algebra generated by a family $\mathcal{C}$ lies in the σ -algebra generated by some countable subfamily, because the sets with this property form a σ -algebra containing $\mathcal{C}$. So Δ_G lies in the σ -algebra generated by countably many rectangles $A_n \times B_n$. Put $\varphi(x) = (1_{A_n}(x))_n$ and $\psi(y) = (1_{B_n}(y))_n$ in $\{0, 1\}^{\mathbb{N}}$. The sets $(\varphi \times \psi)^{-1}(S)$, $S \subseteq \{0, 1\}^{\mathbb{N}} \times \{0, 1\}^{\mathbb{N}}$, form a σ -algebra containing each $A_n \times B_n$, so $\Delta_G = (\varphi \times \psi)^{-1}(S)$ for some S . If $\varphi(x) = \varphi(x')$, then $(x, x') \in \Delta_G$ because $(\varphi(x), \psi(x')) = (\varphi(x'), \psi(x')) \in S$; so $x = x'$. Hence φ is injective and $|G| \leq 2^{\aleph_0}$, a contradiction. So the Radon product (here counting measure on $G \times G$) lives on a σ -algebra strictly larger than the product σ -algebra. This is why Theorem 5.1 and Theorem 6.2 are proved for Borel functions on $X \times Y$, not only for product-measurable ones.

6. Hilbert tensor products and L^2 of a product

Hilbert tensor products. For Hilbert spaces H, K , $H \otimes K$ is the Hilbert tensor product of **Hilbert spaces and compact operators**, Theorem 8.1. It carries a bilinear map $(\xi, \eta) \mapsto \xi \otimes \eta$ with $\langle \xi \otimes \eta, \xi' \otimes \eta' \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle$, and the elementary tensors span a dense subspace.

Lemma 6.1 (Isometric extension). Suppose $B: H \times K \rightarrow \mathcal{K}$ is bilinear, where $\mathcal{K}$ is a third Hilbert space, and $\langle B(\xi, \eta), B(\xi', \eta') \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle$. There is a unique isometry $U: H \otimes K \rightarrow \mathcal{K}$ with $U(\xi \otimes \eta) = B(\xi, \eta)$.

Proof. For finite sums, $\|\sum_i B(\xi_i, \eta_i)\|^2 = \sum_{i,j} \langle \xi_i, \xi_j \rangle \langle \eta_i, \eta_j \rangle = \|\sum_i \xi_i \otimes \eta_i\|^2$. So $\sum_i \xi_i \otimes \eta_i \mapsto \sum_i B(\xi_i, \eta_i)$ is well defined (a sum of tensors that is 0 goes to a vector of norm 0) and isometric on the span of the elementary tensors. It extends by continuity to the closure, which is $H \otimes K$. Uniqueness holds because the elementary tensors are total. $\square$

Theorem 6.2 (L^2 of the Radon product). Let μ, ν be Radon measures on LCH spaces X, Y , and $\pi = \mu \hat{\times} \nu$. For functions ξ on X and η on Y write $(\xi \boxtimes \eta)(x, y) = \xi(x)\eta(y)$.

1. For $\xi \in L^2(\mu)$ and $\eta \in L^2(\nu)$, $\xi \boxtimes \eta \in L^2(\pi)$; its class depends only on the classes of ξ and η ; and $\langle \xi \boxtimes \eta, \xi' \boxtimes \eta' \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle$.
2. There is a unique unitary $U: L^2(\mu) \otimes L^2(\nu) \rightarrow L^2(\pi)$ with $U(\xi \otimes \eta) = \xi \boxtimes \eta$.

3. Every $\zeta \in L^2(\pi)$ satisfies $\zeta(x, \cdot) \in L^2(\nu)$ for μ -almost every x , and $\|\zeta\|^2 = \int \|\zeta(x, \cdot)\|_{L^2(\nu)}^2 d\mu(x)$; likewise in the other variable.
4. For a bounded Borel function a on X , the operator $U^* M_{a \circ \text{pr}_1} U$, where M denotes multiplication on $L^2(\pi)$, is the unique bounded operator on $L^2(\mu) \otimes L^2(\nu)$ that sends every $\xi \otimes \eta$ to $(a\xi) \otimes \eta$. It is the operator $m_a \otimes 1$. The same holds for the second factor.

Proof. (1) Take Borel representatives. The sets $S_\xi = \{\xi \neq 0\}$ and $S_\eta = \{\eta \neq 0\}$ are σ -finite by Proposition 3.1(4). The function $\xi \boxtimes \eta$ is Borel on $X \times Y$ and vanishes outside $S_\xi \times S_\eta$, which is σ -finite by Theorem 5.1(6). Tonelli's theorem 5.1(4) gives $\int |\xi \boxtimes \eta|^2 d\pi = \|\xi\|^2 \|\eta\|^2$. If $\xi = \xi'$ outside a μ -null Borel set N , then $\xi \boxtimes \eta = \xi' \boxtimes \eta$ outside $N \times S_\eta$, which is π -null by Theorem 5.1(6); similarly in the second variable. The function $(\xi \bar{\xi}') \boxtimes (\eta \bar{\eta}')$ is in $L^1(\pi)$ by the Cauchy–Schwarz inequality, and Fubini's theorem 5.1(5) gives the inner product formula. (2) Lemma 6.1, with $B(\xi, \eta) = \xi \boxtimes \eta$, gives the isometry U . Its range is closed, because U is an isometry on a complete space, and it contains $\varphi \boxtimes g$ for $\varphi \in C_c(X)$ and $g \in C_c(Y)$. These span a dense subspace of $L^2(\pi)$. Indeed, let $\zeta \in L^2(\pi)$ and $\varepsilon > 0$. Since C_c is dense in L^2 (Proposition 3.1(4)), there is $F \in C_c(X \times Y)$ with $\|\zeta - F\|_2 < \varepsilon$. Choose an open $O \supseteq K_F$ with compact closure (T1). The tensor approximation (Lemma 4.1(2)), applied with O , gives $F' = \sum_i \varphi_i \boxtimes g_i$ with $|F - F'| \leq \delta$, where F and F' vanish outside the compact set $\bar{O} \times L_F$. Then $\|F - F'\|_2 \leq \delta \pi(\bar{O} \times L_F)^{1/2} = \delta (\mu(\bar{O})\nu(L_F))^{1/2}$ by (4.2), which is less than ε for small δ . So U is onto. Uniqueness holds because elementary tensors are total. (3) Apply Tonelli's theorem 5.1(4) to $|\zeta|^2$. It vanishes outside the set $\{\zeta \neq 0\}$, which is σ -finite by Proposition 3.1(4) applied to π . (4) $M_{a \circ \text{pr}_1}(\xi \boxtimes \eta) = (a\xi) \boxtimes \eta$, so $U^* M_{a \circ \text{pr}_1} U(\xi \otimes \eta) = (a\xi) \otimes \eta$. Two bounded operators that agree on a total set are equal. $\square$

In words: $L^2(\mu) \otimes L^2(\nu) \cong L^2(X \times Y, \mu \hat{\times} \nu)$ **through** $\xi \otimes \eta \mapsto \xi \boxtimes \eta$, **for arbitrary Radon measures, with the Radon product and not the ordinary product measure.** Every element of $L^2(\pi)$ has σ -finite support, so the pathological sets of Examples 13.5 and 13.6 never carry L^2 mass.

7. Topological groups

Proposition 7.1. Let G be a group, topologized so that multiplication $G \times G \rightarrow G$ and inversion are continuous. The Hausdorff property is not assumed in this proposition.

1. Left and right translations and inversion are homeomorphisms of G . If U is open and $A \subseteq G$ is any set, then AU and UA are open.
2. Every neighbourhood U of e contains a neighbourhood V of e with $V = V^{-1}$ and $VV \subseteq U$.
3. The closure of a subgroup is a subgroup. An open subgroup is closed.
4. If A and B are compact, so is AB .
5. Let H be a subgroup, G/H the set of left cosets with the quotient topology, and $q: G \rightarrow G/H$ the quotient map. Then q is open. If H is closed, G/H is Hausdorff. If G is locally compact, every point of G/H has a compact neighbourhood. If H is normal, G/H is a group with continuous multiplication and inversion.
6. If $\{e\}$ is closed, G is Hausdorff. In general $N = \overline{\{e\}}$ is a normal subgroup as well as closed, and G/N is Hausdorff.

Proof. (1) Translation by x has the continuous inverse translation by x^{-1} , and inversion is its own inverse. Also $AU = \bigcup_{a \in A} aU$, and each aU is open. (2) Continuity of multiplication at (e, e) gives neighbourhoods W_1, W_2 of e with $W_1 W_2 \subseteq U$. Put $W = W_1 \cap W_2$ and $V = W \cap W^{-1}$. (3) The continuous map $(x, y) \mapsto xy^{-1}$ sends $H \times H$ into H , hence sends $\bar{H} \times \bar{H} = \overline{H \times H}$ into $\bar{H}$. If H is open, its complement is a union of cosets xH , which are open by (1). (4) Multiplication is continuous and maps the compact set $A \times B$ onto AB . (5) For open $V \subseteq G$, $q^{-1}(q(V)) = VH$ is open by (1), so $q(V)$ is open. Let H be closed and $y \notin xH$. The coset xH is closed, so some neighbourhood U of e has $Uy \cap xH = \emptyset$; by (2) take V symmetric with $VV \subseteq U$. If $q(Vx)$ and $q(Vy)$ met, we would have $v_1 x h_1 = v_2 y h_2$ with $v_i \in V$ and $h_i \in H$, so $v_1^{-1} v_2 y = x h_1 h_2^{-1} \in xH$ with $v_1^{-1} v_2 \in VV \subseteq U$, which is impossible. So $q(Vx)$ and $q(Vy)$ are disjoint neighbourhoods of xH and yH (q is open). If C is a compact neighbourhood of e in G , then $q(xC)$ is compact and contains the open set $q(x \text{ int } C) \ni xH$. If H is normal and W is an open neighbourhood of $q(xy)$, continuity of multiplication in G gives open $V_1 \ni x$ and $V_2 \ni y$ with $V_1 V_2 \subseteq q^{-1}(W)$, so $q(V_1)q(V_2) \subseteq W$ with $q(V_i)$ open. Inversion is

handled the same way. (6) If $\{e\}$ is closed, apply (5) with $H = \{e\}$. In general N is a subgroup by (3), and it lies in every closed subgroup. For $z \in G$, zNz^{-1} is a closed subgroup (conjugation is a homeomorphism and an automorphism), so it contains N . Applying this to z^{-1} gives $zNz^{-1} = N$. Now apply (5) with $H = N$. $\square$

By (6), dividing by $\overline{\{e\}}$ removes any failure of the Hausdorff property, which is why we may assume it throughout.

From now on G is a locally compact group. Two more facts are needed: G splits into σ -compact pieces, and compactly supported continuous functions are uniformly continuous.

Proposition 7.2 (An open σ -compact subgroup). Some subgroup G_0 of G is at once open, closed and σ -compact. Its left cosets xG_0 partition G into open, closed, σ -compact pieces.

Proof. Let C be a compact neighbourhood of e and $U = C \cap C^{-1}$, a compact symmetric neighbourhood of e . The sets $U^n = U \cdots U$ (n factors) are compact by Proposition 7.1(4), and $G_0 = \bigcup_n U^n$ is the subgroup generated by U . It contains the neighbourhood xU of each of its points x , since $xU \subseteq U^{n+1}$ when $x \in U^n$. So it is open, hence closed by Proposition 7.1(3), and it is σ -compact. If G is connected, the open and closed subgroup G_0 is all of G , so every connected locally compact group is σ -compact. $\square$

Proposition 7.3 (Uniform continuity). For every $f \in C_c(G)$, $\|R_y f - f\|_{\sup} \rightarrow 0$ and $\|L_y f - f\|_{\sup} \rightarrow 0$ as $y \rightarrow e$.

Proof. Let $K = \text{supp } f$ and $\varepsilon > 0$. For $x \in K$ choose a neighbourhood U_x of e with $|f(xz) - f(x)| < \varepsilon/2$ for $z \in U_x$, and a symmetric neighbourhood V_x with $V_x V_x \subseteq U_x$; note $V_x \subseteq U_x$. Finitely many sets $x_j V_{x_j}$ cover K ; let $V = \bigcap_j V_{x_j}$, a symmetric neighbourhood of e . Let $y \in V$. If $x \in K$, write $x = x_j u$ with $u \in V_{x_j}$. Then $xy = x_j(uy)$ with $uy \in V_{x_j} V_{x_j} \subseteq U_{x_j}$, so $f(xy)$ and $f(x)$ are both within $\varepsilon/2$ of $f(x_j)$, and $|f(xy) - f(x)| < \varepsilon$. If $xy \in K$, the same argument applied to the point xy and to $y^{-1} \in V$ gives $|f(x) - f(xy)| < \varepsilon$. Otherwise both values are 0. So $\|R_y f - f\|_{\sup} \leq \varepsilon$ for $y \in V$. For left translations, $L_y f(x) = \tilde{f}(x^{-1}y) = R_y \tilde{f}(x^{-1})$, so $\|L_y f - f\|_{\sup} = \|R_y \tilde{f} - \tilde{f}\|_{\sup}$ with $\tilde{f} \in C_c(G)$. $\square$

Proposition 7.3 underlies the continuity of translation in L^p (Theorem 14.2(6)), and with it the strong continuity of the regular representations (Exercise 16.3). A bounded function f with $\|L_y f - f\|_{\sup} \rightarrow 0$ as $y \rightarrow e$ is called *left uniformly continuous*, and one with $\|R_y f - f\|_{\sup} \rightarrow 0$ *right uniformly continuous*. The two names are also used the other way round, so this lesson always writes the condition out.

8. Existence of Haar measure

Definition 8.1. Fix a locally compact group G . We call a Radon measure $\mu \neq 0$ on G a *left (right) Haar measure* when $\mu(xE) = \mu(E)$ ($\mu(Ex) = \mu(E)$) for every Borel set E and every $x \in G$. A *left Haar integral* is a left-invariant positive linear functional $I \neq 0$ on $C_c(G)$, that is, $I(L_y f) = I(f)$ for all f and y .

Proposition 8.2 (Invariance through integrals). Fix a Radon measure μ on G .

1. μ is a left Haar measure exactly when $\tilde{\mu}(E) = \mu(E^{-1})$ is a right Haar measure.
2. $\mu(yE) = \mu(E)$ for all Borel E and all y exactly when $\int L_y f d\mu = \int f d\mu$ for all $f \in C_c(G)$ and all y .
In that case $\int f(yx) d\mu(x) = \int f d\mu$ for all y , all Borel $f \geq 0$, and all $f \in L^1(\mu)$.

So, through the Riesz theorem 2.2, left Haar measures and left Haar integrals are the same objects.

Proof. (1) Inversion ι is a homeomorphism, so $\tilde{\mu} = \iota_* \mu$ is Radon by Proposition 3.1(1). Also $\tilde{\mu}(Ex) = \mu(x^{-1}E^{-1})$, and this equals $\tilde{\mu}(E) = \mu(E^{-1})$ for all E and x exactly when μ is left invariant. (2) Let $\ell_z(x) = zx$ and $\mu_y = (\ell_{y^{-1}})_* \mu$, so $\mu_y(E) = \mu(yE)$. By Proposition 3.1(1), μ_y is Radon and $\int f d\mu_y = \int f(y^{-1}x) d\mu(x) = \int L_y f d\mu$. If $\int L_y f d\mu = \int f d\mu$ on $C_c(G)$, the Radon measures μ_y and μ agree by the uniqueness in Theorem 2.2. The converse and the last sentence follow from the same integral formula, applied to y^{-1} . $\square$

Theorem 8.3 (Existence). Every locally compact group has a left Haar measure.

Historical reading: F. van Doorn's freely accessible paper describes Haar's original construction and the later generalization by Weil. Fremlin supplies a complementary construction on general LCH groups.

The proof builds a left Haar integral as a limit of normalized "covering numbers", which compare a function f with translates of a function φ of small support.

Covering numbers. For $f, \varphi \in C_c^+(G)$, let $(f : \varphi)$ be the infimum of $\sum_j c_j$ over all finite families $c_1, \dots, c_n \geq 0$ and $x_1, \dots, x_n \in G$ with $f \leq \sum_j c_j L_{x_j} \varphi$. This number is finite. Indeed, the open set $W = \{\varphi > \|\varphi\|_{\text{sup}}/2\}$ is nonempty; fix $w \in W$. The open sets $xw^{-1}W$, $x \in \text{supp } f$, cover $\text{supp } f$, so finitely many of them, $x_j w^{-1}W$ for $j \leq n$, do. On $x_j w^{-1}W$ we have $L_{x_j w^{-1}} \varphi > \|\varphi\|_{\text{sup}}/2$. Hence $f \leq \sum_j \frac{2\|f\|_{\text{sup}}}{\|\varphi\|_{\text{sup}}} L_{x_j w^{-1}} \varphi$, and $(f : \varphi) \leq 2n\|f\|_{\text{sup}}/\|\varphi\|_{\text{sup}}$. For $f, f_1, f_2, \varphi, \psi \in C_c^+(G)$, $y \in G$ and $c > 0$:

$$\begin{aligned} (L_y f : \varphi) &= (f : \varphi), \quad (f_1 + f_2 : \varphi) \leq (f_1 : \varphi) + (f_2 : \varphi), \quad (cf : \varphi) = c(f : \varphi), \\ f_1 \leq f_2 &\Rightarrow (f_1 : \varphi) \leq (f_2 : \varphi), \quad (f : \varphi) \geq \frac{\|f\|_{\text{sup}}}{\|\varphi\|_{\text{sup}}}, \quad (f : \psi) \leq (f : \varphi)(\varphi : \psi). \end{aligned} \quad (8.1)$$

For the first rule, $f \leq \sum_j c_j L_{x_j} \varphi$ holds exactly when $L_y f \leq \sum_j c_j L_{yx_j} \varphi$. The next three follow by adding, scaling or comparing dominations. For the fifth, evaluate a domination at a point t_0 where $f(t_0) = \|f\|_{\text{sup}}$: $\|f\|_{\text{sup}} \leq \sum_j c_j \varphi(x_j^{-1}t_0) \leq \|\varphi\|_{\text{sup}} \sum_j c_j$. For the last, if $f \leq \sum_i c_i L_{x_i} \varphi$ and $\varphi \leq \sum_k d_k L_{z_k} \psi$, then $f \leq \sum_{i,k} c_i d_k L_{x_i z_k} \psi$, and $\sum_{i,k} c_i d_k = (\sum_i c_i)(\sum_k d_k)$.

Fix $f_0 \in C_c^+(G)$ and put $I_\varphi(f) = (f : \varphi)/(f_0 : \varphi)$. The last rule of (8.1), used twice, gives

$$(f_0 : f)^{-1} \leq I_\varphi(f) \leq (f : f_0). \quad (8.2)$$

Each I_φ is left invariant, subadditive, positively homogeneous and monotone. It is additive only approximately:

Lemma 8.4 (Almost additivity). Let $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$. There is a neighbourhood V of e such that $I_\varphi(f_1) + I_\varphi(f_2) \leq I_\varphi(f_1 + f_2) + \varepsilon$ whenever $\text{supp } \varphi \subseteq V$.

Proof. By Urysohn's lemma (T2) choose $g \in C_c^+(G)$ with $g = 1$ on $\text{supp}(f_1 + f_2)$. Let $\delta > 0$, to be fixed at the end, and put $h = f_1 + f_2 + \delta g$. Let $h_i = f_i/h$ where $h > 0$, and $h_i = 0$ elsewhere ($i = 1, 2$). Since $h \geq \delta$ on $\text{supp } f_i$, h_i is continuous on the open set $\{h > 0\}$ and vanishes on the open set $G \setminus \text{supp } f_i$; these two open sets cover G . So $h_i \in C_c(G)$, and $h_1 + h_2 \leq 1$. By uniform continuity (Proposition 7.3) there is a neighbourhood V of e with $|h_i(xv) - h_i(x)| < \delta$ for all $x \in G$, $v \in V$ and $i = 1, 2$. Let $\text{supp } \varphi \subseteq V$ and $h \leq \sum_j c_j L_{x_j} \varphi$. If $\varphi(x_j^{-1}x) \neq 0$, then $x_j^{-1}x \in V$, so $|h_i(x) - h_i(x_j)| < \delta$. Hence

$$f_i(x) = h(x)h_i(x) \leq \sum_j c_j \varphi(x_j^{-1}x)h_i(x) \leq \sum_j c_j \varphi(x_j^{-1}x)(h_i(x_j) + \delta).$$

So $(f_i : \varphi) \leq \sum_j c_j (h_i(x_j) + \delta)$, and $(f_1 : \varphi) + (f_2 : \varphi) \leq (1 + 2\delta) \sum_j c_j$. Taking the infimum over dominations of h and using (8.1),

$$(f_1 : \varphi) + (f_2 : \varphi) \leq (1 + 2\delta)(h : \varphi) \leq (1 + 2\delta)((f_1 + f_2 : \varphi) + \delta(g : \varphi)).$$

Divide by $(f_0 : \varphi)$ and use (8.2): $I_\varphi(f_1) + I_\varphi(f_2) \leq I_\varphi(f_1 + f_2) + 2\delta(f_1 + f_2 : f_0) + \delta(1 + 2\delta)(g : f_0)$. Choose δ so that the last two terms add up to less than ε . $\square$

Proof of Theorem 8.3. Let $\Omega = \prod_{f \in C_c^+(G)} [(f_0 : f)^{-1}, (f : f_0)]$. It is compact Hausdorff by Tychonoff's theorem, and by (8.2) each I_φ is a point of Ω . For a neighbourhood V of e , let Φ_V be the closure in Ω of $\{I_\varphi : \varphi \in C_c^+(G), \text{supp } \varphi \subseteq V\}$. It is nonempty by (T2), and $\Phi_{V_1} \cap \dots \cap \Phi_{V_n} \supseteq \Phi_{V_1 \cap \dots \cap V_n} \neq \emptyset$. By compactness some point I lies in every Φ_V . By the definition of the product topology this means: for every neighbourhood V of e , all $f_1, \dots, f_m \in C_c^+(G)$ and all $\varepsilon > 0$, there is φ with $\text{supp } \varphi \subseteq V$ and $|I(f_k) - I_\varphi(f_k)| < \varepsilon$ for $k \leq m$.

Let $f_1, f_2 \in C_c^+(G)$ and $\varepsilon > 0$. Take V from Lemma 8.4 and such a φ for f_1, f_2 and $f_1 + f_2$. Then

$$I(f_1) + I(f_2) < I_\varphi(f_1) + I_\varphi(f_2) + 2\varepsilon \leq I_\varphi(f_1 + f_2) + 3\varepsilon < I(f_1 + f_2) + 4\varepsilon,$$

and $I(f_1 + f_2) < I_\varphi(f_1 + f_2) + \varepsilon \leq I_\varphi(f_1) + I_\varphi(f_2) + \varepsilon < I(f_1) + I(f_2) + 3\varepsilon$. So I is additive on $C_c^+(G)$. The same approximation, without the lemma, gives $I(L_y f) = I(f)$ and $I(cf) = cI(f)$ for $c > 0$, because each I_φ has these properties exactly. Also $I(f) \geq (f_0 : f)^{-1} > 0$ and $I(f_0) = 1$.

Put $I(0) = 0$. A real $f \in C_c(G)$ is $f^+ - f^-$ with $f^\pm \in C_c^+(G) \cup \{0\}$; set $I(f) = I(f^+) - I(f^-)$. If also $f = g - h$ with $g, h \in C_c^+(G) \cup \{0\}$, then $g + f^- = h + f^+$, and additivity gives $I(g) - I(h) = I(f^+) - I(f^-)$. So I is well defined and linear on real functions; for complex f put $I(f) = I(\operatorname{Re} f) + iI(\operatorname{Im} f)$. Then I is a left Haar integral with $I(f_0) = 1$. By the Riesz theorem 2.2 and Proposition 8.2, its Radon measure is a left Haar measure. $\square$

Where the hypotheses enter. Compact supports make every $(f : \varphi)$ finite. Local compactness and the Hausdorff property supply the small functions φ , through Urysohn's lemma (T2), and they make the Riesz theorem available. Continuity of the group operations gives the uniform continuity used in Lemma 8.4. The axiom of choice enters through Tychonoff's theorem; Cartan's proof, mentioned in the introduction, avoids it, but it is not given here. Right Haar measures exist too: by Proposition 8.2(1), $\tilde{\mu}$ is a right Haar measure.

9. Positivity and uniqueness of Haar measure

Proposition 9.1 (Positivity). A left Haar measure μ satisfies $\mu(U) > 0$ for every nonempty open set U , and $\int f d\mu > 0$ for every $f \in C_c^+(G)$.

Proof. Suppose $U \neq \emptyset$ is open and $\mu(U) = 0$. Fix $u \in U$. For compact K , the null open sets $xu^{-1}U$, $x \in K$, cover K , and finitely many suffice; so $\mu(K) = 0$. Then $\mu(G) = 0$ by (R3), which contradicts $\mu \neq 0$. For $f \in C_c^+(G)$, the open set $\{f > \|f\|_{\sup}/2\}$ is nonempty, so $\int f d\mu \geq \frac{1}{2}\|f\|_{\sup} \mu(\{f > \|f\|_{\sup}/2\}) > 0$. $\square$

Theorem 9.2 (Uniqueness). Any two left Haar measures μ, ν on G are proportional: $\mu = c\nu$ for some $c \in (0, \infty)$. Equivalently, every left Haar integral is a positive multiple of $f \mapsto \int f d\mu$.

Historical reading: The accessible original paper [von Neumann 1936] treats uniqueness for second countable groups. Fremlin's freely accessible text proves general LCH uniqueness; the separately licensed supplement gives that argument in full.

Proof. Fix $f, g \in C_c^+(G)$. It suffices to show $\int f d\mu / \int f d\nu = \int g d\mu / \int g d\nu$; the denominators are positive by Proposition 9.1. Fix a compact symmetric neighbourhood V_0 of e , and let $A = (\operatorname{supp} f)V_0 \cup V_0(\operatorname{supp} f)$ and $B = (\operatorname{supp} g)V_0 \cup V_0(\operatorname{supp} g)$, which are compact by Proposition 7.1(4). For $y \in V_0$, the function $x \mapsto f(xy) - f(yx)$ vanishes off A : $f(xy) \neq 0$ forces $x \in (\operatorname{supp} f)y^{-1}$, and $f(yx) \neq 0$ forces $x \in y^{-1}\operatorname{supp} f$. Let $\varepsilon > 0$. Since $|f(xy) - f(yx)| \leq |R_y f(x) - f(x)| + |f(x) - L_{y^{-1}} f(x)|$, uniform continuity (Proposition 7.3) gives a symmetric neighbourhood $V \subseteq V_0$ of e with $|f(xy) - f(yx)| < \varepsilon$ and $|g(xy) - g(yx)| < \varepsilon$ for all $x \in G$ and $y \in V$. Choose $h_1 \in C_c^+(G)$ with $\operatorname{supp} h_1 \subseteq V$ and put $h = h_1 + h_1$. Then $h \in C_c^+(G)$, $\operatorname{supp} h \subseteq V$, and $h(x^{-1}) = h(x)$.

Every integrand below is a continuous function with compact support on $G \times G$, so (4.1) allows the order of integration to be exchanged. By left invariance of μ in x ,

$$\int h d\nu \int f d\mu = \iint h(y)f(yx) d\mu(x) d\nu(y).$$

By left invariance of μ in x , then an exchange and the symmetry of h , then left invariance of ν in y , then another exchange,

$$\begin{aligned} & \int h d\mu \int f d\nu \\ &= \iint h(y^{-1}x)f(y) d\mu(x) d\nu(y) \\ &= \iint h(x^{-1}y)f(y) d\nu(y) d\mu(x) \\ &= \iint h(y)f(xy) d\nu(y) d\mu(x) \\ &= \iint h(y)f(xy) d\mu(x) d\nu(y). \end{aligned}$$

Subtract. For $y \in \operatorname{supp} h \subseteq V$ the inner integrand $h(y)(f(yx) - f(xy))$ is at most $\varepsilon h(y)$ in absolute value

and vanishes for $x \notin A$. Hence

$$\left| \int h d\nu \int f d\mu - \int h d\mu \int f d\nu \right| \leq \varepsilon \mu(A) \int h d\nu.$$

Dividing by $\int h d\nu \int f d\nu$ gives $\left| \frac{\int f d\mu}{\int f d\nu} - \frac{\int h d\mu}{\int h d\nu} \right| \leq \frac{\varepsilon \mu(A)}{\int f d\nu}$. The same holds for g with B . So the two ratios for f and g differ by at most $\varepsilon(\mu(A)/\int f d\nu + \mu(B)/\int g d\nu)$, where A and B do not depend on ε . Hence the ratio is a constant $c > 0$ on $C_c^+(G)$. By linearity $\int f d\mu = c \int f d\nu$ on $C_c(G)$, and $\mu = c\nu$ by the uniqueness in Theorem 2.2. $\square$

Only compactly supported continuous integrands on $G \times G$ occur, so this proof needs nothing beyond (4.1); in particular it needs no σ -finiteness. From now on, μ denotes a fixed left Haar measure on G , and we write $\int f(t) dt = \int f d\mu$.

Fremlin gives a second proof that compares masses of shrinking symmetric neighborhoods. The **complete neighborhood-ratio proof** uses the open-set assertion of Theorem 5.1 and reaches the same conclusion for every LCH group.

10. The modular function

A left Haar measure need not be right invariant. The modular function measures how far it is from right invariance.

Theorem 10.1. Let μ be a left Haar measure on G .

1. For each $x \in G$, $E \mapsto \mu(Ex)$ is a left Haar measure. So there is a unique $\Delta(x) \in (0, \infty)$ with $\mu(Ex) = \Delta(x)\mu(E)$ for every Borel E . Another left Haar measure in place of μ gives the same $\Delta(x)$.
2. Δ is a continuous homomorphism from G to the multiplicative group $(0, \infty)$.
3. For every Borel $f \geq 0$, and for every $f \in L^1(\mu)$,

$$\int f(tx) dt = \Delta(x)^{-1} \int f(t) dt. \quad (10.1)$$

In particular $\|R_x f\|_p = \Delta(x)^{-1/p} \|f\|_p$ for $f \in L^p(\mu)$, $1 \leq p < \infty$.

4. $\Delta = 1$ on every compact subgroup. Every compact, abelian or discrete group is *unimodular* ($\Delta \equiv 1$). G is unimodular exactly when μ is right invariant as well. If $G/[G, G]$ is compact, where $[G, G]$ is the closure of the subgroup generated by the commutators $xyx^{-1}y^{-1}$, then G is unimodular.

Proof. (1) Right translation by x is a homeomorphism, so $\mu_x(E) = \mu(Ex)$ is Radon by Proposition 3.1(1). It is nonzero and left invariant, since $y(Ex) = (yE)x$. By uniqueness (Theorem 9.2), $\mu_x = \Delta(x)\mu$ with $\Delta(x) > 0$. Replacing μ by $c\mu$ does not change Δ . (2) Let E be a compact neighbourhood of e , so $0 < \mu(E) < \infty$ by Proposition 9.1 and (R1). Then $\Delta(xy)\mu(E) = \mu(Exy) = \Delta(y)\mu(Ex) = \Delta(y)\Delta(x)\mu(E)$. So Δ is a homomorphism, and $\Delta(x^{-1}) = \Delta(x)^{-1}$. Continuity is proved after (3). (3) For $f = 1_E$, $\int 1_E(tx) dt = \mu(Ex^{-1}) = \Delta(x)^{-1}\mu(E)$. Linearity, then the layer functions (1.1) with monotone convergence, then splitting into four nonnegative parts extend this to simple functions, to Borel $f \geq 0$, and to L^1 . Apply it to $|f|^p$ for the norm formula. *Continuity of Δ .* Fix $f \in C_c^+(G)$; by (3), $\Delta(x)^{-1} = \int R_x f dt / \int f dt$, where the denominator is positive by Proposition 9.1. Let W be a compact neighbourhood of e and $x \in x_0 W$. Then $R_x f$ vanishes off the compact set $(\text{supp } f)W^{-1}x_0^{-1}$, and $\|R_x f - R_{x_0} f\|_{\text{sup}} = \|R_{x_0}(R_{x_0^{-1}} x f - f)\|_{\text{sup}} = \|R_{x_0^{-1}} x f - f\|_{\text{sup}}$, which tends to 0 as $x \rightarrow x_0$ by uniform continuity (Proposition 7.3). So $\int R_x f dt \rightarrow \int R_{x_0} f dt$. (4) $\Delta(K)$ is a compact subgroup of $(0, \infty)$ for a compact subgroup K . If it contained some $r \neq 1$, it would contain all powers r^n , $n \in \mathbb{Z}$, which form an unbounded set. So $\Delta(K) = \{1\}$; take $K = G$ for compact G . For abelian G , $\mu(Ex) = \mu(xE) = \mu(E)$. On a discrete group, counting measure is invariant on both sides, and it is Radon: compact sets are finite and every set is open. The equivalence with right invariance is the definition of Δ . Finally, Δ takes values in an abelian group, so it is 1 on every commutator, hence on the subgroup they generate, and by continuity on its closure $N = [G, G]$. This N is closed, and it is normal because conjugates of commutators are commutators and conjugation is a homeomorphism. So $\Delta = \bar{\Delta} \circ q$ for a homomorphism $\bar{\Delta}$ on the locally compact group G/N (Proposition 7.1(5)), and $\bar{\Delta}$ is continuous because

q is open: $\bar{\Delta}^{-1}(W) = q(\Delta^{-1}(W))$. If G/N is compact, $\Delta(G) = \bar{\Delta}(G/N)$ is a compact subgroup of $(0, \infty)$, hence $\{1\}$. $\square$

In shorthand, (10.1) reads $d\mu(tx) = \Delta(x) d\mu(t)$.

11. Inversion and right Haar measure

Theorem 11.1 (Inversion). For every Borel $f: G \rightarrow [0, \infty]$,

$$\int f(t^{-1}) dt = \int f(t) \Delta(t)^{-1} dt, \quad (11.1)$$

and the same holds for every Borel f with $\int |f| \Delta^{-1} d\mu < \infty$. Equivalently, the right Haar measure $\tilde{\mu}(E) = \mu(E^{-1})$ equals $\Delta^{-1}\mu$, that is, $\tilde{\mu}(E) = \int_E \Delta(t)^{-1} dt$. Moreover $\int f(t^{-1}) \Delta(t)^{-1} dt = \int f(t) dt$ for every Borel $f \geq 0$.

Proof. For $f \in C_c(G)$ put $f^\natural(t) = f(t^{-1})\Delta(t)^{-1}$, again in $C_c(G)$, and $J(f) = \int f^\natural d\mu$. The functional J is linear, and $J(f) > 0$ for $f \in C_c^+(G)$ by Proposition 9.1. It is left invariant: with $k = f^\natural$ and $y \in G$,

$$(L_y f)^\natural(t) = f((ty)^{-1})\Delta(t)^{-1} = f((ty)^{-1})\Delta(ty)^{-1}\Delta(y) = \Delta(y) k(ty),$$

so $J(L_y f) = \Delta(y)\Delta(y)^{-1} \int k d\mu = J(f)$ by (10.1). So J is a left Haar integral, and by Proposition 8.2 and uniqueness (Theorem 9.2), $J = c \int \cdot d\mu$ on $C_c(G)$ for some $c > 0$. Now $(f^\natural)^\natural(t) = f^\natural(t^{-1})\Delta(t)^{-1} = f(t)\Delta(t)\Delta(t)^{-1} = f(t)$. So for $f \in C_c^+(G)$, $\int f d\mu = J(f^\natural) = c \int f^\natural d\mu = cJ(f) = c^2 \int f d\mu$, and $c = 1$. Apply $J = \int \cdot d\mu$ to $g = f\Delta^{-1} \in C_c(G)$: since $g(t^{-1})\Delta(t)^{-1} = f(t^{-1})$, this gives (11.1) for $f \in C_c(G)$.

The measure $\tilde{\mu} = \iota_* \mu$ is Radon (Proposition 3.1(1)) with $\int f d\tilde{\mu} = \int f(t^{-1}) dt$. The measure $\Delta^{-1}\mu$ is Radon by Proposition 3.1(2), because Δ^{-1} is continuous and positive. By the previous paragraph they integrate every $f \in C_c(G)$ alike, so they are equal by the uniqueness in Theorem 2.2. Integrating a Borel $f \geq 0$ against both (Proposition 3.1(1) and (2)) gives (11.1). For f with $\int |f| \Delta^{-1} d\mu < \infty$, apply this to the four nonnegative parts of f . For the last formula apply (11.1) to the Borel function $t \mapsto f(t)\Delta(t)$, using $\Delta(t^{-1}) = \Delta(t)^{-1}$. $\square$

Formula (11.1) holds for all nonnegative Borel functions, whatever their support.

Corollary 11.2.

1. μ and $\tilde{\mu}$ have the same null sets and the same σ -finite sets.
2. For $1 \leq p < \infty$, $(S_p f)(t) = \Delta(t)^{-1/p} f(t^{-1})$ defines a linear isometry of $L^p(\mu)$ onto itself with $S_p^2 = 1$.
For $p = 2$, $(J\xi)(t) = \Delta(t)^{-1/2} \overline{\xi(t^{-1})}$ is a conjugate-linear isometric involution of $L^2(G)$.
3. $f \mapsto \tilde{f}$ and $f \mapsto \Delta^{1/p} f$ are isometries of $L^p(\mu)$ onto $L^p(\tilde{\mu})$.
4. If G is not unimodular, Δ is unbounded above and below.
5. The measure $(1 + \Delta)\mu$ is a Radon measure, and $C_c(G)$ is dense in $L^2((1 + \Delta)\mu)$. Consequently $C_c(G)$ is a core for multiplication by $\Delta^{1/2}$ on its maximal domain $\{\xi \in L^2(G) : \Delta^{1/2}\xi \in L^2(G)\}$, whose graph norm is $(\int |\xi|^2 (1 + \Delta) d\mu)^{1/2}$.

Proof. (1) Since $\tilde{\mu} = \Delta^{-1}\mu$ with a continuous positive density, this is Proposition 3.1(2). (2) By the last formula of the theorem applied to $|f|^p$, $\int \Delta(t)^{-1} |f(t^{-1})|^p dt = \int |f|^p dt$. Also $S_p(S_p f)(t) = \Delta(t)^{-1/p} \Delta(t^{-1})^{-1/p} f(t) = f(t)$, so S_p is onto. Complex conjugation commutes with S_2 and is a conjugate-linear isometry. (3) $\int |\tilde{f}|^p d\tilde{\mu} = \int |f(t)|^p dt$ by the definition of $\tilde{\mu}$, and $\int \Delta |f|^p d\tilde{\mu} = \int |f|^p d\mu$ by the theorem; the inverse maps have the same form. (4) $\Delta(G)$ is a subgroup of $(0, \infty)$ containing some $r \neq 1$, hence all r^n . (5) Apply Proposition 3.1(2) with the positive continuous density $1 + \Delta$, and then the density of C_c in L^2 (Proposition 3.1(4)). The graph norm identity is immediate, and a function ξ lies in the maximal domain exactly when it lies in $L^2((1 + \Delta)\mu)$. $\square$

Example 11.3 (Explicit Haar measures on $\mathbb{R}^\times$ and on the $ax + b$ group). Integrals of continuous compactly supported functions of one real variable below are Riemann integrals, and substitutions use the change-of-variables rule for Riemann integrals (see "Results used from other lessons"). (a) On the multiplicative group $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$, $I(f) = \int f(x) dx/|x|$ is a Haar integral: the substitution $u = cx$ gives $\int f(cx) dx/|x| =$

$\int f(u) du/|u|$ for $c \neq 0$. The group is abelian, so $\Delta \equiv 1$. (b) Let $G = \{(a, b) : a > 0, b \in \mathbb{R}\}$ with $(a, b)(a', b') = (aa', ab' + b)$, identity $(1, 0)$, inverse $(a, b)^{-1} = (1/a, -b/a)$, and the topology of the open half-plane. Put $I(f) = \int_0^\infty (\int_{\mathbb{R}} f(a, b) db) a^{-2} da$. For $x_0 = (a_0, b_0)$, the substitutions $u = a_0 b + b_0$ (inner) and $v = a_0 a$ (outer) give

$$\int_0^\infty \int_{\mathbb{R}} f(a_0 a, a_0 b + b_0) db \frac{da}{a^2} = \int_0^\infty \frac{1}{a_0} \int_{\mathbb{R}} f(a_0 a, u) du \frac{da}{a^2} = \int_0^\infty \int_{\mathbb{R}} f(v, u) du \frac{dv}{v^2},$$

so I is a left Haar integral, $d\mu = a^{-2} da db$. The substitutions $u = ab_0 + b$ and $v = aa_0$ give $\int f(tx_0) d\mu(t) = a_0 \int f d\mu$, so by (10.1)

$$\Delta(a, b) = 1/a.$$

The inversion formula (11.1) can be checked directly: the substitutions $u = -b/a$ and $v = 1/a$ give $\int f(t^{-1}) d\mu(t) = \int_0^\infty \int_{\mathbb{R}} f(v, u) du v^{-1} dv$, which is $\int f \Delta^{-1} d\mu$. So the right Haar measure is $a^{-1} da db$. The commutators $(a, b)(a', b')(a, b)^{-1}(a', b')^{-1}$ have first coordinate 1, and they fill the normal subgroup $\{1\} \times \mathbb{R}$; the quotient $G/(\{1\} \times \mathbb{R}) \cong (0, \infty)$ is not compact, in line with Theorem 10.1(4). This group is a good test case for the signs in (10.1) and (11.1).

12. Products of groups

The Radon product of Haar measures is a Haar measure. For two copies of G this gives the measure on $G \times G$ behind convolution (Section 14); for infinitely many compact groups it gives the Haar measure of an infinite product.

Proposition 12.1 (The group $G \times G$). Keep G and its left Haar measure μ , and let $\pi = \mu \hat{\times} \mu$ on $G \times G$.

1. On the locally compact group $G \times G$, π is invariant under left translations, so it is a left Haar measure; its modular function is $(s, t) \mapsto \Delta(s)\Delta(t)$. In the same way, if each G_j ($j = 1, \dots, n$) carries a left Haar measure μ_j , the iterated Radon product $(\dots(\mu_1 \hat{\times} \mu_2) \dots) \hat{\times} \mu_n$ is one on $G_1 \times \dots \times G_n$, and it integrates functions in C_c by iterated integrals.
2. For every continuous $a: G \rightarrow G$, the map $\Phi_a(s, t) = (s, a(s)t)$ is a homeomorphism of $G \times G$ that preserves π . So $(W_a \zeta)(s, t) = \zeta(s, a(s)t)$ is a unitary on $L^2(G \times G, \pi)$, with inverse $\zeta \mapsto \zeta(s, a(s)^{-1}t)$.
3. Under the unitary U of Theorem 6.2, the operator $1 \otimes \lambda(g)$ is W_a for the constant $a \equiv g^{-1}$: $\zeta(s, t) \mapsto \zeta(s, g^{-1}t)$. For $a(s) = s$, W_a is the operator $(W\zeta)(s, t) = \zeta(s, st)$, with inverse $\zeta(s, s^{-1}t)$.

Proof. (1) For $F \in C_c(G \times G)$ and $(g, h) \in G \times G$, $\int (\int F(gs, ht) dt) ds = \int (\int F(gs, t) dt) ds = \int (\int F(s, t) dt) ds$, by left invariance applied first to the inner function $t \mapsto F(gs, t)$ and then to the outer function $s \mapsto \int F(s, t) dt$, which lies in $C_c(G)$ by Proposition 4.2(1). So π is a left Haar measure by Proposition 8.2 (it is nonzero by (4.2)). In the same way, (10.1) gives $\iint F(sg, th) dt ds = \Delta(g)^{-1} \Delta(h)^{-1} \iint F dt ds$, which identifies the modular function by Theorem 10.1(3). The statement for n factors follows by induction, using Proposition 4.2 for the iterated integral. (2) Φ_a is continuous, with continuous inverse $(s, t) \mapsto (s, a(s)^{-1}t)$. For $F \in C_c(G \times G)$, also $F \circ \Phi_a \in C_c(G \times G)$, and $\int F \circ \Phi_a d\pi = \int (\int F(s, a(s)t) dt) ds = \int (\int F(s, t) dt) ds$ by left invariance in t for each fixed s . So the Radon measures $(\Phi_a)_* \pi$ (Proposition 3.1(1)) and π agree by the uniqueness in Theorem 2.2. Hence $\zeta \mapsto \zeta \circ \Phi_a$ preserves null sets and the $L^2(\pi)$ norm, and its inverse is composition with Φ_a^{-1} . (3) $U(\xi \otimes L_g \eta)(s, t) = \xi(s) \eta(g^{-1}t) = U(\xi \otimes \eta)(s, g^{-1}t)$. Both operators are bounded and agree on elementary tensors. $\square$

Example 12.2 (Infinite products of compact groups, and $(\mathbb{Z}_2)^\mathbb{N}$ versus $[0, 1]$). Let $(G_\alpha)_{\alpha \in A}$ be compact groups with Haar measures $\mu_\alpha(G_\alpha) = 1$ (compact groups are unimodular by Theorem 10.1(4)). The product $G = \prod_\alpha G_\alpha$, with the product topology and the group law taken coordinate by coordinate, is a compact group by Tychonoff's theorem. Write $C_F(G)$ for the continuous functions that depend on only finitely many coordinates, $f(x) = f_0(x_{\alpha_1}, \dots, x_{\alpha_n})$ with f_0 continuous. It is a self-adjoint algebra, it contains the constant functions, and it separates the points of G : if $x_\alpha \neq x'_\alpha$, (T2) on G_α gives a continuous function of the coordinate x_α that separates them. By the Stone–Weierstrass theorem, $C_F(G)$ is dense in $C(G)$. Put $I(f) = \int f_0 d(\mu_{\alpha_1} \hat{\times} \dots \hat{\times} \mu_{\alpha_n})$. The value is independent of how f is written: by (4.1) the order of the factors is irrelevant, and a coordinate on which f_0 does not depend contributes a factor $\mu_\beta(G_\beta) = 1$. So I is linear and positive on $C_F(G)$, $|I(f)| \leq \|f\|_{\sup}$, $I(1) = 1$, and I is left invariant because each finite product measure is a

left Haar measure (Proposition 12.1(1)). It extends uniquely to a positive, left-invariant, norm-one functional on $C(G)$ (positivity passes to the limit because $\operatorname{Re} f_n + \|f - f_n\|_{\sup} \geq 0$ when $f \geq 0$). Its Radon measure is the Haar measure of G with total mass 1.

For $G = (\mathbb{Z}_2)^\mathbb{N}$, each factor with mass $\frac{1}{2}$ on each point, let $\Phi(a) = \sum_j a_j 2^{-j} \in [0, 1]$. This map is continuous and onto, and it is one-to-one except over the dyadic rationals $j2^{-k}$ in $(0, 1)$, each of which has two preimages. It carries Haar measure to Lebesgue measure λ_1 on $[0, 1]$: $\lambda_1(E) = \mu(\Phi^{-1}(E))$ for every Borel E . Proof: $\Phi_*\mu$ is a finite Borel measure on $[0, 1]$. It is outer regular: by (R2) for μ there is an open $O \supseteq \Phi^{-1}(E)$ with $\mu(O \setminus \Phi^{-1}(E)) < \varepsilon$; then $U = [0, 1] \setminus \Phi(G \setminus O)$ is open (Φ is a closed map, since G is compact), contains E , and satisfies $\Phi^{-1}(U) \subseteq O$, so $\Phi_*\mu(U \setminus E) < \varepsilon$. A finite outer regular measure on a compact space is also inner regular on open sets (take complements), so $\Phi_*\mu$ is Radon. For continuous f on $[0, 1]$, the cylinder sets fixing $a_1, \dots, a_k$ have measure 2^{-k} and are mapped onto the intervals $[j2^{-k}, (j+1)2^{-k}]$, so $\int f \circ \Phi d\mu$ differs from the Riemann sum $\sum_j 2^{-k} f(j2^{-k})$ by at most the oscillation of f on intervals of length 2^{-k} . Letting $k \rightarrow \infty$, $\int f d\Phi_*\mu = \int_0^1 f dx$. By the uniqueness in Theorem 2.2, $\Phi_*\mu = \lambda_1$.

The Heisenberg group and its volume scaling

Example 12.3. Let $N = \mathbb{R}^3$ with multiplication

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy'). \quad (12.3)$$

This is the group of matrices $\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}$, with the ordinary Euclidean topology. Its identity is $(0, 0, 0)$ and its inverse is

$$(x, y, z)^{-1} = (-x, -y, -z + xy).$$

Matrix multiplication proves associativity, or one can check that the third coordinate of either association is $z + z' + z'' + xy' + xy'' + x'y''$. The polynomial multiplication and inverse are continuous, so this is an LCH group.

Lebesgue measure $dx dy dz$ is both left and right Haar measure. Indeed, left and right multiplication by (a, b, c) are respectively

$$(x, y, z) \mapsto (a + x, b + y, c + z + ay), \quad (x, y, z) \mapsto (x + a, y + b, z + c + xb).$$

For $f \in C_c(\mathbb{R}^3)$, integrate in z first. Each added third-coordinate term is a translation depending only on x, y , so it leaves that inner integral unchanged. Translate x and y next. Fubini and the one-variable substitution rule give invariance of the full integral in both cases. The transported Radon measures consequently agree on all Borel sets by Riesz uniqueness. Thus $\Delta_N = 1$. The group is nonabelian: with the commutator convention $[u, v] = uvu^{-1}v^{-1}$,

$$[(a, 0, 0), (0, b, 0)] = (0, 0, ab).$$

For $r > 0$, set $D_r(x, y, z) = (rx, ry, r^2z)$. Substitution in (12.3) proves $D_r(uv) = D_r(u)D_r(v)$, and $D_{1/r}$ is its inverse. Three one-variable substitutions give

$$\int_N f(D_r u) du = r^{-4} \int_N f(u) du, \quad \mu(D_r E) = r^4 \mu(E) \quad (12.4)$$

for every Borel E ; the second identity follows from the first by equality of Radon measures. The powers r, r, r^2 explain why the **homogeneous dimension** is 4, even though the underlying manifold has dimension 3.

Fischer and Ruzhansky use symmetric coordinates $t = z - xy/2$. The coordinate change is a shear preserving Lebesgue measure, by integration in the third coordinate, and converts (12.3) into

$$(x, y, t)(x', y', t') = (x + x', y + y', t + t' + (xy' - yx')/2).$$

These two descriptions therefore have the same Haar normalization. Integer matrix coordinates give the lattice computed in **the arithmetic lesson**; in the symmetric coordinates its third coordinate is $k - mn/2$, rather than always an integer. This distinction prevents a spurious factor of two in a fundamental domain.

13. Groups that are not σ -compact

When G is not σ -compact, its Haar measure is not σ -finite, and several familiar facts need care: inner regularity, the duality between L^1 and L^∞ , and Fubini's theorem. This section describes the measure through the cosets of an open σ -compact subgroup, sets up the right notion of L^∞ , and ends with the examples that show what fails.

Fix an open, closed, σ -compact subgroup G_0 (Proposition 7.2) and a set $Y \subseteq G$ that contains exactly one point of each left coset of G_0 . So the open σ -compact sets yG_0 , $y \in Y$, partition G , and μ restricted to any one of them is σ -finite by (R1). The restriction of μ to the Borel subsets of G_0 is a left Haar measure of the locally compact group G_0 : the restriction of a Radon measure to an open set is Radon there, it is nonzero by Proposition 9.1, and it is invariant under left translation by elements of G_0 .

Proposition 13.1 (Cosets). Let $E \subseteq G$ be Borel.

1. If E lies in countably many cosets $y_j G_0$, then $\mu(E) = \sum_j \mu(E \cap y_j G_0)$.
2. If E meets uncountably many cosets, then $\mu(E) = \infty$.
3. Hence a Borel set is σ -finite exactly when it lies in countably many cosets. In particular every Borel set of finite measure does. Moreover G is σ -compact $\iff \mu$ is σ -finite $\iff Y$ is countable.

Proof. (1) is countable additivity. (2) By (R2) it suffices to show $\mu(U) = \infty$ for every open $U \supseteq E$. Such U meets uncountably many cosets, and each nonempty open set $U \cap yG_0$ has positive measure by Proposition 9.1. For some n , uncountably many of these measures exceed $1/n$; adding k of them gives $\mu(U) \geq k/n$ for every k . (3) A σ -finite set is covered by countably many sets of finite measure, each lying in countably many cosets by (2); conversely each coset is σ -finite. If Y is countable, countably many σ -compact cosets cover G ; if G is σ -compact, μ is σ -finite by (R1); if μ is σ -finite, Y is countable by the first claim. $\square$

Definition 13.2. A set $E \subseteq G$ is *locally Borel* when its intersection with each Borel set of finite measure is Borel, and *locally null* when, in addition, all these intersections are null. A property holds *locally almost everywhere* when the points where it fails form a locally null set. Call $f: G \rightarrow \mathbb{C}$ *locally measurable* when the preimage of each Borel subset of $\mathbb{C}$ is locally Borel. $L^\infty(G)$ consists of the locally measurable f with $|f| \leq c$ locally almost everywhere for some constant c ; two such functions are identified when they agree locally almost everywhere; and $\|f\|_\infty$ is the infimum of those constants c .

Why the local notions? With the ordinary definitions, L^∞ need not be the dual of L^1 when the measure is not σ -finite. For Haar measure on $\mathbb{R} \times \mathbb{R}_d$, the indicator of the locally null set Y of Example 13.4 is a nonzero element of the ordinary L^∞ , yet it integrates every L^1 function to 0. The local versions repair this.

Theorem 13.3.

1. E is locally Borel exactly when every $E \cap yG_0$ is Borel, and locally null exactly when every $E \cap yG_0$ is null. f is locally measurable exactly when every restriction $f|_{yG_0}$ is Borel. A Borel set is locally null exactly when all its compact subsets are null. A locally null Borel set is null exactly when it is σ -finite. Countable unions of locally null sets are locally null, and $|f| \leq \|f\|_\infty$ locally almost everywhere.
2. For $f \in L^\infty(G)$ and $g \in L^p(G)$, $1 \leq p < \infty$, the product fg is Borel, its class depends only on the classes of f and g , and $\|fg\|_p \leq \|f\|_\infty \|g\|_p$.
3. ($L^{1*} = L^\infty$) For every bounded linear functional Φ on $L^1(G)$ there is a unique $f \in L^\infty(G)$ with $\Phi(g) = \int fg d\mu$ for all $g \in L^1(G)$, and $\|\Phi\| = \|f\|_\infty$.
4. The map $f \mapsto m_f$, $m_f \xi = f\xi$, is an isometric $*$ -isomorphism of $L^\infty(G)$ onto a von Neumann algebra on $L^2(G)$ that is maximal abelian.
5. (Radon–Nikodym, restricted form) Let ν and μ' be Radon measures on an LCH space X , with ν σ -finite and $\nu(E) = 0$ whenever $\mu'(E) = 0$. Then $\nu(E) = \int_E h d\mu'$ for all Borel E , for some Borel $h \geq 0$.

Proof. (1) Each coset yG_0 is the union of countably many compact sets yK_n , each of finite measure, so $E \cap yG_0 = \bigcup_n E \cap yK_n$ is Borel (null) if E is locally Borel (locally null). Conversely, a Borel F of finite measure lies in countably many cosets $y_j G_0$, so $E \cap F = \bigcup_j (E \cap y_j G_0) \cap F$ is Borel (null). The statement about functions follows. If every compact subset of a Borel set E is null and $\mu(F) < \infty$, then $\mu(E \cap F) = 0$ by inner regularity on sets of finite measure (Proposition 2.3(1)); conversely, compact sets have finite measure. A

σ -finite locally null Borel set is the union of its intersections with countably many sets of finite measure, each null. If E_n are locally null and $\mu(F) < \infty$, then $\mu(\bigcup_n E_n \cap F) \leq \sum_n \mu(E_n \cap F) = 0$. Finally $\{|f| > \|f\|_\infty\} = \bigcup_n \{|f| > \|f\|_\infty + 1/n\}$ is a countable union of locally null sets. (2) By Proposition 3.1(4), $\{g \neq 0\}$ is a disjoint union of Borel sets F_n of finite measure. Each $f1_{F_n}$ is Borel, so $fg = \sum_n (f1_{F_n})g$ is Borel (at each point at most one term is nonzero). If $f = f'$ locally a.e., the set $\{f \neq f'\} \cap \{g \neq 0\}$ is Borel, σ -finite and locally null, hence null by (1); so $fg = f'g$ a.e. In the same way $|fg| \leq \|f\|_\infty |g|$ a.e. (3) For $y \in Y$, the functions in $L^1(G)$ that vanish off yG_0 form a copy of $L^1(yG_0, \mu)$, and μ is σ -finite there. The duality $L^{1*} = L^\infty$ for σ -finite measures provides a Borel function f_y on yG_0 with $|f_y| \leq \|\Phi\|$ and $\Phi(g) = \int_{yG_0} f_y g d\mu$ for those g . Let $f = f_y$ on yG_0 . By (1), f is locally measurable, and $\|f\|_\infty \leq \|\Phi\|$. Let $g \in L^1(G)$. The set $\{g \neq 0\}$ is σ -finite, so it lies in countably many cosets $y_j G_0$, and $g = \sum_j g1_{y_j G_0}$ in L^1 by dominated convergence. Hence $\Phi(g) = \sum_j \int_{y_j G_0} fg d\mu = \int fg d\mu$, again by dominated convergence, since $|fg| \leq \|\Phi\| |g|$. By (2) with $p = 1$, $\|\Phi\| \leq \|f\|_\infty$. For uniqueness, suppose $\int hg d\mu = 0$ for all $g \in L^1(G)$ but $h \neq 0$ on a set that is not locally null. Then for some $c > 0$ and some Borel F of finite measure, $D = \{|h| > c\} \cap F$ has positive measure; with $g = 1_D \bar{h}/|h|$ we get $\int hg d\mu = \int_D |h| d\mu > 0$, a contradiction. (4) By (2), m_f is bounded with $\|m_f\| \leq \|f\|_\infty$. If $c < \|f\|_\infty$, the set $\{|f| > c\}$ is not locally null, so for some Borel F of finite measure the set $D = \{|f| > c\} \cap F$ has $\mu(D) > 0$; then $\|m_f 1_D\|_2^2 \geq c^2 \mu(D) = c^2 \|1_D\|_2^2$. So m is isometric. Clearly $m_{fg} = m_f m_g$ and $m_{\bar{f}} = m_f^*$. Let T commute with every m_f . Then T commutes with the projections $P_y = m_{1_{yG_0}}$, so it maps each $H_y = P_y L^2(G) \cong L^2(yG_0, \mu)$ into itself. On H_y it commutes with multiplication by every bounded Borel function on yG_0 (extended by 0, such a function lies in $L^\infty(G)$). On a σ -finite measure space the multiplication operators form a maximal abelian algebra (Hilbert spaces and compact operators, Theorem 9.1). Applied to the σ -finite measure space yG_0 , this shows that $T|_{H_y}$ is multiplication by a bounded Borel function f_y on yG_0 , which we may take with $|f_y| \leq \|T\|$ by the norm formula just proved. Let $f = f_y$ on yG_0 ; then $f \in L^\infty(G)$. A vector $\xi \in L^2(G)$ lives on countably many cosets (Proposition 3.1(4) and Proposition 13.1(3)), so $\xi = \sum_j P_{y_j} \xi$ and $T\xi = \sum_j f_{y_j} P_{y_j} \xi = f\xi$. So $T = m_f$. Thus $m(L^\infty)' \subseteq m(L^\infty)$, and the reverse inclusion holds because $m(L^\infty)$ is abelian. Hence $m(L^\infty)' = m(L^\infty)$, which is maximal abelian, and $m(L^\infty)'' = m(L^\infty)$. (5) By Proposition 2.3(3), $X = S \cup N$ with S σ -compact and $\nu(N) = 0$. On the Borel subsets of S , μ' is σ -finite by (R1), and the Radon–Nikodym theorem for σ -finite measures gives a Borel $h_0 \geq 0$ on S with $\nu(E) = \int_E h_0 d\mu'$ for Borel $E \subseteq S$. Put $h = h_0$ on S and $h = 0$ on N . $\square$

Part (4) carries the maximal abelian property of L^∞ from σ -finite measure spaces to Haar measure on every locally compact group, and no lifting theorem is needed.

The following examples show what fails without σ -finiteness. In each of them the bad set is locally null, so it carries no L^p mass for $p < \infty$.

Example 13.4 ($\mathbb{R} \times \mathbb{R}_d$: a Haar measure that is not σ -finite). Let $\mathbb{R}_d$ be the additive group of reals with the discrete topology, and $G = \mathbb{R} \times \mathbb{R}_d$. The subgroup $G_0 = \mathbb{R} \times \{0\}$ is open and σ -compact, and its cosets are the lines $\mathbb{R} \times \{d\}$. A compact subset of G meets only finitely many lines, so $I(f) = \sum_d \int_{\mathbb{R}} f(x, d) dx$ is a finite sum for $f \in C_c(G)$; it is a left Haar integral, by translation invariance of the Riemann integral. On each line, the Haar measure μ is Lebesgue measure: the restriction of a Radon measure to an open set is Radon on that set, and both measures integrate the compactly supported continuous functions on the open set $\mathbb{R} \times \{d\}$ alike. By Proposition 13.1, $\mu(E) = \sum_d \lambda_1(\{x : (x, d) \in E\})$ if E meets countably many lines, and $\mu(E) = \infty$ otherwise.

The closed set $Y = \{0\} \times \mathbb{R}_d$ meets every line, so $\mu(Y) = \infty$. Its compact subsets are finite and null. So μ is not inner regular on Y , and Y is locally null (Theorem 13.3(1)) but not null. No measure that is inner and outer regular on every Borel set can represent the Haar integral of G : it would be Radon, hence equal to μ by the uniqueness in Theorem 2.2, and μ fails inner regularity on Y . Nevertheless every $f \in L^p(G)$, $p < \infty$, lives on countably many lines; $C_c(G)$ is dense in $L^p(G)$ (Proposition 3.1(4)); and $L^\infty(G)$ consists of the functions whose restrictions to the lines are Borel and essentially bounded, uniformly in d ; it is the dual of $L^1(G)$ (Theorem 13.3(3)).

Example 13.5 (The diagonal of $\mathbb{R}_d \times \mathbb{R}$: Tonelli's theorem needs σ -finiteness). Let $X = \mathbb{R}_d$ with counting measure c and $Y = \mathbb{R}$ with Lebesgue measure m . Their Radon product is the Haar measure of the group $\mathbb{R}_d \times \mathbb{R}$ (Proposition 12.1(1)). The diagonal $D = \{(t, t)\}$ is a closed subgroup. Its sections are single points:

$m(D_s) = 0$ for every s , while $c(D^t) = 1$ for every t . So

$$\int m(D_s) dc(s) = 0, \quad \int c(D^t) dm(t) = \infty, \quad (c \hat{\times} m)(D) = \infty,$$

the last because D meets every coset $\{s\} \times \mathbb{R}$ of the open subgroup $\{0\} \times \mathbb{R}$ (Proposition 13.1(2)). Thus Tonelli's theorem 5.1(4) fails for 1_D , whose support is not σ -finite. The compact subsets of D are finite, and each point has measure $c(\{s\})m(\{s\}) = 0$ by (4.2); so D is locally null (Theorem 13.3(1)). The example also shows that the rectangle function $A \times B \mapsto c(A)m(B)$ has no extension to a measure m' on $\mathcal{B}(X \times Y)$ that is inner and outer regular on every Borel set. Such an m' would be Radon. It would integrate $\varphi \boxtimes g$ to $\int \varphi dc \int g dm$ for $\varphi \in C_c(X)$ and $g \in C_c(Y)$, by uniform approximation with simple functions on the compact rectangle $\text{supp } \varphi \times \text{supp } g$, which has finite m' -measure. By the tensor approximation (Lemma 4.1) it would then agree with $c \hat{\times} m$ on $C_c(X \times Y)$, hence equal it by the uniqueness in Theorem 2.2. But inner regularity would force $m'(D) = 0$.

Example 13.6 ($\{e\} \times G$: the rectangle rule needs σ -finite factors). Let $G = \mathbb{R} \times \mathbb{R}_d$ and $\pi = \mu \hat{\times} \mu$. The closed set $\{e\} \times G$ has $\mu(\{e\}) = 0$ but $\pi(\{e\} \times G) = \infty$. Indeed π is a Haar measure on $G \times G$ (Proposition 12.1(1)), $G_0 \times G_0$ is an open σ -compact subgroup of $G \times G$, and $\{e\} \times G$ meets each of the uncountably many cosets $G_0 \times (\mathbb{R} \times \{d\})$; apply Proposition 13.1(2). So Theorem 5.1(6) fails when B is not σ -finite, and " $\pi(A \times B) = \mu(A)\mu(B)$ with $0 \cdot \infty = 0$ " is false for the Radon product. The set is locally null by Theorem 13.3(1): its intersection with a compact $K \subseteq G \times G$ lies in $\{e\} \times \text{pr}_2(K)$, which is null by (4.2). So, like Y and D , it carries no L^2 mass.

Two regularity conventions for Haar measure

Fremlin's *Measure Theory* uses a complete, locally determined measure that is inner regular on every measurable set. Our Borel measure is outer regular on every Borel set and inner regular on open sets. On a sigma-compact group these conventions agree after completion. On an arbitrary group, the following construction explains the difference and the common integration theory.

Proposition 13.7 (The locally determined measure). With the open sigma-compact cosets yG_0 used above, define for Borel $E \subset G$

$$m(E) = \sum_{y \in Y} \mu(E \cap yG_0), \quad \sum_{y \in Y} a_y := \sup_{F \subset Y \text{ finite}} \sum_{y \in F} a_y.$$

Then m is left invariant, locally finite and inner regular on every Borel set. It agrees with μ on open sets, compact sets and sigma-finite Borel sets, and integrates every $C_c(G)$ function in the same way. There is a canonical isometric identification $L^p(\mu) \cong L^p(m)$ for $1 \leq p < \infty$. Nevertheless m can fail outer regularity on Borel sets. Completing each coset measure and allowing sets measurable on every coset gives a Radon measure in Fremlin's convention.

Proof. A sum of nonnegative numbers is the supremum of its finite subsums. The two suprema in a sum over $Y \times \mathbb{N}$ commute: each finite collection of pairs is contained in the product of finite subsets, and the converse products are finite. Applying countable additivity on each coset therefore proves countable additivity of m . Left translation permutes the cosets and preserves μ on their Borel subsets, so it preserves m .

A compact set meets only finitely many cosets, since the cosets form an open cover. Thus m and μ agree on compact sets. They agree on each sigma-finite Borel set by Proposition 13.1, since such a set lies in countably many cosets. Local finiteness follows by taking a relatively compact neighborhood inside a single coset.

The restriction of μ to each coset is sigma-finite and inner regular on every Borel set by Proposition 2.3. Given $a < m(E)$, with $a \geq 0$, choose a finite set of cosets whose contributions exceed a . In each of those cosets choose compact subsets of E whose measures still have sum exceeding a . Their finite union is compact. This proves inner regularity of m on every Borel set. If U is open, inner regularity of both measures on U , and their equality on compact sets, give $m(U) = \mu(U)$. A C_c function is supported on a compact set, so its two integrals agree.

The identity on Borel representatives maps $L^p(\mu)$ isometrically into $L^p(m)$: a μ -integrable function is supported on a sigma-finite set, where the measures agree. For surjectivity, take a Borel f with $\int |f|^p dm < \infty$. The sum of its coset integrals is finite, so only countably many of those integrals are nonzero. Set f to zero outside those cosets. The resulting Borel function f_0 equals f m -almost everywhere, is supported on a sigma-finite set and has the same norm for μ . This also proves independence of representatives and surjectivity.

For the last assertion, let Σ_{loc} consist of the subsets whose intersections with every coset are measurable for its completed sigma-finite measure. Define the measure by the same sum. It is complete because a subset of a null set is null on every coset. It is locally determined: if a set has measurable intersections with every measurable set of finite measure, intersect it with a countable compact exhaustion of each coset to see that its coset sections are measurable. Completion preserves compact inner approximation on each coset: a completed measurable set contains a Borel subset of the same measure, and the latter has compact approximants. The preceding finite-subsum argument now gives compact inner regularity on all of Σ_{loc} . These properties, local finiteness and the Hausdorff topology give Fremlin's Radon convention.

Finally, in Example 13.4 the closed set $Z = \{0\} \times \mathbb{R}_d$ satisfies $m(Z) = 0$, while every open neighborhood of Z has infinite m -measure because m and μ agree on open sets. Thus m is not outer regular there. $\square$

The finite- p spaces for Σ_{loc} have the same identification: an integrable function has nonzero coset integrals on only countably many cosets. Choose a Borel representative on each of those cosets and set it to zero elsewhere; this gives a global Borel representative of its class, with the same integral.

The construction also identifies the local L^∞ of Definition 13.2 with the L^∞ of the completed, locally determined measure: both take essentially bounded measurable functions on each coset, with a common bound, modulo functions null on every coset. Completed coset functions have Borel representatives there, so the completed version gives the same classes. This explains why the duality in Theorem 13.3 agrees with Fremlin's localizable-measure duality, despite the difference between the two global Borel measures.

14. Convolution and continuity of translation

In this section and the next, $\pi = \mu \hat{\times} \mu$ on $G \times G$, and integrals over G are taken with respect to μ . Convolution integrals are iterated integrals over $G \times G$ in disguise, so we first record how π behaves under the changes of variables that occur.

Lemma 14.1 (Changes of variables on $G \times G$). The maps

$$\Theta_1(x, y) = (y, y^{-1}x), \quad \Theta_2(x, y) = (y, xy^{-1}), \quad \Theta_3(x, y) = (y, xy), \quad \sigma(x, y) = (y, x)$$

are homeomorphisms of $G \times G$ with $(\Theta_1)_*\pi = \sigma_*\pi = \pi$, $(\Theta_2)_*\pi = (\Delta \circ \text{pr}_1)\pi$ and $(\Theta_3)_*\pi = (\Delta^{-1} \circ \text{pr}_1)\pi$. Consequently each of them pulls π -null sets back to π -null sets and σ -finite sets back to σ -finite sets, and for every Borel $H \geq 0$ on $G \times G$

$$\begin{aligned} \int H \circ \Theta_1 d\pi &= \int H d\pi, \\ \int H \circ \Theta_2 d\pi &= \int H(y, z) \Delta(y) d\pi(y, z), \\ \int H \circ \Theta_3 d\pi &= \int H(y, z) \Delta(y)^{-1} d\pi(y, z). \end{aligned}$$

Proof. The inverses are $(y, z) \mapsto (yz, y)$, (zy, y) and (zy^{-1}, y) , all continuous. For $F \in C_c(G \times G)$, (4.1) lets us integrate first in x ; left invariance and (10.1) in the inner integral give

$$\begin{aligned} \int F \circ \Theta_1 d\pi &= \iint F(y, y^{-1}x) dx dy = \int F d\pi, \\ \int F \circ \Theta_2 d\pi &= \iint F(y, xy^{-1}) dx dy = \int \Delta(y) \int F(y, x) dx dy, \end{aligned}$$

and similarly $\int F \circ \Theta_3 d\pi = \int \Delta(y)^{-1} \int F(y, x) dx dy$ and $\int F \circ \sigma d\pi = \int F d\pi$. The image measures are Radon by Proposition 3.1(1), and so are $(\Delta^{\pm 1} \circ \text{pr}_1)\pi$ by Proposition 3.1(2). The uniqueness in Theorem 2.2

gives the four identities of measures. Densities that are continuous and positive do not change null sets or σ -finite sets (Proposition 3.1(2)). The integral formulas follow from Proposition 3.1(1) and (2). $\square$

For Borel functions f, g on G , put

$$f * g(x) = \int f(y) g(y^{-1}x) dy \quad (14.1)$$

at every x where the integral converges absolutely. For $f, g \geq 0$ the integral is defined at every x , with values in $[0, \infty]$.

Theorem 14.2.

1. (*Tonelli for convolutions*) If $f, g \geq 0$ are Borel and vanish outside σ -finite sets, then $f * g$ is μ -a.e. Borel and $\int f * g dx = \int f dx \int g dx$.
2. For $f, g \in L^1(G)$, (14.1) converges absolutely for almost every x , $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$, and the class of $f * g$ depends only on the classes of f and g . If one of the following integrals converges absolutely, so do the others, and

$$f * g(x) = \int f(xy) g(y^{-1}) dy = \int f(y^{-1}) g(yx) \Delta(y^{-1}) dy = \int f(xy^{-1}) g(y) \Delta(y^{-1}) dy. \quad (14.2)$$

3. With the involution $f^*(x) = \Delta(x)^{-1} \overline{f(x^{-1})}$, $L^1(G)$ is a Banach $*$ -algebra: convolution is associative, $\|f^*\|_1 = \|f\|_1$, $f^{**} = f$ and $(f * g)^* = g^* * f^*$. Moreover $L_z(f * g) = (L_z f) * g$ and $R_z(f * g) = f * (R_z g)$.
4. Let $1 \leq p < \infty$, $f \in L^1(G)$ and $g \in L^p(G)$. Then $f * g(x)$ converges absolutely for almost every x , and $\|f * g\|_p \leq \|f\|_1 \|g\|_p$. If also $\int |f| \Delta^{-1} dx < \infty$ (for instance if f has compact support, or if G is unimodular), then $g * f(x) = \int g(xy^{-1}) f(y) \Delta(y^{-1}) dy$ converges absolutely for almost every x , and

$$\|g * f\|_p \leq \|\Delta^{-1} f\|_1^{1-1/p} \|f\|_1^{1/p} \|g\|_p. \quad (14.3)$$

For $p = 1$ the condition on f is not needed, and $\|g * f\|_1 \leq \|g\|_1 \|f\|_1$.

5. If G is unimodular, $1 < p < \infty$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in L^p(G)$ and $g \in L^q(G)$, then $f * g(x)$ converges absolutely for every x , $f * g \in C_0(G)$, and $\|f * g\|_{\sup} \leq \|f\|_p \|g\|_q$.
6. (*Continuity of translation*) For $1 \leq p < \infty$ and $f \in L^p(G)$, $\|L_y f - f\|_p \rightarrow 0$ and $\|R_y f - f\|_p \rightarrow 0$ as $y \rightarrow e$.
7. Let $f \in L^1(G)$ and $g \in L^\infty(G)$ (Section 13). Then $f * g(x)$ is defined for every x , $|f * g| \leq \|f\|_1 \|g\|_\infty$, and $\|L_z(f * g) - f * g\|_{\sup} \rightarrow 0$ as $z \rightarrow e$. If also $\Delta^{-1} f \in L^1(G)$, then $g * f(x) = \int g(y) f(y^{-1}x) dy$ is defined for every x , $|g * f| \leq \|g\|_\infty \|\Delta^{-1} f\|_1$, and $\|R_z(g * f) - g * f\|_{\sup} \rightarrow 0$. Without that condition $g * f$ can be infinite everywhere (Exercise 16.4).

Scope warning: The second half of (7) cannot be asserted for every $f \in L^1(G)$, with the bound $\|g\|_\infty \|R_z f - f\|_1$; that version fails on every group that is not unimodular (Remark 14.3 and Exercise 16.4).

Proof. (1) The function $(x, y) \mapsto f(y)g(y^{-1}x)$ is $(f \boxtimes g) \circ \Theta_1$. The function $f \boxtimes g$ vanishes outside a product of σ -finite sets, which is σ -finite by Theorem 5.1(6); so $(f \boxtimes g) \circ \Theta_1$ vanishes outside a σ -finite set by Lemma 14.1. By Tonelli's theorem 5.1(4) and Lemma 14.1, $\int (\int f(y)g(y^{-1}x) dy) dx = \int (f \boxtimes g) \circ \Theta_1 d\pi = \int f \boxtimes g d\pi = \int f \int g$. (2) Apply (1) to $|f|$ and $|g|$: the inner integral is finite for almost every x , and $|f * g| \leq |f| * |g|$ gives the bound. By Fubini's theorem 5.1(5), $f * g$ is μ -a.e. Borel. Changing g on a null set N changes the integrand only on $\Theta_1^{-1}(\{f \neq 0\} \times N)$, which is π -null by Theorem 5.1(6) and Lemma 14.1; by Theorem 5.1(3) this affects, for almost every x , only a null set of y . The same holds for f . For (14.2), fix x . The substitution $y \mapsto xy$, allowed by left invariance (Proposition 8.2(2)), turns (14.1) into the first form. The inversion formula (11.1), applied to $h(y) = f(y^{-1})g(yx)$ and to $h(y) = f(xy^{-1})g(y)$, turns (14.1) and the first form into the second and third forms. Each step preserves absolute convergence. (3) For $f, g \in C_c(G)$, $f * g$ vanishes off $(\text{supp } f)(\text{supp } g)$, and $|f * g(xz) - f * g(x)| \leq \|f\|_1 \|R_z g - g\|_{\sup} \rightarrow 0$ as $z \rightarrow e$ by uniform continuity (Proposition 7.3), so $f * g \in C_c(G)$. For $f, g, h \in C_c(G)$ and fixed x , the substitution $w \mapsto y^{-1}w$ in the inner integral gives

$$f * (g * h)(x) = \int f(y) \left(\int g(y^{-1}z) h(z^{-1}x) dz \right) dy, \quad (f * g) * h(x) = \int \left(\int f(y) g(y^{-1}z) dy \right) h(z^{-1}x) dz,$$

and the integrand is in $C_c(G \times G)$, so (4.1) makes them equal. For $f, g \in C_c(G)$, a direct computation with $\Delta(y)^{-1}\Delta(y^{-1}x)^{-1} = \Delta(x)^{-1}$ and the substitution $y \mapsto xy$ gives $(f * g)^* = g^* * f^*$. By the last formula of Theorem 11.1, $\|f^*\|_1 = \int \Delta(x)^{-1}|f(x^{-1})| dx = \|f\|_1$; and $f^{**}(x) = \Delta(x)^{-1}\Delta(x^{-1})^{-1}f(x) = f(x)$. By (2), convolution is a bounded bilinear map on L^1 , and the involution is a conjugate-linear isometry; since $C_c(G)$ is dense (Proposition 3.1(4)), the identities pass to $L^1(G)$, which is complete. The translation rules follow from the substitution $y \mapsto zy$ in (14.1), and directly for R_z . (4) The case $p = 1$ of the first claim is (2). Let $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. For fixed x , Hölder's inequality for the measure $|f(y)| dy$ gives

$$\left(\int |f(y)||g(y^{-1}x)| dy \right)^p \leq \|f\|_1^{p-1} \int |f(y)||g(y^{-1}x)|^p dy.$$

By (1) for $|f|$ and $|g|^p \in L^1$, the right side has integral $\|f\|_1^{p-1}\|f\|_1\|g\|_p^p$ over x . This gives absolute convergence almost everywhere and the bound. For $g * f$, put $w = |f|\Delta^{-1}$. The function $(x, y) \mapsto |g(xy^{-1})|^p w(y)$ is $(w \boxtimes |g|^p) \circ \Theta_2$, so by Tonelli's theorem and Lemma 14.1,

$$\int \left(\int |g(xy^{-1})|^p w(y) dy \right) dx = \int w(y) \Delta(y) dy \int |g|^p = \|f\|_1 \|g\|_p^p.$$

For $p = 1$ this is the claim. For $p > 1$, Hölder's inequality for the measure $w(y) dy$ of mass $\|\Delta^{-1}f\|_1$ gives $(\int |g(xy^{-1})| w(y) dy)^p \leq \|\Delta^{-1}f\|_1^{p-1} \int |g(xy^{-1})|^p w(y) dy$, and integrating in x gives (14.3). If f vanishes off a compact set K , then $\|\Delta^{-1}f\|_1 \leq (\sup_K \Delta^{-1})\|f\|_1$, and (14.3) gives $\|g * f\|_p \leq (\sup_K \Delta^{(1/p)-1})\|f\|_1\|g\|_p$. (5)

For every x , Hölder's inequality gives $|f * g(x)| \leq \|f\|_p (\int |g(y^{-1}x)|^q dy)^{1/q}$. By (11.1) and unimodularity, $\int |g(y^{-1}x)|^q dy = \int |g(yx)|^q \Delta(y)^{-1} dy = \|g\|_q^q$. For $f, g \in C_c(G)$, $f * g \in C_c(G)$ by (3). For general f, g , take $f_n, g_n \in C_c(G)$ with $f_n \rightarrow f$ in L^p and $g_n \rightarrow g$ in L^q (Proposition 3.1(4)). The bound gives $f_n * g_n \rightarrow f * g$ uniformly, and a uniform limit of functions in $C_c(G)$ lies in $C_0(G)$. The endpoint cases fail on every noncompact G : for $f \geq 0$ with $\int f = 1$ and $g \equiv 1$, $f * g \equiv 1 \notin C_0(G)$. (6) Fix a compact neighbourhood V of e . For $g \in C_c(G)$ and $y \in V$, $L_y g$ and $R_y g$ vanish outside the compact set $K = V(\text{supp } g) \cup (\text{supp } g)V^{-1}$, so $\|L_y g - g\|_p \leq \mu(K)^{1/p} \|L_y g - g\|_{\text{sup}} \rightarrow 0$ by uniform continuity (Proposition 7.3), and the same for R_y . For $f \in L^p$ and $\varepsilon > 0$, choose $g \in C_c(G)$ with $\|f - g\|_p < \varepsilon$. Since $\|L_y\| = 1$, and $\|R_y\| = \Delta(y)^{-1/p} \leq C$ on V by Theorem 10.1(3) and the continuity of Δ , we get $\|R_y f - f\|_p \leq (C + 1)\varepsilon + \|R_y g - g\|_p$, and similarly for L_y . (7) Left and right translations and inversion map Borel sets to Borel sets and preserve null sets and σ -finite sets (Theorem 10.1(1) and Corollary 11.2(1)). Hence they map locally Borel and locally null sets to sets of the same kind: for such a map θ and $\mu(F) < \infty$, the set $\theta^{-1}(F)$ is covered by countably many sets of finite measure, so $\theta(E) \cap F = \theta(E \cap \theta^{-1}(F))$ is Borel (null) when E is locally Borel (locally null). The map $y \mapsto y^{-1}x$ and its inverse $z \mapsto xz^{-1}$ are composites of an inversion and a translation, so $y \mapsto g(y^{-1}x)$ is locally measurable and bounded by $\|g\|_\infty$ locally almost everywhere. As in the proof of Theorem 13.3(2), multiplying by f gives an integrable function with $|f * g(x)| \leq \|f\|_1 \|g\|_\infty$. By (3), $L_z(f * g) - f * g = (L_z f - f) * g$, and $\|L_z f - f\|_1 \rightarrow 0$ by (6). For $g * f$, the substitutions of (2) give $\int |g(y)||f(y^{-1}x)| dy \leq \|g\|_\infty \int |f(y^{-1}x)| dy = \|g\|_\infty \int |f(y)| \Delta(y)^{-1} dy$. Also $R_z(g * f) - g * f = g * (R_z f - f)$, and $\Delta^{-1} R_z f = \Delta(z) R_z(\Delta^{-1} f)$, so

$$\|\Delta^{-1}(R_z f - f)\|_1 \leq |\Delta(z) - 1| \|R_z(\Delta^{-1} f)\|_1 + \|R_z(\Delta^{-1} f) - \Delta^{-1} f\|_1 \rightarrow 0$$

by (6) applied to $\Delta^{-1} f \in L^1$. $\square$

Remark 14.3 (The factor Δ^{-1} in (7) is needed). In the second half of (7) the bound is $|R_z(g * f) - g * f| \leq \|g\|_\infty \|\Delta^{-1}(R_z f - f)\|_1$, with $\Delta^{-1}(R_z f - f)$ where one might expect $R_z f - f$. This cannot be improved. For $h \in L^1$ and fixed x , the third form of (14.2) gives $g * h(x) = \int g(xy^{-1})h(y)\Delta(y)^{-1} dy$. The choice $g(w) = \text{sgn } h(w^{-1}x)$ shows that the supremum of $|g * h(x)|$ over $\|g\|_\infty \leq 1$ is $\|\Delta^{-1}h\|_1$, which exceeds $\|h\|_1$ whenever $h \neq 0$ lives where $\Delta < 1$. Such $h = R_z f - f$ occur on every group that is not unimodular: take $f \in C_c^+(G)$ supported in the nonempty open set $\{\Delta < 1\}$, and z with $\Delta(z) > 1$. Then $R_z f$ is supported in $(\text{supp } f)z^{-1}$, where Δ is smaller still, and $R_z f \neq f$ because $\int R_z f = \Delta(z)^{-1} \int f$. So the bound $\|g\|_\infty \|R_z f - f\|_1$ fails for this f , this z and a suitable g . The extra hypothesis $\Delta^{-1} f \in L^1$ cannot be dropped either: Exercise 16.4 gives $f \in L^1$ on the $ax + b$ group with $1 * f \equiv \infty$. For unimodular groups $\Delta \equiv 1$, and nothing changes. The first half of (7), and (4) for $f * g$, hold for all groups.

The full Young inequality on a unimodular group

Theorem 14.4 (Young). Suppose G is unimodular, $1 \leq p, q, r \leq \infty$, and

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r},$$

where $1/\infty = 0$. For $f \in L^p(G)$ and $g \in L^q(G)$, convolution defines an element of $L^r(G)$, with

$$\|f * g\|_r \leq \|f\|_p \|g\|_q. \quad (14.4)$$

It converges absolutely almost everywhere if $r < \infty$, and at every point if $r = \infty$, using locally bounded representatives at the L^∞ endpoints. Fischer and Ruzhansky state this general inequality in their freely accessible monograph. Here is a direct proof, requiring no interpolation theorem.

Proof. If $p = 1$, the relation gives $q = r$, and Theorem 14.2(4),(7) gives the result. If $q = 1$, then $p = r$; the same theorem applies with its modular factors equal to one. These include $r = 1$ and all cases with an infinite input exponent. If $r = \infty$ and $1 < p, q < \infty$, the exponents are conjugate. Hölder and unimodular inversion give, for every x ,

$$\int_G |f(y)g(y^{-1}x)| dy \leq \|f\|_p \left(\int_G |g(y^{-1}x)|^q dy \right)^{1/q} = \|f\|_p \|g\|_q.$$

These cases are unchanged by alterations on null sets, since translations and inversion preserve null sets.

It remains to consider $1 < p, q < r < \infty$. The exponent relation implies $p < r$ and $q < r$. Apply Hölder with the three exponents

$$r, \quad \frac{p}{1 - p/r}, \quad \frac{q}{1 - q/r},$$

whose reciprocals sum to one, to the product

$$(|f(y)|^p |g(y^{-1}x)|^q)^{1/r} |f(y)|^{1-p/r} |g(y^{-1}x)|^{1-q/r}.$$

This gives

$$|f * g(x)|^r \leq \|f\|_p^{r-p} \|g\|_q^{r-q} \int_G |f(y)|^p |g(y^{-1}x)|^q dy.$$

Integrating in x , left translation gives

$$\int_G \int_G |f(y)|^p |g(y^{-1}x)|^q dy dx = \|f\|_p^p \|g\|_q^q.$$

These interchanges do not impose sigma-finiteness on G . The nonzero supports of f and g are sigma-finite because their finite powers are integrable. Their product is sigma-finite by Theorem 5.1(6), and its inverse image under Θ_1 is sigma-finite by Lemma 14.1. The sigma-finite Tonelli theorem therefore applies to the displayed integrand, and Fubini gives a measurable convolution where it is absolutely convergent. The finite double integral also makes the inner integral finite almost everywhere. The three-factor estimate then proves absolute convergence there and (14.4). If either input norm is zero, apply the same Tonelli calculation to obtain the zero convolution class. $\square$

The unimodular hypothesis belongs to this symmetric form of the inequality. The preceding weighted estimates remain the applicable bounds on a general group; the affine counterexample in Exercise 16.4 explains why their modular factors cannot be discarded.

15. Approximate identities

Theorem 15.1. Let the neighbourhoods U of e be directed by reverse inclusion.

1. For every neighbourhood U of e there is $\psi_U \in C_c(G)$ with $\psi_U \geq 0$, $\text{supp } \psi_U \subseteq U$, $\int \psi_U = 1$ and $\psi_U(x^{-1}) = \psi_U(x)$.

2. Let (ψ_U) be any family with $\psi_U \geq 0$, $\int \psi_U = 1$, and ψ_U vanishing outside a compact subset of U (continuity is not required). For $1 \leq p < \infty$ and $f \in L^p(G)$,

$$\|\psi_U * f - f\|_p \leq \sup_{y \in U} \|L_y f - f\|_p \rightarrow 0. \quad (15.1)$$

If f is bounded with $\|L_y f - f\|_{\sup} \rightarrow 0$ as $y \rightarrow e$, then $\|\psi_U * f - f\|_{\sup} \rightarrow 0$.

3. If moreover $\psi_U(x^{-1}) = \psi_U(x)$, then $\|f * \psi_U - f\|_p \leq \sup_{y \in U} \|R_y f - f\|_p \rightarrow 0$, and $\|f * \psi_U - f\|_{\sup} \rightarrow 0$ for bounded f with $\|R_y f - f\|_{\sup} \rightarrow 0$.
4. (Weak-integral form) For $\psi \in L^1(G)$ and $\xi, \eta \in L^2(G)$,

$$\langle \psi * \xi, \eta \rangle = \int \psi(s) \langle L_s \xi, \eta \rangle ds, \quad (15.2)$$

where $s \mapsto \langle L_s \xi, \eta \rangle$ is continuous and bounded by $\|\xi\| \|\eta\|$. So the operator $\xi \mapsto \psi * \xi$ is the weak integral $\int \psi(s) \lambda(s) ds$ of the left regular representation, its norm is at most $\|\psi\|_1$, and $\psi_U * \xi \rightarrow \xi$ in L^2 for every ξ .

Proof. (1) The set $U' = \text{int } U \cap (\text{int } U)^{-1}$ is an open neighbourhood of e . By (T2) choose φ with $\{e\} \prec \varphi \prec U'$, put $\varphi' = \varphi + \check{\varphi}$, which vanishes off $\text{supp } \varphi \cup (\text{supp } \varphi)^{-1} \subseteq U'$, and $\psi_U = \varphi' / \int \varphi'$; the integral is positive by Proposition 9.1. (2) Let C be a compact set outside of which $\psi = \psi_U$ vanishes. For almost every x , (14.1) converges absolutely by Theorem 14.2(4), and since $\int \psi = 1$, $\psi * f(x) - f(x) = \int \psi(y)(f(y^{-1}x) - f(x)) dy$. Hölder's inequality for the probability measure $\psi(y) dy$ gives $|\psi * f(x) - f(x)|^p \leq \int \psi(y) |f(y^{-1}x) - f(x)|^p dy$. The function $(x, y) \mapsto \psi(y) |f(y^{-1}x) - f(x)|^p$ is Borel and vanishes outside $(\{f \neq 0\} \times C) \cup \Theta_1^{-1}(C \times \{f \neq 0\})$, which is σ -finite by Theorem 5.1(6) and Lemma 14.1. By Tonelli's theorem,

$$\|\psi * f - f\|_p^p \leq \int \psi(y) \left(\int |f(y^{-1}x) - f(x)|^p dx \right) dy = \int \psi(y) \|L_y f - f\|_p^p dy \leq \sup_{y \in U} \|L_y f - f\|_p^p,$$

which tends to 0 by the continuity of translation (Theorem 14.2(6)). For the uniform statement, $|\psi * f(x) - f(x)| \leq \int \psi(y) |L_y f(x) - f(x)| dy \leq \sup_{y \in U} \|L_y f - f\|_{\sup}$ at every x . (3) By the first form of (14.2) and the symmetry of ψ , $f * \psi(x) = \int f(xy) \psi(y) dy$, so $f * \psi(x) - f(x) = \int \psi(y) (R_y f(x) - f(x)) dy$; the integral converges absolutely for almost every x by Theorem 14.2(4), since ψ has compact support. Repeat the argument of (2), with Θ_3 in place of Θ_1 and $\int |f(xy) - f(x)|^p dx = \|R_y f - f\|_p^p$. (4) The function $(x, s) \mapsto \psi(s) \xi(s^{-1}x) \overline{\eta(x)}$ is Borel, vanishes outside a σ -finite set (as in (2)), and by Tonelli's theorem and the Cauchy-Schwarz inequality in x it is π -integrable with integral of its absolute value at most $\int |\psi(s)| \|L_s \xi\| \|\eta\| ds = \|\psi\|_1 \|\xi\| \|\eta\|$. Fubini's theorem 5.1(5) gives (15.2). Continuity: $|\langle L_s \xi - L_{s_0} \xi, \eta \rangle| \leq \|L_{s_0^{-1}s} \xi - \xi\| \|\eta\| \rightarrow 0$ by Theorem 14.2(6). The norm bound is Theorem 14.2(4) with $p = 2$, and the convergence is (2). $\square$

Units. If G is discrete (with counting measure), $\delta = 1_{\{e\}}$ satisfies $\delta * f = f * \delta = f$. If G is not discrete, $L^1(G)$ has no unit, which is why approximate identities are needed. Indeed, suppose $u * f = f$ for all $f \in L^1(G)$. Taking $f = \psi_U$ symmetric, $\psi_U = u * \psi_U \rightarrow u$ in L^1 by (3). For a compact K with $e \notin K$, $\int_K |\psi_U| = 0$ as soon as $U \cap K = \emptyset$, so $\int_K |u| = 0$. By inner regularity on the σ -finite set $\{u \neq 0\}$ (Proposition 2.3(2)), $u = 0$ almost everywhere off $\{e\}$. And $\mu(\{e\}) = 0$: otherwise every point would have the same positive measure (by left invariance), a compact neighbourhood of e would be finite by (R1), and $\{e\}$ would be open. So $u = 0$ almost everywhere, and $u * f = 0 \neq f$ for $f \neq 0$, a contradiction.

A family as in (1) is a *compactly supported approximate identity*. Its translates behave well: for $g \in G$, $(L_g \psi_U) * \xi = L_g(\psi_U * \xi)$ by Theorem 14.2(3), so convolution by $L_g \psi_U$ is $\lambda(g)$ composed with convolution by ψ_U .

16. Exercises

Exercise 16.1 (A density that vanishes on a closed set). On $G = \mathbb{R} \times \mathbb{R}_d$ with Haar measure μ (Example 13.4), let $\varphi(x, d) = |x|$ and $\nu(E) = \int_E \varphi d\mu$. Show that ν is a Borel measure, finite on compact sets, with $\nu(Y) = 0$ for $Y = \{0\} \times \mathbb{R}_d$ but $\nu(U) = \infty$ for every open $U \supseteq Y$. Conclude that the positivity of the density in Proposition 3.1(2) cannot be dropped.

Solution. ν is a measure because φ is a nonnegative Borel function. A compact set K meets finitely many lines and φ is bounded on it, so $\nu(K) \leq \max_K \varphi \cdot \mu(K) < \infty$. Since $\varphi = 0$ on Y , $\nu(Y) = 0$. If $U \supseteq Y$ is open,

then for each d there is $\delta_d > 0$ with $(-\delta_d, \delta_d) \times \{d\} \subseteq U$, and $\nu((-\delta_d, \delta_d) \times \{d\}) = \int_{-\delta_d}^{\delta_d} |x| dx = \delta_d^2 > 0$, because μ is Lebesgue measure on each line. These sets are disjoint and uncountably many, so, as in the proof of Proposition 13.1(2), $\nu(U) = \infty$. So ν fails (R2) at Y .

Exercise 16.2 (Closed ideals of $L^1(G)$). Let $\mathcal{J} \subseteq L^1(G)$ be a closed subspace. Show that $\mathcal{J}$ is a left ideal ($g * f \in \mathcal{J}$ for $g \in L^1$, $f \in \mathcal{J}$) exactly when $L_x \mathcal{J} \subseteq \mathcal{J}$ for all x , and a right ideal exactly when $R_x \mathcal{J} \subseteq \mathcal{J}$ for all x .

Solution. *Left, only if.* For $f \in \mathcal{J}$ and $x \in G$, $L_x(\psi_U * f) = (L_x \psi_U) * f \in \mathcal{J}$ by Theorem 14.2(3), and $L_x(\psi_U * f) \rightarrow L_x f$ in L^1 by Theorem 15.1(2), since L_x is isometric. As $\mathcal{J}$ is closed, $L_x f \in \mathcal{J}$. *Left, if.* Since $C_c(G)$ is dense in L^1 and $\|g * f\|_1 \leq \|g\|_1 \|f\|_1$, it suffices to take $g \in C_c(G)$. Let $K = \text{supp } g$ and $\varepsilon > 0$. By Theorem 14.2(6), $y \mapsto L_y f$ is continuous into L^1 , so each $y \in K$ has an open neighbourhood O_y with $\|L_{y'} f - L_y f\|_1 < \varepsilon$ for $y' \in O_y$. Cover K by $O_{y_1}, \dots, O_{y_n}$, and let $E_j = (K \cap O_{y_j}) \setminus \bigcup_{i < j} O_{y_i}$. Put $h = \sum_j (\int_{E_j} g) L_{y_j} f \in \mathcal{J}$. For almost every x , $g * f(x) - h(x) = \sum_j \int_{E_j} g(y) (f(y^{-1}x) - f(y_j^{-1}x)) dy$. The integrands vanish outside σ -finite sets (Lemma 14.1), so Tonelli's theorem gives $\|g * f - h\|_1 \leq \sum_j \int_{E_j} |g(y)| \|L_y f - L_{y_j} f\|_1 dy \leq \varepsilon \|g\|_1$. Hence $g * f \in \overline{\mathcal{J}} = \mathcal{J}$. *Right.* If $\mathcal{J}$ is a right ideal, take symmetric ψ_U (Theorem 15.1(1)); then $R_x(f * \psi_U) = f * (R_x \psi_U) \in \mathcal{J}$, and $f * \psi_U \rightarrow f$ by Theorem 15.1(3), while R_x is bounded on L^1 . Conversely, for $g \in C_c(G)$ the first form of (14.2) reads $f * g(x) = \int g(y^{-1}) R_y f(x) dy$, and the same Riemann-sum argument, with the continuous map $y \mapsto R_y f$ and Θ_3 for the σ -finiteness, shows $f * g \in \mathcal{J}$.

Exercise 16.3 (The regular representations). Show that $\lambda(g) = L_g$ and $\rho(g) = \Delta(g)^{1/2} R_g$ are unitary representations of G on $L^2(G)$, continuous for the strong operator topology, and that $\lambda(g)\rho(h) = \rho(h)\lambda(g)$.

Solution. $\lambda(g)$ is isometric by left invariance (Proposition 8.2(2)), and $\rho(g)$ is isometric because $\|R_g \xi\|_2 = \Delta(g)^{-1/2} \|\xi\|_2$ (Theorem 10.1(3)). Since $L_{gh} = L_g L_h$, $R_{gh} = R_g R_h$ and Δ is multiplicative, both are homomorphisms, so each $\lambda(g)$, $\rho(g)$ has the inverse $\lambda(g^{-1})$, $\rho(g^{-1})$ and is unitary. They commute: $L_g R_h \xi(t) = \xi(g^{-1}th) = R_h L_g \xi(t)$. At e , $\|\lambda(g)\xi - \xi\| \rightarrow 0$ by Theorem 14.2(6), and $\|\rho(g)\xi - \xi\| \leq |\Delta(g)^{1/2} - 1| \|R_g \xi\| + \|R_g \xi - \xi\| \rightarrow 0$ by the continuity of Δ (Theorem 10.1(2)) and Theorem 14.2(6). At g_0 , $\|\lambda(g)\xi - \lambda(g_0)\xi\| = \|\lambda(g_0^{-1}g)\xi - \xi\|$, and likewise for ρ .

Exercise 16.4 (The $ax + b$ group: left against right, and a failure of $L^\infty * L^1$). On the $ax + b$ group of Example 11.3, with $d\mu = a^{-2} da db$, $d\tilde{\mu} = a^{-1} da db$ and $\Delta(a, b) = 1/a$: (a) find Borel sets E, E' with $\mu(E) < \infty = \tilde{\mu}(E)$ and $\tilde{\mu}(E') < \infty = \mu(E')$, and conclude that neither of $L^2(\mu)$, $L^2(\tilde{\mu})$ contains the other; (b) for $g \equiv 1 \in L^\infty(G)$ and $f = 1_E$ with E from (a), show $f \in L^1(\mu)$ and $g * f(x) = \infty$ for every x .

Solution. By Proposition 3.1(2), μ and $\tilde{\mu}$ are the measures with densities a^{-2} and a^{-1} with respect to the Radon product of Lebesgue measures on $(0, \infty) \times \mathbb{R}$: both sides are Radon and integrate C_c alike. Measures of regions are then iterated integrals by Tonelli's theorem 5.1(4). (a) Let $E = \{a > 1, 0 < b < 1\}$: $\mu(E) = \int_1^\infty a^{-2} da = 1$ and $\tilde{\mu}(E) = \int_1^\infty a^{-1} da = \infty$. Let $E' = \{0 < a < 1, 0 < b < a\}$: $\tilde{\mu}(E') = \int_0^1 a \cdot a^{-1} da = 1$ and $\mu(E') = \int_0^1 a \cdot a^{-2} da = \infty$. So $1_E \in L^2(\mu) \setminus L^2(\tilde{\mu})$ and $1_{E'} \in L^2(\tilde{\mu}) \setminus L^2(\mu)$. Corollary 11.2(3) gives isometries between the two spaces, but they are different spaces. (b) $\|f\|_1 = \mu(E) = 1$. For every x , left invariance and (11.1) give $g * f(x) = \int f(y^{-1}x) dy = \int f(y^{-1}) dy = \int f \Delta^{-1} d\mu = \tilde{\mu}(E) = \infty$. Here $\Delta^{-1} f \notin L^1$, which is exactly the extra hypothesis in Theorem 14.2(7).

Results used from other lessons

Topology, from **The Stone–Weierstrass theorem for functions vanishing at infinity** and from **Weak topologies: Tychonoff, Banach–Alaoglu, Mazur, bipolars, Krein–Milman and Eberlein–Šmulian**.

- **Compact closures** (Proposition 4.2(2) of the Stone–Weierstrass lesson). If K is a compact subset of an LCH space and V is a neighbourhood of K , some open $U \supseteq K$ has compact closure contained in V . This is (T1).
- **Urysohn's lemma** (Corollary 5.2 of the Stone–Weierstrass lesson). If K is compact, B is closed and $K \cap B = \emptyset$ in an LCH space X , some $f \in C_c(X)$ with $0 \leq f \leq 1$ equals 1 on K and 0 on B .

- **The Stone–Weierstrass theorem** (Theorem 10.1 of that lesson). Let X be an LCH space and A a subalgebra of $C_0(X)$ that is closed under complex conjugation, separates the points of X , and contains for each $x \in X$ a function that does not vanish at x . Then A is dense in $C_0(X)$. It is used only in Example 12.2.
- **Compact Tietze extension** (Proposition 16.1(2) of the Stone–Weierstrass lesson). On a compact Hausdorff space, a continuous real function on a closed subset extends continuously with the same supremum bound; real and imaginary extensions give the complex version. Its proof uses the compact Stone–Weierstrass theorem and a uniformly summable correction series. This supplies compact extension whenever needed; no normality of an arbitrary LCH space is assumed.
- **Tychonoff's theorem** (Theorem 2.3 of the lesson on weak topologies). Every product of compact spaces is compact in the product topology.

Measure theory, from **Measure and Hilbert space tools for Haar integration**: Theorem 1.1 proves Carathéodory; Section 2 proves measurability of sequential limits; Theorems 2.1–2.2 prove monotone and dominated convergence; Theorems 3.1–3.2 prove Hölder, completeness and simple approximation; Theorems 4.1–4.2 prove Radon–Nikodym and the sigma-finite dual of L^1 . Fremlin is a complementary human source for these classical statements.

- **Carathéodory's theorem** the preceding tools lesson, Theorem 1.1. For an outer measure μ^* on X , the sets A with $\mu^*(Y) = \mu^*(Y \cap A) + \mu^*(Y \setminus A)$ for every $Y \subseteq X$ form a σ -algebra on which μ^* is countably additive. To check the condition it suffices to show $\mu^*(Y) \geq \mu^*(Y \cap A) + \mu^*(Y \setminus A)$ when $\mu^*(Y) < \infty$: the reverse inequality is subadditivity, and the case $\mu^*(Y) = \infty$ is trivial.
- **Measurable functions** [Fremlin]. The supremum, the infimum and the pointwise limit of a sequence of Borel functions with values in $\mathbb{R}$ or in $[0, \infty]$ are Borel.
- **Convergence theorems**. Monotone convergence for functions with values in $[0, \infty]$ [Fremlin], and for integrable functions [Fremlin]. Dominated convergence [Fremlin]: if $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ with g integrable, then $\int f_n d\mu \rightarrow \int f d\mu$; applied to $|f_n - f| \leq 2g$ it gives $\int |f_n - f| d\mu \rightarrow 0$.
- **L^p spaces**. Hölder's inequality [Fremlin]. For $1 \leq p < \infty$, $L^p(\mu)$ is complete [Fremlin], and the simple functions in $L^p(\mu)$ are dense in it [Fremlin].
- **The dual of L^1** [Fremlin]. If μ is σ -finite, every bounded linear functional Φ on $L^1(\mu)$ has the form $\Phi(g) = \int fg d\mu$ for a unique $f \in L^\infty(\mu)$, and $\|\Phi\| = \|f\|_\infty$.
- **The Radon–Nikodym theorem** [Fremlin]. Let μ and ν be measures on the same σ -algebra, ν finite and μ σ -finite, with $\nu(E) = 0$ whenever $\mu(E) = 0$. Then $\nu(E) = \int_E h d\mu$ for an integrable $h \geq 0$. In Theorem 13.3(5) both measures are σ -finite on the set considered; the set is split into countably many Borel pieces of finite measure for both, and the theorem is applied on each piece.
- **Substitution in Riemann integrals**. If $\phi: [\alpha, \beta] \rightarrow \mathbb{R}$ is continuously differentiable and f is continuous on $\phi([\alpha, \beta])$, then $\int_{\phi(\alpha)}^{\phi(\beta)} f(u) du = \int_\alpha^\beta f(\phi(t)) \phi'(t) dt$. The integral identity is proved in **Measure and Hilbert space tools for Haar integration**, Lemma 6.1, from the elementary differential-calculus prerequisites.

Hilbert spaces, from **Hilbert spaces and compact operators**.

- **Hilbert tensor products** (Theorem 8.1). For Hilbert spaces H and K there is a Hilbert space $H \otimes K$ with a bilinear map $(\xi, \eta) \mapsto \xi \otimes \eta$ such that $\langle \xi \otimes \eta, \xi' \otimes \eta' \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle$ and the elementary tensors span a dense subspace.
- **Multiplication operators on σ -finite spaces** (Theorem 9.1). For a σ -finite measure space (Z, ν) , the operators $m_f \xi = f\xi$ on $L^2(\nu)$ with $f \in L^\infty(\nu)$ form an algebra that is maximal abelian: every bounded operator on $L^2(\nu)$ that commutes with all m_f is itself some m_f .

Where this leads

- The lesson *The Plancherel weight and Fourier coefficients of a locally compact group*, in the course on modular theory and weights, is built on this one and, like it, works for every locally compact group. It uses the normalization (10.1) of the modular function and the inversion formula (11.1); the regular representations of Exercise 16.3, whose strong continuity rests on Proposition 7.3 and Theorem 14.2(6), and the conjugation J of Corollary 11.2(2); the density of $C_c(G)$ in $L^2(G)$ (Proposition 3.1(4)) and the

core of Corollary 11.2(5); the approximate identities and weak integrals of Theorem 15.1; the identification $L^2(G) \otimes L^2(G) \cong L^2(G \times G, \mu \hat{\times} \mu)$ of Theorem 6.2, with the operators $m_a \otimes 1$ of part (4), the unitary W of Proposition 12.1, and Tonelli's theorem 5.1(4) for norm computations on elementary tensors; and the von Neumann algebra $L^\infty(G)$ of Theorem 13.3(4). The unitarity of W can also be checked on compactly supported tensors, one s at a time, and then extended by density; Theorem 6.2(3) is what justifies working with one s at a time.

- Tensor products of $L^\infty(G)$ with other von Neumann algebras are spatial tensor products; see the lesson [Spatial tensor products of von Neumann algebras](#).
- For a Radon measure that is not σ -finite and does not come from a group, the maximal abelian property of L^∞ (Theorem 13.3(4) for Haar measure) needs a decomposition of the space into disjoint compact pieces; for Haar measure, the open σ -compact cosets of Section 13 play that role. The lesson on vector-valued functions and preduals constructs [the decomposition](#) and proves [the maximal abelian property for every Radon measure](#).

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- [Fischer–Ruzhansky] V. Fischer and M. Ruzhansky, [Quantization on Nilpotent Lie Groups](#), Birkhäuser, 2016, freely accessible under CC BY 4.0. Its Heisenberg and homogeneous-group treatment complements Example 12.3; its convolution discussion includes the full Young bound of Theorem 14.4. The calculations and proof here are independently written.

Quotient measures and Weil's integration formula

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Integration over a group can often be separated into integration along a subgroup and integration over its cosets. For a nonabelian group, an invariant measure on the coset space need not exist. The precise obstruction is the mismatch between two modular functions:

$$G/H \text{ has a nonzero invariant Radon measure} \iff \Delta_G|_H = \Delta_H.$$

When the obstruction does not vanish, a positive continuous density still gives a strongly quasi-invariant measure. We construct it, prove its change-of-variables formula and calculate examples where the distinction matters.

The lesson works for every locally compact Hausdorff group G and every closed subgroup H . Neither is assumed second countable, σ -compact or unimodular. All interchanges of integrals in the main construction involve continuous functions with compact support. We will not infer a Fubini theorem for arbitrary nonnegative Borel functions on a non- σ -finite product.

What to know first

The preceding programme lesson **Haar measure on locally compact groups** supplies the following exact results:

- Theorem 2.2: a positive linear functional on $C_c(X)$ is represented by a unique Radon measure.
- Proposition 3.1(2): multiplication by a strictly positive continuous function gives a Radon measure in the same measure class.
- Proposition 4.2: iterated integrals of a continuous compactly supported function on a product agree.
- Proposition 9.1 and Theorem 9.2: positivity on nonempty open sets and uniqueness of Haar measure.
- Theorems 10.1 and 11.1: the modular function and the inversion formula.

Their elementary measure-theory prerequisites are proved in the preceding tools lesson. The topology needed specifically for the quotient construction is proved below, including the cutoff argument sometimes hidden behind a reference to a partition of unity.

Fix left Haar measures dg and dh . Our convention is

$$\int_G f(ga) dg = \Delta_G(a)^{-1} \int_G f(g) dg. \quad (1)$$

Equivalently, the measure of a right translate Ea is $\Delta_G(a)$ times the measure of E . The same convention defines Δ_H . Put $Y = G/H$, the space of cosets gH , and $q(g) = gH$.

As in the preceding lesson, a Radon measure here is a Borel measure finite on compact sets, outer regular on Borel sets and inner regular on open sets. Equality of Radon measures can therefore be checked on C_c . Left and right translations of functions mean $L_a f(g) = f(a^{-1}g)$ and $R_a f(g) = f(ga)$.

Local topology and compactly supported functions

Lemma 1.1. An LCH space has a base of relatively compact open neighborhoods. If $K \subset V$, where K is compact and V is open, there is $u \in C_c(X)$ with $0 \leq u \leq 1$, $u = 1$ on K , and $\text{supp } u \subset V$.

Proof. We first recall the separation argument for a compact Hausdorff space C . For a point x and a closed set A not containing it, separate x from each point of A by disjoint open neighborhoods. A finite subcover of A gives an open neighborhood W of x whose closure misses A . For two disjoint closed sets, do this for each point of the first, then take a finite union of the resulting neighborhoods. Thus whenever $A \subset O$, with A closed and O open, there is open W with

$$A \subset W \subset \overline{W} \subset O. \quad (2)$$

This proves the continuous separation needed here. For disjoint closed $A, B \subset C$, choose open sets U_r , indexed by dyadic rationals $r \in [0, 1]$, with $A \subset U_0$, $U_1 = C \setminus B$, and $\overline{U}_r \subset U_s$ when $r < s$. Start by shrinking A inside $C \setminus B$; at each finite dyadic stage use (2) to insert the new intermediate open sets. Set

$$v(x) = \inf\{r : x \in U_r\},$$

with value 1 when the set is empty. Then $v = 0$ on A and $v = 1$ on B . The identity $\{v < t\} = \bigcup_{r < t} U_r$ and, for $0 \leq t < 1$, $\{v > t\} = \bigcup_{r > t} (C \setminus \overline{U}_r)$ prove continuity. In the second identity, choose a dyadic r strictly between t and $v(x)$; if $x \in \overline{U}_r$, the nesting would put x in every U_s with $s > r$, contradicting $v(x) > r$.

Now let $x \in O \subset X$, with O open and X LCH. Choose a compact neighborhood C of x . Apply (2) within C to x and $O \cap \text{int } C$. The resulting neighborhood lies in $\text{int } C$, so it is open in X , and its closure is compact and lies in O . This proves the neighborhood-base assertion.

For compact $K \subset V$, take finitely many such neighborhoods to obtain open W with $K \subset W \subset \overline{W} \subset V$ and $\overline{W}$ compact. In $\overline{W}$, continuously separate K and $\overline{W} \setminus W$, taking the value 1 on the first and 0 on the second. Extend the function by zero outside W . The two definitions agree on the boundary, so the pasting lemma on the two closed sets $\overline{W}$ and $X \setminus W$ proves continuity. This is the required u . $\square$

Lemma 1.2. The quotient map q is open, Y is LCH, and every compact subset $E \subset Y$ is the image of a compact subset of G .

Proof. For open $O \subset G$, $q^{-1}(q(O)) = OH$ is open, so $q(O)$ is open. If $xH \neq yH$, then $y^{-1}x \notin H$. Since H is closed, continuity gives neighborhoods V of x and W of y with $W^{-1}V \cap H = \emptyset$. Their images under q are disjoint open neighborhoods. Thus Y is Hausdorff. The image of a compact neighborhood of g contains the image of its interior, an open neighborhood of gH , so Y is locally compact.

Choose an identity neighborhood $V \subset G$ with compact closure. Finitely many $q(g_i V)$ cover E . Then

$$K = q^{-1}(E) \cap \bigcup_i g_i \bar{V}$$

is compact and $q(K) = E$. Here $q^{-1}(E)$ is closed because Y is Hausdorff. No continuous section of q is being assumed. $\square$

Averaging along the subgroup

Why not simply push Haar measure forward through q ? Consider $G = \mathbb{R}^2$, with two-dimensional Lebesgue measure, and $H = \mathbb{R} \times \{0\}$, so that $q(x, y) = y$. The inverse image of a nondegenerate compact interval $[a, b]$ contains every rectangle $[-N, N] \times [a, b]$. Its measure is at least $2N(b-a)$ for every N , hence is infinite. The pushforward is therefore not finite on compact intervals and is not the required Radon measure. Nevertheless, the quotient plainly has the invariant measure dy .

The construction below separates these two operations. First integrate a compactly supported function along the fiber, rather than measuring the whole infinite fiber. Then integrate the resulting function over the base. For the example, this is precisely $\int f(x, y) dx$ followed by integration in y . A function constant on every fiber is usually not compactly supported upstairs; the cutoff used below supplies a compactly supported representative without pretending that the fiber has finite mass. This is why surjectivity of the averaging map, rather than existence of a global section or finiteness of subgroup volume, is the useful starting point.

For $f \in C_c(G)$, define

$$Pf(gH) = \int_H f(gh) dh. \quad (3)$$

Left invariance of dh makes this independent of the representative g : replacing g by gr , $r \in H$, replaces h by rh .

Lemma 2.1. The map $P : C_c(G) \rightarrow C_c(Y)$ is positive and surjective. For each compact $E \subset Y$, there is $b \in C_c(G)$, $b \geq 0$, with $Pb = 1$ on E . A nonnegative function in $C_c(Y)$ has a nonnegative lift through P .

Proof. Positivity is immediate. For g in a compact neighborhood A of g_0 , only

$$h \in H \cap A^{-1} \text{supp } f$$

can contribute to (3). This set is compact. The continuous function $(g, h) \mapsto f(gh)$ varies uniformly in h on this compact set as $g \rightarrow g_0$: a finite cover of that set by continuity neighborhoods proves the assertion. Its integral therefore varies continuously. The resulting continuous function on G is constant on fibers, so descends continuously through the quotient map. Its support lies in $q(\text{supp } f)$, a compact set.

Lift E to compact $K \subset G$ by Lemma 1.2, and choose $a \geq 0$ in $C_c(G)$ with $a = 1$ on K , by Lemma 1.1. On every fiber over E , a is positive on a relatively open H -neighborhood of a point of K ; hence $Pa > 0$ there by positivity of Haar measure. Choose $\chi \in C_c(Y)$, equal to 1 on E , with $0 \leq \chi \leq 1$ and $\text{supp } \chi \subset \{Pa > 0\}$. Set

$$b(g) = a(g) \frac{\chi(q(g))}{Pa(q(g))}, \quad (4)$$

where the fraction is zero outside $\{Pa > 0\}$. The numerator has support compactly inside that open set, so this extension is continuous. Thus $b \geq 0$, $b \in C_c(G)$ and $Pb = \chi$.

For $F \in C_c(Y)$, choose this b for $E = \text{supp } F$. Then $f = (F \circ q)b$ has compact support and $Pf = F$. If $F \geq 0$, so is f . $\square$

The map respects left translation and has a simple right-subgroup rule:

$$P(L_a f)(y) = Pf(a^{-1}y), \quad P(R_h f) = \Delta_H(h)^{-1}Pf \quad (h \in H). \quad (5)$$

Both follow directly from (3), with the modular convention (1).

Exactly when the quotient measure is invariant

Theorem 3.1. The space G/H has a nonzero G -invariant Radon measure if and only if $\Delta_G(h) = \Delta_H(h)$ for every $h \in H$. When this holds, it is unique up to a positive scalar. For the fixed measures dg, dh , there is exactly one normalization $d\mu$ for which

$$\int_G f(g) dg = \int_Y Pf(y) d\mu(y) \quad (f \in C_c(G)). \quad (6)$$

Proof of necessity. If μ is invariant, the functional

$$I(f) = \int_Y Pf d\mu$$

is positive, left invariant and nonzero. Nonzeroness follows from the positive surjectivity of P and from Radon representation on Y . By Haar uniqueness, $I(f) = c \int_G f dg$, for some $c > 0$. For $h \in H$, (5) gives

$$I(R_h f) = \Delta_H(h)^{-1}I(f),$$

while (1) and the description of I give the same equation with $\Delta_G(h)^{-1}$. Choose nonnegative nonzero f ; then $I(f) > 0$, forcing the two modular functions to agree on H .

Proof of sufficiency. Assume the modular functions agree on H . Fix $f \in C_c(G)$, and choose $b \geq 0$ in $C_c(G)$ with $Pb = 1$ on $q(\text{supp } f)$. Then

$$\begin{aligned} \int_G f(t) dt &= \int_G \int_H f(t)b(th) dh dt \\ &= \int_G b(s) \int_H f(sh^{-1})\Delta_G(h)^{-1} dh ds \\ &= \int_G b(s)Pf(sH) ds. \end{aligned} \quad (7)$$

For the second line use $s = th$ and (1). For the third use $\Delta_G|_H = \Delta_H$ and the inversion formula in H . These interchanges are justified by compact support: initially $t \in \text{supp } f$ and $h \in H \cap (\text{supp } f)^{-1} \text{supp } b$. Thus the integrand is a continuous compactly supported function on $G \times H$, where the preceding lesson's compact-product theorem applies.

In particular $Pf = 0$ implies $\int f = 0$. Consequently

$$\Lambda(Pf) = \int_G f dg$$

is a well-defined linear functional on all of $C_c(Y)$. It is positive because nonnegative functions have nonnegative lifts; it is nonzero and left invariant by (5). Radon representation supplies μ , and equality of the integrals of $C_c(Y)$ functions proves that its translates equal it. This gives (6).

Finally, any invariant Radon measure produces a scalar multiple of Haar integration in the necessity argument. Surjectivity of P then makes that measure a scalar multiple of μ . Equation (6) fixes the scalar. $\square$

Two useful consequences require no additional construction. A compact subgroup always satisfies the criterion: its modular function is one, and the continuous homomorphism $\Delta_G : H \rightarrow (0, \infty)$ is one on a compact

group. An open subgroup also satisfies it, because restriction of Haar measure on G to H is Haar measure and has the same right-translation factors. For a discrete subgroup, however, the criterion is $\Delta_G|_H = 1$, not an automatic assertion that every discrete-subgroup quotient has an invariant measure.

A compact cutoff in this proof must equal one on $q(\text{supp } f)$, not just on the support of Pf . For a signed function, subgroup integration can cancel to zero on a coset while f is nonzero there. The choice above covers those cosets as well.

A cutoff over all cosets

For a general subgroup the modular obstruction need not vanish. To construct a compensating density, we need a continuous nonnegative k on G such that

$$Pk = 1, \quad \text{supp } k \cap q^{-1}(E) \text{ is compact for each compact } E \subset Y. \quad (8)$$

It will not generally have compact support on all of G .

Lemma 4.1. A function k with properties (8) exists.

Proof. Choose a relatively compact symmetric open identity neighborhood V in G . The subgroup $L = \bigcup_{n \geq 1} V^n$ is open and σ -compact: $V^n \subset (\overline{V})^n \subset L$, with the latter compact, since $\overline{V} \subset V^2$. For the last inclusion, if $x \in \overline{V}$, the open set xV meets V , so $x \in VV^{-1} = V^2$.

The L -orbits in Y are disjoint open sets, because q is open; their complements are unions of other orbits, so they are also closed. Each orbit Y_0 is the continuous image of L , hence is σ -compact and LCH.

On such an orbit choose a compact exhaustion $K_n \subset \text{int } K_{n+1}$, with union Y_0 . To construct it, start with a countable compact cover; at each step cover the preceding compact set and the next compact set in that cover by finitely many relatively compact open neighborhoods, and take the union of their compact closures. Put $K_0 = K_{-1} = \emptyset$. The compact shells

$$S_n = K_n \setminus \text{int } K_{n-1}$$

are contained in the open sets $W_n = \text{int } K_{n+1} \setminus K_{n-2}$, and cover Y_0 . By Lemma 1.1 choose $\chi_n \in C_c(Y_0)$, nonnegative, equal to one on S_n , with support in W_n . The family is locally finite: a neighborhood contained in $\text{int } K_N$ misses W_n for $n \geq N + 2$. Therefore

$$\psi_n = \frac{\chi_n}{\sum_j \chi_j}$$

are continuous, compactly supported, locally finite, nonnegative and sum to one on Y_0 .

Perform this construction in every orbit and extend the functions by zero to Y . The resulting family (ψ_i) is still locally finite because the orbits are open, and each ψ_i has compact support. By Lemma 2.1 choose $b_i \geq 0$ in $C_c(G)$ with $Pb_i = 1$ on $\text{supp } \psi_i$. Set

$$k(g) = \sum_i \psi_i(q(g))b_i(g). \quad (9)$$

Local finiteness makes this continuous. Averaging gives $Pk = \sum_i \psi_i = 1$. A compact subset of Y meets only finitely many supports of this locally finite family; over that compact set, the support of k lies in the union of the corresponding finitely many compact supports of b_i . It is closed there, so is compact. This proves (8). $\square$

The construction uses neither a countable base for G nor a global continuous section of $G \rightarrow G/H$.

The modular density and weighted integration

Write

$$\chi(h) = \frac{\Delta_G(h)}{\Delta_H(h)}.$$

This is a positive continuous character on H . From the cutoff of Lemma 4.1 define

$$\rho(g) = \int_H k(gh)\chi(h) dh. \quad (10)$$

Lemma 5.1. The function ρ is continuous, strictly positive and finite, and

$$\rho(gh) = \frac{\Delta_H(h)}{\Delta_G(h)}\rho(g) \quad (g \in G, h \in H). \quad (11)$$

Proof. For g in a compact neighborhood A , property (8) puts all gh that contribute into one compact subset B of G . Then h lies in the compact set $H \cap A^{-1}B$. The integrand is continuous and varies uniformly on this set; its integral is therefore finite and continuous in g . Since $Pk = 1$, k is not identically zero on any fiber; $\chi > 0$, so $\rho > 0$. For $r \in H$, substitute $t = rh$ using left invariance:

$$\rho(gr) = \int_H k(gt)\chi(r^{-1}t) dt = \chi(r)^{-1}\rho(g).$$

This is (11). $\square$

A continuous positive function satisfying (11) is called a ρ -function. It is a density on the group, not usually a function on the quotient.

Theorem 5.2. For any ρ -function there is a unique Radon measure μ_ρ on Y satisfying

$$\int_Y Pf d\mu_\rho = \int_G f(g)\rho(g) dg \quad (f \in C_c(G)). \quad (12)$$

It is positive on nonempty open subsets of Y . The construction is independent of the cutoff.

Proof. Use a cutoff k as in (8), and define a positive functional by

$$\Lambda(F) = \int_G F(q(g)) k(g)\rho(g) dg \quad (F \in C_c(Y)). \quad (13)$$

The integrand has compact support by (8), so this is finite and Radon representation applies.

For $F = Pf$, exchange the integrals in (13) and put $t = gh$. The translation factor is $\Delta_G(h)^{-1}$, while (11) gives $\rho(th^{-1}) = \chi(h)\rho(t)$. Thus

$$\begin{aligned} \Lambda(Pf) &= \int_G f(t)\rho(t) \left(\int_H k(th^{-1})\Delta_H(h)^{-1} dh \right) dt \\ &= \int_G f(t)\rho(t)Pk(tH) dt = \int_G f(t)\rho(t) dt. \end{aligned} \quad (14)$$

The first integral before substitution is compactly supported on $G \times H$: the g 's lie in $\text{supp } k \cap q^{-1}(q(\text{supp } f))$, which is compact, and the h 's then lie in a compact set as in Lemma 2.1. Hence the compact-product theorem justifies the exchange. Inversion on H gives the second line.

Uniqueness follows from surjectivity of P and uniqueness in Radon representation; consequently no choice of cutoff changes the measure. A nonempty open subset of Y has a nonzero nonnegative $F \in C_c(Y)$ supported there, by Lemma 1.1. Lift it to nonnegative nonzero $f \in C_c(G)$. Then (12), $\rho > 0$ and Haar positivity give $\int F d\mu_\rho > 0$. $\square$

If ρ_1, ρ_2 are two ρ -functions, their ratio is constant on each coset by (11). It descends to a strictly positive continuous function r on Y , because q is a quotient map. Applying (12) to $r(q(g))f(g)$ and using surjectivity of P proves

$$d\mu_{\rho_2} = r d\mu_{\rho_1}. \quad (15)$$

Thus all the measures constructed from continuous positive ρ -functions have the same null sets. No assertion about arbitrary non-regular measures is needed.

Quasi-invariance and the direction of translation

Let $T_a(y) = ay$. The ratio

$$c(a, gH) = \frac{\rho(ag)}{\rho(g)} \quad (16)$$

is well-defined and continuous. Replacing g by gh multiplies both numerator and denominator by the same factor from (11). Direct cancellation gives

$$c(ab, y) = c(a, by)c(b, y). \quad (17)$$

Theorem 6.1. The measure μ_ρ is strongly quasi-invariant. More precisely,

$$\frac{d[E \mapsto \mu_\rho(aE)]}{d\mu_\rho}(gH) = \frac{\rho(ag)}{\rho(g)}, \quad \frac{d(T_a)_*\mu_\rho}{d\mu_\rho}(gH) = \frac{\rho(a^{-1}g)}{\rho(g)}. \quad (18)$$

Both densities are strictly positive and continuous.

Proof. For $F \in C_c(Y)$, substitute $g = at$ in (13), using left Haar invariance:

$$\begin{aligned} \int_Y F(a^{-1}y) d\mu_\rho(y) &= \int_G F(tH)k(at)\rho(at) dt \\ &= \int_G F(tH)c(a, tH)k(at)\rho(t) dt. \end{aligned}$$

The function $k_a(t) = k(at)$ is another cutoff: $Pk_a(tH) = Pk(atH) = 1$, and it has the compact-support-over-compacts property because a acts homeomorphically on Y . Cutoff independence in Theorem 5.2 therefore identifies the last integral with $\int_Y F(y)c(a, y) d\mu_\rho(y)$.

Both sides represent Radon measures, so equality on $C_c(Y)$ is equality of those measures. On the left the measure of E is $\mu_\rho(aE)$. This proves the first formula; replace a by a^{-1} for the ordinary pushforward convention. A strictly positive density preserves null sets in both directions: if its integral over E vanishes, intersect E with the sets where the density is at least $1/n$, then take their countable union. This proves quasi-invariance. $\square$

Notice the inverse in the second formula. The two expressions describe different measures, not an ambiguity in the modular convention.

Relative invariance and a character

The distinction between invariance and quasi-invariance has a useful intermediate case. Following the freely accessible treatment by Bekka, de la Harpe and Valette, call a nonzero Radon measure μ **relatively invariant with character** χ if $\chi : G \rightarrow \mathbb{R}_{>0}$ is a continuous homomorphism and

$$\mu(aE) = \chi(a)\mu(E)$$

for every Borel $E \subset Y$.

Theorem 6.2. Such a measure exists if and only if

$$\chi(h) = \frac{\Delta_H(h)}{\Delta_G(h)} \quad (h \in H). \quad (6.2)$$

For a fixed χ , it is unique up to a positive scalar. If it has finite positive total mass, then $\chi = 1$.

Proof. If (6.2) holds, $\rho = \chi$ is a rho-function. Theorem 5.2 constructs μ_χ , and Theorem 6.1 gives $c(a, y) = \chi(a)$, proving relative invariance.

Conversely, suppose μ has this property. Define

$$L(f) = \int_Y P(f/\chi)(y) d\mu(y) \quad (f \in C_c(G)).$$

This is a positive nonzero functional: multiplication by χ^{-1} preserves $C_c(G)$, and positive surjectivity of P supplies a nonnegative test with positive integral against the nonzero Radon measure. Also

$$P((L_a f)/\chi)(y) = \chi(a)^{-1} P(f/\chi)(a^{-1}y).$$

Relative invariance cancels these two factors, so $L(L_a f) = L(f)$. Riesz representation and Haar uniqueness give $L(f) = C \int_G f dg$, with $C > 0$. Substituting χf yields

$$\int_Y P f d\mu = C \int_G \chi(g) f(g) dg.$$

For $h \in H$, the left side with $R_h f$ is multiplied by $\Delta_H(h)^{-1}$. Right translation on the right side gives the factor $\Delta_G(h)^{-1} \chi(h)^{-1}$. A positive test with nonzero integral equates the two factors and proves (6.2). The displayed identity and surjectivity of P also prove uniqueness up to C . Finally $\mu(aY) = \mu(Y)$, so finite positive total mass forces $\chi(a) = 1$ for all a . $\square$

The quasi-regular representation

Theorem 6.3. For the rho-function measure, define

$$(\pi_\rho(a)\xi)(y) = c(a^{-1}, y)^{1/2} \xi(a^{-1}y), \quad \xi \in L^2(Y, \mu_\rho). \quad (6.3)$$

Then π_ρ is a strongly continuous unitary representation of G . If $\mu_{\rho_2} = r\mu_{\rho_1}$ as in (15), multiplication by $r^{-1/2}$ gives a unitary intertwiner from π_{ρ_1} to π_{ρ_2} .

Proof. Quasi-invariance makes composition with a^{-1} well-defined on equivalence classes. Theorem 6.1, first for nonnegative simple functions and then by monotone convergence, gives the change-of-variables rule

$$\int_Y \phi(y) d\mu_\rho(y) = \int_Y \phi(az) c(a, z) d\mu_\rho(z).$$

The cocycle identity says $c(a^{-1}, az)c(a, z) = 1$. Apply the rule to $\phi(y) = c(a^{-1}, y)|\xi(a^{-1}y)|^2$; it proves $\|\pi_\rho(a)\xi\|_2 = \|\xi\|_2$. The same identity gives

$$c(a^{-1}, y)c(b^{-1}, a^{-1}y) = c((ab)^{-1}, y),$$

so $\pi_\rho(a)\pi_\rho(b) = \pi_\rho(ab)$. Its inverse is $\pi_\rho(a^{-1})$, hence it is unitary.

For $\xi \in C_c(Y)$, fix a compact identity neighborhood $A \subset G$, with $e \in A$. All functions $\pi_\rho(a)\xi$, $a \in A$, vanish off the compact set $A \text{ supp } \xi$. The continuous function $(a, y) \mapsto c(a^{-1}, y)^{1/2} \xi(a^{-1}y)$ converges uniformly on this compact set to $\xi(y)$ as $a \rightarrow e$: continuity at every (e, y) and a finite cover of the compact set give a common identity neighborhood for each prescribed error. Radon finiteness on that compact set proves convergence in L^2 . The density of $C_c(Y)$ in $L^2(Y, \mu_\rho)$, proved by the Haar lesson's Radon-measure Proposition 3.1(4), and the unitary norm bound extend convergence to every ξ . The representation law gives continuity at every a .

For the last assertion, $U\xi = r^{-1/2}\xi$ satisfies $\int |U\xi|^2 d\mu_{\rho_2} = \int |\xi|^2 d\mu_{\rho_1}$ and has inverse multiplication by $r^{1/2}$. Since

$$c_2(a^{-1}, y) = \frac{r(a^{-1}y)}{r(y)} c_1(a^{-1}, y),$$

substitution in (6.3) gives $U\pi_{\rho_1}(a) = \pi_{\rho_2}(a)U$. $\square$

For the affine action on the real line in the following examples, $c((a, b), t) = a$. Formula (6.3) consequently becomes

$$(\pi(a, b)\xi)(t) = a^{-1/2} \xi((t - b)/a).$$

The square root of the inverse scaling is exactly the factor needed for unitarity on $L^2(\mathbb{R}, dt)$.

Finite covolume forces unimodularity

Proposition 6.4. If H is a closed unimodular subgroup and G/H has a nonzero invariant Radon measure of finite total mass, then G is unimodular. In particular, every group admitting a lattice is unimodular.

Proof. Theorem 3.1 gives $\Delta_G|_H = 1$, so $d(gH) = \Delta_G(g)$ is a well-defined continuous positive function on Y . Suppose $\Delta_G(a) = s > 1$; if a nontrivial modular value is smaller than one, replace that element by its inverse. The Borel sets

$$E_j = \{y : s^j \leq d(y) < s^{j+1}\}, \quad j \in \mathbb{Z},$$

are disjoint and cover Y . Since $d(ay) = s d(y)$, multiplication by a sends E_j onto E_{j+1} . Invariance makes their measures equal. Positive total mass forces at least one to have positive measure, whereas infinitely many disjoint sets of that same positive measure contradict finite total mass. Thus $\Delta_G = 1$. A discrete subgroup is unimodular for counting measure. It is also closed: choose an open identity neighborhood V with $V^{-1}V \cap \Gamma = \{e\}$. Every translate xV meets Γ in at most one point. If x lies in its closure, choose $\gamma \in xV \cap \Gamma$; if $x \neq \gamma$, the open neighborhood $xV \setminus \{\gamma\}$ would miss Γ , a contradiction. Hence $x \in \Gamma$. This proves the closedness needed to apply the proposition and completes the lattice assertion. $\square$

Corollary 6.5. If H is closed and unimodular and G/H is compact, then G is unimodular and G/H has a finite nonzero invariant Radon measure. In particular, a closed discrete cocompact subgroup is a lattice, often called a **uniform lattice**.

Proof. The character $\chi = \Delta_G^{-1}$ satisfies (6.2) because $\Delta_H = 1$. Theorem 6.2 gives a nonzero relatively invariant Radon measure. Compactness makes its total mass finite; its last assertion therefore gives $\chi = 1$, or $\Delta_G = 1$. The measure is invariant and has positive mass. $\square$

Integration of quotient densities

Vogan's freely accessible treatment gives an intrinsic formulation even when an invariant scalar measure does not exist. It regards the modular correction as part of the function being integrated. Here is the formulation and a full derivation from the weighted Weil formula.

Corollary 6.6. Let $\mathcal{D}_c(G/H)$ be the continuous functions $F : G \rightarrow \mathbb{C}$, with support compact modulo H , satisfying

$$F(gh) = \frac{\Delta_H(h)}{\Delta_G(h)} F(g). \quad (6.6)$$

Here support compact modulo H means that the support of the descended function F/ρ on Y is compact; this condition is independent of the positive rho-function. The map

$$Af(g) = \int_H f(gh) \frac{\Delta_G(h)}{\Delta_H(h)} dh$$

sends $C_c(G)$ onto $\mathcal{D}_c(G/H)$. There is a unique positive linear integral J on this space with

$$J(Af) = \int_G f(g) dg. \quad (6.7)$$

It is independent of the choice of rho-function and invariant under $F(g) \mapsto F(a^{-1}g)$.

Proof. Equation (11) gives

$$Af(g) = \rho(g)P(f/\rho)(gH).$$

Thus Af has the covariance (6.6) and compact quotient support. Conversely, F/ρ descends to a member $\phi \in C_c(Y)$. Choose $u \in C_c(G)$ with $Pu = \phi$ by Lemma 2.1. Then $f = \rho u$ is compactly supported and $Af = F$. Define

$$J(F) = \int_Y \frac{F(g)}{\rho(g)} d\mu_\rho(gH).$$

The quotient integrand is well-defined, continuous and compactly supported. Theorem 5.2 gives $J(Af) = \int (f/\rho)\rho dg = \int f dg$. This also proves uniqueness and positivity, using a positive lift when $F \geq 0$.

If $\rho_2 = r\rho_1$ on the quotient, then $\mu_{\rho_2} = r\mu_{\rho_1}$, and the factors cancel in this definition of J . Finally $A(L_a f)(g) = Af(a^{-1}g)$. Surjectivity of A and left Haar invariance in (6.7) prove the claimed invariance of J . The integral on these covariant functions therefore survives even when Theorem 3.1 rules out an invariant scalar measure. $\square$

Examples that test the formula

Translations inside the affine group. Write

$$G = \{(a, b) : a > 0, b \in \mathbb{R}\}, \quad (a, b)(a', b') = (aa', b + ab').$$

The preceding Haar lesson, Example 11.3, calculates $dg = a^{-2} da db$ and $\Delta_G(a, b) = a^{-1}$. For $H = \{(1, t) : t \in \mathbb{R}\}$, $dh = dt$ and $\Delta_H = 1$. The restriction of Δ_G to H is one, so the quotient has an invariant measure. It is parametrized by a , and (6) becomes

$$\int_0^\infty \int_{\mathbb{R}} f(a, b) \frac{db da}{a^2} = \int_0^\infty \left(\int_{\mathbb{R}} f(a, at) dt \right) \frac{da}{a}. \quad (19)$$

The substitution $b = at$ verifies the formula and identifies the quotient measure da/a .

Dilations inside the same group. Instead take $H = \{(a, 0) : a > 0\}$, with $dh = da/a$. It is abelian, so $\Delta_H = 1$, but $\Delta_G|_H = a^{-1}$. There is no invariant quotient measure. The quotient is parametrized by b , and $\rho(a, b) = a$ satisfies (11). Formula (12) gives ordinary db on Y :

$$\int_{\mathbb{R}} \int_0^\infty f(a, b) \frac{da}{a} db = \int_G f(a, b) \rho(a, b) dg.$$

The element (a_0, b_0) acts by $b \mapsto b_0 + a_0 b$. Its pullback-on-sets density is a_0 , and its pushforward density is a_0^{-1} , precisely as in (18).

Compact subgroups. If H is compact, normalize $dh(H) = 1$. Then $\rho = 1$ is allowed and averaging is ordinary fiberwise probability averaging. The quotient measure need not itself be finite: $\mathbb{R}/\{0\}$ is already a counterexample. Compactness of the fiber does not mean compactness of the base.

The upper half-plane. Let $G = \mathrm{SL}_2(\mathbb{R})$ and $H = \mathrm{SO}(2)$. The action

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}$$

preserves $\mathbb{H} = \{x + iy : y > 0\}$, since direct subtraction of the complex conjugate gives

$$\mathrm{Im}(gz) = \frac{y}{|cz + d|^2}.$$

The denominator never vanishes on $\mathbb{H}$. The element $s(x + iy) = \begin{pmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{pmatrix}$ sends i to $x + iy$. Solving $gi = i$ gives $a = d$, $b = -c$, $a^2 + b^2 = 1$, so the stabilizer is exactly H . Thus $gH \mapsto gi$ is a continuous bijection $G/H \rightarrow \mathbb{H}$, and its inverse is the continuous map $z \mapsto s(z)H$. This proves the claimed identification, including its topology.

Here is a direct integral proof of invariance of

$$dA(z) = y^{-2} dx dy. \quad (20)$$

This positive continuous density defines a Radon measure. Upper triangular matrices act by translations $z \mapsto z + t$ and positive dilations $z \mapsto a^2 z$. Iterated one-variable substitution proves that both preserve dA . The substitution and integrated-derivative rules are proved in [Measure and Hilbert space tools for Haar integration](#), Lemma 6.1.

For rotations $k_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$, their action has infinitesimal vector field

$$v(x, y) = (1 + x^2 - y^2, 2xy), \quad \partial_x(y^{-2}v_1) + \partial_y(y^{-2}v_2) = 0.$$

If $F \in C_c^1(\mathbb{H})$, put $J(\theta) = \int F(k_\theta z) dA(z)$. On every compact parameter interval, the supports of these functions and their derivatives lie in one compact subset of $\mathbb{H}$: it is the image of that interval times $\text{supp } F$ under $(\theta, w) \mapsto k_{-\theta}w$. The derivatives are bounded there, and y is bounded away from zero. Dominated convergence permits differentiation. The flow identity gives

$$J'(\theta) = \int v(z) \cdot \nabla_z(F(k_\theta z)) y^{-2} dx dy = 0.$$

For the last equality, integrate each coordinate derivative of the compactly supported product $y^{-2}v_j F(k_\theta z)$. Its integral is zero by the integrated-derivative rule and Fubini. The remaining term is minus the displayed divergence, hence zero. Therefore rotations preserve the integral.

This extends to $C_c(\mathbb{H})$ without assuming a smoothing theorem. On a rectangle strictly inside $\mathbb{H}$, polynomials uniformly approximate a continuous function by **the compact Stone–Weierstrass theorem**, Theorem 8.2. Multiply them by a C^1 cutoff equal to one on the function's support and supported in that rectangle. Such a cutoff is a product of one-variable plateaus joined to zero by the cubic $3t^2 - 2t^3$, whose endpoint derivatives vanish. The approximants have one compact support and converge uniformly; both their original and rotated integrals converge because the relevant compact sets have finite area.

Finally every $g \in G$ is $s(z)k$, as proved by its action on i ; $s(z)$ is upper triangular. Translations, positive dilations and rotations therefore generate the action of G . This proves invariance of (20) under every g . By Theorem 3.1 it is the quotient measure up to Haar normalization. The next lesson calculates the modular lattice volume from this measure.

An integer lattice. For $G = \mathbb{R}^n$, $H = \mathbb{Z}^n$, Lebesgue and counting measures give

$$\int_{\mathbb{R}^n} f(x) dx = \int_{[0,1)^n} \sum_{m \in \mathbb{Z}^n} f(x + m) dx. \quad (21)$$

For $f \in C_c(\mathbb{R}^n)$, only finitely many lattice translates meet its support while x ranges over the unit cube. Translate each summand and use the disjoint half-open tiling of $\mathbb{R}^n$. This proves (21); the boundary faces have Lebesgue measure zero. The resulting quotient measure has total mass one.

Exercises with solutions

1. Rescaling the measures. If dg is multiplied by $A > 0$ and dh by $B > 0$, how does the normalized invariant quotient measure change?

Solution. The new averaging operator is BP . To keep (6), the quotient measure must be multiplied by A/B . The same rule holds in (12) when the function ρ is kept fixed. The modular functions do not change under Haar rescaling.

2. A discrete subgroup with no invariant quotient measure. In the affine group let $H = \{(2^n, 0) : n \in \mathbb{Z}\}$. Prove that H is closed and discrete, and decide whether G/H has an invariant measure.

Solution. Any compact interval in the a -coordinate bounded away from zero meets $\{2^n\}$ finitely. Around each group element one may choose such an interval, so there is no accumulation point in G , and each point of H is isolated. Thus H is closed and discrete. Its modular function is one, whereas $\Delta_G(2^n, 0) = 2^{-n}$. Theorem 3.1 rules out a nonzero invariant quotient measure. This shows why “discrete” cannot replace the modular criterion.

3. Two choices of density. Suppose ρ_1, ρ_2 satisfy (11). Prove both (15) and the transformation rule

$$c_2(a, y) = \frac{r(ay)}{r(y)} c_1(a, y).$$

Solution. Their ratio $r(q(g)) = \rho_2(g)/\rho_1(g)$ is unchanged by right multiplication by H , hence is continuous on the quotient. For $F = Pf$, the integral of F against $r\mu_{\rho_1}$ is, by (12), $\int_G f(g)r(q(g))\rho_1(g) dg = \int_G f(g)\rho_2(g) dg$. Surjectivity of P and Radon uniqueness give (15). Substitution in (16) gives the cocycle formula. In particular, changing the density changes the cocycle by this explicit ratio, without changing the measure class.

4. Quotient averaging is equivariant. Prove (5) and use it to show that two invariant quotient measures normalized by (6) coincide.

Solution. Directly,

$$P(L_a f)(gH) = \int_H f(a^{-1}gh) dh = Pf(a^{-1}gH).$$

For $r \in H$, right translation in the integration variable gives $\int_H f(ghr) dh = \Delta_H(r)^{-1}Pf(gH)$. If two measures satisfy (6), they integrate Pf equally for every f . Lemma 2.1 says every member of $C_c(Y)$ is such a Pf ; Radon uniqueness proves equality.

5. No global compact cutoff on a noncompact base. Show that a cutoff k satisfying $Pk = 1$ cannot have compact support in G when Y is noncompact.

Solution. If k had compact support, then $\text{supp } Pk \subset q(\text{supp } k)$ would be compact. But $Pk = 1$ has support all of Y . Therefore Y would be compact. The weaker compactness condition in (8) is exactly what permits the global construction.

6. Track one modular factor. In (14), explain why the product of the right-translation factor and the density-change factor is $\Delta_H(h)^{-1}$, not $\Delta_H(h)$.

Solution. Writing $t = gh$ in an integral over g contributes $\Delta_G(h)^{-1}$. Equation (11) gives $\rho(th^{-1}) = \Delta_G(h)\Delta_H(h)^{-1}\rho(t)$. Multiplying cancels $\Delta_G(h)$, leaving $\Delta_H(h)^{-1}$. That is exactly the factor in the inversion formula $\int_H a(h^{-1})\Delta_H(h)^{-1} dh = \int_H a(h) dh$.

What this enables

The invariant criterion is available to the programme's subgroup and quotient, adèle and automorphic courses. Theorem 6.3 turns the density and cocycle formulas into a strongly continuous quasi-regular representation, including its unitary equivalence under a change of density. Proposition 6.4 and Corollary 6.5 explain the unimodularity and compactness constraints on lattices. Corollary 6.6 gives an invariant integral on quotient densities. The two inverse-translation conventions in (18) remain explicit in these action formulas.

The principal formulas above are stated and proved on $C_c(G)$. Extending them to arbitrary measurable functions requires the precise regularity and support conventions of the preceding measure-theory lessons. The abelian Fourier course already proves its L^1 extension. This lesson does not quietly replace the general non- σ -finite measure theory by a second-countable argument.

Sources and credit

- D. H. Fremlin, *Measure Theory, Volume 4*, develops positive subgroup averaging, quotient integration and the same modular criterion. His complete, locally determined measure convention is related to ours by **Proposition 13.7 of the Haar lesson**. The compactly supported kernel argument above explicitly places its cutoff on the projection of the original support, including cosets where signed integration cancels.
- David Vogan, *Invariant measures on homogeneous spaces*, gives a complementary treatment of averaging and the invariant-measure criterion, and the quotient-density viewpoint developed in Corollary 6.6.
- *Haar measure on locally compact groups*, programme lesson originally by Claude Opus 5.5, supplies the Haar, Radon-product and modular proofs named above. Retained programme expression is CC0.
- *Quotient measure and the subgroup modular correction*, OA-FLOW lesson 43, credits OpenAI Codex. Its density construction and sign convention inform the rho-function argument here. Its historical exact model is not recorded.

- *Subgroups, quotients and annihilators*, HA-LCA lesson by GPT-6.1 Sol, Ultra, supplies the earlier abelian compact-lift and positive-averaging arguments. The present lesson proves their general-group versions in full.
- B. Bekka, P. de la Harpe and A. Valette, *Kazhdan's Property (T)*, freely accessible author preprint dated 23 February 2007. Its homogeneous-space and lattice treatment motivates Theorems 6.2–6.3 and the finite-covolume results. The proofs here are independently written using this lesson's earlier constructions.

Finite covolume and arithmetic quotients

Written by GPT-6.1 Sol (OpenAI), Ultra reasoning effort, October 2026. Self-checked by the writing AI. Independent AI review remains separate; no human review is claimed. Original exposition and exercises are dedicated under CC0.

A quotient measure can exist and still have infinite total mass. A **lattice** in a locally compact group is a discrete subgroup whose invariant quotient measure has finite positive mass. That mass is its **covolume**, once the Haar measures on the group and subgroup have been fixed. We calculate three arithmetic examples and a nonabelian Heisenberg lattice, keeping the normalizations visible: the modular group, the additive rational adèles, and the norm-one rational idèles.

What to know first

The preceding **quotient lesson**, Theorem 3.1, proves the modular criterion and the normalized Weil formula. Its hyperbolic example proves that $dx dy/y^2$ is invariant under $\mathrm{SL}_2(\mathbb{R})$. **Haar measure on locally compact groups**, Theorems 2.2, 5.1, 10.1 and 11.1, supplies Riesz representation, sigma-finite Tonelli, the modular function and inversion.

For the last two examples, read these exact programme proofs:

- **Completions, the p-adic numbers and complete discretely valued fields**, Theorem 2.1 and Proposition 2.2: the field $\mathbb{Q}_p$, its compact open ring $\mathbb{Z}_p$, and its residue quotients. Only the rational case of the construction is needed.
- **Restricted products and profinite completions**, Propositions 1.1–1.2: restricted product topology and its Haar measure; the Chinese remainder argument and the ring/residue part of Proposition 1.4. Character duality in that lesson is not used here.
- **The adèle ring of a number field**, the rational proof of Theorem 2.2: a half-open additive fundamental domain. We apply it to integration below.
- **Idèles and the idèle class group**, Proposition 3.1: the idèle topology. We give the rational norm-one reduction explicitly, without using the class number or unit theorem.

These are specific internal proofs. The present lesson supplies the integration and normalization arguments; it does not assume the general number-field compactness theorem. Elementary integer arithmetic used in the reductions is recalled with proof next.

1. Two integer reductions

Lemma 1.1. If $c, d \in \mathbb{Z}$ are relatively prime, they are the bottom row of a matrix in $\mathrm{SL}_2(\mathbb{Z})$. Every nonzero rational number has a unique expression $\varepsilon \prod_p p^{e_p}$, with $\varepsilon \in \{1, -1\}$, integer exponents, and only finitely many nonzero exponents. Consequently

$$|r|_\infty \prod_p |r|_p = 1. \quad (1.1)$$

Proof. Repeated division with remainder strictly decreases nonnegative remainders, so the Euclidean algorithm terminates. Back substitution expresses the greatest common divisor as an integer combination. For coprime c, d , choose a, b with $ad - bc = 1$. This proves the first assertion and Bézout's identity.

For completeness, factor a positive integer by induction: a composite integer is a product of smaller positive integers. If a prime p divides uv and does not divide u , Bézout gives $sp + tu = 1$; multiplying by v shows that p

divides v . This prime-divisor rule proves uniqueness by cancelling a prime from two proposed factorizations and inducting. Applying the result to numerator and denominator gives the rational factorization. Its real absolute value is $\prod p^{e_p}$, and $|r|_p = p^{-e_p}$, proving (1.1). This also proves directly that these rational absolute values are multiplicative and ultrametric: over a common denominator, the valuation of a sum is at least the lesser valuation. ■

Borel fundamental domains and covolume

The general quotient facts now have full proofs in [the preceding lesson](#): a lattice forces its ambient group to be unimodular, and a closed discrete cocompact subgroup is a lattice. To calculate its covolume, the following result turns a Borel set of representatives into a measure calculation. Bekka, de la Harpe and Valette give a complementary treatment of fundamental domains; the proof below explicitly handles Borel representatives.

Theorem 1.2. Let G be a sigma-compact unimodular LCH group and Γ a closed discrete subgroup, equipped with counting measure. Then Γ is countable and there exists a Borel set $D \subset G$ meeting each right coset $g\Gamma$ exactly once. For the quotient measure ν normalized by the Weil formula,

$$\nu(G/\Gamma) = \mu(D), \quad (1.2)$$

allowing infinite values. Any Borel set of unique right-coset representatives gives the same equality.

Proof. A compact set meets Γ in finitely many points: its intersection with the closed discrete subgroup is a compact discrete space, whose singleton cover has a finite subcover. A countable compact cover of G therefore makes Γ countable.

Choose a relatively compact open identity neighborhood V with $V^{-1}V \cap \Gamma = \{e\}$. It exists by continuity of multiplication and inversion and discreteness. Sigma-compactness gives a countable cover $G = \bigcup_{n \geq 1} U_n$, with $U_n = g_n V$: each set of a countable compact cover has a finite subcover by translates of V . The quotient map q is open, and its restriction to each U_n is injective, because $u^{-1}u' \in \Gamma$ for two points of one coset. It is consequently a homeomorphism onto the open set $q(U_n)$.

Define

$$D_n = U_n \setminus \bigcup_{m < n} U_m \Gamma, \quad D = \bigcup_{n \geq 1} D_n.$$

The saturated sets $U_m \Gamma$ are open, so D is Borel. Each coset has a first chart which meets it, and in that chart exactly one point; this point lies in D_n , and later charts contribute none. Hence G is the disjoint union of $D\gamma$, $\gamma \in \Gamma$.

The same integration argument works for any Borel transversal D . Put $\nu_0(B) = \mu(D \cap q^{-1}(B))$ for Borel $B \subset G/\Gamma$. This is a Borel measure. For a Borel $B \subset q(U_n)$, write $U_B = (q|_{U_n})^{-1}(B)$. The sets

$$D \cap U_B \gamma^{-1}, \quad \gamma \in \Gamma,$$

partition $D \cap q^{-1}(B)$, and their right translates by γ partition U_B . Right Haar invariance and countable additivity therefore give

$$\nu_0(B) = \mu(U_B). \quad (1.3)$$

In particular, ν_0 is finite and Radon on every chart, as the transport of $\mu|_{U_n}$. For completeness, one can establish its global Radon property without assuming a regularity theorem for arbitrary Borel transversals. The functional $F \mapsto \int_D F(q(g)) d\mu(g)$ is positive and finite on $C_c(G/\Gamma)$: a finite chart cover of its support bounds its absolute integral by $\|F\|_\infty$ times the sum of the finite $\mu(U_n)$ in that cover. Riesz representation gives a Radon measure ν_1 . For tests supported in a chart it agrees with the transported measure in (1.3), so local Radon uniqueness identifies ν_1 and ν_0 on that chart. Partitioning a Borel set among the first chart containing each point, using the countable cover $q(U_n)$, proves equality on every Borel set. Thus ν_0 itself is Radon.

Finally, for $f \in C_c(G)$, countable Tonelli and right invariance give

$$\begin{aligned} \int_{G/\Gamma} \sum_{\gamma \in \Gamma} f(g\gamma) d\nu_0(g\Gamma) &= \int_D \sum_{\gamma \in \Gamma} f(d\gamma) d\mu(d) \\ &= \sum_{\gamma \in \Gamma} \int_{D\gamma} f(u) d\mu(u) = \int_G f(u) d\mu(u). \end{aligned}$$

For complex f , absolute summability follows by applying the same calculation to $|f|$, whose integral is finite. The quotient lesson's normalized uniqueness identifies $\nu_0 = \nu$. Taking the measure of the whole quotient proves (1.2). $\square$

2. A fundamental region in the upper half-plane

Write $G = \mathrm{SL}_2(\mathbb{R})$, $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, and $K = \mathrm{SO}(2)$. Integer matrices form a closed discrete subset of real matrix space, and a compact set meets them in finitely many points. Hence Γ is a closed discrete subgroup. Counting measure on Γ is unimodular.

We also need unimodularity of G , rather than an unproved general assertion about Lie groups. Put

$$U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad L(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \quad D(a) = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

Conjugating $U(t)$ by $D(2)$ gives $U(4t) = U(t)^4$. A homomorphism into an abelian group is constant on conjugacy classes. Thus its positive modular value satisfies $\Delta(U(t)) = \Delta(U(t))^4$, so it is one. Conjugation by $D(1/2)$ gives the same conclusion for $L(t)$. These matrices generate G . Indeed,

$$W(a) = U(a)L(-1/a)U(a) = \begin{pmatrix} 0 & a \\ -1/a & 0 \end{pmatrix}, \quad D(a) = W(a)W(1)^{-1}$$

for $a \neq 0$, and a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $a \neq 0$ equals $L(c/a)D(a)U(b/a)$. If $a = 0$, multiplication on the left by $U(1)$ makes the top-left entry nonzero. Therefore $\Delta_G = 1$. The quotient criterion now applies.

Let $\mathbb{H} = \{x + iy : y > 0\}$ and

$$\mathcal{F} = \{z \in \mathbb{H} : |\mathrm{Re} z| \leq 1/2, |z| \geq 1\}. \quad (2.1)$$

The action is $gz = (az + b)/(cz + d)$, with $\mathrm{Im}(gz) = y/|cz + d|^2$.

Proposition 2.1. Every Γ -orbit meets $\mathcal{F}$. If two points of its interior lie in the same orbit, they are equal, and the transforming matrix is I or $-I$.

Proof. For fixed $z = x + iy$, there are only finitely many integer pairs (c, d) with $|cz + d| \leq R$: the inequality bounds $|c|y$, and then $|cx + d|$. Among relatively prime pairs choose one minimizing the positive number $|cz + d|$; the pair $(0, 1)$ supplies an initial bound. Lemma 1.1 extends the minimizing pair to $\gamma \in \Gamma$. The point $w = \gamma z$ has greatest imaginary part in its orbit. An integer translation moves its real part into $[-1/2, 1/2]$ without changing its imaginary part. If $|w| < 1$, the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ increases that imaginary part, a contradiction. Thus $w \in \mathcal{F}$.

For an interior point, $|x| < 1/2$ and $y^2 > 1 - x^2 > 3/4$. If $c \neq 0$, then $|cz + d| > 1$. For $|c| \geq 2$, its square is at least $c^2 y^2 > 3$. For $|c| = 1, d = 0$, use $|z| > 1$. For $|c| = 1, |d| \geq 1$, its square is at least

$$(1 - |x|)^2 + y^2 > 2 - 2|x| > 1.$$

Consequently a matrix with $c \neq 0$ strictly decreases the imaginary part of an interior point. If it sends one interior point to another, its inverse has nonzero bottom-left entry too, and would strictly decrease it back. This is impossible. If $c = 0$, the determinant and integrality give $a = d = \pm 1$; the action is an integer translation. Two real parts strictly between $-1/2$ and $1/2$ force that translation to be zero, leaving $\gamma = \pm I$. $\blacksquare$

The boundary consists of two vertical rays and a circular arc. Each has two-dimensional Lebesgue measure zero: their vertical sections are finite except at two abscissae, and sigma-finite Tonelli applies. Their hyperbolic area is also zero because the density y^{-2} is locally bounded. Countably many Γ -translates of this boundary still have area zero. Removing them leaves a full-measure invariant subset on which the region's interior gives unique orbit representatives modulo $\pm I$.

Its area is finite and explicit:

$$\text{area}(\mathcal{F}) = \int_{-1/2}^{1/2} \int_{\sqrt{1-x^2}}^{\infty} \frac{dy dx}{y^2} = \int_{-1/2}^{1/2} \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{3}. \quad (2.2)$$

The last step uses the one-variable substitution $x = \sin t$, with $-\pi/6 \leq t \leq \pi/6$. The integral identities used here are proved in the tools lesson, Lemma 6.1.

3. From area to group covolume

Area in $\mathbb{H}$ and Haar volume in G have different normalizations. Let

$$s(x + iy) = \begin{pmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{pmatrix}, \quad k_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Every $g \in G$ has a unique expression $s(z)k$: set $z = gi$, and then $s(z)^{-1}g$ fixes i , so belongs to K . These operations are continuous, giving a homeomorphism $\mathbb{H} \times K \rightarrow G$. Fix the Haar normalization

$$dg = \frac{dx dy}{y^2} \frac{d\theta}{2\pi}. \quad (3.1)$$

This really is left Haar measure. For $a \in G$, $as(z) = s(az)\kappa(a, z)$ for a continuous K -valued function κ . Left multiplication by a therefore sends (z, k) to $(az, \kappa(a, z)k)$. Integrate first over Haar probability on K , then over invariant hyperbolic area. The integral of every $C_c(G)$ function is unchanged. The transported Radon product is consequently Haar.

Theorem 3.1. With (3.1) and counting measure on Γ , both $\Gamma \backslash G$ and G/Γ have covolume $\pi/6$. The hyperbolic quotient has area $\pi/3$. In particular, Γ is a lattice.

Proof. On the full-measure set of regular base points from Proposition 2.1, each left orbit first selects a unique $z \in \mathcal{F}^\circ$, then selects k_θ with $0 \leq \theta < \pi$. The remaining ambiguity $-I \in \Gamma$ changes θ by π . Thus the Borel set

$$\mathcal{D} = \{s(z)k_\theta : z \in \mathcal{F}^\circ, 0 \leq \theta < \pi\}$$

and its Γ -translates partition a full-Haar-measure subset of G . Omitted fibers lie over the null boundary translates; Tonelli and (3.1) show they have Haar measure zero. The group is sigma-compact, since $\mathbb{H} \times K$ is, so ordinary sigma-finite Tonelli is available. Hence for $f \in C_c(G)$,

$$\int_G f(g) dg = \int_{\mathcal{D}} \sum_{\gamma \in \Gamma} f(\gamma g) dg. \quad (3.2)$$

First prove this for nonnegative f by the countable partition; apply it to $|f|$ to justify complex summation. The discrete averaging sum is locally finite: on compact subsets, only finitely many γ can move one compact set to meet another, since those γ lie in a compact set intersected with Γ .

Define a positive functional on $C_c(\Gamma \backslash G)$ by integrating its pullback over $\mathcal{D}$. It is finite because

$$\mu_G(\mathcal{D}) = \text{area}(\mathcal{F}) \cdot \frac{\pi}{2\pi} = \frac{\pi}{6}. \quad (3.3)$$

Riesz representation gives a Radon measure. Formula (3.2), positive surjectivity of discrete averaging, and right invariance of dg show that this measure is right G -invariant and has the normalized Weil formula. Its total mass is (3.3): an increasing family of compactly supported quotient cutoffs tends to one, since the

quotient is sigma-compact; monotone convergence on $\mathcal{D}$ proves the assertion. Such cutoffs are obtained from the compact exhaustion and bump functions of the preceding lesson.

Finally, inversion identifies $\Gamma \backslash G$ with G/Γ . The inversion formula has no density factor because $\Delta_G = 1$, so it preserves the normalization and covolume. Equation (2.2) gives the hyperbolic area. ■

Normalization example. The measure $d\theta$, in place of $d\theta/(2\pi)$, multiplies the group covolume by 2π , giving $\pi^2/3$. Passing instead to $\mathrm{PSL}_2(\mathbb{R})$ and normalizing its compact stabilizer to probability gives covolume $\pi/3$. Indeed $-I$ has disappeared there, so the full compact fiber is used. A claimed number for a lattice volume is incomplete without its group and Haar normalization.

4. Additive rational adèles

Put

$$\mathbb{A} = \mathbb{R} \times \prod_p' (\mathbb{Q}_p, \mathbb{Z}_p), \quad Z = \prod_p \mathbb{Z}_p.$$

Use Lebesgue measure at the real place and additive Haar measures of mass one on each $\mathbb{Z}_p$. The restricted product provider constructs their compatible Haar measure. On the open subgroup $\mathbb{R} \times Z$ it is Lebesgue measure times Haar probability on Z .

Theorem 2.2 of the adèle lesson proves, with rational principal parts and integer reduction, that

$$D = [0, 1) \times Z \quad (4.1)$$

is a Borel fundamental domain for $\mathbb{A}/\mathbb{Q}$, with unique representatives, and that this quotient is compact and Hausdorff. Its proof uses only the local residue construction: subtract each of the finitely many negative local digit strings, then subtract an integer. A rational integral at every prime is an integer; two representatives in (4.1) can therefore differ only by an integer whose real absolute value is less than one, hence by zero. This also shows why integer reduction must change the finite components as well as the real one.

Theorem 4.1. The normalized invariant measure on $\mathbb{A}/\mathbb{Q}$ is probability, and

$$\int_{\mathbb{A}} f(x) dx = \int_D \sum_{r \in \mathbb{Q}} f(x+r) dx \quad (f \in C_c(\mathbb{A})). \quad (4.2)$$

Proof. The translates $D+r$ are a disjoint Borel partition. The group is sigma-compact: the compact sets $[-n, n] \times (n!)^{-1}Z$, $n \geq 1$, exhaust it. Countable additivity, translation invariance and Tonelli give (4.2), with absolute convergence for complex f . Moreover only finitely many summands occur uniformly on D : a contributing rational lies in $\mathrm{supp} f - \overline{D}$, a compact set; a compact set meets the closed discrete $\mathbb{Q}$ finitely.

The positive functional $F \mapsto \int_D F(x + \mathbb{Q}) dx$ on the compact quotient is represented by a Radon measure. Equation (4.2) and the quotient theorem identify it with the normalized invariant quotient measure. Its total mass is $\mu(D) = 1$, since the real interval has length one and Z has mass one. ■

Worked reduction. Take real component $11/10$, local components $1/4$ at 2 , $2/9$ at 3 , and zero at all other primes. Subtract $r = 1/4 + 2/9 = 17/36$. At 2 the remainder is $-2/9 \in \mathbb{Z}_2$; at 3 it is $-1/4 \in \mathbb{Z}_3$; every other coordinate is integral. The real remainder is $113/180$, so it already lies in (4.1). This is an additive representative, not an idèle reduction.

5. Norm-one rational idèles

Let

$$J = \mathbb{R}^\times \times \prod_p' (\mathbb{Q}_p^\times, \mathbb{Z}_p^\times), \quad |x| = |x_\infty| \prod_p |x_p|_p, \quad J^1 = \ker |\cdot|.$$

Use the restricted multiplicative topology, as proved in Proposition 3.1 of the idèle lesson. The norm is a continuous homomorphism: on each open chart it is a finite product of continuous local absolute values. Thus J^1 is a closed LCA group. Lemma 1.1 puts the diagonal $\mathbb{Q}^\times$ inside it.

Theorem 5.1. Put $U = \prod_p \mathbb{Z}_p^\times$, embedded in J^1 with real component one. Multiplication gives a topological group isomorphism

$$\mathbb{Q}_{\text{discrete}}^\times \times U \longrightarrow J^1. \quad (5.1)$$

Consequently $J^1/\mathbb{Q}^\times \simeq U$ is compact. With counting measure on $\mathbb{Q}^\times$ and Haar probability on U , its covolume is one and

$$\int_{J^1} f(x) dx = \int_U \sum_{r \in \mathbb{Q}^\times} f(ru) du. \quad (5.2)$$

Proof. For $x \in J^1$, set $e_p = v_p(x_p)$, a finitely supported integer family. Let $r_0 = \prod_p p^{e_p} > 0$. Then each x_p/r_0 is a unit. The norm-one equation gives $|x_\infty| = r_0$. Taking $r = \text{sign}(x_\infty)r_0$ makes the real component of x/r one. Thus $x = ru$ for $u \in U$. If a diagonal rational belongs to U , all its valuations vanish, so it is ± 1 ; its real component is one, forcing it to be one. This proves uniqueness and algebraic bijectivity.

The subgroup U is compact and open in J^1 . Compactness follows from compactness of each $\mathbb{Z}_p^\times$, a closed subset of $\mathbb{Z}_p$, and arbitrary-product compactness. For openness, intersect J^1 with the open idèle set whose real component is positive and whose finite components are all units. The norm forces its real component to be exactly one, so that intersection is U . Its cosets rU are therefore open and closed and have their transported compact product topologies. This proves the topology assertion in (5.1), including discreteness and closedness of the diagonal factor.

Transport counting measure times Haar probability through (5.1). It is Haar on J^1 ; on the open compact subgroup U it has mass one. Compact supports meet only finitely many open cosets rU . Finite summation on such supports proves (5.2). Its quotient measure is Haar probability on U , establishing the covolume. ■

Worked reduction. Suppose $x_\infty = -12/5$, $x_2 = 4$, $x_3 = 3$, $x_5 = 1/5$, and $x_p = 1$ elsewhere. The norm is $(12/5)(1/4)(1/3)5 = 1$. The rational factor in (5.1) is $-12/5$. Division by it gives real component one and finite components $-5/3, -5/4, -1/12$ at 2, 3, 5, respectively, all local units. At the remaining primes the component is $-5/12$, also a unit. The entire tuple, not just its exceptional coordinates, changes.

Heisenberg lattice volume

Example 5.2. In the Heisenberg group N of [Example 12.3 of the Haar lesson](#), the subgroup $\Gamma = \{(m, n, k) : m, n, k \in \mathbb{Z}\}$ is a uniform lattice. For Lebesgue Haar measure and counting measure on Γ , its covolume is one.

Proof. The group law and inverse formula preserve integer coordinates, so Γ is a subgroup. It is closed and discrete in $\mathbb{R}^3$. Let $D = [0, 1]^3$. For $g = (x, y, z)$, define

$$\begin{aligned} m &= \lfloor x \rfloor, & n &= \lfloor y \rfloor, \\ u &= x - m, & v &= y - n. \\ k &= \lfloor z - un \rfloor, & w &= z - un - k. \end{aligned}$$

Then $d = (u, v, w) \in D$, $\gamma = (m, n, k) \in \Gamma$, and

$$d\gamma = (u + m, v + n, w + k + un) = (x, y, z).$$

The half-open conditions first force m, n , then k , so the factorization is unique. Thus D is a Borel right-coset transversal. Its closure is compact and its quotient image covers N/Γ ; this proves cocompactness. The group is sigma-compact and unimodular by the Haar example, so Theorem 1.2 gives

$$\nu(N/\Gamma) = \mu(D) = \int_0^1 \int_0^1 \int_0^1 dz dy dx = 1.$$

There is also a left transversal with the same cube: in the factorization $g = \gamma d$, replace the third-coordinate reduction by $k = \lfloor z - mv \rfloor$ and $w = z - mv - k$. The product has third coordinate $k + w + mv$, proving existence and uniqueness in that direction too. This checks the side of translation explicitly. □

The upper triangular model and its lattice appear in Bekka, de la Harpe and Valette; Fischer and Ruzhansky explain its relation to homogeneous groups. The floor reductions above supply the exact representatives and normalization without a general theorem about nilpotent lattices.

6. Exercises with complete solutions

1 (easy): a nonunit normalization. Replace counting measure on a discrete subgroup by b times counting measure. Keeping group Haar fixed, find the covolumes in Theorems 3.1, 4.1 and 5.1.

Solution. Averaging is multiplied by b ; Weil's formula therefore divides the quotient measure by b . The covolumes become $\pi/(6b), 1/b, 1/b$, respectively. Discreteness does not impose a preferred scale until counting measure is chosen.

2 (medium): another additive reduction. Take real component $-1/5$, components $3/8$ at 2 , $1/5$ at 5 , and zero elsewhere. Find its representative in D .

Solution. Subtract the principal parts $q_0 = 3/8 + 1/5 = 23/40$. The real remainder is $-31/40$; subtracting the integer -1 moves it to $9/40$. Hence the total rational subtraction is $q = -17/40$. The finite remainders are $4/5$ at 2 , $5/8$ at 5 , and $17/40$ elsewhere. All are integral at their respective primes. Uniqueness follows from the rational-integral intersection and interval length in (4.1).

3 (medium): finite-index lattice volumes. If $\Gamma' \leq \Gamma$ has finite index m , prove that its covolume in G , with counting measure, is $m\pi/6$.

Solution. Choose representatives $t_1, \dots, t_m$ with $\Gamma = \bigsqcup_i \Gamma' t_i$. The sets $t_i \mathcal{D}$ form a fundamental domain for the left Γ' -action on the same full-measure set: an equality of two representatives would contradict the uniqueness of the Γ -representative and then the distinct cosets. Left Haar invariance gives total volume $m\mu(\mathcal{D})$. The proof of Theorem 3.1 applies to their union, and inversion gives the right quotient too. No assumption that $-I \in \Gamma'$ is needed; the index already accounts for any central difference.

4 (hard): splitting all rational idèles. Prove $J \simeq \mathbb{R}_{>0} \times \mathbb{Q}_{\text{discrete}}^\times \times U$, where the first coordinate is the idèle norm. Identify $J/\mathbb{Q}^\times$ and decide whether it is compact.

Solution. Let $s(t)$ be the idèle with real component $t > 0$ and all finite components one. It is a continuous homomorphism with norm t . Every $x \in J$ is uniquely $s(|x|)y$ with $y \in J^1$; both directions are continuous, by norm continuity and group operations. Apply (5.1) to y . The rational diagonal has norm one and occupies exactly the middle factor, so the quotient is $\mathbb{R}_{>0} \times U$. It is not compact: its continuous norm image is the noncompact $\mathbb{R}_{>0}$. This distinguishes norm-one compactness from compactness of the full idèle class group.

References and relation to the programme

The adèle and local-field lessons named above are CC0 programme sources, written by GPT-6.1 Sol at Ultra and self-checked by their writing AI. Their local constructions and rational fundamental-domain proof are used at the exact locators stated. This lesson's reductions, volume calculations and exercises were independently authored; their self-check does not claim a fresh independent review of those providers.

J. S. Milne's [Modular Functions and Modular Forms, version 1.31](#), treats the same modular region and its boundary identifications. K. Conrad's [The character group of \$\mathbb{Q}\$](#) gives the same half-open rational adèle representatives through a pole-part reduction. Here the integer/CRT reductions, unfolding, central factor and Haar normalizations are proved explicitly. B. Bekka, P. de la Harpe and A. Valette's [Kazhdan's Property \(T\)](#), author preprint dated 23 February 2007, gives a complementary treatment of lattices, fundamental domains and the Heisenberg example. V. Fischer and M. Ruzhansky's [Quantization on Nilpotent Lie Groups](#), 2016, develops the Heisenberg group and its homogeneous scaling. General number-field adèles and class groups remain in the adèle course; Fourier duality and Poisson summation remain in the abelian harmonic-analysis course.

Weak topologies: Tychonoff, Banach–Alaoglu, Mazur, bipolars, Krein–Milman and Eberlein–Šmulian

Originally written and self-checked by Claude Opus 5.5 (Anthropic), October 2026. GPT-6.1 Sol (OpenAI), at the Ultra setting, read and self-checked the full lesson and all four solutions, corrected the zero-space cases and supplied the complex-duality and extreme-point details, October 2026. Public domain (CC0).

This lesson proves the facts about weak topologies that the operator-algebra lessons use:

- the continuous functionals of a weak topology are exactly the functionals that define it;
- Tychonoff's theorem, and from it the Banach–Alaoglu theorem;
- Mazur's theorem: convex sets have the same weak and norm closures, and more generally the same closures in all locally convex topologies with the same continuous functionals;
- annihilators and bipolars, Goldstine's theorem, and the dual of a quotient;
- second adjoints;
- the Krein–Milman theorem and Milman's converse;
- the Eberlein–Šmulian theorem, which says that weak compactness of a subset of a Banach space can be tested with sequences.

It builds on the lesson [Hahn–Banach, Baire and the basic theorems on Banach spaces](#), cited below as *the previous lesson*.

Conventions

As in the previous lesson. For a set S of linear functionals, $\bigcap_{f \in S} \ker f$ is written $\ker S$. A *net* is a family indexed by a directed set. A subset of a topological space is *relatively compact* if its closure is compact.

1. Weak topologies

Let X be a vector space and Y a vector space of linear functionals on X . The *weak topology* $\sigma(X, Y)$ is the locally convex topology given by the seminorms $x \mapsto |y(x)|$, $y \in Y$. It is the coarsest topology making every $y \in Y$ continuous. It is Hausdorff exactly when Y separates the points of X .

For a normed space E :

- the *weak topology* on E is $\sigma(E, E^*)$;
- the *weak* topology* on E^* is $\sigma(E^*, j(E))$, where $j(x)(\varphi) = \varphi(x)$.

A net $x_i \rightarrow x$ weakly when $\varphi(x_i) \rightarrow \varphi(x)$ for every $\varphi \in E^*$, and $\varphi_i \rightarrow \varphi$ weak* when $\varphi_i(x) \rightarrow \varphi(x)$ for every $x \in E$.

Lemma 1.1. Let $f, f_1, \dots, f_n$ be linear functionals on a vector space X with $\ker\{f_1, \dots, f_n\} \subseteq \ker f$. Then f is a linear combination of $f_1, \dots, f_n$.

Proof. The map $\pi : X \rightarrow \mathbb{K}^n$, $x \mapsto (f_1(x), \dots, f_n(x))$, has kernel inside $\ker f$. So $g(\pi(x)) = f(x)$ defines a linear functional g on the subspace $\pi(X) \subseteq \mathbb{K}^n$. Extend g linearly to $\mathbb{K}^n$; then $g(t) = \sum_i c_i t_i$, and $f = \sum_i c_i f_i$. $\square$

Theorem 1.2. The linear functionals on X that are continuous for $\sigma(X, Y)$ are exactly the elements of Y .

Proof. Elements of Y are continuous by definition. Let f be continuous. Then $\{x : |f(x)| < 1\}$ contains a basic neighbourhood $\{x : |y_i(x)| < \varepsilon, i \leq n\}$ with $y_i \in Y$. If $y_i(x) = 0$ for all i , then tx lies in this neighbourhood for every scalar t , so $|f(tx)| < 1$ for all t , and $f(x) = 0$. By Lemma 1.1, f is a linear combination of the y_i , so $f \in Y$. $\square$

Corollary 1.3. The weak topology of a normed space has continuous dual E^* . The weak* topology of E^* has continuous dual $j(E)$. Both topologies are Hausdorff: E^* separates points of E by the previous lesson, Corollary 2.3, and $j(E)$ separates points of E^* trivially.

2. Tychonoff's theorem

A *filter* on a set S is a nonempty family $\mathcal{F}$ of subsets with three properties:

- $\emptyset \notin \mathcal{F}$;
- $A \cap B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$;
- $B \in \mathcal{F}$ whenever $B \supseteq A \in \mathcal{F}$.

An *ultrafilter* is a filter not properly contained in another. A filter on a topological space *converges* to x if it contains every neighbourhood of x .

Lemma 2.1.

1. Every filter is contained in an ultrafilter.
2. A filter $\mathcal{U}$ is an ultrafilter iff for every $A \subseteq S$, either $A \in \mathcal{U}$ or $S \setminus A \in \mathcal{U}$.
3. If $f : S \rightarrow T$ and $\mathcal{U}$ is an ultrafilter on S , then $f_*\mathcal{U} = \{B \subseteq T : f^{-1}(B) \in \mathcal{U}\}$ is an ultrafilter on T .

Proof. 1. Filters containing the given one, ordered by inclusion, have unions of chains as upper bounds. Zorn's lemma (previous lesson, Theorem 1.1) gives a maximal one.

2. Suppose $\mathcal{U}$ is an ultrafilter and $A \notin \mathcal{U}$. If $A \cap U \neq \emptyset$ for all $U \in \mathcal{U}$, then the sets containing some $A \cap U$ form a filter that contains $\mathcal{U}$ and A , contradicting maximality. So $A \cap U = \emptyset$ for some $U \in \mathcal{U}$, and then $S \setminus A \supseteq U$ lies in $\mathcal{U}$. Conversely, a filter with this property is maximal: a larger filter would contain some $A \notin \mathcal{U}$ together with $S \setminus A \in \mathcal{U}$, hence $\emptyset$.

3. $f_*\mathcal{U}$ is a filter, and 2 applies because $f^{-1}(T \setminus B) = S \setminus f^{-1}(B)$. $\square$

Lemma 2.2. A topological space K is compact iff every ultrafilter on K converges.

Proof. *Compact implies convergence.* Suppose an ultrafilter $\mathcal{U}$ converges to no point. Every x then has an open neighbourhood $V_x \notin \mathcal{U}$, so $K \setminus V_x \in \mathcal{U}$ by Lemma 2.1(2). Finitely many $V_{x_1}, \dots, V_{x_n}$ cover K . Then $\bigcap_j (K \setminus V_{x_j}) = \emptyset$ lies in $\mathcal{U}$, which is impossible.

Convergence implies compact. Let $\mathcal{O}$ be an open cover with no finite subcover. The complements of finite unions of members of $\mathcal{O}$ are nonempty and closed under finite intersections. So they generate a filter, contained in an ultrafilter $\mathcal{U}$ (Lemma 2.1(1)). Let $\mathcal{U}$ converge to x , and pick $O \in \mathcal{O}$ with $x \in O$. Then $O \in \mathcal{U}$ and $K \setminus O \in \mathcal{U}$, so $\emptyset \in \mathcal{U}$, a contradiction. $\square$

Theorem 2.3 (Tychonoff). A product $\prod_{i \in I} K_i$ of compact spaces is compact in the product topology.

Proof. Let $\mathcal{U}$ be an ultrafilter on the product and π_i the projections. By Lemma 2.1(3) each $(\pi_i)_*\mathcal{U}$ is an ultrafilter on K_i . By Lemma 2.2 it converges to some x_i ; this uses the axiom of choice to choose the limits.

The point $x = (x_i)$ is a limit of $\mathcal{U}$. A basic neighbourhood of x has the form $\bigcap_{i \in F} \pi_i^{-1}(V_i)$, with F finite and V_i a neighbourhood of x_i . Each $\pi_i^{-1}(V_i) \in \mathcal{U}$, since $V_i \in (\pi_i)_*\mathcal{U}$, and $\mathcal{U}$ is closed under finite intersections. By Lemma 2.2 the product is compact. $\square$

3. The Banach–Alaoglu theorem

Theorem 3.1 (Banach–Alaoglu). For every normed space E , the closed unit ball B^* of E^* is weak* compact.

Proof. For $x \in E$ let $D_x = \{\lambda \in \mathbb{K} : |\lambda| \leq \|x\|\}$, and $P = \prod_{x \in E} D_x$, compact by Theorem 2.3. The map $\Phi : B^* \rightarrow P$, $\varphi \mapsto (\varphi(x))_x$, is injective.

Φ is a homeomorphism onto its image, with B^* carrying the weak* topology: both topologies are the topology of pointwise convergence.

The image is closed in P . It is the set of points (λ_x) satisfying $\lambda_{x+y} = \lambda_x + \lambda_y$ and $\lambda_{tx} = t\lambda_x$ for all x, y, t , and each of these conditions defines a closed set: such a point is a linear functional bounded by $\|x\|$ at x , hence an element of B^* .

A closed subset of a compact space is compact. $\square$

Proposition 3.2. If E is separable, the weak* topology on B^* is metrizable.

Proof. Let (x_n) be dense in the unit ball of E . Put $d(\varphi, \psi) = \sum_n 2^{-n} |\varphi(x_n) - \psi(x_n)|$ on B^* .

- d is a metric: if $d(\varphi, \psi) = 0$, then $\varphi = \psi$ on a dense subset of the ball, so $\varphi = \psi$.
- The identity map from (B^*, weak^*) to (B^*, d) is continuous: each term is continuous, and the series converges uniformly because $|\varphi(x_n) - \psi(x_n)| \leq 2$.
- A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. $\square$

4. Mazur's theorem

Theorem 4.1 (Mazur). A convex subset C of a normed space has the same closure in the weak and in the norm topology. In particular, norm-closed convex sets are weakly closed. If $x_n \rightarrow x$ weakly, then some sequence of convex combinations of the x_n converges to x in norm.

Proof. The weak topology is coarser, so the norm closure $\overline{C}$ lies in the weak closure. Conversely, if $x \notin \overline{C}$, the previous lesson, Corollary 6.4(2), gives $\varphi \in E^*$ and t with $\operatorname{Re} \varphi \leq t$ on $\overline{C}$ and $\operatorname{Re} \varphi(x) > t$. The weakly open set $\{\operatorname{Re} \varphi > t\}$ contains x and misses C , so x is not in the weak closure.

For the last statement, apply this to the convex hull of $\{x_n : n \geq 1\}$. The point x lies in its weak closure, hence in its norm closure. $\square$

The same argument works in every locally convex space.

Theorem 4.2 (Closures of convex sets). Let (X, τ) be a locally convex space with continuous dual X^* . A convex set $C \subseteq X$ has the same closure for τ and for $\sigma(X, X^*)$. Consequently, two locally convex topologies on X with the same continuous linear functionals have the same closed convex sets.

Proof. Every $\varphi \in X^*$ is τ -continuous, so $\sigma(X, X^*) \subseteq \tau$, and the τ -closure $\overline{C}$ lies in the $\sigma(X, X^*)$ -closure. Let $x \notin \overline{C}$. The set $\overline{C}$ is closed and convex. The previous lesson, Corollary 6.4(2), gives $\varphi \in X^*$ and t with $\operatorname{Re} \varphi \leq t$ on $\overline{C}$ and $\operatorname{Re} \varphi(x) > t$. The $\sigma(X, X^*)$ -open set $\{\operatorname{Re} \varphi > t\}$ contains x and misses C . For the last claim, both topologies give the closure of C that $\sigma(X, X^*)$ gives. $\square$

5. Annihilators, bipolars and Goldstine's theorem

For $L \subseteq E$ let $L^\perp = \{\varphi \in E^* : \varphi|_L = 0\}$. For $N \subseteq E^*$ let $N_\perp = \{x \in E : \varphi(x) = 0 \ \forall \varphi \in N\}$.

Theorem 5.1. Let E be a normed space.

1. For a subspace $L \subseteq E$, the norm closure of L is $(L^\perp)_\perp$.
2. For a subspace $N \subseteq E^*$, the weak* closure of N is $(N_\perp)^\perp$.

Proof. 1. The set $(L^\perp)_\perp$ is norm closed and contains L . If $x \notin \overline{L}$, the previous lesson, Corollary 2.3(3), gives $\varphi \in L^\perp$ with $\varphi(x) \neq 0$, so $x \notin (L^\perp)_\perp$.

2. The set $(N_\perp)^\perp$ is weak* closed and contains N . Let ψ lie outside the weak* closure $\overline{N}$, a weak* closed subspace. The weak* topology is locally convex. By the previous lesson, Corollary 6.4(2), there is a weak* continuous functional F with $\operatorname{Re} F(\psi) > t \geq \operatorname{Re} F(\overline{N})$ for some t .

Since $\overline{N}$ is a subspace, F vanishes on it: if $F(\varphi) \neq 0$ for some $\varphi \in \overline{N}$, suitable multiples $s\varphi$ would make $\operatorname{Re} F(s\varphi)$ as large as we like. So $t \geq 0$ and $\operatorname{Re} F(\psi) > 0$.

By Corollary 1.3, $F = j(x)$ for some $x \in E$. Then $x \in N_\perp$ and $\psi(x) \neq 0$, so $\psi \notin (N_\perp)^\perp$. $\square$

Theorem 5.2 (Goldstine). The image $j(B)$ of the closed unit ball B of E is weak* dense in the closed unit ball of E^{**} .

Proof. Let K be the weak* closure of $j(B)$. It is convex and lies in the unit ball of E^{**} , which is weak* closed.

Suppose Φ is in the unit ball of E^{**} but not in K . The space $(E^{**}, \sigma(E^{**}, E^*))$ is locally convex with continuous dual E^* (Theorem 1.2). The previous lesson, Corollary 6.4(2), gives $\varphi \in E^*$ and t with $\operatorname{Re} \Phi(\varphi) > t \geq \operatorname{Re} j(x)(\varphi) = \operatorname{Re} \varphi(x)$ for all $x \in B$.

Taking the supremum over B , and using that B is balanced, gives $\|\varphi\| \leq t$. Then $\operatorname{Re} \Phi(\varphi) > t \geq \|\varphi\| \geq \|\Phi\| \|\varphi\|$, which is impossible. $\square$

Proposition 5.3 (second adjoints). For $T \in B(E, F)$ let $T^* \in B(F^*, E^*)$, $T^*\psi = \psi \circ T$, and $T^{**} = (T^*)^*$.

1. $\|T^*\| = \|T\|$ and $\|T^{**}\| = \|T\|$.
2. $T^{**} \circ j_E = j_F \circ T$.
3. $(ST)^{**} = S^{**}T^{**}$.
4. T^{**} is weak*-weak* continuous.

Proof. 1. $\|T^*\psi\| = \sup_{\|x\| \leq 1} |\psi(Tx)| \leq \|\psi\|\|T\|$. Conversely, for $\|x\| \leq 1$ choose ψ of norm at most 1 with $\psi(Tx) = \|Tx\|$ (previous lesson, Corollary 2.3(2)); then $\|Tx\| \leq \|T^*\|$. Apply the same to T^* .

2.

$$\begin{aligned} T^{**}(j_E x)(\psi) \\ &= j_E x(T^*\psi) \\ &= \psi(Tx) \\ &= j_F(Tx)(\psi). \end{aligned}$$

3. $(ST)^* = T^*S^*$, so $(ST)^{**} = S^{**}T^{**}$.

4. If $\Phi_i \rightarrow \Phi$ weak*, then

$$\begin{aligned} T^{**}\Phi_i(\psi) \\ &= \Phi_i(T^*\psi) \rightarrow \Phi(T^*\psi) \\ &= T^{**}\Phi(\psi) \end{aligned}$$

for every $\psi \in F^*$. $\square$

Proposition 5.4 (The dual of a quotient). Let M be a closed subspace of a normed space E , with quotient map $q : E \rightarrow E/M$ and quotient norm $\|x + M\| = \text{dist}(x, M)$. Then $g \mapsto g \circ q$ is an isometric isomorphism of $(E/M)^*$ onto $M^\perp$.

Proof.

- $g \circ q$ is bounded and vanishes on M .
- *Isometry.* q maps the open unit ball of E onto the open unit ball of E/M : if $\|x + M\| < 1$, some $m \in M$ has $\|x - m\| < 1$. Hence $\|g \circ q\| = \|g\|$.
- *Onto $M^\perp$.* Let $\varphi \in M^\perp$. Then $g(x + M) = \varphi(x)$ is well defined and linear. For every $m \in M$, $|g(x + M)| = |\varphi(x - m)| \leq \|\varphi\| \|x - m\|$. Taking the infimum over m gives $|g(x + M)| \leq \|\varphi\| \|x + M\|$, so $g \in (E/M)^*$ and $g \circ q = \varphi$. $\square$

6. The Krein–Milman theorem

A point e of a convex set K is *extreme* if $e = tx + (1 - t)y$ with $x, y \in K$ and $0 < t < 1$ forces $x = y = e$. A nonempty closed subset $F \subseteq K$ is a *face* (an *extreme subset*) if whenever $tx + (1 - t)y \in F$ with $x, y \in K$ and $0 < t < 1$, both $x, y \in F$.

Theorem 6.1 (Krein–Milman). Let X be a Hausdorff locally convex space and $K \subseteq X$ a nonempty compact convex set. Then K has an extreme point, and K is the closed convex hull of its extreme points.

Proof. *Faces contain extreme points.* Faces of K , ordered by reverse inclusion, satisfy the hypothesis of Zorn's lemma. The intersection of a chain is nonempty, by compactness and the finite intersection property, and it is closed and a face. So there is a minimal face F .

Suppose F has two points $x \neq y$. Choose a continuous φ with $\text{Re } \varphi(x) \neq \text{Re } \varphi(y)$; this exists by the previous lesson, Corollary 6.4(1), replacing φ by $i\varphi$ if needed. The set $F' = \{z \in F : \text{Re } \varphi(z) = \max_F \text{Re } \varphi\}$ is then:

- nonempty, because F is compact;
- closed, and a face of F , hence of K : if $tx' + (1 - t)y' \in F'$ with $x', y' \in K$, then $x', y' \in F$ because F is a face, and both attain the maximum, since a strict average of two values at most the maximum equals the maximum only if both do;
- smaller than F , since it cannot contain both x and y .

This contradicts minimality. So $F = \{e\}$, and e is an extreme point of K , because $\{e\}$ is a face.

The closed convex hull. Let C be the closed convex hull of the extreme points; it lies in K . If some $x \in K \setminus C$ existed, the previous lesson, Theorem 6.3, would give a continuous φ and t with $\text{Re } \varphi < t$ on C and $\text{Re } \varphi(x) > t$.

The set $F = \{z \in K : \operatorname{Re} \varphi(z) = \max_K \operatorname{Re} \varphi\}$ is a face of K disjoint from C , because its points have $\operatorname{Re} \varphi \geq \operatorname{Re} \varphi(x) > t$. By the first part, applied to the compact convex set F , it contains an extreme point of F , which is an extreme point of K because F is a face, and so lies in C . This is a contradiction. $\square$

Theorem 6.2 (Milman). Let X be a Hausdorff locally convex space, $Q \subseteq X$ compact, and suppose that the closed convex hull K of Q is compact. Then every extreme point of K lies in Q .

Proof. Let $e \in K$ be extreme, and suppose $e \notin Q$.

A neighbourhood that keeps e away from Q . For each $q \in Q$, $e - q \neq 0$. Since X is Hausdorff and locally convex, there is a convex balanced open neighbourhood U_q of 0 with $e - q \notin U_q + U_q$. Finitely many sets $q_j + U_{q_j}$ cover Q . Put $W = \bigcap_j U_{q_j}$. Then $e \notin Q + W$: if $e = q + w$ with $q \in q_j + U_{q_j}$ and $w \in W$, then $e - q_j \in U_{q_j} + U_{q_j}$.

- Choose a convex balanced open neighbourhood W' of 0 with $W' + W' \subseteq W$, and let V be the closure of W' . Then V is closed and convex, and $V \subseteq W' + W' \subseteq W$, because every point of the closure of W' lies in $y + W'$ for some $y \in W'$. So $e \notin Q + V$.

Splitting K . Finitely many sets $p_k + W'$, $p_k \in Q$, cover Q . Let K_k be the closed convex hull of $Q \cap (p_k + V)$.

- $K_k \subseteq p_k + V$, because $p_k + V$ is closed and convex. And $K_k \subseteq K$ is closed, hence compact.
- The convex hull K' of $K_1 \cup \dots \cup K_n$ is the image of the compact set $K_1 \times \dots \times K_n \times \Delta_n$, where Δ_n is the simplex of weights, under the continuous map $(x_1, \dots, x_n, t) \mapsto \sum_k t_k x_k$. So K' is compact, hence closed. It is convex and contains Q , so it contains K .

Conclusion. So $e = \sum_k t_k x_k$ with $x_k \in K_k$ and weights t_k . Since e is extreme in K , $e = x_k$ for every k with $t_k > 0$: if $0 < t_1 < 1$, write $e = t_1 x_1 + (1 - t_1)y$ with $y \in K$, so $x_1 = y = e$, and continue with y . Hence $e \in K_k \subseteq p_k + V \subseteq Q + V$ for some k , a contradiction. $\square$

7. The Eberlein–Šmulian theorem

Lemma 7.1. Let E be a normed space and $M \subseteq E^{**}$ a finite-dimensional subspace. Then there are finitely many $\varphi_1, \dots, \varphi_m \in E^*$ of norm 1 with $\max\{0, |\Phi(\varphi_1)|, \dots, |\Phi(\varphi_m)|\} \geq \frac{1}{2} \|\Phi\|$ for every $\Phi \in M$. The family may be empty when $M = \{0\}$.

Proof. If $M = \{0\}$, the empty family works. Otherwise the unit sphere S_M of M is nonempty and compact (previous lesson, Theorem 7.1). For each $\Phi \in S_M$, choose φ_Φ of norm 1 with $|\Phi(\varphi_\Phi)| > 3/4$. The open sets $\{\Psi \in S_M : \|\Psi - \Phi\| < 1/4\}$ cover S_M , so finitely many centres $\Phi_1, \dots, \Phi_m$ suffice. For $\Psi \in S_M$ with $\|\Psi - \Phi_k\| < 1/4$, $|\Psi(\varphi_{\Phi_k})| \geq |\Phi_k(\varphi_{\Phi_k})| - \|\Psi - \Phi_k\| > 1/2$. Scale for general $\Phi \in M$. $\square$

Theorem 7.2 (Eberlein–Šmulian). For a subset A of a Banach space E , the following are equivalent:

1. A is relatively weakly compact;
2. every sequence in A has a subsequence that converges weakly to a point of E ;
3. every sequence in A has a weak cluster point in E .

Proof. If $E = \{0\}$ or $A = \emptyset$, all three assertions are immediate. Suppose otherwise. 2 implies 3. The limit of a weakly convergent subsequence is a weak cluster point.

1 implies 3. A sequence in the compact set $\overline{A}^w$, the weak closure of A , has a cluster point there. The terms $\{x_n : n \geq m\}$ form a decreasing family of sets with the finite intersection property, so their closures meet.

3 implies 1.

Step 1: A is bounded. Otherwise there is $\varphi \in E^*$ with $\sup_{x \in A} |\varphi(x)| = \infty$, by the previous lesson, Corollary 4.3(2). So there are $x_n \in A$ with $|\varphi(x_n)| \geq n$. A weak cluster point x of (x_n) would make $\varphi(x)$ a cluster point of $(\varphi(x_n))$, which has none.

Step 2: the weak* closure of $j(A)$ lies in $j(E)$. The weak* closure $\overline{j(A)}^{w*}$ is weak* compact by Theorem 3.1, since A is bounded. Let Φ be in it. We build $x_n \in A$ and functionals of norm 1 inductively.

- Let φ_1 be any functional of norm 1, and choose $x_1 \in A$ with $|(\Phi - jx_1)(\varphi_1)| < 1$.
- Given $x_1, \dots, x_n$, let $M_n = \text{span}\{\Phi, \Phi - jx_1, \dots, \Phi - jx_n\}$. Lemma 7.1 gives finitely many functionals of norm 1 that norm M_n up to the factor $\frac{1}{2}$. Append them to the list $\varphi_1, \varphi_2, \dots$.
- Since Φ lies in the weak* closure of $j(A)$, choose $x_{n+1} \in A$ with $|(\Phi - jx_{n+1})(\varphi_k)| < 1/(n+1)$ for all functionals φ_k listed so far.

Let x be a weak cluster point of (x_n) .

Step 3: $\Phi = j(x)$.

- $\Psi = \Phi - jx$ lies in $\overline{\bigcup_n M_n}$. The point x is in the weak closure of $\{x_n\}$, hence in the norm-closed linear span of $\{x_n\}$ by Theorem 4.1. So jx is a norm limit of combinations $\sum c_k jx_k$. Then Ψ is a norm limit of $\Phi - \sum c_k jx_k$. Each of these lies in M_n for large n : write

$$\begin{aligned} & \Phi - \sum c_k jx_k \\ &= (1 - \sum c_k) \Phi + \sum c_k (\Phi - jx_k). \end{aligned}$$

- $\|\Psi\| \leq 2 \sup_k |\Psi(\varphi_k)|$. For $G \in M_n$, $\|G\| \leq 2 \max_k |G(\varphi_k)|$, with k running over the functionals chosen for M_n . Approximating Ψ by such G gives $\|\Psi\| \leq 2 \sup_k |\Psi(\varphi_k)| + 3\|\Psi - G\|$, and the last term tends to 0.
- $\Psi(\varphi_k) = 0$ for every k . For n large, $|\Phi(\varphi_k) - \varphi_k(x_n)| < 1/n$. Since $\varphi_k(x)$ is a cluster point of $(\varphi_k(x_n))$, we get $\Phi(\varphi_k) = \varphi_k(x)$.

Hence $\Psi = 0$, that is, $\Phi = jx \in j(E)$.

Step 4: conclusion. The map j is a homeomorphism from (E, weak) onto $(j(E), \text{weak}^*)$: both topologies are pointwise convergence on E^* . By Step 2, $\overline{j(A)}^{\text{weak}^*} = j(K)$ for some $K \subseteq E$. This K is weakly compact and contains A , so A is relatively weakly compact.

1 implies 2. Let (x_n) be a sequence in A .

- Let E_0 be the norm-closed span of $\{x_n\}$, a separable closed subspace. By Theorem 4.1 it is weakly closed. The weak closure K of $\{x_n\}$ is weakly compact by 1 and lies in E_0 .
- Let (y_m) be dense in E_0 . Choose $\psi_m \in E^*$ of norm 1 with $\psi_m(y_m) = \|y_m\|$. The family (ψ_m) separates the points of E_0 : if $z \neq 0$ in E_0 , choose y_m with $\|z - y_m\| < \|z\|/3$; then $|\psi_m(z)| \geq \|y_m\| - \|z - y_m\| > \|z\|/3$.
- By 1 implies 3 and Step 1, A is norm bounded. The weakly closed norm ball containing it also contains K , so this series is uniformly convergent on $K \times K$. The topology on K induced by $d(u, v) = \sum_m 2^{-m} |\psi_m(u - v)|$ is Hausdorff and coarser than the weak topology. A compact topology admits no strictly coarser Hausdorff topology, so the weak topology on K is this metric topology.
- In a compact metric space the closures of the sequence tails have a common point x , by the finite intersection property. Every ball about x contains arbitrarily late terms. Choose successively increasing indices whose terms lie in the balls of radii $1/k$; the subsequence converges in the metric. Thus (x_n) has a subsequence converging weakly in $K \subseteq E$. $\square$

Exercises

Exercise 1 (medium). Show that in an infinite-dimensional normed space the weak closure of the unit sphere $\{\|x\| = 1\}$ is the closed unit ball.

Solution.

- *The closure lies in the ball:* the closed unit ball is convex and norm closed, so weakly closed (Theorem 4.1).
- *Every point with $\|x_0\| < 1$ is in the closure.* A basic weak neighbourhood $U = \{x : |\varphi_i(x - x_0)| < \varepsilon, i \leq n\}$ contains $x_0 + \ker\{\varphi_1, \dots, \varphi_n\}$. This kernel is a nonzero subspace, because a linear map from an infinite-dimensional space to $\mathbb{K}^n$ has nonzero kernel. Pick $y \neq 0$ in it. The function $t \mapsto \|x_0 + ty\|$ is continuous, equals $\|x_0\| < 1$ at $t = 0$, and tends to infinity. Choose $T > 0$ with $\|x_0 + Ty\| > 1$.

Bisect the interval $[0, T]$, retaining endpoint values that bracket one. Its nested endpoints converge to a common real number t ; continuity gives $\|x_0 + ty\| = 1$. This produces a point of the sphere in U .

- Every point with $\|x_0\| = 1$ is on the sphere itself. $\square$

Exercise 2 (medium). Let $E = c_0$, the null sequences with the supremum norm. Show that $\delta_n \rightarrow 0$ weakly but not in norm, and find convex combinations of the δ_n that converge to 0 in norm.

Solution.

- *Weak convergence:* put $a_k = \varphi(\delta_k)$ for $\varphi \in c_0^*$. On a finite set F , choose $\theta_k = \overline{a_k}/|a_k|$ when $a_k \neq 0$, and zero otherwise. Then $\|\sum_{k \in F} \theta_k \delta_k\|_\infty \leq 1$, so $\sum_{k \in F} |a_k| \leq \|\varphi\|$. Hence $a \in \ell^1$. Finite cutoffs approximate each $x \in c_0$ in supremum norm, giving $\varphi(x) = \sum_k a_k x_k$. Conversely this series defines a bounded functional for every $a \in \ell^1$, with norm at most $\|a\|_1$; the same finite phase tests give the reverse inequality. Thus $c_0^* = \ell^1$ isometrically, and $\varphi_a(\delta_n) = a_n \rightarrow 0$.
- *No norm convergence:* $\|\delta_n\| = 1$.
- *Convex combinations:* $\frac{1}{N}(\delta_1 + \cdots + \delta_N)$ has norm $1/N \rightarrow 0$, as Theorem 4.1 predicts. $\square$

Exercise 3 (medium). Show that the closed unit ball of ℓ^1 has extreme points $\{\lambda \delta_n : |\lambda| = 1\}$, but that the closed unit ball of c_0 has none. Deduce that c_0 is not isometrically isomorphic to the dual of any normed space.

Solution.

- ℓ^1 : if $\|x\|_1 < 1$, choose $0 < \varepsilon < 1 - \|x\|_1$; the distinct vectors $x \pm \varepsilon \delta_1$ belong to the ball and average to x . If $\|x\|_1 = 1$ and $x_i, x_j \neq 0$ with $i \neq j$, put $u_i = x_i/|x_i|$, $u_j = x_j/|x_j|$, and choose $0 < \varepsilon < \min(|x_i|, |x_j|)$. The two vectors $x \pm \varepsilon(u_i \delta_i - u_j \delta_j)$ both have norm one, since their two altered coordinate magnitudes add to $|x_i| + |x_j|$. They are distinct and average to x . The remaining points are $\lambda \delta_n$, $|\lambda| = 1$, and these are extreme: if $\lambda \delta_n = (x + y)/2$ with $\|x\|_1, \|y\|_1 \leq 1$, multiply the n th coordinate by $\bar{\lambda}$. The real parts of both resulting numbers are at most one and average to one, so both numbers equal one. Hence $x_n = y_n = \lambda$, and the norm bounds force every other coordinate of x, y to vanish.
- c_0 : given x in the unit ball, $|x_k| < 1/2$ for some k . Then $x \pm \frac{1}{2} \delta_k$ are in the ball and average to x .
- *Deduction:* the unit ball of a dual space is weak* compact and convex (Theorem 3.1), so by Theorem 6.1 it has extreme points. A linear isometric isomorphism preserves convex combinations and therefore extreme points of balls. $\square$

Exercise 4 (easy). Show that a reflexive Banach space (one with $j(E) = E^{**}$) has a weakly compact closed unit ball, and deduce that every bounded sequence in it has a weakly convergent subsequence.

Solution.

- *Weak compactness:* the ball of $E^{**} = j(E)$ is weak* compact by Theorem 3.1, applied to E^* . By Step 4 of the proof of Theorem 7.2, j is a homeomorphism from the weak topology to the weak* topology. So the unit ball of E is weakly compact.
- *Sequences:* Theorem 7.2, $1 \Rightarrow 2$, gives the weakly convergent subsequence. $\square$

Where this leads

The lesson **Hilbert spaces and compact operators** uses these results for operators on Hilbert space. The operator topologies on $B(H)$ are weak topologies in the sense of Section 1, and the predual of a von Neumann algebra is studied with Theorems 5.1 and 7.2 in **Polar decomposition of functionals and weak compactness in preduals**.

References

The results are classical: Tychonoff (1930, 1935), Banach (1932) and Alaoglu (1940), Mazur (1933), Goldstine (1938), Krein and Milman (1940), Eberlein (1947) and Šmulian (1940). The proof of the Eberlein–Šmulian theorem follows R. Whitley, "An elementary proof of the Eberlein–Šmulian theorem", *Mathematische Annalen* 172 (1967) 116–118, written here in our own words.

Hilbert spaces and compact operators

Originally written by Claude Opus 5.5 (Anthropic), October 2026, and self-checked by that writing AI. GPT-6.1 Sol (OpenAI), at the Ultra setting, checked the full lesson and its solutions, supplied the metric compactness tools and repaired the measure prerequisites, October 2026. Public domain (CC0). GPT-6.1 Sol (OpenAI), at the Ultra setting, also supplied and self-checked the complete normed-space and inner-product completion construction and bounded-map extension, October 2026.

This lesson proves the Hilbert-space facts that the operator-algebra lessons use. It covers:

- the Cauchy–Schwarz inequality for positive semidefinite forms;
- the projection theorem and the Riesz–Fréchet theorem;
- bounded sesquilinear forms and adjoints;
- orthonormal bases: Bessel's inequality, Parseval's identity, existence and Hilbert dimension;
- compact operators: they form a closed ideal, closed under adjoints, in which the finite-rank operators are dense;
- the spectral theorem for compact self-adjoint operators;
- the Riesz theory of compact operators: for $\lambda \neq 0$, $T - \lambda$ has finite-dimensional kernel and closed range; it is invertible when injective; and nonzero points of the spectrum are isolated eigenvalues;
- Hilbert tensor products;
- multiplication operators on a σ -finite measure space, which form an algebra equal to its own commutant.

It builds on [Hahn–Banach, Baire and the basic theorems on Banach spaces](#), cited as *the first lesson*.

Conventions

Hilbert spaces are complex unless a statement says otherwise. No separability is assumed. Inner products are linear in the first variable. $B(H)$ is the algebra of bounded operators. The *spectrum* $\sigma(T)$ of $T \in B(H)$ is the set of $\lambda \in \mathbb{C}$ for which $T - \lambda$ has no inverse in $B(H)$. For $\xi, \eta \in H$, $\theta_{\xi, \eta}$ is the rank-one operator $\zeta \mapsto \langle \zeta, \eta \rangle \xi$. $L^\perp = \{\xi : \langle \xi, \eta \rangle = 0 \ \forall \eta \in L\}$.

1. Positive semidefinite forms

A *sesquilinear form* on a complex vector space V is a map $B : V \times V \rightarrow \mathbb{C}$, linear in the first variable and conjugate-linear in the second. It is *Hermitian* if $B(\eta, \xi) = \overline{B(\xi, \eta)}$, and *positive semidefinite* if $B(\xi, \xi) \geq 0$ for all ξ .

Proposition 1.1.

1. (Polarization) $4B(\xi, \eta) = \sum_{k=0}^3 i^k B(\xi + i^k \eta, \xi + i^k \eta)$.
2. A form with $B(\xi, \xi) \in \mathbb{R}$ for all ξ is Hermitian.
3. (Cauchy–Schwarz) If B is positive semidefinite, then $|B(\xi, \eta)|^2 \leq B(\xi, \xi)B(\eta, \eta)$. Consequently $\{\xi : B(\xi, \xi) = 0\}$ is a subspace, and $\xi \mapsto B(\xi, \xi)^{1/2}$ is a seminorm.

Proof.

1. Expand each term by sesquilinearity: the terms $B(\xi, \xi)$ and $B(\eta, \eta)$ cancel, and the cross terms give $4B(\xi, \eta)$.
2. Apply 1 to $B(\eta, \xi)$ and to $\overline{B(\xi, \eta)}$, and compare the expressions term by term, using that the diagonal values are real.
3. By 2, B is Hermitian. For $t \in \mathbb{R}$ and θ with $e^{-i\theta} B(\xi, \eta) = |B(\xi, \eta)|$, the quantity

$$\begin{aligned} & 0 \\ & \leq B(\xi + te^{i\theta}\eta, \xi + te^{i\theta}\eta) \\ & = B(\xi, \xi) + 2t|B(\xi, \eta)| + t^2B(\eta, \eta) \end{aligned}$$

is a nonnegative quadratic polynomial in t . If $B(\eta, \eta) > 0$, its discriminant is ≤ 0 . If $B(\eta, \eta) = 0$, the linear polynomial $B(\xi, \xi) + 2t|B(\xi, \eta)|$ is nonnegative for all t , so $|B(\xi, \eta)| = 0$.

For the consequence, Cauchy–Schwarz gives $B(\xi + \eta, \xi + \eta) \leq (B(\xi, \xi)^{1/2} + B(\eta, \eta)^{1/2})^2$. $\square$

An inner product is a positive definite Hermitian form. A *Hilbert space* is an inner product space that is complete for $\|\xi\| = \langle \xi, \xi \rangle^{1/2}$. The *parallelogram law* $\|\xi + \eta\|^2 + \|\xi - \eta\|^2 = 2\|\xi\|^2 + 2\|\eta\|^2$ follows by expanding.

Completing normed and inner product spaces

Lemma (Completion). A normed space E embeds linearly and isometrically as a dense subspace of a Banach space $\widehat{E}$. If its norm comes from an inner product, that product extends to $\widehat{E}$, which is then a Hilbert space. Every bounded linear map $T : E \rightarrow F$, with F a Banach space, has a unique bounded linear extension to $\widehat{E}$, of the same norm. In particular the Hilbert construction applies to a positive semidefinite form after taking its quotient by the null subspace.

Proof. Let $\mathcal{S}$ be the vector space of Cauchy sequences in E , with termwise operations. Identify $a = (a_n)$ and $b = (b_n)$ when $\|a_n - b_n\| \rightarrow 0$. The triangle inequality shows that this is an equivalence relation compatible with the vector operations. The reverse triangle inequality gives

$$|\|a_n\| - \|a_m\|| \leq \|a_n - a_m\|.$$

So these norms have a limit, unchanged on replacing a by an equivalent sequence. Set $\widehat{E} = \mathcal{S}/\sim$ and $\|[a]\| = \lim_n \|a_n\|$. Homogeneity and the triangle inequality pass to this limit. Its value is zero exactly when a is equivalent to the zero sequence, so this is a norm. Constant sequences give the asserted linear isometry $E \rightarrow \widehat{E}$.

The constants a_n converge to $[a]$: if $\|a_n - a_m\| < \varepsilon$ for all sufficiently large n, m , taking the limit in m gives $\|a_n - [a]\| \leq \varepsilon$. This proves density. If (z_k) is Cauchy in $\widehat{E}$, choose $y_k \in E$ with $\|y_k - z_k\| < 1/k$. Then

$$\|y_k - y_l\| \leq 1/k + \|z_k - z_l\| + 1/l.$$

So (y_k) is Cauchy in E , and represents $z = [(y_k)]$. The density argument gives $y_k \rightarrow z$; therefore $\|z_k - z\| \leq 1/k + \|y_k - z\| \rightarrow 0$. This proves completeness.

If E has an inner product, its Cauchy sequences are bounded: a sufficiently late tail lies within distance 1 of one term, and the remaining initial block is finite. Cauchy–Schwarz therefore shows that the inner products have limits, using

$$\begin{aligned} |\langle a_n, b_n \rangle - \langle a_m, b_m \rangle| &\leq \|a_n - a_m\| \|b_n\| \\ &\quad + \|a_m\| \|b_n - b_m\|. \end{aligned}$$

The same estimate makes the limit independent of the representatives. Sesquilinearity and symmetry pass to the limit; its diagonal value is $\lim_n \|a_n\|^2 = \|[a]\|^2$, so it is a positive definite inner product inducing the complete norm.

For the extension put $\widehat{T}[a] = \lim_n T a_n$. The limit exists in F because $\|T a_n - T a_m\| \leq \|T\| \|a_n - a_m\|$. Equivalent sequences give the same limit by that bound. Linearity passes to limits, and $\|\widehat{T}[a]\| \leq \|T\| \|[a]\|$. Restriction to the constant sequences gives the reverse norm inequality. Density gives uniqueness. Finally Proposition 1.1 shows that a null vector for a positive semidefinite form is orthogonal to every vector, so the form descends to a positive definite inner product on its quotient. The construction just proved then applies. $\square$

2. The projection theorem and the Riesz–Fréchet theorem

Theorem 2.1. Let C be a nonempty closed convex subset of a real or complex Hilbert space H , and $\xi \in H$. There is exactly one $c_0 \in C$ with $\|\xi - c_0\| = \text{dist}(\xi, C)$.

Proof. Let $d = \text{dist}(\xi, C)$ and $c_n \in C$ with $\|\xi - c_n\| \rightarrow d$. The parallelogram law applied to $\xi - c_n$ and $\xi - c_m$, together with $\frac{c_n + c_m}{2} \in C$, gives

$$\begin{aligned} & \|c_n - c_m\|^2 \\ &= 2\|\xi - c_n\|^2 + 2\|\xi - c_m\|^2 \\ &\quad - 4\left\|\xi - \frac{c_n + c_m}{2}\right\|^2 \\ &\leq 2\|\xi - c_n\|^2 + 2\|\xi - c_m\|^2 - 4d^2 \rightarrow 0. \end{aligned}$$

So (c_n) is Cauchy. Its limit $c_0 \in C$ attains the distance. Two minimizers form a minimizing sequence when alternated, so they are equal. $\square$

Theorem 2.2 (projection theorem). Let L be a closed subspace of a real or complex Hilbert space H . Then:

1. $H = L \oplus L^\perp$;
2. the map $P : \xi \mapsto c_0$ of Theorem 2.1 is the orthogonal projection onto L , a bounded self-adjoint idempotent with $\|P\| \leq 1$;
3. for every subspace Y , $\bar{Y} = (Y^\perp)^\perp$.

Proof. Let c_0 be the nearest point of L to ξ . For $\eta \in L$ and scalars t , $\|\xi - c_0 - t\eta\|^2 \geq \|\xi - c_0\|^2$. Expanding gives $-2\text{Re}(t\langle \xi - c_0, \eta \rangle) + |t|^2\|\eta\|^2 \geq 0$. Small t of suitable phase force $\langle \xi - c_0, \eta \rangle = 0$. So $\xi = c_0 + (\xi - c_0)$ with $\xi - c_0 \in L^\perp$. The decomposition is unique because $L \cap L^\perp = \{0\}$.

The projection is linear and idempotent, with $\|P\xi\|^2 + \|(1-P)\xi\|^2 = \|\xi\|^2$. It is self-adjoint because $\langle P\xi, \eta \rangle = \langle P\xi, P\eta \rangle = \langle \xi, P\eta \rangle$.

For 3: $(Y^\perp)^\perp$ is closed and contains Y . By 1, applied to $\bar{Y}$, we have $H = \bar{Y} \oplus \bar{Y}^\perp$ and $Y^\perp = \bar{Y}^\perp$. An element of $(Y^\perp)^\perp$ has zero component in $\bar{Y}^\perp$. $\square$

Theorem 2.3 (Riesz–Fréchet). Every bounded linear functional f on H is $f(\xi) = \langle \xi, \eta \rangle$ for exactly one $\eta \in H$, and $\|f\| = \|\eta\|$.

Proof. If $f = 0$, take $\eta = 0$. Otherwise $\ker f$ is a closed proper subspace. By Theorem 2.2 there is a unit vector $u \perp \ker f$. For every ξ , the vector $f(\xi)u - f(u)\xi$ lies in $\ker f$, so it is orthogonal to u . This gives $f(\xi) = f(u)\langle \xi, u \rangle = \langle \xi, f(u)u \rangle$.

Uniqueness: $\langle \xi, \eta - \eta' \rangle = 0$ for all ξ forces $\eta = \eta'$. The norm equality is Cauchy–Schwarz together with $\xi = \eta$. $\square$

3. Sesquilinear forms and adjoints

Theorem 3.1. Let B be a sesquilinear form on H with $|B(\xi, \eta)| \leq C\|\xi\|\|\eta\|$. There is exactly one $t \in B(H)$ with $B(\xi, \eta) = \langle t\xi, \eta \rangle$, and $\|t\| = \sup\{|B(\xi, \eta)| : \|\xi\|, \|\eta\| \leq 1\} \leq C$. If $B(\xi, \xi) \geq 0$ for all ξ , then $\langle t\xi, \xi \rangle \geq 0$.

Proof. For fixed ξ , the map $\eta \mapsto \overline{B(\xi, \eta)}$ is a bounded linear functional. By Theorem 2.3 it is $\langle \eta, t\xi \rangle$ for a unique vector $t\xi$. Uniqueness makes t linear. The norm formula follows from $\|t\xi\| = \sup_{\|\eta\| \leq 1} |\langle t\xi, \eta \rangle|$. $\square$

Corollary 3.2. Every $T \in B(H)$ has a unique adjoint $T^* \in B(H)$ with $\langle T\xi, \eta \rangle = \langle \xi, T^*\eta \rangle$. Moreover:

- $T^{**} = T$, $\|T^*\| = \|T\|$ and $\|T^*T\| = \|T\|^2$;
- $(ST)^* = T^*S^*$, and $T \mapsto T^*$ is conjugate-linear;
- (complex scalars) if $\langle T\xi, \xi \rangle = 0$ for all ξ , then $T = 0$;
- (complex scalars) if $\langle T\xi, \xi \rangle \in \mathbb{R}$ for all ξ , then $T = T^*$;
- for self-adjoint T , $\|T\| = \sup_{\|\xi\|=1} |\langle T\xi, \xi \rangle|$.

Proof. Apply Theorem 3.1 to $B(\xi, \eta) = \langle \xi, T\eta \rangle$ to obtain T^* . The algebraic rules follow from uniqueness. $\|T\xi\|^2 = \langle T^*T\xi, \xi \rangle \leq \|T^*T\|\|\xi\|^2$ gives $\|T\|^2 \leq \|T^*T\| \leq \|T^*\|\|T\|$. So $\|T\| \leq \|T^*\|$, and by symmetry the two norms are equal. The two complex-scalar statements follow from Proposition 1.1(1) and (2), applied to $B(\xi, \eta) = \langle T\xi, \eta \rangle$.

The last statement. Let $m = \sup_{\|\xi\|=1} |\langle T\xi, \xi \rangle|$, so $|\langle T\xi, \xi \rangle| \leq m\|\xi\|^2$. For unit ξ, η , expanding and using self-adjointness gives

$$\begin{aligned} & 4 \operatorname{Re} \langle T\xi, \eta \rangle \\ &= \langle T(\xi + \eta), \xi + \eta \rangle \\ &\quad - \langle T(\xi - \eta), \xi - \eta \rangle \\ &\leq m(\|\xi + \eta\|^2 + \|\xi - \eta\|^2) \\ &= 4m. \end{aligned}$$

Replacing η by $e^{i\theta}\eta$ gives $|\langle T\xi, \eta \rangle| \leq m$, so $\|T\| \leq m$. The reverse inequality is Cauchy–Schwarz. $\square$

4. Orthonormal bases

An *orthonormal family* $(e_i)_{i \in I}$ satisfies $\langle e_i, e_j \rangle = \delta_{ij}$. It is an *orthonormal basis* if its closed linear span is H .

Theorem 4.1. Let $(e_i)_{i \in I}$ be an orthonormal family in H .

1. (Bessel) $\sum_i |\langle \xi, e_i \rangle|^2 \leq \|\xi\|^2$. In particular $\langle \xi, e_i \rangle \neq 0$ for at most countably many i .
2. For $(c_i) \in \ell^2(I)$, the sum $\sum_i c_i e_i$ converges, as the net of finite partial sums, and $\|\sum_i c_i e_i\|^2 = \sum_i |c_i|^2$.
3. The following are equivalent:
 - the family is a basis;
 - it is maximal among orthonormal families;
 - $\xi = \sum_i \langle \xi, e_i \rangle e_i$ for every ξ ;
 - Parseval's identity $\langle \xi, \eta \rangle = \sum_i \langle \xi, e_i \rangle \langle e_i, \eta \rangle$ holds for all ξ, η .
4. Every orthonormal family is contained in an orthonormal basis. A separable Hilbert space has a finite or countable orthonormal basis.
5. Any two orthonormal bases of H have the same cardinality, the *Hilbert dimension* of H .

Proof.

1. For finite $F \subseteq I$, $\xi - \sum_{i \in F} \langle \xi, e_i \rangle e_i$ is orthogonal to each e_i with $i \in F$. So $\|\xi\|^2 \geq \sum_{i \in F} |\langle \xi, e_i \rangle|^2$. The set $\{i : |\langle \xi, e_i \rangle| > 1/n\}$ is finite for every n .
2. Only countably many c_i are nonzero. The partial sums form a Cauchy net, because $\|\sum_{i \in F \setminus G} c_i e_i\|^2 = \sum_{F \setminus G} |c_i|^2$.
3. *Basis implies expansion.* The vector $\xi - \sum_i \langle \xi, e_i \rangle e_i$ is orthogonal to every e_j , hence to their closed span H , so it is zero.
 - *Expansion implies Parseval:* take inner products of the expansions.
 - *Parseval implies maximality:* a unit vector e orthogonal to all e_i would have $\|e\|^2 = \sum |\langle e, e_i \rangle|^2 = 0$.
 - *Maximality implies basis:* if the closed span L were not H , Theorem 2.2 would give a unit vector in $L^\perp$, which could be added to the family.
4. Orthonormal families containing the given one, ordered by inclusion, have unions of chains as upper bounds. Zorn's lemma (first lesson, Theorem 1.1) gives a maximal one, a basis by 3. For separable H , apply the Gram–Schmidt process to a dense sequence.
5. If one basis is finite, H is finite-dimensional, and both bases have $\dim H$ elements. Suppose both are infinite, $(e_i)_{i \in I}$ and $(f_j)_{j \in J}$. Each e_i has at most countably many j with $\langle e_i, f_j \rangle \neq 0$, by 1. Every j occurs for some i , since $f_j \neq 0$ is the sum of its expansion in the e_i . So J is a union of countable sets indexed by I , and $|J| \leq |I|$ (first lesson, Theorem 8.4(3)). Symmetrically $|I| \leq |J|$, and $|I| = |J|$ by the Cantor–Schröder–Bernstein theorem (first lesson, Theorem 8.1). $\square$

5. Compact operators

$T \in B(H)$ is *compact* if the image of the unit ball is relatively compact. $K(H)$ is the set of compact operators and $F(H)$ the set of finite-rank operators.

Metric compactness tools

A metric space is *totally bounded* if, for every $\varepsilon > 0$, finitely many balls of radius ε cover it.

Lemma 5.0. A metric space is compact if and only if it is complete and totally bounded. A compact metric space is sequentially compact and separable.

Proof. In a compact space, every sequence has a point whose every neighbourhood contains infinitely many sequence terms, counted with their indices: otherwise a finite subcover of neighbourhoods containing only finitely many terms gives a contradiction. Choosing increasing indices in the balls of radius $1/k$ at that point gives a convergent subsequence. A Cauchy sequence with a convergent subsequence converges to the same limit, so the space is complete. Compactness also gives finite covers by balls of any prescribed radius.

Conversely, in a totally bounded space, repeatedly choose an infinite set of remaining sequence indices lying in one ball of radius 2^{-k} . Take these sets nested and choose increasing indices from them. The resulting subsequence is Cauchy, since its terms from stage k onward are at mutual distance at most 2^{1-k} . Completeness gives its limit.

To obtain compactness, consider an open cover. There is a number $\delta > 0$ such that every ball of radius δ is contained in a member of the cover. If there were no such number, choose x_n whose ball of radius $1/n$ is contained in no member. A convergent subsequence tends to some x in a member U . Choose $r > 0$ with $B(x, 2r) \subseteq U$. For large subsequence indices, $d(x_n, x) < r$ and $1/n < r$, so $B(x_n, 1/n) \subseteq U$, a contradiction. A finite cover by δ -balls now gives a finite subcover of the original cover. Finally choose finite $1/n$ -nets for all n ; their union is countable and dense. $\square$

In a finite-dimensional Hilbert space, bounded sets are totally bounded: express them in an orthonormal basis, bound each coordinate by Cauchy–Schwarz, and use a finite grid in the resulting bounded coordinate box. A finite-dimensional subspace is closed, because the coordinates of a convergent sequence converge and its limit is their finite basis expansion. Its bounded closed subsets are therefore complete and compact by the lemma.

Compactness and finite-rank approximation

Theorem 5.1.

1. $K(H)$ is a norm-closed two-sided ideal of $B(H)$ that contains $F(H)$.
2. T is compact iff T^* is compact.
3. $K(H)$ is the norm closure of $F(H)$.
4. A compact operator maps weakly convergent sequences to norm convergent sequences.
5. Let (P_i) be a net in $B(H)$ with $\|P_i\| \leq 1$ and $P_i\xi \rightarrow \xi$ for every $\xi \in H$. Then $\|T - P_iT\| \rightarrow 0$ for every compact T . This applies to the projections P_n onto $\text{span}\{e_1, \dots, e_n\}$ for an orthonormal basis (e_k) of a separable H (Theorem 4.1(3)).

Proof. 5. The closure K of $T(\text{ball})$ is compact. Given $\varepsilon > 0$, cover K by finitely many balls $B(y_k, \varepsilon)$, and choose i_0 with $\|(1 - P_i)y_k\| < \varepsilon$ for every k and every $i \geq i_0$. For $y \in K$ with $\|y - y_k\| < \varepsilon$,

$$\begin{aligned} & \|(1 - P_i)y\| \\ & \leq \|(1 - P_i)y_k\| + \|1 - P_i\| \|y - y_k\| < 3\varepsilon. \end{aligned}$$

So $\|T - P_iT\| = \sup_{y \in K} \|(1 - P_i)y\| \leq 3\varepsilon$ for $i \geq i_0$.

1. A finite-rank bounded operator maps the ball into a bounded subset of a finite-dimensional space, which is relatively compact. Sums of compact operators are compact, since the image of the ball lies in the sum of two relatively compact sets. Products with bounded operators are compact, since bounded operators are continuous.

Closedness. Let $T_n \rightarrow T$ in norm with T_n compact, and $\varepsilon > 0$. Choose n with $\|T - T_n\| < \varepsilon/3$, and cover $T_n(\text{ball})$ by finitely many $\varepsilon/3$ -balls. The ε -balls with the same centres cover $T(\text{ball})$. So $T(\text{ball})$ is totally bounded, hence relatively compact, since H is complete.

2. If T is compact, so is TT^* . For $\|\xi_n\| \leq 1$, choose a subsequence along which $TT^*\xi_n$ converges. Then

$$\begin{aligned} & \|T^*(\xi_n - \xi_m)\|^2 \\ &= \langle TT^*(\xi_n - \xi_m), \xi_n - \xi_m \rangle \\ &\leq 2\|TT^*(\xi_n - \xi_m)\| \rightarrow 0, \end{aligned}$$

so $T^*\xi_n$ converges along the subsequence. Apply this to T^* for the converse.

3. Let T be compact. The closure K of $T(\text{ball})$ is a compact metric space, so it has a countable dense subset. The range of T is the union of the sets $nT(\text{ball})$, so its closure L is separable. Let (e_k) be an orthonormal basis of L , and P_n the projection onto $\text{span}\{e_1, \dots, e_n\}$. Then $P_n y \rightarrow y$ for $y \in L$ (Theorem 4.1(3)), and $P_n = 0$ on $L^\perp$. The argument of 5, applied with $K \subseteq L$, gives $\|T - P_n T\| \rightarrow 0$. Each $P_n T$ has finite rank.

4. Let $\xi_n \rightarrow \xi$ weakly. Then (ξ_n) is bounded (first lesson, Corollary 4.3(2)), and $T\xi_n \rightarrow T\xi$ weakly. If $\|T\xi_n - T\xi\| \not\rightarrow 0$, a subsequence stays at distance $\geq \varepsilon$. A further subsequence converges in norm, by compactness, and its limit must be the weak limit $T\xi$, a contradiction. $\square$

6. The spectral theorem for compact self-adjoint operators

Lemma 6.1. Let T be compact and self-adjoint, $T \neq 0$. Then $\|T\|$ or $-\|T\|$ is an eigenvalue of T .

Proof. By Corollary 3.2, there are unit vectors ξ_n with $\langle T\xi_n, \xi_n \rangle \rightarrow \lambda$, where $|\lambda| = \|T\|$. Then

$$\begin{aligned} & \|T\xi_n - \lambda\xi_n\|^2 \\ &= \|T\xi_n\|^2 - 2\lambda\langle T\xi_n, \xi_n \rangle + \lambda^2 \\ &\leq 2\lambda^2 - 2\lambda\langle T\xi_n, \xi_n \rangle \rightarrow 0. \end{aligned}$$

A subsequence has $T\xi_n \rightarrow \eta$, by compactness. Then $\lambda\xi_n \rightarrow \eta$, so $\|\eta\| = |\lambda| \neq 0$, and $T\eta = \lim \lambda T\xi_n = \lambda\eta$. $\square$

Theorem 6.2. Let T be compact and self-adjoint, $T \neq 0$. There are an orthonormal family $(e_n)_{n \in N}$, with $N = \{1, \dots, r\}$ or $N = \mathbb{N}$, and real numbers $\lambda_n \neq 0$ with $|\lambda_1| \geq |\lambda_2| \geq \dots$, tending to 0 if $N = \mathbb{N}$, such that

$$T = \sum_n \lambda_n \theta_{e_n, e_n},$$

with convergence in norm. The nonzero eigenvalues of T are the λ_n . The eigenspace of an eigenvalue $\mu \neq 0$ is spanned by the e_n with $\lambda_n = \mu$ and is finite-dimensional. $T \geq 0$ iff all $\lambda_n > 0$.

Proof. *The eigenvectors.* Put $H_1 = H$. Lemma 6.1 gives a unit eigenvector e_1 with eigenvalue λ_1 , $|\lambda_1| = \|T\|$. The space $H_2 = \{e_1\}^\perp$ is invariant under T , because T is self-adjoint, and $T|_{H_2}$ is compact and self-adjoint. Repeat. The process stops if $T|_{H_{r+1}} = 0$. Otherwise it produces (e_n, λ_n) for all n , with $|\lambda_n|$ nonincreasing.

The eigenvalues tend to 0. If not, $|\lambda_n| \geq \delta > 0$ for all n . Then $\|Te_n - Te_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\delta^2$, so (Te_n) has no convergent subsequence, contradicting compactness.

The expansion. $T - \sum_{n \leq k} \lambda_n \theta_{e_n, e_n}$ vanishes on $\text{span}\{e_1, \dots, e_k\}$ and equals T on H_{k+1} . So its norm is $\|T|_{H_{k+1}}\| = |\lambda_{k+1}| \rightarrow 0$.

Eigenvalues and eigenspaces. If $T\xi = \mu\xi$ with $\mu \neq 0$, then $\mu\xi = \sum \lambda_n \langle \xi, e_n \rangle e_n$. So ξ lies in the closed span of the e_n , and $\langle \xi, e_n \rangle = 0$ unless $\lambda_n = \mu$. Only finitely many λ_n equal μ , because $\lambda_n \rightarrow 0$.

Positivity. $\langle T\xi, \xi \rangle = \sum \lambda_n |\langle \xi, e_n \rangle|^2$. $\square$

7. The Riesz theory of compact operators

Theorem 7.1. Let $T \in K(H)$ and $\lambda \in \mathbb{C} \setminus \{0\}$, and put $S = T - \lambda$.

1. $\ker S$ is finite-dimensional.

2. $S(H)$ is closed.
3. If S is injective, it is surjective, hence invertible in $B(H)$.
4. Consequently, every nonzero point of the spectrum $\sigma(T)$ is an eigenvalue of finite multiplicity.
5. The nonzero eigenvalues of T have no accumulation point other than 0. Each nonzero point of $\sigma(T)$ is isolated.

Proof. 1. On $\ker S$, $T = \lambda$. So the unit ball of $\ker S$ equals $\lambda^{-1}T(\text{ball of } \ker S)$, which is relatively compact. An infinite orthonormal sequence in $\ker S$ would have mutual distances $\sqrt{2}$. So $\ker S$ is finite-dimensional.

2. Let $Sx_n \rightarrow y$. Write $x_n = u_n + v_n$ with $u_n \in \ker S$ and $v_n \perp \ker S$, so $Sw_n = Sx_n$.

(v_n) is bounded. Otherwise, along a subsequence with $\|v_n\| \rightarrow \infty$, the unit vectors $w_n = v_n/\|v_n\|$ have $Sw_n \rightarrow 0$. A further subsequence has $Tw_n \rightarrow z$. Then $\lambda w_n = Tw_n - Sw_n \rightarrow z$, so $w_n \rightarrow z/\lambda$, a unit vector orthogonal to $\ker S$ with $S(z/\lambda) = 0$, which is impossible.

Conclusion. A subsequence of the bounded sequence has $Tv_n \rightarrow z'$. Then $\lambda v_n = Tv_n - Sv_n \rightarrow z' - y$, so $v_n \rightarrow v = (z' - y)/\lambda$, and $Sv = y$.

3. Suppose S is injective but $S(H) \neq H$. Each S^k is of the form $(-\lambda)^k + T_k$ with T_k compact, so $S^k(H)$ is closed by 2. The sequence $H \supsetneq S(H) \supsetneq S^2(H) \supsetneq \dots$ is strictly decreasing: if $S^k(H) = S^{k+1}(H)$, take $x \notin S(H)$. Then $S^k x = S^{k+1}y$ for some y , so $S^k(x - Sy) = 0$, and injectivity gives $x = Sy$, a contradiction.

Choose unit vectors $x_k \in S^k(H) \ominus S^{k+1}(H)$. For $k < m$,

$$Tx_k - Tx_m = \lambda x_k + (Sx_k - Sx_m - \lambda x_m),$$

and the bracket lies in $S^{k+1}(H)$, which is orthogonal to x_k . So $\|Tx_k - Tx_m\| \geq |\lambda|$, and (Tx_k) has no convergent subsequence, contradicting compactness. Hence S is surjective, and its inverse is bounded by the inverse mapping theorem (first lesson, Corollary 5.2).

4. If $\lambda \neq 0$ is not an eigenvalue, then S is injective, hence invertible by 3, so $\lambda \notin \sigma(T)$. The multiplicity is finite by 1.
5. Suppose $\lambda_n \rightarrow \lambda \neq 0$ are distinct eigenvalues with eigenvectors e_n . Eigenvectors for distinct eigenvalues are linearly independent. Let $M_n = \text{span}\{e_1, \dots, e_n\}$ and choose unit vectors $y_n \in M_n \ominus M_{n-1}$. Then $(T - \lambda_n)M_n \subseteq M_{n-1}$. For $m < n$,

$$\begin{aligned} & \frac{Ty_n}{\lambda_n} - \frac{Ty_m}{\lambda_m} \\ &= y_n - \left(\frac{(\lambda_n - T)y_n}{\lambda_n} + \frac{Ty_m}{\lambda_m} \right), \end{aligned}$$

and the bracket lies in $M_{n-1} \perp y_n$. So the left side has norm ≥ 1 . Since $\|y_n/\lambda_n\|$ is bounded, this contradicts compactness. Hence the nonzero eigenvalues accumulate only at 0, and by 4 every nonzero point of $\sigma(T)$ is isolated. $\square$

8. Hilbert tensor products

For a set S , $\ell^2(S)$ is the space of functions $c : S \rightarrow \mathbb{C}$ with $\sum_s |c_s|^2 < \infty$, with $\langle c, d \rangle = \sum_s c_s \bar{d}_s$. It is a Hilbert space. Completeness: a Cauchy sequence $(c^{(n)})$ converges at each s to some c_s . For every finite $F \subseteq S$,

$$\begin{aligned} & \sum_{s \in F} |c_s - c_s^{(n)}|^2 \\ &= \lim_m \sum_{s \in F} |c_s^{(m)} - c_s^{(n)}|^2 \\ &\leq \sup_{m \geq n} \|c^{(m)} - c^{(n)}\|^2 \end{aligned}$$

. So $c - c^{(n)} \in \ell^2(S)$ and $\|c - c^{(n)}\| \rightarrow 0$. The functions δ_s form an orthonormal basis.

Theorem 8.1 (Hilbert tensor products). Let H and K be Hilbert spaces.

1. There are a Hilbert space $H \otimes K$ and a bilinear map $(\xi, \eta) \mapsto \xi \otimes \eta$ from $H \times K$ to $H \otimes K$ with

$$\langle \xi \otimes \eta, \xi' \otimes \eta' \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle,$$

such that the elementary tensors $\xi \otimes \eta$ span a dense subspace.

2. If $(e_i)_{i \in I}$ and $(f_j)_{j \in J}$ are orthonormal bases of H and K , then $(e_i \otimes f_j)$ is an orthonormal basis of $H \otimes K$.
3. (*Universal property*.) Let $B : H \times K \rightarrow \mathcal{K}$ be a bilinear map into a Hilbert space with $\langle B(\xi, \eta), B(\xi', \eta') \rangle = \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle$. There is exactly one isometry $U : H \otimes K \rightarrow \mathcal{K}$ with $U(\xi \otimes \eta) = B(\xi, \eta)$. It is unitary if the vectors $B(\xi, \eta)$ span a dense subspace. So $H \otimes K$ is unique up to a unitary that matches the elementary tensors.

Proof. (1) and (2). Fix orthonormal bases as in (2), and put $H \otimes K = \ell^2(I \times J)$ and $(\xi \otimes \eta)(i, j) = \langle \xi, e_i \rangle \langle \eta, f_j \rangle$.

- By Parseval's identity (Theorem 4.1(3)), $\sum_{i,j} |\langle \xi, e_i \rangle|^2 |\langle \eta, f_j \rangle|^2 = \|\xi\|^2 \|\eta\|^2$, so $\xi \otimes \eta \in \ell^2(I \times J)$.
- The map is bilinear, and by Parseval's identity

$$\begin{aligned} & \langle \xi \otimes \eta, \xi' \otimes \eta' \rangle \\ &= \sum_i \langle \xi, e_i \rangle \langle e_i, \xi' \rangle \sum_j \langle \eta, f_j \rangle \langle f_j, \eta' \rangle \\ &= \langle \xi, \xi' \rangle \langle \eta, \eta' \rangle. \end{aligned}$$

- $e_i \otimes f_j = \delta_{(i,j)}$. These form an orthonormal basis, so the elementary tensors span a dense subspace.

(3) For finite sums,

$$\begin{aligned} & \left\| \sum_k c_k B(\xi_k, \eta_k) \right\|^2 \\ &= \sum_{k,l} c_k \bar{c}_l \langle \xi_k, \xi_l \rangle \langle \eta_k, \eta_l \rangle \\ &= \left\| \sum_k c_k \xi_k \otimes \eta_k \right\|^2. \end{aligned}$$

So $\sum_k c_k \xi_k \otimes \eta_k \mapsto \sum_k c_k B(\xi_k, \eta_k)$ is well defined, since a combination equal to 0 goes to a vector of norm 0, and it is isometric. It extends by continuity to $H \otimes K$. Uniqueness holds because the elementary tensors are total. An isometry has closed range, so if its range is dense, it is onto. $\square$

9. Multiplication operators

Let (Z, Σ, ν) be a σ -finite measure space. $L^\infty(\nu)$ is the space of essentially bounded measurable functions modulo null functions, with the essential supremum norm $\|f\|_\infty$. For $f \in L^\infty(\nu)$, m_f is the operator $\xi \mapsto f\xi$ on $L^2(\nu)$. The **measure-tools lesson**, Theorems 3.1–3.2, proves Hölder, completeness of L^2 , and density of simple functions on arbitrary measure spaces. Its proof uses elementary integration and convergence, not the operator theorem here. Thus the use of Hilbert representation in the later Radon–Nikodym theorem creates no circular dependence.

Theorem 9.1.

1. $f \mapsto m_f$ is an isometric unital $*$ -homomorphism from $L^\infty(\nu)$ into $B(L^2(\nu))$.
2. An operator $T \in B(L^2(\nu))$ commutes with every m_f if and only if $T = m_g$ for some $g \in L^\infty(\nu)$. So the algebra $\{m_f : f \in L^\infty(\nu)\}$ equals its own commutant: it is *maximal abelian*.

Proof. (1) $\|f\xi\|_2 \leq \|f\|_\infty \|\xi\|_2$, and $\langle f\xi, \eta \rangle = \langle \xi, \bar{f}\eta \rangle$. For $\varepsilon > 0$, the set $\{|f| > \|f\|_\infty - \varepsilon\}$ has positive measure. Since ν is σ -finite, it contains a set E with $0 < \nu(E) < \infty$. Then $\|f1_E\|_2 \geq (\|f\|_\infty - \varepsilon)\|1_E\|_2$. So $\|m_f\| = \|f\|_\infty$.

(2) *Reduction to a finite measure.* Write Z as a disjoint union of sets $Z_n \in \Sigma$ of finite measure. Let $w = \sum_n 2^{-n}(1 + \nu(Z_n))^{-1}1_{Z_n}$. Then $w > 0$ everywhere, and $\lambda = w\nu$ is a finite measure with the same null sets as ν .

- So $L^\infty(\lambda) = L^\infty(\nu)$.
- $V\xi = w^{-1/2}\xi$ is a unitary from $L^2(\nu)$ onto $L^2(\lambda)$, with inverse $\xi \mapsto w^{1/2}\xi$.
- $Vm_fV^{-1} = m_f$.

So we may assume $\nu(Z) < \infty$. Then $1 \in L^2(\nu)$, and $L^\infty(\nu) \subseteq L^2(\nu)$.

The function g . Let T commute with every m_f , and put $g = T1 \in L^2(\nu)$. For $f \in L^\infty(\nu)$,

$$Tf = Tm_f1 = m_fT1 = fg.$$

- g is essentially bounded. For $c > \|T\|$ let $E = \{|g| > c\}$. Then $c^2\nu(E) \leq \|1_Eg\|_2^2 = \|T1_E\|_2^2 \leq \|T\|^2\nu(E)$, so $\nu(E) = 0$. Hence $\|g\|_\infty \leq \|T\|$.
- $T = m_g$. The two operators agree on $L^\infty(\nu)$, which contains the simple functions. These are dense in $L^2(\nu)$ **Theorem 3.2 of the measure-tools lesson**.

The converse holds because the algebra is commutative. $\square$

Exercises

Exercise 1 (easy; diagonal operators). Let (e_n) be an orthonormal basis of H and $T = \sum_n \mu_n \theta_{e_n, e_n}$ for a bounded sequence (μ_n) . Show that T is compact iff $\mu_n \rightarrow 0$.

Solution. If $\mu_n \rightarrow 0$, the finite-rank truncations converge to T in norm, so T is compact by Theorem 5.1(1). If $|\mu_{n_k}| \geq \delta > 0$ along a subsequence, then $\|Te_{n_k} - Te_{n_l}\|^2 \geq 2\delta^2$, so T is not compact. $\square$

Exercise 2 (medium; Hilbert–Schmidt operators). Show that an operator T with $\sum_n \|Te_n\|^2 < \infty$ for some orthonormal basis (e_n) is compact.

Solution. Let P_k be the projection onto $\text{span}\{e_1, \dots, e_k\}$. Then

$$\begin{aligned} & \|T(1 - P_k)\xi\| \\ & \leq \sum_{n>k} |\langle \xi, e_n \rangle| \|Te_n\| \\ & \leq \|\xi\| \left(\sum_{n>k} \|Te_n\|^2 \right)^{1/2} \end{aligned}$$

by Cauchy–Schwarz. So $TP_k \rightarrow T$ in norm, and each TP_k has finite rank. $\square$

Exercise 3 (hard; the Volterra operator). On $L^2[0, 1]$, let $(V\xi)(s) = \int_0^s \xi(t) dt$. Show that V is compact and that $\sigma(V) = \{0\}$.

Solution. First V is bounded: Cauchy–Schwarz gives $|V\xi(s)|^2 \leq s\|\xi\|_2^2$, hence $\|V\xi\|_2^2 \leq \frac{1}{2}\|\xi\|_2^2$. Also $|V\xi(s) - V\xi(t)| \leq |s - t|^{1/2}\|\xi\|_2$, so $V\xi$ is continuous.

- *Compactness:* with an orthonormal basis (e_n) ,

$$\begin{aligned} & \sum_n \|Ve_n\|^2 \\ & = \int_0^1 \sum_n |\langle 1_{[0,s]}, \overline{e_n} \rangle|^2 ds \\ & = \int_0^1 \|1_{[0,s]}\|^2 ds \\ & = \frac{1}{2} \end{aligned}$$

, by Parseval applied to the basis $(\overline{e_n})$ and **monotone convergence**. By Exercise 2, V is compact.

- *Spectrum:* by Theorem 7.1(4), a nonzero point of $\sigma(V)$ would be an eigenvalue. If $V\xi = \lambda\xi$, then $\xi = \lambda^{-1}V\xi$ is continuous, then C^1 by the fundamental theorem of calculus proved in **Lemma 6.1 of the measure-tools lesson**, with $\xi = \lambda\xi'$ and $\xi(0) = 0$. The function $e^{-s/\lambda}\xi(s)$ has zero derivative and is zero at $s = 0$, so $\xi = 0$. Finally $0 \in \sigma(V)$, because V is not invertible in infinite dimensions: a compact invertible operator would make the identity compact. $\square$

Exercise 4 (medium; closed range of $1 - T$). Let T be compact. Show that $\dim \ker(1 - T) = \dim \ker(1 - T^*)$.

Solution.

- Both kernels are finite-dimensional (Theorem 7.1(1), applied to T and to T^* , which is compact by Theorem 5.1(2)).
- Let $S = 1 - T$. Since $S(H)$ is closed, $\ker S^* = S(H)^\perp$, and $H = S(H) \oplus \ker S^*$.
- Suppose $\dim \ker S < \dim \ker S^*$. Choose an injective linear map A from $\ker S$ into $\ker S^*$ that is not onto, and extend it by 0 on $(\ker S)^\perp$. This gives a finite-rank operator. Then $S' = S + A = 1 - (T - A)$ is injective with range $S(H) + A(\ker S) \neq H$, contradicting Theorem 7.1(3).
- Exchanging T and T^* gives the reverse inequality. $\square$

Where this leads

The continuous functional calculus and the operator algebra $B(H)$ are developed in [C*-algebras: continuous functional calculus, automatic continuity, positive cones, approximate identities and quotients](#). The spectral theorem for bounded self-adjoint operators with projection-valued measures, and Calkin's theorem on the ideals of $B(H)$, are proved in [The spectral theorem for bounded self-adjoint operators](#). The trace class and the operator topologies are in [Compact and trace-class operators, the predual of \$B\(H\)\$, and the operator topologies](#). Hilbert tensor products are used for L^2 of a product of Radon measures in [Haar measure on locally compact groups](#), and multiplication operators are the starting point of [Abelian operator algebras](#).

References

Hilbert (1904–1910), Riesz (1907, 1918) and Fréchet (1907). The Riesz theory of compact operators is from F. Riesz, "Über lineare Funktionalgleichungen", *Acta Mathematica* 41 (1918) 71–98. The proofs are written here in our own words.

The Stone–Weierstrass theorem for functions vanishing at infinity

Written by Claude Opus 5.5 (Anthropic), September 2026, revised October 2026. The September text was spot-checked by Claude Opus 5.5 in a separate session. The October revisions correct two points found by an AI reviewer in a separate OpenAI Codex session and add proofs of the background facts that the core courses leave as exercises; they are self-checked by the writing AI. Public domain (CC0).

This lesson proves the Stone–Weierstrass theorem for continuous functions that vanish at infinity. Let X be a locally compact Hausdorff space, and let A be a subalgebra of $C_0(X)$ that is closed under complex conjugation, separates the points of X , and has no common zero. Then every function in $C_0(X)$ is a uniform limit of functions in A ([Theorem 10.1](#)). The algebra need not contain the constant functions, and X need not be compact, σ -compact or metrizable. Commutative C*-algebras and harmonic analysis on locally compact groups need the theorem in this form.

The proof follows the classical route. A recursion for polynomials without constant term approximates $|t|$ (Section 6). With interpolation at two points (Section 7) and Stone's lattice lemma (Section 8), it gives the real theorem on compact spaces (Section 9). The one-point compactification (Section 3) carries the result over to $C_0(X)$, and real and imaginary parts give the complex theorem (Section 10). Sections 11–13 give worked examples, show that each hypothesis is needed, and describe the closure of an arbitrary self-adjoint subalgebra. Sections 2–5 develop the function spaces and the topology that surround the proof: the one-point compactification, locally compact spaces, Urysohn's lemma and metrizability. Sections 14–18 add classical companions: the Weierstrass theorem with Bernstein's constructive proof, Korovkin's theorem, further density theorems, the Tietze extension theorem, the disc algebra, and approximation by polynomials with integer coefficients.

The lesson assumes the core courses [Real Analysis I](#), [Real Analysis II](#) and [Point-Set Topology](#): open and closed sets, continuity, compactness defined by open covers, metric spaces and uniform convergence. A few facts about numbers, the Riemann integral and the exponential function are taken from the core courses; they are

listed near the end, under "Results used from other lessons", with the place of each proof. No other lesson is needed.

Weierstrass proved in 1885 that every continuous function on a closed bounded interval is a uniform limit of polynomials [Weierstrass 1885]. Stone proved the general theorem in 1937 and gave a simpler proof, by lattices of functions, in 1948 [Stone 1937], [Stone 1948] P. The one-point compactification is due to Alexandroff [Alexandroff 1924]. The core course *Real Analysis II* proves the Weierstrass theorem and the real and complex Stone–Weierstrass theorems for compact metric spaces (Lebl, Section 11.7). Free texts that treat the theorems are [van Neerven] and [Blackadar]; see also, e.g., [Rudin, Theorems 7.26, 7.32 and 7.33].

1. Conventions

Spaces. A *neighbourhood* of a point is a set that contains an open set containing the point. *Compact* means that every open cover has a finite subcover; the Hausdorff property is not part of the word. A space is *locally compact Hausdorff* (LCH) if it is Hausdorff and every point has a compact neighbourhood. Five elementary facts are used throughout.

- (K1) A closed set C contained in a compact set K is compact. Add $X \setminus C$ to an open cover of C , take a finite subcover of K , and discard $X \setminus C$.
- (K2) A continuous image of a compact set is compact, and a finite union of compact sets is compact.
- (K3) Let X be Hausdorff, $K \subseteq X$ compact and $x \notin K$. Then x and K have disjoint open neighbourhoods. For each $y \in K$ choose disjoint open sets $U_y \ni x$ and $V_y \ni y$. Finitely many sets $V_{y_1}, \dots, V_{y_m}$ cover K ; take $U_{y_1} \cap \dots \cap U_{y_m}$ and $V_{y_1} \cup \dots \cup V_{y_m}$. In particular, compact subsets of Hausdorff spaces are closed.
- (K4) Let a continuous map from a compact space onto a Hausdorff space be one-to-one. Then its inverse is continuous, because the map sends closed sets to closed sets: a closed set is compact by (K1), its image is compact by (K2), and a compact subset of a Hausdorff space is closed by (K3).
- (K5) A continuous $u : K \rightarrow \mathbb{R}$ on a nonempty compact space is bounded and attains its maximum and minimum. The open sets $\{u < n\}$ cover K , so u is bounded above. If $u < s = \sup u$ everywhere, the open sets $\{u < s - 1/n\}$ cover K , and a finite subcover gives $u \leq s - 1/N$, which is absurd. Apply this to $-u$ for the minimum.

Functions. Functions are complex-valued unless they are called real. For a bounded function f on a set S , put $\|f\| = \sup_S |f|$, and $\|f\|_T = \sup_T |f|$ for $T \subseteq S$. On a space X , $C(X)$ and $C(X; \mathbb{R})$ are the continuous complex and real functions, and $C_b(X)$ the bounded continuous ones. The *support* of f is the closure of $\{f \neq 0\}$, and $C_c(X)$ consists of the continuous functions with compact support. $C_0(X)$ consists of the continuous f with

$$\{|f| \geq \varepsilon\} = \{x \in X : |f(x)| \geq \varepsilon\} \text{ compact for every } \varepsilon > 0. \quad (1.1)$$

Equivalently, for every $\varepsilon > 0$ there is a compact set K with $|f| < \varepsilon$ off K . Indeed, if there is such a K , then the closed set $\{|f| \geq \varepsilon\}$ lies in K and is compact by (K1); conversely, $K = \{|f| \geq \varepsilon\}$ will do. Real versions carry the suffix $;\mathbb{R}$, as in $C_0(X; \mathbb{R})$.

Algebras. On a set S , an *algebra of functions* is a complex linear space of functions $S \rightarrow \mathbb{C}$ that is closed under pointwise products. A *real algebra* is the same with real functions and real scalars. No unit is assumed. An algebra A is *self-adjoint* if $f \in A$ implies $\bar{f} \in A$. It *separates points* if for $x \neq y$ some $f \in A$ has $f(x) \neq f(y)$. It *vanishes nowhere* if for every x some $f \in A$ has $f(x) \neq 0$. Its *common zero set* is

$$Z(A) = \{x \in S : f(x) = 0 \text{ for all } f \in A\},$$

so A vanishes nowhere exactly when $Z(A) = \emptyset$. $\bar{A}$ is the closure of A for $\|\cdot\|$. For real functions, $f \vee g = \max(f, g)$ and $f \wedge g = \min(f, g)$, pointwise.

2. Spaces of bounded and continuous functions

Proposition 2.1. Let S be a set and X a topological space.

1. (*Completeness*) The bounded functions on S form a Banach space under $\|\cdot\|$. The bounded continuous functions on X and the bounded Borel functions on X form closed subspaces of it. If X is compact, then $C(X) = C_b(X)$.
2. (*Closures of algebras*) The closure of an algebra (or real algebra) of bounded functions is again one. The closure of a self-adjoint algebra is self-adjoint.
3. (*Vanishing at infinity*) $C_c(X) \subseteq C_0(X) \subseteq C_b(X)$. $C_0(X)$ is a closed self-adjoint subalgebra of $C_b(X)$, hence a Banach space. $C_c(X)$ is a self-adjoint subalgebra of $C_0(X)$, and it is dense in $C_0(X)$.

Proof. (1) Let (f_n) be a Cauchy sequence. For each s , $(f_n(s))$ is Cauchy in $\mathbb{C}$; let $f(s)$ be its limit. Given $\varepsilon > 0$, choose N with $\|f_n - f_m\| \leq \varepsilon$ for $m, n \geq N$. Letting $m \rightarrow \infty$ gives $|f_n(s) - f(s)| \leq \varepsilon$ for all s and all $n \geq N$. So f is bounded and $f_n \rightarrow f$. Suppose the f_n are continuous. Fix $x \in X$, fix $n \geq N$, and take an open $U \ni x$ on which $|f_n - f_n(x)| < \varepsilon$. On U , $|f - f(x)| \leq |f - f_n| + |f_n - f_n(x)| + |f_n(x) - f(x)| < 3\varepsilon$, so f is continuous. Suppose the f_n are Borel. Then $\{\operatorname{Re} f > a\} = \bigcup_k \bigcup_N \bigcap_{n \geq N} \{\operatorname{Re} f_n > a + 1/k\}$ is a Borel set, and so is the analogous set for $\operatorname{Im} f$; so f is Borel. The last claim is (K5) applied to $|f|$.

(2) Let $f_n \rightarrow f$ and $g_n \rightarrow g$ with f_n, g_n in the algebra. Then $\|f_n\| \leq \|f\| + \|f_n - f\|$ is bounded, and $\|f_n g_n - f g\| \leq \|f_n\| \|g_n - g\| + \|g\| \|f_n - f\| \rightarrow 0$. Sums, scalar multiples, complex conjugates and real values pass to uniform limits in the same way.

(3) If $f \in C_c(X)$, then $\{|f| \geq \varepsilon\}$ is a closed subset of the compact support, hence compact by (K1). If $f \in C_0(X)$, then $|f| < 1$ off the compact set $\{|f| \geq 1\}$, and $|f|$ is bounded on that set by (K5); so f is bounded. For $f, g \in C_0(X)$, $\varepsilon > 0$ and a scalar $c \neq 0$,

$$\{|f + g| \geq \varepsilon\} \subseteq \{|f| \geq \frac{\varepsilon}{2}\} \cup \{|g| \geq \frac{\varepsilon}{2}\}, \quad \{|fg| \geq \varepsilon\} \subseteq \{|f| \geq \frac{\varepsilon}{\|g\|+1}\}, \quad \{|cf| \geq \varepsilon\} = \{|f| \geq \frac{\varepsilon}{|c|}\}.$$

The sets on the left are closed. The sets on the right are compact, since a finite union of compact sets is compact (K2). So the sets on the left are compact by (K1), and $f + g, fg, cf \in C_0(X)$. Clearly $\bar{f} \in C_0(X)$. If $f_n \in C_0(X)$ and $\|f_n - f\| \leq \varepsilon/2$, then $\{|f| \geq \varepsilon\} \subseteq \{|f_n| \geq \varepsilon/2\}$. So $C_0(X)$ is closed in $C_b(X)$, and complete by (1). The supports of $f + g, fg, cf$ and $\bar{f}$ are closed subsets of $\operatorname{supp} f \cup \operatorname{supp} g$, so $C_c(X)$ is a self-adjoint subalgebra. For density, let $f \in C_0(X)$ and $\varepsilon > 0$. Define $\psi_\varepsilon : \mathbb{C} \rightarrow \mathbb{C}$ by $\psi_\varepsilon(z) = 0$ for $|z| \leq \varepsilon$ and $\psi_\varepsilon(z) = z - \varepsilon z/|z|$ for $|z| \geq \varepsilon$. It is continuous, and $|\psi_\varepsilon(z) - z| \leq \varepsilon$. So $\psi_\varepsilon \circ f$ is continuous and within ε of f . It is nonzero only on $\{|f| > \varepsilon\}$, whose closure lies in the compact set $\{|f| \geq \varepsilon\}$. So $\psi_\varepsilon \circ f \in C_c(X)$. $\square$

3. The one-point compactification

Definition 3.1 (One-point compactification). Let X be a topological space and ∞ a point not in X . Put $X_\infty = X \cup \{\infty\}$. Let τ_∞ consist of the open subsets of X together with the sets $X_\infty \setminus C$, where $C \subseteq X$ is closed and compact.

If X is Hausdorff, its compact subsets are closed by (K3), so the word "closed" can then be omitted. For a general space X , it makes parts (1) and (2) below true.

Proposition 3.2.

1. τ_∞ is a topology on X_∞ . X is open in X_∞ , and the subspace topology on X is its original topology.
2. X_∞ is compact.
3. If X is locally compact Hausdorff, then X_∞ is Hausdorff.
4. ∞ is an isolated point of X_∞ exactly when X is compact. So X is dense in X_∞ exactly when X is not compact.
5. (*Uniqueness*) Let Y be compact Hausdorff, $q \in Y$, and φ a homeomorphism of X onto $Y \setminus \{q\}$. Extending φ by $\infty \mapsto q$ gives a homeomorphism of X_∞ onto Y .

Proof. (1) $\emptyset$ is open in X , and $X_\infty = X_\infty \setminus \emptyset$. For open $U \subseteq X$ and closed compact $C, D \subseteq X$, the set $U \cap (X_\infty \setminus C) = U \setminus C$ is open in X , and $(X_\infty \setminus C) \cap (X_\infty \setminus D) = X_\infty \setminus (C \cup D)$ with $C \cup D$ closed and compact. Consider a union of sets U_i open in X and sets $X_\infty \setminus C_j$, with at least one j . It equals $X_\infty \setminus D$, where $D = \bigcap_j C_j \cap \bigcap_i (X \setminus U_i)$. D is closed and lies in one C_j , so it is compact by (K1). A union of open

subsets of X is open in X . The traces on X of the members of τ_∞ are the sets U and $X \setminus C$, all open in X ; so the subspace topology is the original one. Finally $X \in \tau_\infty$.

(2) Let $\mathcal{W}$ be an open cover of X_∞ . Some $W_0 \in \mathcal{W}$ contains ∞ , so $W_0 = X_\infty \setminus C_0$ with C_0 compact. The traces $W \cap X$, $W \in \mathcal{W}$, are open in X and cover C_0 , so finitely many of them do. Those members and W_0 cover X_∞ .

(3) Two points of X are separated by open subsets of X , which are open in X_∞ . Let $x \in X$, with a compact neighbourhood N and an open V such that $x \in V \subseteq N$. N is closed by (K3), so $X_\infty \setminus N$ is an open neighbourhood of ∞ disjoint from V .

(4) If X is compact, then $\{\infty\} = X_\infty \setminus X$ is open. If $\{\infty\}$ is open, it is $X_\infty \setminus C$ with $C = X$, so X is compact. X is dense exactly when every open neighbourhood of ∞ meets X , that is, when $\{\infty\}$ is not open.

(5) Call the extension Φ ; it is a bijection. Let $V \subseteq Y$ be open. If $q \notin V$, then $\Phi^{-1}(V) = \varphi^{-1}(V)$ is open in X . If $q \in V$, then $Y \setminus V$ is compact by (K1) and lies in $Y \setminus \{q\}$. So $C = \varphi^{-1}(Y \setminus V)$ is compact by (K2), and closed in X , since its complement is $\varphi^{-1}(V \setminus \{q\})$. Then $\Phi^{-1}(V) = X_\infty \setminus C$ is open. So Φ is continuous, and by (K4) its inverse is continuous too. $\square$

Examples 3.3. (a) Let $\mathbb{T} = \{w \in \mathbb{C} : |w| = 1\}$. The Cayley map $\kappa(x) = (x - i)/(x + i)$ is a homeomorphism of $\mathbb{R}$ onto $\mathbb{T} \setminus \{1\}$. Indeed $|x - i| = |x + i|$ for real x , $\kappa(x) \neq 1$, and the inverse is $w \mapsto i(1 + w)/(1 - w)$, which is real on $\mathbb{T} \setminus \{1\}$ because $(1 + w)/(1 - w) = (w - \bar{w})/|1 - w|^2$ there. $\mathbb{T}$ is compact by Lemma 4.1(b) below, so (5) gives $\mathbb{R}_\infty \cong \mathbb{T}$ with $\infty \mapsto 1$. In the same way, $y \mapsto (2y, |y|^2 - 1)/(|y|^2 + 1)$ is a homeomorphism of $\mathbb{R}^n$ onto the unit sphere $S^n \subseteq \mathbb{R}^{n+1}$ with the point $(0, \dots, 0, 1)$ removed; its inverse is $(u, t) \mapsto u/(1 - t)$. So $(\mathbb{R}^n)_\infty \cong S^n$.

(b) If Y is compact Hausdorff and $q \in Y$, then (5), applied to the identity map of $Y \setminus \{q\}$, identifies $(Y \setminus \{q\})_\infty$ with Y .

(c) $\mathbb{N}_\infty$ is homeomorphic to $Y = \{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$, with $\infty \mapsto 0$. Y is compact, because an open set containing 0 contains all but finitely many of the points $1/n$.

(d) If X is compact, X_∞ is X together with an isolated point, by (4). Then X is not dense in X_∞ , so X_∞ is not a compactification of X in the usual sense, which asks for X to be dense.

The functions in $C_0(X)$ are the continuous functions on X_∞ that vanish at ∞ , as the next proposition shows. For a function f on X , let $\tilde{f}$ be the function on X_∞ that equals f on X and 0 at ∞ . Put

$$I_\infty = \{g \in C(X_\infty) : g(\infty) = 0\}.$$

Proposition 3.4. Let X be any topological space.

1. $f \in C_0(X)$ if and only if $\tilde{f}$ is continuous on X_∞ .
2. The map $f \mapsto \tilde{f}$ is a bijection of $C_0(X)$ onto I_∞ that preserves sums, scalar multiples, products, complex conjugation and the norm $\|\cdot\|$. Its inverse is restriction to X . It maps $C_0(X; \mathbb{R})$ onto the real functions in I_∞ .
3. Every $g \in C(X_\infty)$ has the unique form $g = g(\infty)1 + h$ with $h \in I_\infty$. So $C(X_\infty)$ is $C_0(X)$ with a unit adjoined.

Proof. (1) Let $f \in C_0(X)$. $\tilde{f}$ is continuous at points of X , because X is open in X_∞ and carries its own topology there (Proposition 3.2(1)). At ∞ : given $\varepsilon > 0$, the set $C = \{|f| \geq \varepsilon\}$ is compact, and closed because $|f|$ is continuous. So $X_\infty \setminus C$ is an open neighbourhood of ∞ on which $|\tilde{f}| < \varepsilon$. Conversely, let $\tilde{f}$ be continuous. Then f is continuous, and for $\varepsilon > 0$ the set $\{x \in X : |f(x)| \geq \varepsilon\} = \{t \in X_\infty : |\tilde{f}(t)| \geq \varepsilon\}$ is closed in the compact space X_∞ . It is compact by (K1), and it lies in X . Since X carries the subspace topology, it is compact in X .

(2) The operations are pointwise, and 0 at ∞ is preserved by all of them. $\sup_{X_\infty} |\tilde{f}| = \sup_X |f|$. Restriction inverts the map by (1).

(3) Put $h = g - g(\infty)1$. If also $g = c1 + h'$ with $h' \in I_\infty$, evaluating at ∞ gives $c = g(\infty)$, so the form is unique. $\square$

Part (3) says that adjoining a unit to $C_0(X)$ yields $C(X_\infty)$.

4. Compact and locally compact spaces

Lemma 4.1 (Compactness tools).

- (a) Every closed bounded interval $[a, b]$ is compact.
- (b) (*Tube lemma*) Let Y be compact, $W \subseteq X \times Y$ open, and $\{x_0\} \times Y \subseteq W$. Then $U \times Y \subseteq W$ for some open $U \ni x_0$. Consequently, a product of finitely many compact spaces is compact, and closed bounded subsets of $\mathbb{R}^n$ are compact. Examples are $\mathbb{T}$, the closed unit disc and the spheres S^n .
- (c) A compact Hausdorff space is regular and normal: a point and a closed set not containing it, and two disjoint closed sets, have disjoint open neighbourhoods. Equivalently, if C is closed, O is open and $C \subseteq O$, then some open G satisfies $C \subseteq G \subseteq \overline{G} \subseteq O$.
- (d) In a compact metric space every sequence has a convergent subsequence, so the space is complete. It is separable and second countable, and every continuous map from it to a metric space is uniformly continuous. Conversely, a metric space in which every sequence has a convergent subsequence is compact.

Proof. (a) Let $\mathcal{O}$ be an open cover, and let S be the set of $s \in [a, b]$ such that finitely many members cover $[a, s]$. Then $a \in S$. Let $\sigma = \sup S$, and let $O \in \mathcal{O}$ contain σ , so that $O \supseteq (\sigma - \eta, \sigma + \eta) \cap [a, b]$ for some $\eta > 0$. Some $s \in S$ exceeds $\sigma - \eta$. Adding O to a finite cover of $[a, s]$ covers $[a, \min(\sigma + \eta/2, b)]$. If $\sigma < b$, this contradicts $\sigma = \sup S$. So $\sigma = b$, and $b \in S$.

(b) For each $y \in Y$ choose open sets $U_y \ni x_0$ and $V_y \ni y$ with $U_y \times V_y \subseteq W$. Finitely many V_{y_j} cover Y ; put $U = \bigcap_j U_{y_j}$. Now let X and Y be compact and $\mathcal{W}$ an open cover of $X \times Y$. Let $\mathcal{U}$ be the family of open $U \subseteq X$ such that finitely many members of $\mathcal{W}$ cover $U \times Y$. Each x lies in a member of $\mathcal{U}$: the compact set $\{x\} \times Y$ is covered by finitely many members, and the tube lemma applies to their union. Finitely many members of $\mathcal{U}$ cover X . Induction handles finite products. A closed bounded subset of $\mathbb{R}^n$ lies in a box $[-M, M]^n$, which is compact by (a) and the product statement, so it is compact by (K1).

(c) Let $p \notin C$ with C closed. C is compact by (K1), and (K3) separates p from C . Let E, F be disjoint and closed. For each $e \in E$, (K3) gives disjoint open sets $U_e \ni e$ and $V_e \supseteq F$. Finitely many U_e cover the compact set E ; take their union and the intersection of the corresponding V_e . For the last form, separate C and $X \setminus O$ by disjoint open sets $G \supseteq C$ and $H \supseteq X \setminus O$; then $\overline{G} \subseteq X \setminus H \subseteq O$.

(d) Let (x_j) be a sequence in a compact metric space K with no convergent subsequence. Then each $p \in K$ has a ball containing x_j for only finitely many j ; otherwise indices $j_1 < j_2 < \dots$ with $d(x_{j_i}, p) < 1/i$ could be chosen. Finitely many such balls cover K , which is absurd. A Cauchy sequence with a convergent subsequence converges, so K is complete. For each n , finitely many balls of radius $1/n$ cover K . Their centres, over all n , form a countable dense set C , and the balls $B(c, 1/m)$, $c \in C$, $m \geq 1$, form a countable base: if $B(x, 2/m) \subseteq O$, some $c \in C$ has $d(c, x) < 1/m$, and then $x \in B(c, 1/m) \subseteq O$. Conversely, let every sequence in a metric space M have a convergent subsequence, and let $\mathcal{O}$ be an open cover. Some $\delta > 0$ has the property that every ball $B(x, \delta)$ lies in a member of $\mathcal{O}$. Otherwise there are balls $B(x_n, 1/n)$ lying in no member; a subsequence of the centres converges to some x in a member O with $B(x, 2r) \subseteq O$, and then $B(x_n, 1/n) \subseteq O$ for large n in the subsequence. Also finitely many balls $B(x, \delta)$ cover M . Otherwise points with mutual distances at least δ can be chosen one after another, and they form a sequence with no convergent subsequence. So finitely many members of $\mathcal{O}$ cover M . Finally, let ϕ be a continuous map from a compact metric space K to a metric space, and $\varepsilon > 0$. By the first part, every sequence in K has a convergent subsequence. So the argument above gives, for the open cover by the sets $\{x : d(\phi(x), \phi(p)) < \varepsilon/2\}$, $p \in K$, a $\delta > 0$ such that every ball $B(x, \delta)$ lies in one of them. If $d(x, y) < \delta$, then x and y lie in one such set, so $d(\phi(x), \phi(y)) < \varepsilon$. $\square$

Proposition 4.2 (Locally compact spaces).

1. (*Examples.*) The following are locally compact Hausdorff: compact Hausdorff spaces, $\mathbb{R}^n$, discrete spaces, closed subspaces of LCH spaces, open subspaces of compact Hausdorff spaces, and open subspaces of LCH spaces. The spaces $\mathbb{Q}$, $\mathbb{N}^{\mathbb{N}}$ (a countable product of discrete copies of $\mathbb{N}$) and $\mathbb{R}^{\mathbb{N}}$ (product topology) are Hausdorff, but no point in them has a compact neighbourhood.
2. (*Compact closures.*) Let X be LCH, $A \subseteq X$ compact and V a neighbourhood of A . Then some open $U \supseteq A$ has compact closure contained in V .
3. (*Exhaustion.*) If X is LCH and σ -compact, that is, a countable union of compact sets, then there are open sets $U_1 \subseteq U_2 \subseteq \dots$ with $\overline{U_n}$ compact, $\overline{U_n} \subseteq U_{n+1}$ and $\bigcup_n U_n = X$. Every compact subset of X lies in some U_n .
4. (*Countability at infinity.*) For LCH X , the following are equivalent: (i) ∞ has a countable neighbourhood base in X_∞ ; (ii) $\{\infty\}$ is a countable intersection of open subsets of X_∞ ; (iii) X is σ -compact; (iv) X is Lindelöf, that is, every open cover has a countable subcover.
5. (*Variants of local compactness.*) For a topological space consider: (i) every point has a compact neighbourhood; (ii) every neighbourhood of a point contains a compact neighbourhood of that point; (iii) every point has an open neighbourhood with compact closure; (iii') every open neighbourhood V of a point p contains an open $U \ni p$ with compact closure; (iv) every open neighbourhood V of p contains an open $U \ni p$ with $\overline{U}$ compact and $\overline{U} \subseteq V$. Then (iv) implies (ii) and (iii); (ii) implies (i); and (iii) and (iii') are equivalent and imply (i). In a Hausdorff space all five are equivalent.

Proof. (1) A compact Hausdorff space is a compact neighbourhood of each of its points. In $\mathbb{R}^n$ closed balls are compact by Lemma 4.1(b). In a discrete space, $\{x\}$ is a compact neighbourhood of x . If $Y \subseteq X$ is closed and N is a compact neighbourhood of $p \in Y$ in X , then $N \cap Y$ is a compact neighbourhood of p in Y , by (K1). If U is open in a compact Hausdorff space and $p \in U$, Lemma 4.1(c) gives an open $V \ni p$ with $\overline{V} \subseteq U$, and $\overline{V}$ is compact by (K1). An open subspace of an LCH space X is open in X_∞ , which is compact Hausdorff by Proposition 3.2(2)–(3); so the previous case applies. Subspaces and products of Hausdorff spaces are Hausdorff.

For $\mathbb{Q}$: suppose N is a compact neighbourhood of $q \in \mathbb{Q}$. It contains $\mathbb{Q} \cap (q - 2r, q + 2r)$ for some $r > 0$, so it contains $P = \mathbb{Q} \cap [q - r, q + r]$, which is closed and hence compact by (K1). Pick an irrational $\xi \in (q - r, q + r)$. The open sets $\{t \in P : |t - \xi| > 1/m\}$, $m \geq 1$, cover P with no finite subcover. For $\mathbb{N}^{\mathbb{N}}$ and $\mathbb{R}^{\mathbb{N}}$: a neighbourhood of a point contains a basic open set that leaves the coordinates beyond some index m free. The projection onto coordinate $m + 1$ is continuous, so it maps a compact neighbourhood onto a compact set (K2). Compact subsets of $\mathbb{N}$ are finite and compact subsets of $\mathbb{R}$ are bounded, but the image of the basic open set is all of $\mathbb{N}$ or $\mathbb{R}$.

(2) First let $A = \{p\}$. The interior of V in X is open in X_∞ , so $F = X_\infty \setminus \text{int } V$ is closed, and $p \notin F$. X_∞ is compact Hausdorff (Proposition 3.2(2)–(3)), so by Lemma 4.1(c) there is an open $W \ni p$ whose closure in X_∞ misses F , that is, lies in $\text{int } V \subseteq X$. Put $U = W \cap X$. Its closure in X_∞ is compact by (K1) and lies in X . So it is also the closure of U in X , and it is compact in X and contained in V . For general compact A , take such a U_p for each $p \in A$, a finite subcover $U_{p_1}, \dots, U_{p_k}$ of A , and $U = \bigcup_i U_{p_i}$. Then $\overline{U} = \bigcup_i \overline{U_{p_i}}$ is compact and lies in V .

(3) Let $X = \bigcup_n K_n$ with each K_n compact. By (2), choose an open $U_1 \supseteq K_1$ with compact closure, and inductively an open U_{n+1} with compact closure that contains the compact set $\overline{U_n} \cup K_{n+1}$. Then $\overline{U_n} \subseteq U_{n+1}$ and $\bigcup U_n = X$. A compact set is covered by the increasing open sets U_n , so it lies in one of them.

(4) (i) $\Rightarrow$ (ii): let $W_1, W_2, \dots$ form a neighbourhood base at ∞ . Their interiors are open, contain ∞ , and again form a neighbourhood base. The intersection of the interiors is $\{\infty\}$, because X_∞ is Hausdorff (Proposition 3.2(3)). (ii) $\Rightarrow$ (iii): each open $W_n \ni \infty$ has the form $X_\infty \setminus C_n$ with C_n compact, and $X = \bigcup C_n$. (iii) $\Rightarrow$ (i): with U_n from (3), the sets $X_\infty \setminus \overline{U_n}$ form a neighbourhood base at ∞ , because a neighbourhood $X_\infty \setminus C$ of ∞ contains $X_\infty \setminus \overline{U_n}$ as soon as $C \subseteq U_n$. (iii) $\Rightarrow$ (iv): each K_n is covered by finitely many members of an open cover. (iv) $\Rightarrow$ (iii): by (2) the open sets with compact closure cover X ; a countable subcover (V_n) gives $X = \bigcup_n \overline{V_n}$.

(5) (iv) $\Rightarrow$ (iii), (iii') $\Rightarrow$ (iii) and (ii) $\Rightarrow$ (i): take $V = X$. (iii) $\Rightarrow$ (iii'): if $W \ni p$ is open with compact closure and $V \ni p$ is open, then $U = W \cap V$ has closure inside $\overline{W}$, which is compact by (K1). (iii) $\Rightarrow$ (i): $\overline{W}$ is a

compact neighbourhood. (iv) $\Rightarrow$ (ii): $\overline{U}$ is a compact neighbourhood of p inside V , and every neighbourhood of p contains an open one. In a Hausdorff space, (i) says that the space is LCH, and then (2) gives (iv). $\square$

Without the Hausdorff property the conditions in (5) need not agree. For example, $\mathbb{Q}_\infty$ is compact by [Proposition 3.2\(2\)](#), so (i) holds at each of its points. But (ii) fails at every rational point q . Indeed, $\mathbb{Q}$ is an open neighbourhood of q in $\mathbb{Q}_\infty$ that carries its own topology there ([Proposition 3.2\(1\)](#)), so a compact neighbourhood of q inside $\mathbb{Q}$ would be a compact neighbourhood of q in $\mathbb{Q}$, and by (1) there is none. The open database [\[\$\pi\$ -Base\]](#) collects many more examples of this kind, with the properties of each space ([spaces without compact neighbourhoods](#)); see also, e.g., [Steen–Seebach].

5. Urysohn's lemma and metrizability

Theorem 5.1 (Urysohn's lemma). Let X be a space in which any two disjoint closed sets have disjoint open neighbourhoods, and let $E, F \subseteq X$ be disjoint closed sets. Then there is a continuous $f : X \rightarrow [0, 1]$ with $f = 0$ on E and $f = 1$ on F . By [Lemma 4.1\(c\)](#), this applies to every compact Hausdorff space.

Reference: [Urysohn 1925].

Proof. By the argument at the end of the proof of [Lemma 4.1\(c\)](#), which uses only the separation of disjoint closed sets, every closed set C inside an open set O has an open G with $C \subseteq G \subseteq \overline{G} \subseteq O$. Let D be the set of dyadic rationals in $[0, 1]$. We build open sets U_r , $r \in D$, with

$$E \subseteq U_0, \quad U_1 = X \setminus F, \quad \overline{U_r} \subseteq U_s \text{ whenever } r < s. \quad (5.1)$$

Put $U_1 = X \setminus F$, and choose U_0 open with $E \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$. Suppose U_r has been built for all $r = j/2^n$, $0 \leq j \leq 2^n$, with (5.1). For $r = (2j+1)/2^{n+1}$, its neighbours $r_- = j/2^n$ and $r_+ = (j+1)/2^n$ satisfy $\overline{U_{r_-}} \subseteq U_{r_+}$; choose an open U_r with $\overline{U_{r_-}} \subseteq U_r \subseteq \overline{U_r} \subseteq U_{r_+}$. For $r < s$ at level $n+1$, (5.1) follows by passing through the neighbours of r and s at level n . Now define $f(x) = \inf\{r \in D : x \in U_r\}$, with $\inf \emptyset = 1$. Then $0 \leq f \leq 1$, $f = 0$ on E , and $f = 1$ on F , since F misses $U_1 \supseteq U_r$ for every r . For $0 < \alpha \leq 1$ and $0 \leq \beta < 1$,

$$\{f < \alpha\} = \bigcup_{r < \alpha} U_r, \quad \{f > \beta\} = \bigcup_{s > \beta} (X \setminus \overline{U_s}),$$

with $r, s \in D$. For the first: if $f(x) < \alpha$, some $r < \alpha$ has $x \in U_r$, and conversely $x \in U_r$ gives $f(x) \leq r$. For the second: if $f(x) > \beta$, pick $s, s' \in D$ with $\beta < s < s' < f(x)$; then $x \notin U_{s'} \supseteq \overline{U_s}$. Conversely, if $x \notin \overline{U_s}$, then $x \notin U_r$ for every $r \leq s$, so $f(x) \geq s$. For other values of α and β the sets $\{f < \alpha\}$ and $\{f > \beta\}$ are $\emptyset$ or X . So all these sets are open, and f is continuous. $\square$

Corollary 5.2 (Urysohn's lemma, locally compact form). Let X be LCH, $A \subseteq X$ compact and $B \subseteq X$ closed with $A \cap B = \emptyset$. Then some $f \in C_c(X)$ with $0 \leq f \leq 1$ equals 1 on A and 0 on B . In particular, X is completely regular (take $A = \{p\}$), and $C_c(X)$, hence $C_0(X)$, separates points and vanishes nowhere.

[Theorem 10.1](#) then gives a second proof that $C_c(X)$ is dense in $C_0(X)$.

Proof. By [Proposition 4.2\(2\)](#) with $V = X \setminus B$, choose an open $U \supseteq A$ whose closure K is compact and lies in $X \setminus B$. K is compact Hausdorff, so Urysohn's lemma gives a continuous $g : K \rightarrow [0, 1]$ with $g = 1$ on A and $g = 0$ on $K \setminus U$. Put $f = g$ on K and $f = 0$ on $X \setminus K$; so $f = 0$ on $X \setminus U$. For closed $F \subseteq [0, 1]$, $f^{-1}(F)$ is $g^{-1}(F)$ if $0 \notin F$, and $g^{-1}(F) \cup (X \setminus U)$ if $0 \in F$. Both are closed in X , since K is closed in X . So f is continuous, its support lies in K , and $f = 0$ on B . $\square$

Theorem 5.3 (Metrizability). For an LCH space X , the following are equivalent.

- (i) X is σ -compact and metrizable.
- (ii) Some countable family $\mathcal{K}$ of compact sets has this property: every neighbourhood of every point p contains a member of $\mathcal{K}$ that is a neighbourhood of p .
- (iii) X is second countable.
- (iv) X is Polish: separable and completely metrizable.
- (v) X_∞ is metrizable.

Lemma 5.4. A compact Hausdorff space K with a countable base is metrizable.

Proof. Let (B_n) be a countable base. For each pair (m, n) with $\overline{B_m} \subseteq B_n$, Urysohn's lemma gives a continuous $f_{mn} : K \rightarrow [0, 1]$ that is 1 on $\overline{B_m}$ and 0 off B_n . List these functions as $h_1, h_2, \dots$, and put $d(x, y) = \sum_k 2^{-k} |h_k(x) - h_k(y)|$. Then d is symmetric, satisfies the triangle inequality, and $y \mapsto d(x, y)$ is continuous, as a uniform limit of continuous functions. Let $x \neq y$. Since $\{y\}$ is closed, some B_n contains x but not y . By Lemma 4.1(c), there is an open $V \ni x$ with $\overline{V} \subseteq B_n$, and some B_m satisfies $x \in B_m \subseteq V$. Then $\overline{B_m} \subseteq B_n$ and $f_{mn}(x) = 1 \neq 0 = f_{mn}(y)$, so $d(x, y) > 0$. So d is a metric. Its open balls are open in K , so the identity map from K to (K, d) is a continuous bijection onto a Hausdorff space, and a homeomorphism by (K4). $\square$

Lemma 5.5. Let (M, d) be a complete metric space and $O \subseteq M$ open. Then O is completely metrizable.

Proof. If $O = M$ there is nothing to prove. Otherwise let $F = M \setminus O$ and $\phi(x) = 1/d(x, F)$ for $x \in O$; it is finite and continuous, since F is closed. Put $d'(x, y) = d(x, y) + |\phi(x) - \phi(y)|$. This is a metric on O , and it gives the same topology: $d \leq d'$, and $d'(x, y) \rightarrow 0$ as $y \rightarrow x$ in d , by continuity of ϕ . Let (x_j) be Cauchy for d' . It is Cauchy for d , so it converges to some $y \in M$. The numbers $\phi(x_j)$ form a Cauchy sequence, so they are bounded by some c ; then $d(x_j, F) \geq 1/c$, and $d(y, F) \geq 1/c > 0$. So $y \in O$, and $d'(x_j, y) \rightarrow 0$ by continuity of ϕ at y . $\square$

Proof of Theorem 5.3. (ii) $\Rightarrow$ (iii): the interiors of the members of $\mathcal{K}$ form a countable base.

(iii) $\Rightarrow$ (ii): let (B_n) be a countable base, and $\mathcal{K}$ the family of those $\overline{B_n}$ that are compact. Given p and a neighbourhood V , Proposition 4.2(2) gives an open $U \ni p$ with compact closure inside V . Choose B_n with $p \in B_n \subseteq U$. Then $\overline{B_n} \subseteq \overline{U}$ is compact, lies in V , and is a neighbourhood of p .

(iii) $\Rightarrow$ (v): a second countable space is Lindelöf, because each member of an open cover is a union of base sets, and each base set that lies in some member can be assigned to one such member. So X is σ -compact by Proposition 4.2(4), and ∞ has a countable neighbourhood base (W_n) . Replacing each W_n by its interior, we may take the W_n open. The sets B_n and W_n together form a countable base of X_∞ : an open $O \ni \infty$ contains some W_m , and each of its points in X lies in some $B_n \subseteq O \cap X$. X_∞ is compact Hausdorff (Proposition 3.2(2)–(3)), so Lemma 5.4 applies.

(v) $\Rightarrow$ (i): X is a subspace of the metrizable space X_∞ , so it is metrizable. X_∞ is compact metrizable, hence second countable by Lemma 4.1(d). So ∞ has a countable neighbourhood base, and X is σ -compact by Proposition 4.2(4).

(i) $\Rightarrow$ (iii): $X = \bigcup K_n$ with each K_n a compact metric space, which is separable by Lemma 4.1(d). So X has a countable dense set, and the balls of rational radius about its points form a countable base.

(v) $\Rightarrow$ (iv): X_∞ is a compact metric space, hence complete and second countable (Lemma 4.1(d)). X is open in X_∞ , so it is completely metrizable by Lemma 5.5. It is second countable as a subspace, hence separable.

(iv) $\Rightarrow$ (iii): as in (i) $\Rightarrow$ (iii), the balls of rational radius about the points of a countable dense set form a countable base. $\square$

Because X_∞ is compact, the proof needs a metrization argument only for compact spaces (Lemma 5.4), not the general Urysohn metrization theorem for regular second countable spaces. An uncountable discrete space is metrizable and LCH but satisfies none of (i)–(v) (Example 11.2).

6. The polynomial step: absolute values without constant terms

Lemma 6.1. Define real polynomials by

$$P_0 = 0, \quad P_{n+1}(t) = P_n(t) + \frac{1}{2}(t^2 - P_n(t)^2). \quad (6.1)$$

1. $P_n(t) = Q_n(t^2)$ for real polynomials Q_n with $Q_n(0) = 0$.
2. For $|t| \leq 1$: $0 \leq P_n(t) \leq P_{n+1}(t) \leq |t|$, and

$$0 \leq |t| - P_n(t) \leq |t|(1 - \frac{|t|}{2})^n \leq \frac{2}{n} \quad (n \geq 1). \quad (6.2)$$

Proof. (1) Put $Q_0 = 0$ and $Q_{n+1}(s) = Q_n(s) + \frac{1}{2}(s - Q_n(s)^2)$. By induction $P_n(t) = Q_n(t^2)$, and $Q_{n+1}(0) = Q_n(0) - \frac{1}{2}Q_n(0)^2 = 0$.

(2) Fix t and write $a = |t| \in [0, 1]$ and $p_n = P_n(t)$. Since $t^2 = a^2$, (6.1) gives

$$a - p_{n+1} = (a - p_n) \left(1 - \frac{a + p_n}{2}\right).$$

Suppose $0 \leq p_n \leq a$. Then $0 \leq \frac{a+p_n}{2} \leq a \leq 1$, so $0 \leq a - p_{n+1} \leq a - p_n$; that is, $p_n \leq p_{n+1} \leq a$. Since $p_0 = 0$, induction gives $0 \leq p_n \leq p_{n+1} \leq a$ for all n . Using $p_n \geq 0$, the second factor above is at most $1 - a/2$, so $a - p_n \leq a(1 - a/2)^n$. For the last bound, let $0 < a \leq 1$ and $s = a/2 \leq \frac{1}{2}$. By Bernoulli's inequality $(1 + u)^n \geq 1 + nu$ ($u \geq 0$),

$$(1 - s)^{-n} = \left(1 + \frac{s}{1-s}\right)^n \geq 1 + \frac{ns}{1-s} \geq 1 + ns,$$

so $a(1 - a/2)^n \leq a/(1 + na/2) = 2a/(2 + na) < 2/n$. $\square$

Proposition 6.2. Let A be a real algebra of bounded real functions on a set S .

1. If p is a real polynomial with $p(0) = 0$ and $f \in A$, then $p \circ f \in A$. If such polynomials p_k converge uniformly to a function φ on an interval containing $f(S)$, then $\varphi \circ f \in \bar{A}$.
2. Let $f \in A$ and $r > 0$ with $\|f\| \leq r$. Then $g_n = r Q_n(f^2/r^2)$ lies in A , and $0 \leq |f| - g_n \leq 2r/n$. The polynomials $t \mapsto mP_n(t/m)$ have no constant term and converge to $|t|$ uniformly on $[-m, m]$.
3. $\bar{A}$ is a real algebra, and $f, g \in \bar{A}$ imply $|f|$, $f \vee g$, $f \wedge g \in \bar{A}$. So $\bar{A}$ is closed under maxima and minima of finitely many elements.

Proof. (1) If $p(t) = \sum_{k=1}^d a_k t^k$, then $p \circ f = \sum_{k=1}^d a_k f^k \in A$, since A is closed under products and real linear combinations. No constant term is needed, so no unit is needed. Also $\|p_k \circ f - \varphi \circ f\|$ is at most the supremum of $|p_k - \varphi|$ on the interval.

(2) g_n is a real combination of powers f^{2k} with $k \geq 1$, so it lies in A . By Lemma 6.1 applied to $t = f(s)/r$, $r P_n(f/r) = g_n$ satisfies $0 \leq |f| - g_n \leq 2r/n$. The statement about $[-m, m]$ is (6.2) after the substitution $t \mapsto t/m$.

(3) $\bar{A}$ is a real algebra by **Proposition 2.1(2)**. Apply (2) to $f \in \bar{A}$: $|f|$ is a uniform limit of elements of $\bar{A}$, so $|f| \in \bar{A}$. Then $f \vee g = \frac{1}{2}(f + g + |f - g|)$ and $f \wedge g = \frac{1}{2}(f + g - |f - g|)$, and induction handles finitely many functions. $\square$

The recursion (6.1) keeps the constant term zero at every step. So the approximation of $|f|$ uses neither the constant function 1 nor the Weierstrass approximation theorem, and this is what the nonunital theorem needs.

7. Interpolation at two points

Lemma 7.1. Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$, and let A be an algebra of $\mathbb{F}$ -valued functions on a set S that separates points.

1. $Z(A)$ has at most one point.
2. Let $x, y \in S$ and $a, b \in \mathbb{F}$. Assume $a = 0$ if $x \in Z(A)$, $b = 0$ if $y \in Z(A)$, and $a = b$ if $x = y$. Then some $h \in A$ has $h(x) = a$ and $h(y) = b$.

Proof. First let $x = y$. If $x \in Z(A)$, take $h = 0$. Otherwise pick $u \in A$ with $u(x) \neq 0$ and put $h = (a/u(x))u$.

Now let $x \neq y$. The set

$$V = \{(h(x), h(y)) : h \in A\} \subseteq \mathbb{F}^2$$

is a linear subspace that is closed under coordinatewise multiplication, because $h \mapsto (h(x), h(y))$ respects sums, scalars and products. Such a subspace is one of $\{0\}$, $\mathbb{F}(1, 0)$, $\mathbb{F}(0, 1)$, $\mathbb{F}(1, 1)$ and $\mathbb{F}^2$. Indeed, if $V = \mathbb{F}v$ with $v = (v_1, v_2) \neq 0$, then $(v_1^2, v_2^2) = cv$ for some c . If $v_1 \neq 0 \neq v_2$, this forces $v_1 = c = v_2$, so $V = \mathbb{F}(1, 1)$. If $v_1 = 0$ or $v_2 = 0$, then V is $\mathbb{F}(0, 1)$ or $\mathbb{F}(1, 0)$.

Separation of x and y rules out $\{0\}$ and $\mathbb{F}(1, 1)$, which would give $h(x) = h(y)$ for every h . If $V = \mathbb{F}(0, 1)$, every h vanishes at x , so $x \in Z(A)$, $a = 0$, and $(a, b) \in V$. The case $V = \mathbb{F}(1, 0)$ is symmetric. If $V = \mathbb{F}^2$ there is nothing to prove. This proves (2). If $x \neq y$ both lay in $Z(A)$, then $V = \{0\}$; so (1) holds. $\square$

When A vanishes nowhere, (2) says that every pair of values is attained at every pair of distinct points. The proof looks only at the image of A in $\mathbb{F}^2$. It also shows what happens at the one possible common zero, which the nonunital case needs.

8. Stone's lattice lemma

Lemma 8.1 (Stone's lattice lemma, pointwise form). Let K be a nonempty compact space and $L \subseteq C(K; \mathbb{R})$ a set that is closed under $\vee$ and $\wedge$. Let $f \in C(K; \mathbb{R})$ and $\varepsilon > 0$. Suppose that for all $x, y \in K$ some $h \in L$ satisfies

$$|h(x) - f(x)| < \varepsilon \quad \text{and} \quad |h(y) - f(y)| < \varepsilon. \quad (8.1)$$

Then some $h \in L$ satisfies $\|h - f\| < \varepsilon$.

Proof. *Step 1: approximation from above, anchored at one point.* Fix $x \in K$. Let $\mathcal{U}_x$ be the family of open sets $U \subseteq K$ for which some $h \in L$ satisfies $h(x) > f(x) - \varepsilon$ and $h < f + \varepsilon$ on U . Every $y \in K$ lies in a member of $\mathcal{U}_x$: take h from (8.1) for the pair (x, y) and $U = \{h < f + \varepsilon\}$. Choose a finite subcover $U_1, \dots, U_m$ and, for each U_j , one function h_j as in the definition. Put $g_x = h_1 \wedge \dots \wedge h_m \in L$. Each point of K lies in some U_j , where $g_x \leq h_j < f + \varepsilon$. So $g_x < f + \varepsilon$ on K , and $g_x(x) > f(x) - \varepsilon$.

Step 2: approximation from below. Let $\mathcal{V}$ be the family of open sets $V \subseteq K$ for which some $g \in L$ satisfies $g < f + \varepsilon$ on K and $g > f - \varepsilon$ on V . By Step 1, every x lies in the member $\{g_x > f - \varepsilon\}$ of $\mathcal{V}$. Choose a finite subcover $V_1, \dots, V_n$, one function g_i for each V_i , and put $h = g_1 \vee \dots \vee g_n \in L$. Then $h < f + \varepsilon$ everywhere, because every g_i is, and $h > f - \varepsilon$ everywhere, because every point lies in some V_i , where $h \geq g_i > f - \varepsilon$. By (K5) the continuous function $|h - f|$ attains its maximum, so $\|h - f\| < \varepsilon$. $\square$

The proof makes only finitely many choices at each step. The covers are families of open sets defined by the existence of suitable functions, and no function is chosen for each point of K . So the argument does not use the axiom of choice.

Theorem 8.2 (Lattice versions). Let K be a compact space and $L \subseteq C(K; \mathbb{R})$ a linear subspace closed under $\wedge$ (then also under $\vee$, since $f \vee g = -((-f) \wedge (-g))$).

1. Suppose that (i) for $x \neq y$ in K and $a, b \in \mathbb{R}$, some $h \in L$ has $h(x) = a$ and $h(y) = b$, and (ii) for $x \in K$ and $a \in \mathbb{R}$, some $h \in L$ has $h(x) = a$. Then L is dense in $C(K; \mathbb{R})$. If K has at least two points, (i) implies (ii).
2. If L separates points and every constant function lies in L , then (i) and (ii) hold, so L is dense.
3. Let $Y \subseteq C(K)$ be a complex linear subspace with $1 \in Y$ that separates points and satisfies $\bar{g} \in Y$ and $|g| \in Y$ for every $g \in Y$. Then Y is dense in $C(K)$.
4. (Nonunital form) Let X be a topological space and $L \subseteq C_0(X; \mathbb{R})$ a linear subspace closed under $\wedge$. If (i) and (ii) hold for points of X , then L is dense in $C_0(X; \mathbb{R})$. If X is locally compact Hausdorff, the converse also holds.

Reference: [Blackadar, XX.10.2.14] assumes only condition (i) in (1). That version fails on a one-point space, where $L = \{0\}$ satisfies (i) vacuously and is not dense.

Proof. (1) If $K = \emptyset$ there is nothing to prove. Otherwise, given f and x, y , conditions (i)–(ii) give $h \in L$ with $h(x) = f(x)$ and $h(y) = f(y)$, and Lemma 8.1 applies. If K has a second point $y \neq x$, then (i) with the pair (x, y) gives (ii).

(2) For $x \neq y$ pick $f \in L$ with $f(x) \neq f(y)$. The vectors $(1, 1)$ and $(f(x), f(y))$ in $\mathbb{R}^2$ are linearly independent, and both lie in the image of the linear map $h \mapsto (h(x), h(y))$. So the image is $\mathbb{R}^2$, which is (i). Constants give (ii).

(3) For $g \in Y$, $\operatorname{Re} g = \frac{1}{2}(g + \bar{g})$ and $\operatorname{Im} g = \frac{1}{2i}(g - \bar{g})$ lie in Y . Let $Y_{\mathbb{R}}$ be the real functions in Y . It is a real subspace that contains 1 and separates points, since $g(x) \neq g(y)$ forces a difference in the real or the imaginary part. It is closed under $|\cdot|$, hence under $\wedge$, because $f \wedge g = \frac{1}{2}(f + g - |f - g|)$ (Proposition 6.2(3)). By (2), $Y_{\mathbb{R}}$ is dense in $C(K; \mathbb{R})$. Approximating the real and imaginary parts of $f \in C(K)$ gives density of Y .

(4) Suppose (i) and (ii) hold. The set $\tilde{L} = \{\tilde{h} : h \in L\} \subseteq C(X_{\infty}; \mathbb{R})$ of Proposition 3.4 is closed under $\vee$ and $\wedge$, since these operations keep the value 0 at ∞ . Let $f \in C_0(X; \mathbb{R})$. For a pair of points of X , (i)–(ii) give an

h that agrees with f there. For a pair (x, ∞) , (ii) gives h with $h(x) = f(x)$, and $\tilde{h}(\infty) = 0 = \tilde{f}(\infty)$. For the pair (∞, ∞) , take $h = 0$. Lemma 8.1, applied on the compact space X_∞ (Proposition 3.2(2)), puts $\tilde{f}$ in the closure of $\tilde{L}$. Since $h \mapsto \tilde{h}$ preserves the norm (Proposition 3.4(2)), f lies in the closure of L . Conversely, let X be locally compact Hausdorff and L dense. For $x \neq y$, the evaluation $h \mapsto (h(x), h(y))$ is continuous, and it maps $C_0(X; \mathbb{R})$ onto $\mathbb{R}^2$ by Urysohn's lemma (Corollary 5.2). So it maps L onto a dense linear subspace of $\mathbb{R}^2$, which is all of $\mathbb{R}^2$. The one-point case is the same. Without local compactness the converse fails: for $X = \mathbb{Q}$ one has $C_0(X) = \{0\}$ (Proposition 12.1), so $L = \{0\}$ is dense while (ii) fails. $\square$

Lemma 8.1 needs the interpolation (8.1) only for the one function f that is being approximated, not for all target values. Part (4) of Theorem 8.2 is the lattice counterpart of Theorem 9.2.

9. The real Stone–Weierstrass theorem

Theorem 9.1 (Stone–Weierstrass theorem, real form). Let K be a compact space and $A \subseteq C(K; \mathbb{R})$ a real algebra that separates points. Then $Z(A)$ has at most one point, and

$$\bar{A} = \{f \in C(K; \mathbb{R}) : f = 0 \text{ on } Z(A)\}. \quad (9.1)$$

In particular:

- (a) if A vanishes nowhere, for example if A contains the constant functions, then A is dense in $C(K; \mathbb{R})$;
- (b) if every function in A vanishes at a point z , then $\bar{A} = \{f : f(z) = 0\}$.

Reference: [Stone 1937], [Stone 1948].

Proof. $Z(A)$ has at most one point by Lemma 7.1(1). Evaluation at a point is continuous for $\|\cdot\|$, so $\bar{A}$ lies in the right side of (9.1). Conversely, let $f \in C(K; \mathbb{R})$ vanish on $Z(A)$, and assume $K \neq \emptyset$ (otherwise both sides are $\{0\}$). Put $B = \bar{A}$. By Proposition 6.2(3), B is closed under $\vee$ and $\wedge$. For $x, y \in K$, apply Lemma 7.1(2) to A with $a = f(x)$ and $b = f(y)$. Its conditions hold, because $f(x) = 0$ whenever $x \in Z(A)$, and likewise for y . So some $h \in A \subseteq B$ agrees with f at x and y . Lemma 8.1 gives, for each $\varepsilon > 0$, an $h \in B$ with $\|h - f\| < \varepsilon$. So $f \in \bar{B} = B$. $\square$

Where the hypotheses enter. Products are used in Proposition 6.2 (the functions g_n) and in Lemma 7.1 (the set V is closed under products). Separation is used in Lemma 7.1, to rule out $V = \{0\}$ and $V = \mathbb{R}(1, 1)$. Compactness is used in Lemma 8.1, for the two finite subcovers, and in (K5). The Hausdorff property is not used. It follows anyway from separation: if $h(x) \neq h(y)$, the sets $\{|h - h(x)| < r\}$ and $\{|h - h(y)| < r\}$ with $r = |h(x) - h(y)|/2$ are disjoint open neighbourhoods.

For a closed algebra A that separates points, (9.1) gives a dichotomy: either $A = C(K; \mathbb{R})$, or A consists of all the functions that vanish at one point. Exercise 2 proves the case with a common zero in another way, by adjoining the constants.

The passage to $C_0(X)$ goes through the one-point compactification.

Theorem 9.2 (Real form for $C_0(X)$). Let X be locally compact Hausdorff, and let $A \subseteq C_0(X; \mathbb{R})$ be a real algebra that separates the points of X and vanishes nowhere. Then A is dense in $C_0(X; \mathbb{R})$.

Proof. By Proposition 3.4, $\tilde{A} = \{\tilde{f} : f \in A\}$ is a real algebra of continuous functions on the compact space X_∞ (Proposition 3.2(2)). It separates the points of X_∞ . Two points of X are separated by A . A point $x \in X$ and ∞ are separated by any $f \in A$ with $f(x) \neq 0$, since $\tilde{f}(\infty) = 0$; such an f exists because A vanishes nowhere. The same fact gives $Z(\tilde{A}) = \{\infty\}$. By Theorem 9.1(b), the closure of $\tilde{A}$ is the set of real functions in I_∞ . Since $f \mapsto \tilde{f}$ is an isometric bijection of $C_0(X; \mathbb{R})$ onto that set (Proposition 3.4(2)), A is dense in $C_0(X; \mathbb{R})$. $\square$

The proof uses neither local compactness nor the Hausdorff property of X . Proposition 12.1 explains why: both follow from the hypotheses on A .

10. The complex Stone–Weierstrass theorem

Theorem 10.1 (Stone–Weierstrass theorem for $C_0(X)$). Let X be locally compact Hausdorff, and let $A \subseteq C_0(X)$ be a complex subalgebra that is self-adjoint, separates the points of X , and vanishes nowhere. Then

$$\bar{A} = C_0(X) : \quad (10.1)$$

every $f \in C_0(X)$ can be approximated uniformly on X by elements of A .

Compact form. Let K be a compact space and $A \subseteq C(K)$ a self-adjoint subalgebra that separates points. Then $Z(A)$ has at most one point, and $\bar{A} = \{f \in C(K) : f = 0 \text{ on } Z(A)\}$. In particular A is dense in $C(K)$ if it vanishes nowhere, for example if it contains the constants.

Proof. Let $A_{\mathbb{R}}$ be the set of real functions in A . For $f \in A$, self-adjointness and complex linearity give

$$\operatorname{Re} f = \frac{1}{2}(f + \bar{f}) \in A_{\mathbb{R}}, \quad \operatorname{Im} f = \frac{1}{2i}(f - \bar{f}) \in A_{\mathbb{R}}.$$

$A_{\mathbb{R}}$ is a real algebra. It separates points: if $f(x) \neq f(y)$, then $\operatorname{Re} f$ or $\operatorname{Im} f$ takes different values at x and y . For the same reason $Z(A_{\mathbb{R}}) = Z(A)$. In the setting of the theorem, **Theorem 9.2** shows that $A_{\mathbb{R}}$ is dense in $C_0(X; \mathbb{R})$. Let $f \in C_0(X)$ and $\varepsilon > 0$. Since $|\operatorname{Re} f| \leq |f|$ and $|\operatorname{Im} f| \leq |f|$, for every $\delta > 0$ the sets $\{|\operatorname{Re} f| \geq \delta\}$ and $\{|\operatorname{Im} f| \geq \delta\}$ are closed subsets of the compact set $\{|f| \geq \delta\}$, hence compact by (K1). So both parts lie in $C_0(X; \mathbb{R})$. Choose $u, v \in A_{\mathbb{R}}$ within $\varepsilon/2$ of them. Then $u + iv \in A$ and $\|f - (u + iv)\| < \varepsilon$. In the compact form, **Theorem 9.1** applies to $A_{\mathbb{R}}$ in the same way: a function f vanishing on $Z(A)$ has real and imaginary parts vanishing on $Z(A_{\mathbb{R}})$. $\square$

Where each hypothesis is used.

- *Subalgebra.* Closure under products enters in **Proposition 6.2** and **Lemma 7.1**. Complex linearity gives $\operatorname{Re} f, \operatorname{Im} f \in A$ and $u + iv \in A$. **Example 12.2(d)** shows that products cannot be dropped.
- *Self-adjoint.* Used once, to put $\operatorname{Re} f$ and $\operatorname{Im} f$ into A . **Example 12.2(a)** and Exercise 1 show that it cannot be dropped.
- *Separates points.* Used in **Lemma 7.1**, for pairs of points of X . **Example 12.2(b)** shows that it cannot be dropped.
- *Vanishes nowhere.* Used in **Theorem 9.2**, to separate each $x \in X$ from ∞ and to make ∞ the only common zero on X_{∞} . **Example 12.2(c)** shows that it cannot be dropped.
- $A \subseteq C_0(X)$. Makes the functions $\tilde{f}$ continuous on X_{∞} (**Proposition 3.4**).
- *Locally compact Hausdorff.* Not used in the proof. It is the natural setting because the other hypotheses force it (**Proposition 12.1**).
- *Compactness of X_{∞} .* This is where the topology does its work, in the finite subcovers of **Lemma 8.1**.

No countability is needed. The proof uses no sequences of points and no countable bases, only finite subcovers. So the theorem applies to spaces that are neither σ -compact nor metrizable, such as uncountable discrete spaces (**Example 11.2**) and locally compact groups of this kind.

11. Worked examples

Several examples here and in Section 12 use two functions on $\mathbb{R}$:

$$\varphi(x) = \frac{1}{1+x^2}, \quad \psi(x) = \frac{x}{1+x^2}.$$

Both lie in $C_0(\mathbb{R})$: since $|\psi(x)| \leq (1+x^2)^{-1/2}$, the sets $\{|\varphi| \geq \varepsilon\}$ and $\{|\psi| \geq \varepsilon\}$ are closed and lie in $\{1+x^2 \leq \varepsilon^{-2}\}$, so they are compact (**Lemma 4.1(a)**). Also $\varphi > 0$, φ is even, ψ is odd, and $\psi/\varphi = x$.

Example 11.1 (A compact space without the constants). On $[0, 1]$, let $A = \{p(x+1) : p \text{ a real polynomial, } p(0) = 0\}$. A is a real algebra. It vanishes nowhere, because $x+1 \geq 1$, and $x+1$ separates points. So A is dense in $C([0, 1]; \mathbb{R})$ by **Theorem 9.1(a)**. Yet $1 \notin A$: if $p(0) = 0$ and $p(y) = 1$ for $y \in [1, 2]$, the polynomial $p - 1$ would have infinitely many roots and constant term -1 . An explicit approximation is $e_n(x) = 1 - (1 - \frac{x+1}{2})^n \in A$; since $0 \leq 1 - \frac{x+1}{2} \leq \frac{1}{2}$ on $[0, 1]$, $|1 - e_n| \leq 2^{-n}$. On a compact space, “vanishes nowhere” is strictly weaker than “contains the constants”.

Example 11.2 (An uncountable discrete space). Let X be an uncountable set with the discrete topology. Its compact subsets are finite, because an infinite subset has an open cover by singletons with no finite subcover. So $C_0(X)$ consists of the f for which $\{|f| \geq \varepsilon\}$ is finite for every $\varepsilon > 0$. The finitely supported functions form a self-adjoint subalgebra that separates points and vanishes nowhere (it contains the indicator of each point), so it is dense. Here X is not σ -compact, and ∞ has no countable neighbourhood base in X_∞ (**Proposition 4.2(4)**). The proof of **Theorem 10.1** never uses countability, so this does not matter.

Example 11.3 (Degenerate spaces). If $X = \emptyset$, then $C_0(X) = \{0\}$, and $A = \{0\}$ satisfies all hypotheses vacuously; the conclusion holds. If X is one point, $C_0(X) = \mathbb{C}$; an algebra that vanishes nowhere contains a nonzero constant, hence equals $\mathbb{C}$.

Example 11.4 (Compact X). If X is compact, then $C_0(X) = C(X)$, because $\{|f| \geq \varepsilon\}$ is closed in X and so compact by (K1). The point ∞ is isolated in X_∞ (**Proposition 3.2(4)**), and **Theorem 10.1** reduces to its compact form.

Example 11.5 (The real line). The complex algebra generated by φ and ψ consists of the complex polynomials without constant term in φ and ψ . It is self-adjoint, because φ and ψ are real. It separates points, because $\psi/\varphi = x$, and it vanishes nowhere, because $\varphi > 0$. So it is dense in $C_0(\mathbb{R})$.

Example 11.6 (One common zero, seen twice). On $[0, 1]$, the complex polynomials without constant term have closure $\{f \in C([0, 1]) : f(0) = 0\}$, by the compact form of **Theorem 10.1**; for real coefficients this is **Theorem 9.1(b)**. Restricted to $X = (0, 1]$ they form a self-adjoint subalgebra of $C_0(X)$ that separates points and vanishes nowhere, so **Theorem 10.1** makes them dense in $C_0((0, 1])$. The two statements agree: by **Proposition 3.2(5)** with $Y = [0, 1]$ and $q = 0$, and by **Proposition 3.4**, $C_0((0, 1])$ is the space of continuous functions on $[0, 1]$ that vanish at 0.

12. The role of each hypothesis

The topological hypothesis of **Theorem 10.1** comes for free, and each hypothesis on the algebra is needed.

Proposition 12.1 (The topological hypotheses are automatic). Let X be a topological space.

1. If some set of functions in $C_0(X)$ separates the points of X , then X is Hausdorff.
2. If $f \in C_0(X)$ and $f(x) \neq 0$, then x has a compact neighbourhood. So if some $A \subseteq C_0(X)$ vanishes nowhere, every point of X has a compact neighbourhood. If no point of X has a compact neighbourhood, then $C_0(X) = \{0\}$; this is the case for $X = \mathbb{Q}$ (**Proposition 4.2(1)**).
3. Consequently **Theorem 9.2** and **Theorem 10.1** hold for every topological space X : whenever their hypotheses on A hold, X is locally compact Hausdorff.

Proof. (1) If $h(x) \neq h(y)$, put $r = |h(x) - h(y)|/2$. The open sets $\{|h - h(x)| < r\}$ and $\{|h - h(y)| < r\}$ are disjoint and contain x and y .

(2) Put $c = |f(x)|/2 > 0$. The set $\{|f| \geq c\}$ is compact by (1.1), and it contains the open set $\{|f| > c\}$, which contains x .

(3) By (1) and (2), X is Hausdorff and every point has a compact neighbourhood. $\square$

Part (2) gives condition (i) of **Proposition 4.2(5)**, one of several conditions that serve as definitions of local compactness for spaces that need not be Hausdorff. Together with the Hausdorff property from (1), all of them agree. So nothing is lost by assuming that X is locally compact Hausdorff. Conversely, on every locally compact Hausdorff space the algebra $C_0(X)$ itself separates points and vanishes nowhere, by Urysohn's lemma (**Corollary 5.2**).

Examples 12.2. In examples (a)–(d), all hypotheses of **Theorem 10.1** hold except one, and the conclusion fails. Examples (e)–(h) test the lattice versions, the hypothesis $A \subseteq C_0(X)$, the topological hypothesis, and the form of the conclusion. Several of them use the functions φ and ψ of **Section 11**.

(a) Self-adjointness. Let $\mathbb{T} = \{w \in \mathbb{C} : |w| = 1\}$, a compact space (**Lemma 4.1(b)**), and let A be the complex polynomials in the coordinate w , restricted to $\mathbb{T}$. A is a subalgebra. It contains 1, so it vanishes nowhere, and w separates points. But $\bar{w} \notin \bar{A}$. To see this, fix $N \geq 2$ and let $\omega = E(2\pi i/N)$, where E is the complex

exponential. By the addition formula $E(z + w) = E(z)E(w)$, $\omega^m = E(2\pi im/N)$. So $|\omega| = 1$, because $|E(it)| = 1$ for real t ; $\omega^N = E(2\pi i) = 1$; and $\omega^m \neq 1$ for $0 < m < N$, because $E(it) \neq 1$ for $0 < t < 2\pi$. These properties of the exponential function are recalled in (B2) near the end of the lesson. For $g \in C(\mathbb{T})$ put

$$\Lambda_N(g) = \frac{1}{N} \sum_{k=0}^{N-1} g(\omega^k) \omega^k.$$

Then $|\Lambda_N(g)| \leq \|g\|$. For $0 \leq j \leq N-2$, the number $\rho = \omega^{j+1}$ satisfies $\rho \neq 1$ and $\rho^N = 1$, so $\Lambda_N(\omega^j) = \frac{1}{N} \sum_{k=0}^{N-1} \rho^k = \frac{1}{N} \frac{\rho^N - 1}{\rho - 1} = 0$. On $\mathbb{T}$, $\bar{w} = w^{-1}$, so $\Lambda_N(\bar{w}) = \frac{1}{N} \sum_k \omega^{-k} \omega^k = 1$. Given $p \in A$ of degree d , take $N \geq d+2$. Then $\Lambda_N(\bar{w} - p) = 1$, so $\|\bar{w} - p\| \geq 1$. In particular A is not self-adjoint. **Exercise 1** transfers this example to $\mathbb{R}$ with the Cayley map, so self-adjointness is needed for noncompact X as well.

The number $2\pi\Lambda_N(g)$ is a Riemann sum for the integral $\int_0^{2\pi} g(e^{it})e^{it} dt$, and the finite sums avoid integration. **Proposition 17.3(c)–(d)** gives the version with integrals.

(b) Separation. The even functions in $C_0(\mathbb{R})$ form a closed self-adjoint subalgebra A_{ev} . It vanishes nowhere, since $\varphi \in A_{\text{ev}}$, and it does not separate x from $-x$. For $f \in A_{\text{ev}}$, $1 = \psi(1) - \psi(-1) = (\psi - f)(1) - (\psi - f)(-1) \leq 2\|\psi - f\|$, so ψ has distance at least $\frac{1}{2}$ from A_{ev} . On $[-1, 1]$, the even polynomials form an algebra that does not separate x from $-x$. By **Theorem 13.2**, their closure is the set of even continuous functions, not $C([-1, 1])$.

(c) Vanishing nowhere. $A_0 = \{f \in C_0(\mathbb{R}) : f(0) = 0\}$ is a closed self-adjoint subalgebra. It separates points. Let $x \neq y$; by symmetry we may take $x \neq 0$. Put $r = \min(|x|, |x - y|) > 0$ and $\tau(t) = \max(0, 1 - |t - x|/r)$. Then $\tau \in C_c(\mathbb{R})$, $\tau(0) = 0$, $\tau(x) = 1$ and $\tau(y) = 0$. But $\|\varphi - f\| \geq |\varphi(0) - f(0)| = 1$ for every $f \in A_0$, so A_0 is not dense. On $[0, 1]$, the polynomials without constant term separate points and vanish at 0, and their closure is $\{f : f(0) = 0\}$ by **Theorem 9.1(b)**.

(d) Closure under products. $V = \text{span}_{\mathbb{C}}\{\varphi, \psi\}$ is closed under conjugation, since φ and ψ are real. It separates points: if $\varphi(x) = \varphi(y)$ and $\psi(x) = \psi(y)$, then $x = \psi(x)/\varphi(x) = y$. It vanishes nowhere. But V has dimension 2, so it is closed, while $C_0(\mathbb{R})$ has infinite dimension (it contains infinitely many nonzero functions with disjoint supports). So V is not dense. It is not an algebra: $\varphi^2 = a\varphi + b\psi$ would give $\varphi(x) = a + bx$ for all x , which fails at $x = 0, \pm 1$.

(e) Lattices. For the lattice versions of **Theorem 8.2**, separation and nowhere vanishing are not enough, and the pointwise lattice operations matter.

- (i) On $K = [0, 1]$, $L = \{f \in C(K; \mathbb{R}) : f(0) = 0\}$ is a closed subalgebra, closed under $\vee$ and $\wedge$, that separates points (by (c)). It fails condition (ii) of **Theorem 8.2(1)** at 0, and it is not dense. The same holds for any compact Hausdorff K and $p \in K$, with Urysohn's lemma (**Theorem 5.1**) supplying separation.
- (ii) Let $K = \{a, b\}$ be discrete, so $C(K; \mathbb{R}) = \mathbb{R}^2$. For $\lambda > 0$, $\lambda \neq 1$, let $L_\lambda = \{(s, \lambda s) : s \in \mathbb{R}\}$. Since $\lambda > 0$, $(s, \lambda s) \wedge (t, \lambda t) = (s \wedge t, \lambda(s \wedge t))$, so L_λ is a lattice subspace. It separates the two points and vanishes at neither, but $(1, 1) \notin L_\lambda$, so it fails (i) and is not dense. By contrast, an algebra with these two properties is all of $\mathbb{R}^2$ (**Lemma 7.1**).
- (iii) Let $K = \{a, b, c\}$ be discrete, $0 < \theta < 1$, and $L = \{f : f(c) = \theta f(a) + (1 - \theta)f(b)\}$. L contains the constants and satisfies (i): at any two points, the value at the third point can be solved for. In its own order, L is a lattice, isomorphic to $\mathbb{R}^2$ through $(f(a), f(b))$. But it is not closed under pointwise $\wedge$: $f = (1, 0, \theta)$ and $g = (0, 1, 1 - \theta)$ lie in L , while $f \wedge g = (0, 0, \min(\theta, 1 - \theta))$ does not. L is a proper subspace, hence closed and not dense.

(f) Bounded functions instead of C_0 . Let $X = \mathbb{N}$, so $C_b(X) = \ell^\infty$. The convergent sequences c form a closed self-adjoint subalgebra that contains 1 and separates points ($n \mapsto 1/n$ is injective). For $a \in c$ with limit l , the sequence $e_n = (-1)^n$ satisfies $\|e - a\| \geq \max(|1 - l|, |1 + l|) \geq 1$. So c is not dense in $C_b(\mathbb{N})$. **Proposition 16.1(3)** proves that this failure occurs on every noncompact metrizable space.

(g) The topological hypothesis. No counterexample exists: by **Proposition 12.1**, the other hypotheses force X to be locally compact Hausdorff.

(h) **Density, not equality.** The conclusion is $\bar{A} = C_0(X)$, not $A = C_0(X)$. For example $C_c(\mathbb{R})$ is dense in $C_0(\mathbb{R})$ (Proposition 2.1(3)) and $\varphi \notin C_c(\mathbb{R})$. The polynomials are dense in $C([0, 1])$, and $|x - \frac{1}{2}|$ is not a polynomial.

13. The closure of an arbitrary self-adjoint subalgebra

Dropping separation and nowhere vanishing does not end the story. The closure can still be described exactly.

Lemma 13.1 (Separation quotient). Let K be a compact space and S a set of continuous functions on K . Write $x \sim_S y$ if $s(x) = s(y)$ for every $s \in S$. Let K_S be the set of classes, $q : K \rightarrow K_S$ the quotient map, and give K_S the quotient topology: $O \subseteq K_S$ is open when $q^{-1}(O)$ is open.

1. A function f on K is continuous and constant on each class if and only if $f = f' \circ q$ for a continuous f' on K_S . Then $\|f\| = \|f'\|$.
2. The functions $s', s \in S$, separate the points of K_S . K_S is compact and Hausdorff.

Proof. (1) If $f = f' \circ q$, then f is continuous and constant on classes. Conversely, f' is well defined, and for an open set O of scalars, $q^{-1}(f'^{-1}(O)) = f^{-1}(O)$ is open; so $f'^{-1}(O)$ is open. Since q is onto, the norms agree. (2) Two distinct classes $q(x) \neq q(y)$ differ at some $s \in S$, which means $s'(q(x)) \neq s'(q(y))$. $K_S = q(K)$ is compact by (K2), and Hausdorff by Proposition 12.1(1), whose proof applies to any separating set of continuous functions. $\square$

Theorem 13.2. Let X be a topological space, and let A be a self-adjoint subalgebra of $C_0(X)$ or a real subalgebra of $C_0(X; \mathbb{R})$. Write $x \sim_A y$ if $f(x) = f(y)$ for all $f \in A$. Then

$$\bar{A} = \{f \in C_0(X) : f(x) = f(y) \text{ whenever } x \sim_A y, \text{ and } f = 0 \text{ on } Z(A)\}, \quad (13.1)$$

with $C_0(X; \mathbb{R})$ in place of $C_0(X)$ in the real case. The same formula holds for a compact space K , with $C(K)$ or $C(K; \mathbb{R})$ in place of $C_0(X)$.

Proof. First the compact case. Let R be the right side. It contains A , and it is closed, since each of its conditions involves the values at one or two points. So $\bar{A} \subseteq R$. Let $f \in R$. By Lemma 13.1 with $S = A$, $f = f' \circ q$. The map $a \mapsto a'$ respects sums, scalars, products and conjugation, so $A' = \{a' : a \in A\}$ is an algebra of the same kind on K_A . It separates points, and its common zero set is $q(Z(A))$, where f' vanishes. By the compact form of Theorem 10.1, or by Theorem 9.1 in the real case, f' lies in the closure of A' . Since $\|f - a\| = \|f' - a'\|$ for $a \in A$, $f \in \bar{A}$.

For $C_0(X)$, apply the compact case to $\tilde{A}$ on X_∞ (Proposition 3.4). On X , the relation $\sim_{\tilde{A}}$ is $\sim_A$. A point $x \in X$ is equivalent to ∞ exactly when $x \in Z(A)$, and $Z(\tilde{A}) = Z(A) \cup \{\infty\}$. So $\tilde{f}$ satisfies the conditions of (13.1) for $\tilde{A}$ exactly when f satisfies them for A . Since $f \mapsto \tilde{f}$ preserves the norm (Proposition 3.4(2)), the closure transfers to X . $\square$

Corollary 13.3.

1. Theorem 10.1 is the case in which $\sim_A$ is equality and $Z(A) = \emptyset$.
2. (One common zero.) If A separates points and $Z(A) = \{z\}$, then $\bar{A} = \{f \in C_0(X) : f(z) = 0\}$.
3. (Closed subalgebras.) For any equivalence relation $\sim$ on X and any $Z \subseteq X$, the functions in $C_0(X)$ that are constant on classes and vanish on Z form a closed self-adjoint subalgebra. By (13.1) every closed self-adjoint subalgebra B has this form, with $\sim_B$ and $Z(B)$.
4. (Dichotomy.) Let K be a compact space, and let B be a closed self-adjoint subalgebra of $C(K)$, or a closed real subalgebra of $C(K; \mathbb{R})$, that separates points. Then B is either the whole space or the set of all its functions that vanish at one point z . Indeed, $Z(B)$ has at most one point (Lemma 7.1(1)). If it is empty, apply Theorem 9.1(a) or the compact form of Theorem 10.1; if $Z(B) = \{z\}$, apply (2).

Self-adjointness cannot be dropped from Theorem 13.2. The closure of the polynomials in w on $\mathbb{T}$ (Example 12.2(a)) separates points and contains 1, yet it is not $C(\mathbb{T})$. Proposition 17.3 describes it.

14. The Weierstrass theorem: Bernstein polynomials and Korovkin's theorem

The polynomial step of [Section 6](#) does not need the Weierstrass approximation theorem, and nothing above uses it. This section proves the classical theorem directly and constructively, with a quantitative form.

Theorem 14.1 (Korovkin's theorem). Let $I = [a, b]$, and let $L_n : C(I; \mathbb{R}) \rightarrow C(I; \mathbb{R})$ be linear maps that are positive: $u \geq 0$ implies $L_n u \geq 0$. Put $e_k(t) = t^k$. If $L_n e_k \rightarrow e_k$ uniformly for $k = 0, 1, 2$, then $L_n f \rightarrow f$ uniformly for every $f \in C(I; \mathbb{R})$.

Reference: [Korovkin 1953].

Proof. Positivity makes each L_n monotone: $u \leq v$ implies $L_n u \leq L_n v$. Fix f and $\varepsilon > 0$. By uniform continuity ([Lemma 4.1\(d\)](#)) there is $\delta > 0$ with $|f(s) - f(t)| < \varepsilon$ when $|s - t| < \delta$. If $|s - t| \geq \delta$, then $|f(s) - f(t)| \leq 2\|f\| \leq 2\|f\|(s - t)^2/\delta^2$. So for all $s, t \in I$,

$$|f(s) - f(t)| \leq \varepsilon + c(s - t)^2, \quad c = 2\|f\|/\delta^2. \quad (14.1)$$

Fix t and read (14.1) as an inequality between functions of s : with $q_t = e_2 - 2te_1 + t^2e_0$, $-\varepsilon e_0 - cq_t \leq f - f(t)e_0 \leq \varepsilon e_0 + cq_t$. Apply L_n and evaluate at t :

$$|(L_n f)(t) - f(t)(L_n e_0)(t)| \leq \varepsilon(L_n e_0)(t) + c(L_n q_t)(t).$$

Here $(L_n q_t)(t) = (L_n e_2)(t) - 2t(L_n e_1)(t) + t^2(L_n e_0)(t)$, which tends to $t^2 - 2t^2 + t^2 = 0$ uniformly in $t \in I$, because $|t|$ is bounded on I . Hence

$$\|L_n f - f\| \leq \|f\| \|L_n e_0 - e_0\| + \varepsilon \|L_n e_0\| + c \sup_{t \in I} |(L_n q_t)(t)|,$$

and the limit superior of the right side is at most ε . $\square$

Proposition 14.2 (Bernstein polynomials). For a bounded $f : [0, 1] \rightarrow \mathbb{R}$ and $n \geq 1$ put

$$b_{n,k}(x) = \binom{n}{k} x^k (1 - x)^{n-k}, \quad (B_n f)(x) = \sum_{k=0}^n f(k/n) b_{n,k}(x).$$

1. $B_n e_0 = e_0$, $B_n e_1 = e_1$ and $B_n e_2 = e_2 + (e_1 - e_2)/n$. Hence

$$\sum_{k=0}^n \left(\frac{k}{n} - x\right)^2 b_{n,k}(x) = \frac{x(1-x)}{n}. \quad (14.2)$$

2. $B_n f \rightarrow f$ uniformly for every continuous f .
3. For every bounded f and every $x \in [0, 1]$, $|f(x) - (B_n f)(x)| \leq \frac{3}{2} \omega(f, n^{-1/2})$, where $\omega(f, \delta) = \sup\{|f(s) - f(t)| : |s - t| \leq \delta\}$.
4. If f is bounded and continuous at x_0 , then $(B_n f)(x_0) \rightarrow f(x_0)$. Boundedness cannot be dropped.
5. (*Bernoulli trials*) For $x \in [0, 1]$ and $\delta > 0$, $\sum_{k: |k/n - x| \geq \delta} b_{n,k}(x) \leq x(1-x)/(n\delta^2)$.

Reference: [Bernstein 1912]. Version 4 of [van Neerven] (April 2022) states (4) in its Remark 2.4 for every function that is continuous at x_0 ; the example in the proof of (4) shows that boundedness is needed, and version 7 (July 2025) adds it.

Proof. (1) The binomial theorem gives $\sum_k b_{n,k}(x) = (x + 1 - x)^n = 1$. Since $\frac{k}{n} \binom{n}{k} = \binom{n-1}{k-1}$ for $k \geq 1$, we get $\sum_k \frac{k}{n} b_{n,k}(x) = x \sum_{j=0}^{n-1} \binom{n-1}{j} x^j (1-x)^{n-1-j} = x$. For $n \geq 2$ and $k \geq 1$, $\left(\frac{k}{n}\right)^2 \binom{n}{k} = \frac{n-1}{n} \binom{n-2}{k-2} + \frac{1}{n} \binom{n-1}{k-1}$ (with $\binom{n-1}{-1} = 0$), which gives $B_n e_2 = \frac{n-1}{n} x^2 + \frac{x}{n}$, as claimed; for $n = 1$, $B_1 e_2(x) = x = x^2 + x(1-x)$ directly. Expanding $(k/n - x)^2$ and using the three identities gives (14.2).

(2) B_n is linear and positive, and $\|B_n e_2 - e_2\| \leq \frac{1}{4n}$ by (1). Apply Korovkin's theorem, Theorem 14.1.

(3) First, $\omega(f, \lambda\delta) \leq (1 + \lambda) \omega(f, \delta)$ for $\lambda > 0$: if $|s - t| \leq \lambda\delta$, split the segment between s and t into m equal pieces, where m is the least integer $\geq \lambda$; each piece has length at most δ , so $|f(s) - f(t)| \leq m \omega(f, \delta) \leq (1 +$

$\lambda) \omega(f, \delta)$. With $\lambda = |s - t|/\delta$ this gives $|f(s) - f(t)| \leq (1 + |s - t|/\delta) \omega(f, \delta)$ for all s, t . Since the weights $b_{n,k}(x)$ are nonnegative with sum 1,

$$|f(x) - (B_n f)(x)| \leq \sum_k b_{n,k}(x) |f(x) - f(k/n)| \leq \omega(f, \delta) \left(1 + \frac{1}{\delta} \sum_k b_{n,k}(x) \left|x - \frac{k}{n}\right|\right).$$

By the Cauchy–Schwarz inequality for these weights and (14.2), the last sum is at most $(x(1-x)/n)^{1/2} \leq \frac{1}{2\sqrt{n}}$. Take $\delta = n^{-1/2}$.

(4) Given $\varepsilon > 0$, take $\delta > 0$ with $|f(t) - f(x_0)| < \varepsilon$ for $|t - x_0| < \delta$. Splitting the sum in (3) according to whether $|k/n - x_0| < \delta$, and using (5), $|f(x_0) - (B_n f)(x_0)| \leq \varepsilon + 2\|f\| x_0(1-x_0)/(n\delta^2)$. For the second claim, let $f(1/m) = 2^{m^2}$ for $m \geq 3$, and $f = 0$ at all other points of $[0, 1]$. Then $f = 0$ on $(\frac{1}{3}, 1]$, so f is continuous at $\frac{1}{2}$. All terms of $(B_n f)(\frac{1}{2})$ are nonnegative, and for $n \geq 3$ the term with $k = 1$ is $n 2^{-n} f(1/n) = n 2^{n^2-n}$. So $(B_n f)(\frac{1}{2}) \rightarrow \infty$, while $f(\frac{1}{2}) = 0$.

(5) Where $|k/n - x| \geq \delta$, $(k/n - x)^2/\delta^2 \geq 1$. Sum against $b_{n,k}(x)$ and use (14.2). $\square$

Probabilistic reading. $b_{n,k}(x)$ is the probability of exactly k successes in n independent trials that each succeed with probability x . So $(B_n f)(x)$ is the expected value of f at the observed frequency of success. Part (5) is the weak law of large numbers for such trials: the frequency lies within δ of x with probability at least $1 - x(1-x)/(n\delta^2)$.

Corollary 14.3 (Weierstrass approximation theorem). Every continuous complex function on $[a, b]$ is a uniform limit of polynomials, with real coefficients when the function is real.

Reference: [Weierstrass 1885].

Proof. The substitution $t = a + (b - a)x$ carries polynomials to polynomials. Apply Proposition 14.2(2) to the real and imaginary parts. A second proof: on the compact space $[a, b]$, the polynomials with complex coefficients form a self-adjoint algebra (for real t , the conjugate of $\sum c_k t^k$ is $\sum \bar{c}_k t^k$) that contains 1 and separates points; apply the compact form of **Theorem 10.1**, or **Theorem 9.1** for real functions. $\square$

15. Further density theorems on compact spaces

Proposition 15.1.

1. (*Polynomials in several variables.*) Let $X \subseteq \mathbb{R}^n$ be compact. The real polynomials in $x_1, \dots, x_n$ are dense in $C(X; \mathbb{R})$, and the complex ones in $C(X)$. Every continuous $F : X \rightarrow \mathbb{R}^m$ is a uniform limit of maps whose m coordinates are such polynomials.
2. (*Polynomials in z and $\bar{z}$.*) Let $X \subseteq \mathbb{C}$ be compact. The functions $z \mapsto p(z, \bar{z})$, with p a complex polynomial in two variables, are dense in $C(X)$.
3. (*Separability.*) If K is a compact metric space, $C(K)$ is separable.
4. (*Trigonometric polynomials.*) The trigonometric polynomials $\sum_{|k| \leq n} c_k w^k$ form a dense subspace of $C(\mathbb{T})$. Equivalently, the functions $\theta \mapsto \sum_{|k| \leq n} c_k e^{ik\theta}$ are dense, uniformly on $\mathbb{R}$, in the continuous 2π -periodic functions.
5. (*Products.*) Let $X_1, \dots, X_k$ be compact spaces, not necessarily Hausdorff. The linear combinations of the functions $f_1(x_1) \cdots f_k(x_k)$, with $f_j \in C(X_j; \mathbb{R})$, are dense in $C(X_1 \times \cdots \times X_k; \mathbb{R})$. Every complex continuous function on the product is a uniform limit of finite sums of functions $\lambda f_1(x_1) \cdots f_k(x_k)$ with $\lambda \in \{1, i\}$ and real f_j .
6. (*Iterated integrals.*) Let X, Y be compact spaces, and $\mu : C(X; \mathbb{R}) \rightarrow \mathbb{R}, \nu : C(Y; \mathbb{R}) \rightarrow \mathbb{R}$ positive linear maps. For $h \in C(X \times Y; \mathbb{R})$, the functions $x \mapsto \nu(h(x, \cdot))$ and $y \mapsto \mu(h(\cdot, y))$ are continuous, and

$$\mu(x \mapsto \nu(h(x, \cdot))) = \nu(y \mapsto \mu(h(\cdot, y))). \quad (15.1)$$

In particular, for continuous h on $[a, b] \times [c, d]$, the two iterated Riemann integrals of h agree, and the Riemann sums of h over grid partitions converge to their common value.

7. (*Moments.*) Let $a < b$. If $f \in C([a, b])$ and $\int_a^b f(x) x^n dx = 0$ for all $n \geq 0$, then $f = 0$. The condition $a < b$ is needed: on $[a, a]$ every integral is 0, so the constant function 1 has all its moments equal to 0.

Proof. (1) The coordinate functions and 1 generate an algebra, which separates points because the coordinates do; apply [Theorem 9.1](#). The complex polynomials form a self-adjoint algebra, since the coordinates are real; apply the compact form of [Theorem 10.1](#). For F , approximate each coordinate.

(2) The algebra contains 1 and z , which separates points. It is self-adjoint: the conjugate of $p(z, \bar{z})$ is $\bar{p}(\bar{z}, z)$, where $\bar{p}$ has the conjugate coefficients. Apply the compact form of [Theorem 10.1](#).

(3) Choose a countable dense set $C \subseteq K$ ([Lemma 4.1\(d\)](#)), and put $d_c(x) = d(x, c)$. The functions d_c and 1 generate a real algebra A_0 that contains the constants. It separates points: if $x \neq y$ and $r = d(x, y)$, choose $c \in C$ with $d(x, c) < r/2$; then $d_c(x) < r/2 < d_c(y)$. By [Theorem 9.1](#), A_0 is dense in $C(K; \mathbb{R})$. The finite products of the functions d_c and 1 form a countable set, and its linear combinations with rational coefficients are dense in A_0 , hence in $C(K; \mathbb{R})$. Combinations with Gaussian rational coefficients give a countable set that is dense in $C(K)$.

(4) On $\mathbb{T}$, $\bar{w} = w^{-1}$, so the trigonometric polynomials form the algebra generated by w and $\bar{w}$. It is self-adjoint, contains 1, and separates points through w . Apply the compact form of [Theorem 10.1](#) on the compact space $\mathbb{T}$. For the periodic form: the map $\theta \mapsto E(i\theta)$ from $[0, 2\pi]$ onto $\mathbb{T}$ (it is onto by (B2)) is continuous from a compact space onto a Hausdorff space, so it maps closed sets to closed sets. A continuous 2π -periodic function takes equal values on each fibre, so it equals $g(E(i\theta))$ for a function g on $\mathbb{T}$, and g is continuous by the closed-map argument used in (5) below. Conversely each $g \in C(\mathbb{T})$ gives a continuous periodic function, and $E(ik\theta) = E(i\theta)^k$.

(5) Apply [Lemma 13.1](#) with $S = C(X_j; \mathbb{R})$ to get compact Hausdorff quotients $q_j : X_j \rightarrow X'_j$ on which continuous functions separate points. Put $X = \prod X_j$, $X' = \prod X'_j$ and $q = q_1 \times \cdots \times q_k$. Let $h \in C(X; \mathbb{R})$. If $q(x) = q(y)$, then $h(x) = h(y)$: change one coordinate at a time, and note that with the other coordinates fixed, h is a continuous function of the remaining one, so it takes equal values at equivalent points. The map q is continuous from the compact space X ([Lemma 4.1\(b\)](#)) onto the Hausdorff space X' . So it maps closed sets to closed sets: a closed set is compact by (K1), its image is compact by (K2), and compact subsets of X' are closed by (K3). So $h = h' \circ q$, and h' is continuous, because $h'^{-1}(F) = q(h^{-1}(F))$ for closed $F \subseteq \mathbb{R}$. On X' , the linear combinations of products $f'_1(x_1) \cdots f'_k(x_k)$ form an algebra with 1 in it, and it separates points, since points that differ in coordinate j are separated by a function of that coordinate. By [Theorem 9.1](#) this algebra is dense in $C(X'; \mathbb{R})$. Composing with q keeps norms and turns each f'_j into $f'_j \circ q_j \in C(X_j; \mathbb{R})$. The complex statement follows by approximating real and imaginary parts.

(6) Since $-\|u\| \leq u \leq \|u\|$, positivity gives $|\mu(u)| \leq \mu(1)\|u\|$, and likewise for ν . Fix x_0 and $\varepsilon > 0$. For each y , continuity of h at (x_0, y) gives open sets $U_y \ni x_0$ and $V_y \ni y$ with $|h - h(x_0, y)| < \varepsilon/2$ on $U_y \times V_y$. Finitely many V_{y_j} cover Y . On $U = \bigcap_j U_{y_j}$ we get $\|h(x, \cdot) - h(x_0, \cdot)\| < \varepsilon$, since both (x, y) and (x_0, y) lie in some $U_{y_j} \times V_{y_j}$. So $x \mapsto \nu(h(x, \cdot))$ is continuous, and so is the other function. The two sides of (15.1) are linear in h and bounded by $\mu(1)\nu(1)\|h\|$. For $h(x, y) = f(x)g(y)$, both equal $\mu(f)\nu(g)$. By (5) such functions span a dense subspace, so (15.1) holds for all h .

For the rectangle, $\mu(u) = \int_a^b u$ and $\nu(v) = \int_c^d v$ are positive linear maps, by (B1). Take a grid of cells $[x_{i-1}, x_i] \times [y_{j-1}, y_j]$ with points (s_i, t_j) in the cells. Additivity over subintervals (B1) writes the iterated integral as the sum over cells of $\int_{x_{i-1}}^{x_i} \int_{y_{j-1}}^{y_j} h \, dy \, dx$. By monotonicity of the integral, each term differs from $h(s_i, t_j)(x_i - x_{i-1})(y_j - y_{j-1})$ by at most the area of the cell times the oscillation of h on it. h is uniformly continuous on the compact rectangle ([Lemma 4.1\(d\)](#)), so the oscillations tend to 0 with the mesh. So the grid Riemann sums converge to the iterated integral.

(7) By the Weierstrass theorem ([Corollary 14.3](#)) choose polynomials p_k with $p_k \rightarrow \bar{f}$ uniformly on $[a, b]$. By hypothesis and linearity $\int_a^b f p_k = 0$. Since $|\int_a^b u| \leq (b-a) \sup |u|$ for continuous u (B1), $|\int_a^b f|^2 - \int_a^b f p_k| \leq (b-a)\|f\|\|\bar{f} - p_k\| \rightarrow 0$, so $\int_a^b |f|^2 = 0$. If $f(x_0) \neq 0$, then by continuity $|f|^2 > |f(x_0)|^2/2$ on $[a, b] \cap (x_0 - \delta, x_0 + \delta)$ for some $\delta > 0$, a subinterval of positive length because $a < b$, and the integral would be positive, by monotonicity and additivity of the integral (B1). So $f = 0$. $\square$

Part (5) needs neither the Hausdorff property nor Urysohn's lemma. The continuous functions on a compact space need not separate its points, and then the products $f_1(x_1) \cdots f_k(x_k)$ do not separate the points of the

product; the quotients X_j' remove this problem. In (6), positive linear functionals are the form in which Radon measures act on continuous functions, so (15.1) also gives the equality of iterated integrals of a continuous function for Radon measures on compact Hausdorff spaces.

16. Compact sets, extension of functions, and bounded functions

Proposition 16.1.

1. (*Uniform approximation on compact sets.*) Let X be any topological space and $A \subseteq C(X; \mathbb{R})$ a real algebra that separates points and vanishes nowhere. For every $f \in C(X; \mathbb{R})$, every compact $C \subseteq X$ and every $\varepsilon > 0$, some $a \in A$ satisfies $|a - f| < \varepsilon$ on C . Suppose that X has compact sets $L_1 \subseteq L_2 \subseteq \dots$ such that every compact set lies in some L_n ; by Proposition 4.2(3) this holds when X is LCH and σ -compact. Then some sequence in A converges to f uniformly on every compact set. The complex analogue holds for self-adjoint A .
2. (*Tietze extension theorem, compact case.*) Let K be compact Hausdorff, $Z \subseteq K$ closed, and $g \in C(Z; \mathbb{R})$. Then $g = G|_Z$ for some $G \in C(K; \mathbb{R})$ with $\|G\| = \|g\|$.
3. (*A converse.*) Let X be a metrizable space that is not compact. Then $C_b(X; \mathbb{R})$ has a closed subalgebra with 1 in it that separates the points of X but is not dense.

Proof. (1) The restrictions of the elements of A to C form a real algebra on the compact space C . It separates the points of C and vanishes nowhere on C , so it is dense in $C(C; \mathbb{R})$ by Theorem 9.1(a); and $f|_C \in C(C; \mathbb{R})$. For the sequence, choose $a_n \in A$ with $|a_n - f| < 1/n$ on L_n . A compact set lies in some L_m , and the L_n increase, so $|a_n - f| < 1/n$ on it for $n \geq m$. In the complex case use the compact form of Theorem 10.1.

(2) If $Z = \emptyset$, take $G = 0$. Otherwise let $R = \{G|_Z : G \in C(K; \mathbb{R})\}$. It is a real algebra on the compact space Z that contains the constants. It separates the points of Z : points of K are closed, so Urysohn's lemma (Theorem 5.1) separates any two of them by a continuous function. By Theorem 9.1(a), R is dense in $C(Z; \mathbb{R})$. It is also closed. Let $g \in \bar{R}$, and choose $G_n \in C(K; \mathbb{R})$ with $\|G_n|_Z - g\| \leq 2^{-n}$. The differences $D_n = G_{n+1} - G_n$ satisfy $\|D_n|_Z\| < 2^{1-n}$. The truncations $E_n = (D_n \wedge 2^{1-n}) \vee (-2^{1-n})$ are continuous on K , agree with D_n on Z , and satisfy $\|E_n\| \leq 2^{1-n}$. So the series $G = G_1 + \sum_n E_n$ converges uniformly on K , G is continuous (Proposition 2.1(1)), and $G|_Z = \lim_n G_n|_Z = g$. Hence $R = C(Z; \mathbb{R})$. Finally, $(G \wedge \|g\|) \vee (-\|g\|)$ still extends g and has norm $\|g\|$.

(3) Let d be a metric for X . Since X is not compact, Lemma 4.1(d) gives a sequence (x_n) with no convergent subsequence. A value repeated infinitely often would give a constant convergent subsequence, so after passing to a subsequence the x_n are distinct. Each point p has a ball that contains x_n for only finitely many n , and after shrinking it contains no x_n other than p itself. So the sets $E_0 = \{x_n : n \text{ even}\}$ and $E_1 = \{x_n : n \text{ odd}\}$ are closed and disjoint. The function $u = d(\cdot, E_0)/(d(\cdot, E_0) + d(\cdot, E_1))$ is continuous with values in $[0, 1]$, and it is 0 on E_0 and 1 on E_1 . Let A be the set of $f \in C_b(X; \mathbb{R})$ for which $f(x_n)$ has a limit along the even n and a limit along the odd n , and the two limits are equal. A is a subalgebra containing the constants. It is closed: if $f_k \in A$ have common limits l_k and $f_k \rightarrow f$, then $|l_k - l_j| \leq \|f_k - f_j\|$, so $l_k \rightarrow l$, and $f(x_n) \rightarrow l$ along both parities. A separates points: for $x \neq y$ choose $0 < r < d(x, y)$ such that the ball $B(x, r)$ contains no x_n other than x , and put $f(z) = \max(0, 1 - d(z, x)/r)$. Then $f(x) = 1$, $f(y) = 0$, and $f(x_n) = 0$ whenever $x_n \neq x$, so $f \in A$. But if $f \in A$ has common limit l , then $\|u - f\| \geq \max(|l|, |1 - l|) \geq \frac{1}{2}$. So A is not dense. $\square$

In (2), Urysohn's lemma is used only to separate points; the closedness of R comes from the truncation argument. Example 12.2(f) is the case $X = \mathbb{N}$ of (3). The conclusion of (3) holds for every completely regular Hausdorff space that is not compact; that version is proved with the Stone-Ćech compactification [Blackadar], which this lesson does not develop.

17. Convex functions, singular functions, and the disc algebra

Proposition 17.1 (Differences of convex functions). Let K be a compact space and P a nonempty set of nonnegative continuous real functions on K , closed under sums, under multiplication by positive scalars, and under $\wedge$ or under $\vee$. Then $L = P - P = \{f - g : f, g \in P\}$ is a linear subspace closed under $\wedge$ and $\vee$. If

P separates points and some nonzero constant function lies in P , then L is dense in $C(K; \mathbb{R})$. In particular, every continuous function on $[a, b]$ is a uniform limit of differences of continuous convex functions.

Proof. L is closed under sums and positive multiples, $L = -L$, and $0 = f - f \in L$; so L is a linear subspace. For $f_1 - g_1$ and $f_2 - g_2$ in L ,

$$(f_1 - g_1) \wedge (f_2 - g_2) = ((f_1 + g_2) \wedge (f_2 + g_1)) - (g_1 + g_2),$$

because subtracting the same function from two functions commutes with $\wedge$. The same identity holds with $\vee$. So if P is closed under $\wedge$, so is L , and then L is closed under $\vee$ because $f \vee g = -((-f) \wedge (-g))$ and $L = -L$. The case of $\vee$ is symmetric. If P contains a constant $c > 0$ and separates points, then L contains all constants and separates points, so it is dense by [Theorem 8.2\(2\)](#). On $[a, b]$, take for P the nonnegative continuous convex functions. Sums, positive multiples and maxima of convex functions are convex, $1 \in P$, and $x \mapsto x - a + 1$ lies in P and separates points. The nonnegative concave functions, with $\wedge$, work in the same way. $\square$

A set $N \subseteq \mathbb{R}$ is *null* if, for every $\varepsilon > 0$, countably many intervals of total length less than ε cover it. A union of two null sets is null, and so is the image of a null set under a map $t \mapsto \alpha + \beta t$.

Proposition 17.2 (Singular functions). Let $C_{\text{sing}}([a, b])$ be the set of continuous $f : [a, b] \rightarrow \mathbb{R}$ that are differentiable with derivative 0 outside a null set. Then $C_{\text{sing}}([a, b])$ is a subalgebra of $C([a, b]; \mathbb{R})$; it contains the constants, and it separates points. So it is dense in $C([a, b]; \mathbb{R})$.

Proof. *Subalgebra.* If $f' = 0$ off a null set N_f and $g' = 0$ off a null set N_g , then off the null set $N_f \cup N_g$ the functions $f + g$, λf and fg are differentiable with derivative 0. For fg , write $f(s)g(s) - f(t)g(t) = f(s)(g(s) - g(t)) + g(t)(f(s) - f(t))$ and use continuity of f .

The Cantor function. Let Φ be the set of continuous nondecreasing $\phi : [0, 1] \rightarrow [0, 1]$ with $\phi(0) = 0$ and $\phi(1) = 1$. For $\phi \in \Phi$ define $T\phi$ to be $\phi(3x)/2$ on $[0, \frac{1}{3}]$, $\frac{1}{2}$ on $[\frac{1}{3}, \frac{2}{3}]$, and $\frac{1}{2} + \phi(3x - 2)/2$ on $[\frac{2}{3}, 1]$. Then $T\phi \in \Phi$, and $\|T\phi - T\chi\| \leq \frac{1}{2}\|\phi - \chi\|$. Starting from $c_0(x) = x$, the functions $c_{n+1} = Tc_n$ satisfy $\|c_{n+1} - c_n\| \leq 2^{-n}$. So they converge uniformly to some $c \in \Phi$ with $Tc = c$. Let $G_1 = (\frac{1}{3}, \frac{2}{3})$ and $G_{n+1} = G_1 \cup \frac{1}{3}G_n \cup (\frac{2}{3} + \frac{1}{3}G_n)$. By induction on n : $G_n \subseteq G_{n+1}$; $[0, 1] \setminus G_n$ is a union of 2^n closed intervals of length 3^{-n} ; and $c = Tc$ is constant on each component interval of G_n . So c is locally constant on the open set $G = \bigcup_n G_n$, and $[0, 1] \setminus G$ is null, since for each n it is covered by intervals of total length $(2/3)^n$.

Separation. Let $a \leq x < y \leq b$. Put $s(t) = 0$ for $t \leq x$, $s(t) = c((t - x)/(y - x))$ for $x \leq t \leq y$, and $s(t) = 1$ for $t \geq y$. Then s is continuous, $s(x) = 0$ and $s(y) = 1$. It is locally constant, so differentiable with derivative 0, off the null set $\{x, y\} \cup (x + (y - x)([0, 1] \setminus G))$. So $s \in C_{\text{sing}}([a, b])$, and [Theorem 9.1\(a\)](#) gives density. $\square$

For the next result, let $\mathcal{P}$ be the algebra of polynomials in w , restricted to $\mathbb{T}$. For $g \in C(\mathbb{T})$ and $n \in \mathbb{Z}$ put $\hat{g}(n) = \frac{1}{2\pi} \int_0^{2\pi} g(e^{i\theta}) e^{-in\theta} d\theta$, and let $\bar{D} = \{z : |z| \leq 1\}$.

Proposition 17.3 (The disc algebra).

- (a) If $p \in \mathcal{P}$ and $\bar{p} \in \mathcal{P}$, then p is constant. So $\mathcal{P}$ is not self-adjoint.
- (b) As functions of θ , the elements of $\mathcal{P}$ are the sums $\sum_{k=0}^n c_k e^{ik\theta}$.
- (c)–(d) $\hat{g}(-1) = 0$ for every g in the closure $\bar{\mathcal{P}}$, while $\bar{w}$ has $\hat{w}(-1) = 1$. So $\bar{w} \notin \bar{\mathcal{P}}$.
- (e) $\bar{\mathcal{P}} = \{G|_{\mathbb{T}} : G \in C(\bar{D}), G(z) = \sum_{n \geq 0} a_n z^n \text{ for } |z| < 1\}$.
- (f) $\bar{\mathcal{P}} = \{g \in C(\mathbb{T}) : \hat{g}(n) = 0 \text{ for all } n < 0\}$.
- (g) Trigonometric polynomials approximate every $g \in C(\mathbb{T})$ uniformly ([Proposition 15.1\(4\)](#)).

Proof. For integers $m \neq 0$, $\int_0^{2\pi} e^{im\theta} d\theta = 0$, by the fundamental theorem of calculus (B1) applied to $e^{im\theta}/(im)$, whose derivative is $e^{im\theta}$ and which takes the same value at 0 and 2π (B2). So if $t = \sum_k c_k e^{ik\theta}$ is a finite sum, then $\hat{t}(n) = c_n$. Also $|\hat{g}(n)| \leq \|g\|$, since $|\int u| \leq \int |u|$ (B1).

(a) Write $p = \sum_{k \geq 0} c_k w^k$. On $\mathbb{T}$, $\bar{p} = \sum_k \bar{c}_k w^{-k}$, so $\hat{\bar{p}}(-k) = \bar{c}_k$. If $\bar{p} \in \mathcal{P}$, its coefficients at negative n vanish, so $c_k = 0$ for $k \geq 1$.

(b) $e^{ik\theta} = (e^{i\theta})^k$ by the addition formula.

(c)–(d) $g \mapsto \hat{g}(-1)$ is continuous and vanishes on $\mathcal{P}$, hence on $\bar{\mathcal{P}}$, while $\bar{w} = e^{-i\theta}$ gives the value 1. (Compare [Example 12.2\(a\)](#), which gives $\|\bar{w} - p\| \geq 1$ without integrals.)

(f) If $g \in \bar{\mathcal{P}}$, then $\hat{g}(n) = 0$ for $n < 0$, by continuity as in (c). Conversely, let $\hat{g}(n) = 0$ for $n < 0$. For $0 \leq r < 1$ put

$$P_r(t) = \sum_{n \in \mathbb{Z}} r^{|n|} e^{int} = \frac{1 - r^2}{|1 - re^{it}|^2} > 0.$$

The series converges absolutely and uniformly in t , and summing its two geometric halves gives the closed form: with $z = re^{it}$, $\frac{1}{1-z} + \frac{\bar{z}}{1-\bar{z}} = \frac{1-|z|^2}{|1-z|^2}$. Integrating term by term, which uniform convergence allows (B1), gives $\frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - t) dt = 1$ and

$$(P_r g)(e^{i\theta}) := \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - t) g(e^{it}) dt = \sum_{n \in \mathbb{Z}} r^{|n|} \hat{g}(n) e^{in\theta}.$$

Here $\hat{g}(n) = 0$ for $n < 0$, so $P_r g = \sum_{n \geq 0} r^n \hat{g}(n) w^n$, a series of elements of $\mathcal{P}$ that converges uniformly on $\mathbb{T}$ because $|r^n \hat{g}(n)| \leq r^n \|g\|$. So $P_r g \in \bar{\mathcal{P}}$. It remains to show $\|P_r g - g\| \rightarrow 0$ as $r \rightarrow 1$. Given $\varepsilon > 0$, uniform continuity on the compact set $\mathbb{T}$ ([Lemma 4.1\(d\)](#)) gives $\eta > 0$ with $|g(u) - g(v)| < \varepsilon$ when $u, v \in \mathbb{T}$ and $|u - v| < \eta$. Since P_r has average 1,

$$(P_r g - g)(e^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} P_r(\theta - t) (g(e^{it}) - g(e^{i\theta})) dt.$$

Let $r \geq 1 - \eta/2$. If $|e^{it} - e^{i\theta}| \geq \eta$, then $|1 - re^{i(\theta-t)}| = |e^{it} - re^{i\theta}| \geq \eta - (1 - r) \geq \eta/2$, so $P_r(\theta - t) \leq 4(1 - r^2)/\eta^2$. Hence, for every t , the integrand has absolute value at most $\varepsilon P_r(\theta - t) + 8\|g\|(1 - r^2)/\eta^2$. Integrating this bound, by monotonicity of the integral (B1), gives $\|P_r g - g\| \leq \varepsilon + 8\|g\|(1 - r^2)/\eta^2$, which is less than 2ε for r close to 1.

(e) Let $g \in \bar{\mathcal{P}}$. By (f), $\hat{g}(n) = 0$ for $n < 0$, and $|\hat{g}(n)| \leq \|g\|$. Put $G(z) = \sum_{n \geq 0} \hat{g}(n) z^n$ for $|z| < 1$, and $G = g$ on $\mathbb{T}$. For $z = re^{i\theta}$ with $r < 1$, $G(z) = (P_r g)(e^{i\theta})$. The series converges uniformly on each disc $|z| \leq \rho < 1$, so G is continuous on the open disc. Let $w_0 \in \mathbb{T}$ and $z \rightarrow w_0$ in $\bar{D}$. For $|z| < 1$ we have $|G(z) - g(w_0)| \leq \|P_{|z|} g - g\| + |g(z/|z|) - g(w_0)|$, and for $|z| = 1$ we have $|G(z) - g(w_0)| = |g(z) - g(w_0)|$. Both tend to 0, by (f) and the continuity of g . So $G \in C(\bar{D})$. Conversely, let $G \in C(\bar{D})$ with $G(z) = \sum a_n z^n$ for $|z| < 1$. For $r < \rho < 1$, the terms $a_n \rho^n$ tend to 0, so $|a_n| r^n \leq C(r/\rho)^n$ for some C . So $G_r(w) = G(rw) = \sum a_n r^n w^n$ converges uniformly on $\mathbb{T}$, and $G_r|_{\mathbb{T}} \in \bar{\mathcal{P}}$. G is uniformly continuous on the compact set $\bar{D}$ ([Lemma 4.1\(d\)](#)), and $|rw - w| = 1 - r$, so $G_r \rightarrow G$ uniformly on $\mathbb{T}$. Hence $G|_{\mathbb{T}} \in \bar{\mathcal{P}}$. $\square$

The proof of (e) uses the dilations G_r , not the partial sums of the Taylor series of G : those partial sums need not converge uniformly on the circle. By [Cauchy's theorem for cycles and its consequences, Lemma 3.1 and Theorem 3.2](#), the functions given by power series on the open disc are exactly the holomorphic ones. So (e) says that $\bar{\mathcal{P}}$ consists of the restrictions to $\mathbb{T}$ of the continuous functions on $\bar{D}$ that are holomorphic on the open disc.

18. Integer coefficients and simultaneous approximation of derivatives

Proposition 18.1 (Integer coefficients on $[0, 1]$). A function $f \in C([0, 1]; \mathbb{R})$ can be approximated uniformly on $[0, 1]$ by polynomials with integer coefficients if and only if $f(0)$ and $f(1)$ are integers.

Proof. Let $\mathcal{G}$ be the closure of $\mathbb{Z}[x]$ in $C([0, 1]; \mathbb{R})$. It is closed under addition, negation and multiplication, by the argument of [Proposition 2.1\(2\)](#). *Necessity:* $p(0), p(1) \in \mathbb{Z}$ for $p \in \mathbb{Z}[x]$, and a limit of integers is an integer.

Step 1 (iterates). Let $g(x) = 2x - 3x^2 + 2x^3 = x + x(1 - x)(1 - 2x)$. Then $g(0) = 0$, $g(\frac{1}{2}) = \frac{1}{2}$, $g(1) = 1$, and $g'(x) = 6(x - \frac{1}{2})^2 + \frac{1}{2} > 0$. So g is an increasing bijection of $[0, 1]$, with $g(x) > x$ on $(0, \frac{1}{2})$ and $g(x) < x$ on $(\frac{1}{2}, 1)$. Let g_m be the m -th iterate of g ; then $g_m \in \mathbb{Z}[x]$ and $0 \leq g_m \leq 1$ on $[0, 1]$. For $0 < x \leq \frac{1}{2}$, the sequence $g_m(x)$ increases and is bounded by $\frac{1}{2}$, so it converges to a fixed point of g in $(0, \frac{1}{2}]$. The fixed points of g are

0, $\frac{1}{2}$ and 1, so the limit is $\frac{1}{2}$. The case $[\frac{1}{2}, 1)$ is symmetric. For $0 < \varepsilon < \frac{1}{2}$ and $x \in [\varepsilon, 1 - \varepsilon]$, monotonicity gives $g_m(\varepsilon) \leq g_m(x) \leq g_m(1 - \varepsilon)$, so $g_m \rightarrow \frac{1}{2}$ uniformly on $[\varepsilon, 1 - \varepsilon]$.

Step 2 (dyadic constants). For $n \geq 0$ and $0 \leq k \leq 2^n$ there are $\phi_m \in \mathbb{Z}[x]$ with $0 \leq \phi_m \leq 1$ on $[0, 1]$ and $\phi_m \rightarrow k/2^n$ uniformly on $[\varepsilon, 1 - \varepsilon]$ for every $\varepsilon \in (0, \frac{1}{2})$. For $k = 0$ and $k = 2^n$ take the constants 0 and 1; this settles $n = 0$. Suppose the claim holds for n . If k is even, $k/2^{n+1} = (k/2)/2^n$. If $k = 2j + 1$, take sequences $\phi_m \rightarrow j/2^n$ and $\chi_m \rightarrow (j + 1)/2^n$, and put $\psi_m = g_m \phi_m + (1 - g_m) \chi_m \in \mathbb{Z}[x]$. It takes values in $[0, 1]$, as a convex combination, and

$$\psi_m - \frac{k}{2^{n+1}} = g_m \left(\phi_m - \frac{j}{2^n} \right) + (1 - g_m) \left(\chi_m - \frac{j+1}{2^n} \right) - \left(g_m - \frac{1}{2} \right) \frac{1}{2^n} \rightarrow 0$$

uniformly on $[\varepsilon, 1 - \varepsilon]$.

Step 3 (real multiples). Let $p \in \mathbb{Z}[x]$ with $p(0) = p(1) = 0$. The set $\Lambda_p = \{\lambda \in \mathbb{R} : \lambda p \in \mathcal{G}\}$ is a closed additive subgroup of $\mathbb{R}$ that contains $\mathbb{Z}$. Let $\lambda = k/2^n \in [0, 1]$ and $\eta > 0$. Choose ε with $|p| \leq \eta$ on $[0, \varepsilon] \cup [1 - \varepsilon, 1]$, and then m with $|\phi_m - \lambda| \leq \eta/(1 + \|p\|)$ on $[\varepsilon, 1 - \varepsilon]$. Since $|\phi_m - \lambda| \leq 1$ everywhere, $\|\phi_m p - \lambda p\| \leq \eta$, and $\phi_m p \in \mathbb{Z}[x]$. So $\lambda \in \Lambda_p$. Adding integers, Λ_p contains all dyadic rationals, and being closed, $\Lambda_p = \mathbb{R}$.

Step 4 (conclusion). A real polynomial r with $r(0) = r(1) = 0$ has the form $x(1 - x)s(x) = \sum_j s_j x^{j+1}(1 - x)$ with real s_j , and each term lies in $\mathcal{G}$ by Step 3; so $r \in \mathcal{G}$. These polynomials form a real algebra. On $X = (0, 1)$ it lies in $C_0(X; \mathbb{R})$, separates points (the ratio of $x^2(1 - x)$ to $x(1 - x)$ is x), and vanishes nowhere. By **Theorem 9.2** it is dense in $C_0(X; \mathbb{R})$. Directly from (1.1), $C_0(X; \mathbb{R})$ consists of the restrictions of the $h \in C([0, 1]; \mathbb{R})$ with $h(0) = h(1) = 0$. So every such h lies in $\mathcal{G}$. Finally, if $f(0), f(1) \in \mathbb{Z}$, then $q(x) = f(0) + (f(1) - f(0))x \in \mathbb{Z}[x]$, and $f - q$ vanishes at 0 and 1; so $f = q + (f - q) \in \mathcal{G}$. $\square$

Proposition 18.2 (Integer coefficients on $[-1, 1]$). A function $f \in C([-1, 1]; \mathbb{R})$ can be approximated uniformly on $[-1, 1]$ by polynomials with integer coefficients if and only if $f(-1), f(0), f(1)$ are integers and $f(-1) \equiv f(1) \pmod{2}$.

Proof. Let $\mathcal{G}'$ be the closure of $\mathbb{Z}[x]$ in $C([-1, 1]; \mathbb{R})$. *Necessity:* for $p = \sum a_k x^k \in \mathbb{Z}[x]$, $p(1) - p(-1) = 2 \sum_{k \text{ odd}} a_k$ is even. Limits of integers are integers, and a convergent sequence of integers is eventually constant, so the parity condition passes to limits. *Sufficiency:* let $p \in \mathbb{Z}[x]$ vanish at $-1, 0$ and 1 , and let ϕ_m be as in Step 2 of the proof of Proposition 18.1. The polynomials $\psi_m(x) = \phi_m(1 - x^2)$ have integer coefficients and values in $[0, 1]$ on $[-1, 1]$. They converge to $k/2^n$ uniformly where $\varepsilon \leq 1 - x^2 \leq 1 - \varepsilon$, that is, where $\sqrt{\varepsilon} \leq |x| \leq \sqrt{1 - \varepsilon}$. Near 0 and ± 1 , p is small. Step 3 of that proof, with these changes, gives $\lambda p \in \mathcal{G}'$ for all real λ . Every real polynomial vanishing at $-1, 0, 1$ is $x(1 - x^2)s(x) = \sum_j s_j x^{j+1}(1 - x^2)$, so it lies in $\mathcal{G}'$. These polynomials form a real algebra which, on $X = (-1, 0) \cup (0, 1)$, separates points (the ratio of $x^2(1 - x^2)$ to $x(1 - x^2)$ is x) and vanishes nowhere. By **Theorem 9.2** it is dense in $C_0(X; \mathbb{R})$, which by (1.1) consists of the restrictions of the continuous functions on $[-1, 1]$ that vanish at $-1, 0$ and 1 . Finally, $\beta = (f(1) - f(-1))/2$ and $\gamma = (f(1) + f(-1))/2 - f(0)$ are integers by the parity condition, and $q(x) = f(0) + \beta x + \gamma x^2 \in \mathbb{Z}[x]$ agrees with f at $-1, 0$ and 1 . So $f = q + (f - q) \in \mathcal{G}'$. $\square$

Proposition 18.3 (Smooth functions). Let $I = [a, b]$ and $r \geq 0$ an integer, and let f be r times continuously differentiable on I , with one-sided derivatives at the ends.

1. For every $\varepsilon > 0$ there is a polynomial p with $\|f^{(k)} - p^{(k)}\| < \varepsilon$ for $0 \leq k \leq r$.
2. If f is infinitely differentiable, there are polynomials p_n with $p_n^{(k)} \rightarrow f^{(k)}$ uniformly for every $k \geq 0$.
3. So the polynomials are dense in $C^r(I)$ for the norm $\max_{k \leq r} \|f^{(k)}\|$, and in $C^\infty(I)$ for the topology of uniform convergence of each derivative.

Proof. (1) Put $c = \max(1, b - a)$. By the Weierstrass theorem (**Corollary 14.3**) choose a polynomial q_r with $\|f^{(r)} - q_r\| < \varepsilon/c^r$. Going down, put $q_{k-1}(x) = f^{(k-1)}(a) + \int_a^x q_k(t) dt$ for $k = r, \dots, 1$, and $p = q_0$. Each q_{k-1} is a polynomial with $q'_{k-1} = q_k$, because the indefinite integral of a continuous function is differentiable with that function as its derivative (B1); so $p^{(k)} = q_k$. By the fundamental theorem of calculus (B1), $f^{(k-1)}(x) = f^{(k-1)}(a) + \int_a^x f^{(k)}(t) dt$. So $\|f^{(k-1)} - q_{k-1}\| \leq (b - a) \|f^{(k)} - q_k\| \leq c \|f^{(k)} - q_k\|$, by the bound $|\int_a^x u| \leq (x - a) \sup |u|$ of (B1). Induction gives $\|f^{(k)} - q_k\| < c^{r-k} \varepsilon/c^r \leq \varepsilon$. (2) Apply (1) with $r = n$ and $\varepsilon = 1/n$. For fixed k and $n \geq k$, $\|f^{(k)} - p_n^{(k)}\| < 1/n$. (3) restates (1) and (2). $\square$

Exercises

Exercise 1 (self-adjointness is needed on the line). Let $\kappa(x) = (x - i)/(x + i)$ be the Cayley map of [Example 3.3\(a\)](#), and $u = \kappa - 1$. Show that $A = \{q \circ u : q \text{ a complex polynomial, } q(0) = 0\}$ is a subalgebra of $C_0(\mathbb{R})$ that separates points and vanishes nowhere, but that $\|\bar{u} - a\| \geq 1$ for every $a \in A$.

Solution. $u(x) = -2i/(x + i)$, so $|u(x)| = 2(1 + x^2)^{-1/2}$. The sets $\{|u| \geq \varepsilon\}$ are closed and bounded, hence compact, so $u \in C_0(\mathbb{R})$, and $A \subseteq C_0(\mathbb{R})$ by [Proposition 2.1\(3\)](#). The condition $q(0) = 0$ survives sums, scalar multiples and products, so A is a subalgebra. u is injective because κ is, so A separates points; and $u(x) \neq 0$ because $\kappa(x) \neq 1$, so A vanishes nowhere. By [Proposition 3.2\(5\)](#) and [Proposition 3.4](#), the rule $G(\kappa(x)) = g(x)$, $G(1) = 0$, defines a norm-preserving bijection $g \mapsto G$ of $C_0(\mathbb{R})$ onto $\{G \in C(\mathbb{T}) : G(1) = 0\}$. For $g = q \circ u$, $G(w) = q(w - 1)$, also at $w = 1$, since $q(0) = 0$. For $g = \bar{u}$, $G(w) = \bar{w} - 1$. Take Λ_N from [Example 12.2\(a\)](#) with $N \geq \deg q + 2$. Then Λ_N vanishes on the polynomial $q(w - 1)$ and on 1, and $\Lambda_N(\bar{w}) = 1$. So $\|\bar{u} - q \circ u\| = \|\bar{w} - 1 - q(w - 1)\| \geq |\Lambda_N(\bar{w} - 1 - q(w - 1))| = 1$.

Exercise 2 (the route through the constants). Let K be compact and $A \subseteq C(K; \mathbb{R})$ a real algebra that separates points and has $Z(A) = \{z\}$. Let $B = \{a + s1 : a \in \bar{A}, s \in \mathbb{R}\}$. Show that B is a closed algebra, that $1 \in B$, and that B separates points. Deduce $\bar{A} = \{f : f(z) = 0\}$ from the case of algebras with constants.

Solution. $\bar{A}$ is an algebra ([Proposition 2.1\(2\)](#)), and $(a + s)(b + t) = (ab + ta + sb) + st$, so B is an algebra. Let $a_n + s_n \rightarrow f$ with $a_n \in \bar{A}$. Evaluating at z gives $s_n \rightarrow f(z)$, so $a_n \rightarrow f - f(z)1$, which then lies in $\bar{A}$; so $f \in B$, and B is closed. B contains 1 and A , so it separates points. By [Proposition 6.2\(3\)](#), B is closed under $\wedge$, so [Theorem 8.2\(2\)](#) makes it dense, and $B = C(K; \mathbb{R})$. If $f(z) = 0$, write $f = a + s1$ with $a \in \bar{A}$. Then $0 = a(z) + s = s$, so $f = a \in \bar{A}$. The reverse inclusion is clear. [Theorem 9.1](#) avoids the detour through B .

Exercise 3 (C_0 of a product). Let X and Y be LCH. Show that the linear span of the functions $(f \otimes g)(x, y) = f(x)g(y)$, $f \in C_0(X)$, $g \in C_0(Y)$, is dense in $C_0(X \times Y)$.

Solution. $X \times Y$ is Hausdorff, and a product of compact neighbourhoods is a compact neighbourhood ([Lemma 4.1\(b\)](#)); so it is LCH. If $|f(x)g(y)| \geq \varepsilon$, then $|f(x)| \geq \varepsilon/(\|g\| + 1)$ and $|g(y)| \geq \varepsilon/(\|f\| + 1)$. So $\{|f \otimes g| \geq \varepsilon\}$ is closed and lies in a product of two compact sets, and $f \otimes g \in C_0(X \times Y)$ by (K1). The span is a self-adjoint subalgebra, since $(f \otimes g)(f' \otimes g') = (ff') \otimes (gg')$ and $\overline{f \otimes g} = \bar{f} \otimes \bar{g}$. Let $(x, y) \neq (x', y')$, say $x \neq x'$. By the locally compact form of Urysohn's lemma ([Corollary 5.2](#)), choose $f \in C_c(X)$ with $f(x) = 1$, $f(x') = 0$, and $g \in C_c(Y)$ with $g = 1$ on $\{y, y'\}$. Then $f \otimes g$ takes the values 1 and 0 at the two points. The case $y \neq y'$ is symmetric, and the same functions show that the span vanishes nowhere. Apply [Theorem 10.1](#).

Exercise 4 (separability). Let X be LCH and second countable. Show that $C_0(X)$ is separable.

Solution. By the metrizable theorem, [Theorem 5.3](#), X_∞ is a compact metrizable space. By [Proposition 15.1\(3\)](#), $C(X_\infty)$ is separable. A subset of a separable metric space is separable, since the space has a countable base of balls. So I_∞ is separable, and so is $C_0(X)$, which is isometric to it ([Proposition 3.4](#)).

Where this leads

- *Commutative C^* -algebras.* In [the lesson on \$C^*\$ -algebras and the continuous functional calculus](#), [Theorem 10.1](#) shows that the Gelfand transform of a commutative C^* -algebra maps onto C_0 of its character space. Urysohn's lemma and the density theorems of Section 15 are used there as well.
- *States on C^* -algebras.* The lesson "Integral representations of states" uses Stone's lattice lemma and the real and complex theorems to show that certain spaces of continuous functions on compact convex sets and on state spaces are dense.
- *Locally compact groups.* In the course "Modular theory and weights", the lesson "The Plancherel weight and Fourier coefficients of a locally compact group" shows that the Fourier algebra $A(G)$ of a locally compact group G lies in $C_0(G)$, is closed under pointwise products and complex conjugation, separates points and vanishes nowhere. [Theorem 10.1](#) then makes $A(G)$ dense in $C_0(G)$. Since the proof uses no countability, this holds for every locally compact group.
- *Algebras that are not self-adjoint.* For these the conclusion can fail, as the disc algebra of [Proposition 17.3](#) shows. Bishop's theorem [Bishop 1961] reduces approximation by a closed subalgebra of $C(K)$ to its maximal antisymmetric sets.

Results used from other lessons

The following facts are proved in the core courses [Real Analysis I](#) and [Real Analysis II](#), whose text is [Lebl], and [Proof, Logic, and Discrete Structures](#), whose text is [Levin]. Each item gives the place of the proof. A few steps that these texts leave to the reader are proved here.

- (B1) *The Riemann integral.* Every continuous function $u : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable ([Real Analysis I, Lemma 5.2.7](#)). The integral is additive over subintervals: $\int_a^b u = \int_a^c u + \int_c^b u$ for $a \leq c \leq b$ ([Proposition 5.2.2](#)). It is monotone: if $u \leq v$, then $\int_a^b u \leq \int_a^b v$ ([Proposition 5.2.6](#)). If $m \leq u \leq M$, then $m(b-a) \leq \int_a^b u \leq M(b-a)$ ([Proposition 5.1.10](#)). And $\int_a^b \alpha u = \alpha \int_a^b u$ for $\alpha \geq 0$ ([Proposition 5.2.4](#)). The remaining cases of linearity are proved as follows. For a bounded function u and a partition P , the infimum of $-u$ on each cell of P is minus the supremum of u there, so the lower and upper sums satisfy $L(P, -u) = -U(P, u)$ and $U(P, -u) = -L(P, u)$. Taking the supremum and the infimum over P shows that $-u$ is integrable with $\int_a^b (-u) = -\int_a^b u$ when u is. For bounded u and v , on each cell the infimum of $u+v$ is at least the sum of the infima of u and v , and the supremum of $u+v$ is at most the sum of the suprema, so $L(P, u) + L(P, v) \leq L(P, u+v)$ and $U(P, u+v) \leq U(P, u) + U(P, v)$. Let u and v be integrable, let P_1 and P_2 be partitions, and put $P = P_1 \cup P_2$. Refining a partition does not lower its lower sums or raise its upper sums ([Real Analysis I, Proposition 5.1.7](#)), and the lower integral is at most the upper integral ([Proposition 5.1.8](#)), so

$$L(P_1, u) + L(P_2, v) \leq L(P, u+v) \leq \int_a^b (u+v) \leq \overline{\int_a^b (u+v)} \leq U(P, u+v) \leq U(P_1, u) + U(P_2, v).$$

Taking the supremum of the left side and the infimum of the right side over P_1 and P_2 gives $\int_a^b u + \int_a^b v \leq \int_a^b (u+v) \leq \overline{\int_a^b (u+v)} \leq \int_a^b u + \int_a^b v$. So $u+v$ is integrable and $\int_a^b (u+v) = \int_a^b u + \int_a^b v$. For a continuous complex function $u = u_1 + iu_2$ one puts $\int_a^b u = \int_a^b u_1 + i \int_a^b u_2$. By the real case, applied to real and imaginary parts, this integral is complex linear. Moreover $|\int_a^b u| \leq \int_a^b |u| \leq (b-a) \sup_{[a,b]} |u|$. The second inequality is Proposition 5.1.10 applied to the continuous function $|u|$. For the first, let $\int_a^b u \neq 0$ and put $c = \int_a^b u / \int_a^b u$. Then $|c| = 1$, and $\int_a^b cu = c \int_a^b u = \int_a^b |u|$ is real, so it equals its real part $\int_a^b \operatorname{Re}(cu)$. Since $\operatorname{Re}(cu) \leq |cu| = |u|$, monotonicity gives $|\int_a^b u| \leq \int_a^b |u|$. A uniformly convergent series of continuous functions can be integrated term by term: apply [Real Analysis II, Theorem 6.2.4](#) to the partial sums, which are continuous and so integrable, and to their real and imaginary parts. If u is continuous, then $x \mapsto \int_a^x u$ is differentiable on $[a, b]$ with derivative u ([Real Analysis I, Theorem 5.3.3](#)). If F is differentiable on $[a, b]$ and F' is continuous, then $\int_a^b F' = F(b) - F(a)$ ([Theorem 5.3.1](#), the fundamental theorem of calculus). For complex functions, apply both to real and imaginary parts. See also, e.g., [Rudin, Theorems 6.8, 6.12, 6.13, 6.20, 6.21, 6.25 and 7.16].

- (B2) *The exponential function.* Put $E(z) = \sum_{n \geq 0} z^n/n!$ for $z \in \mathbb{C}$; the series converges for every z . Write $e^z = E(z)$. The law of exponents $E(z+w) = E(z)E(w)$ is [Real Analysis II, Proposition 11.4.1](#). [Proposition 11.4.2](#) and its proof give the following for real x : $E(ix) = \cos x + i \sin x$ with $\cos x$ and $\sin x$ real, and $|E(ix)| = 1$; $E(i\pi/2) = i$ and $E(2\pi i) = 1$, so E has period $2\pi i$; $t \mapsto E(it)$ is one-to-one on $[0, 2\pi)$, so $E(it) \neq 1 = E(0)$ for $0 < t < 2\pi$; and every $z \in \mathbb{T}$ with $\operatorname{Re} z \geq 0$ and $\operatorname{Im} z \geq 0$ is $E(ix)$ for some $x \in [0, \pi/2]$. Two further facts are proved here. *Every point of $\mathbb{T}$ is $E(it)$ for some $t \in [0, 2\pi)$.* Multiplication by $-i$ maps $a+ib$ to $b-ia$. So for $z \in \mathbb{T}$, one of $z, -iz, -z$ and $iz = (-i)^3 z$ has nonnegative real and imaginary parts: $(-i)^k z = E(ix)$ for some $k \in \{0, 1, 2, 3\}$ and $x \in [0, \pi/2]$. Then $z = i^k E(ix) = E(i(x+k\pi/2))$ by the law of exponents, and $x+k\pi/2 \in [0, 2\pi]$. If $x+k\pi/2 = 2\pi$, then $z = E(2\pi i) = 1 = E(0)$. *For real m , the function $t \mapsto E(imt)$ on $\mathbb{R}$ is differentiable with derivative $imE(imt)$.* For complex w with $|w| \leq 1$,

$$|E(w) - 1 - w| \leq \sum_{k \geq 2} \frac{|w|^k}{k!} \leq |w|^2 \sum_{k \geq 2} \frac{1}{k!} \leq |w|^2,$$

by the triangle inequality for the partial sums, and since $k! \geq 2^{k-1}$ gives $\sum_{k \geq 2} 1/k! \leq \sum_{k \geq 2} 2^{1-k} = 1$. For real $h \neq 0$ with $|mh| \leq 1$, the law of exponents gives $E(im(t+h)) - E(imt) - imhE(imt) = E(imt)(E(imh) - 1 - imh)$, and $|E(imt)| = 1$. Hence

$$\left| \frac{E(im(t+h)) - E(imt)}{h} - imE(imt) \right| \leq \frac{|mh|^2}{|h|} = m^2|h|,$$

which tends to 0 as $h \rightarrow 0$. See also, e.g., [Rudin, (8.25)–(8.28), (8.48) and Theorem 8.7].

- (B3) *Numbers*. The real numbers form an ordered field with the least upper bound property; *Real Analysis I* starts from this (Theorem 1.2.1), and a construction is in, e.g., [Rudin, Theorem 1.19 and the appendix to Chapter 1]. The geometric sum $\sum_{k=0}^{N-1} \rho^k = (1 - \rho^N)/(1 - \rho)$ for $\rho \neq 1$ is proved by induction in Real Analysis I, Example 0.3.8. The induction uses only the field operations, so the identity holds for complex ρ . If $|\rho| \leq r < 1$, then

$$\left| \sum_{k=0}^{N-1} \rho^k - \frac{1}{1-\rho} \right| = \frac{|\rho|^N}{|1-\rho|} \leq \frac{r^N}{1-r},$$

which tends to 0 as $N \rightarrow \infty$ (Real Analysis I, Proposition 2.2.11). So $\sum_{k \geq 0} \rho^k = 1/(1-\rho)$, uniformly for $|\rho| \leq r$. The binomial theorem is Proof, Logic, and Discrete Structures, Theorem 3.1.9, and the formula $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is Theorem 3.4.9 there. Bernoulli's inequality $(1+u)^n \geq 1+nu$ for $u \geq 0$ follows from the binomial theorem, because all its terms are nonnegative. The Cauchy–Schwarz inequality $(\sum_k w_k a_k)^2 \leq (\sum_k w_k)(\sum_k w_k a_k^2)$ for finitely many real numbers a_k and weights $w_k \geq 0$ is Real Analysis II, Lemma 7.1.4, applied to the vectors $(\sqrt{w_k})_k$ and $(\sqrt{w_k} a_k)_k$; see also, e.g., [Rudin, Theorem 1.35].

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Restricted products and profinite completions

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A finite adèle records one number in each field $\mathbb{Q}_p$, with an integrality condition at all but finitely many primes. Its topology must allow those finitely many exceptions while retaining a compact neighbourhood. Restricted products provide exactly this construction. They also explain how local integrals and local characters combine into global ones.

We first build the topology, then integration and characters. We apply the construction to the profinite integers and finite adèles, and finish with test functions and the topology of finite idèles. The prerequisites are point-set topology and measure and integration, together with [Haar measure on locally compact groups](#), *The Pontryagin duality theorem*, *Subgroups, quotients and annihilators*, and *Totally disconnected groups, the p -adic numbers and the adèles*. The precise background statements used without proof are listed below.

Basic references are [Getz–Hahn], [Bost–Connes], [Deitmar] and [Halter-Koch].

A topology with finitely many exceptional coordinates

Let V be any index set. Each G_v is a locally compact Hausdorff group. Fix a finite set $E \subseteq V$, and for $v \notin E$ choose a compact open subgroup $K_v \leq G_v$. Coordinates in E are unrestricted; this permits real or complex factors. Throughout, “almost all” means all but finitely many.

The **restricted product** is the group

$$G = \prod'_{v \in V} (G_v, K_v) = \left\{ x \in \prod_{v \in V} G_v : x_v \in K_v \text{ for almost all } v \notin E \right\}.$$

For finite $S \supseteq E$, set

$$G_S = \prod_{v \in S} G_v \times \prod_{v \notin S} K_v.$$

Give G the topology with basis

$$B(S; U) = \prod_{v \in S} U_v \times \prod_{v \notin S} K_v, \quad U_v \subseteq G_v \text{ open.} \quad (1)$$

The finite set can always be enlarged: a newly included coordinate is assigned the open set K_v . This observation proves that intersections of basic sets are basic sets, allowing empty factors.

Proposition 1.1. The restricted product is a locally compact Hausdorff topological group. It is the upward-directed union of the open subgroups G_S , and each G_S has its product topology. Every compact subset of G lies in some G_S .

Proof. Every tuple satisfies the defining restriction outside a finite set, so $G = \bigcup_S G_S$. Coordinatewise multiplication and inversion preserve the condition because the K_v are subgroups. Thus G is an algebraic subgroup of the full product.

The sets G_S are open by (1). Intersecting (1) with G_S imposes open conditions on only finitely many factors of its product decomposition. Conversely, every product neighbourhood in G_S is obtained this way, since its finitely many conditions on tail factors are open in the corresponding G_v . Hence the induced topology is the product topology.

For $x, y \in G$, choose one G_S containing both. Multiplication is continuous on $G_S \times G_S$, and inversion is continuous on G_S . These are open neighbourhoods of the relevant points in the domain, proving continuity on G .

Each coordinate projection is continuous: at a point, enlarge S to include that coordinate. Distinct tuples therefore have disjoint neighbourhoods pulled back from a Hausdorff factor. This proves the Hausdorff property.

For $x \in G_S$, choose a compact neighbourhood C_v of x_v in each of the finitely many factors indexed by S . The set

$$\prod_{v \in S} C_v \times \prod_{v \notin S} K_v$$

is a compact neighbourhood of x , by the compact-product theorem. This proves local compactness. Finally, the open cover $(G_S)_S$ of a compact subset has a finite subcover. The union of the finitely many indexing sets gives a single chart containing that subset. $\square$

The inclusion $G \hookrightarrow \prod_v G_v$ is continuous. It generally fails to be a topological embedding: (1) constrains every coordinate outside S , whereas a product neighbourhood constrains only finitely many coordinates. Exercise 1 proves the exact criterion. Changing the chosen subgroups at finitely many indices changes neither the underlying restricted product nor its topology; include those indices in every chart.

Integrating local data

Choose left Haar measures μ_v on G_v . Enlarge E , if necessary, so that

$$\mu_v(K_v) = 1 \quad (v \notin E). \quad (2)$$

The enlargement retains any exceptional local normalization as an actual finite factor. An infinite product of arbitrary normalization constants is not part of the construction.

On the compact group $L_S = \prod_{v \notin S} K_v$, let m_S be Haar probability. Its projection to any finite product of tail factors has the product of their normalized Haar measures: the projected measure is invariant, has total mass one, and Haar uniqueness applies. This describes the infinite product as a Radon measure on its compact product topology.

Use the **Radon product** for finite products of locally compact spaces: it is the measure determined by iterated integrals of compactly supported continuous functions. It avoids imposing a countability assumption on the Borel sets of a product.

Proposition 1.2. The measures

$$\mu_S = \left(\widehat{\bigotimes_{v \in S} \mu_v} \right) \widehat{\otimes} m_S \quad \text{on } G_S \quad (3)$$

are compatible under restriction and determine a unique left Haar measure μ on G . If f_v are Borel integrable functions and $f_v = 1_{K_v}$ for all $v \notin S$, where $S \supseteq E$ is finite, then

$$f(x) = \bigotimes_v f_v(x) := \prod_v f_v(x_v)$$

is integrable and

$$\int_G f d\mu = \prod_v \int_{G_v} f_v d\mu_v. \quad (4)$$

The product on the right has only finitely many factors different from one. In particular, (4) applies to compactly supported continuous local functions.

Proof. Suppose $S \subseteq T$. Restricting μ_T to G_S replaces each μ_v , for $v \in T \setminus S$, by its restriction to K_v , of mass one. Their finite product with m_T is Haar probability on L_S , hence is m_S . This proves compatibility in (3).

For $h \in C_c(G)$, Proposition 1.1 places its support in some G_S . Define

$$I(h) = \int_{G_S} h d\mu_S.$$

Compatibility makes this independent of S , and a common larger chart proves linearity and positivity. Given $g \in G$, a chart containing both g and $\text{supp } h$ also contains $g \text{supp } h$. Its measure is a product of left Haar measures, so $I(h(g^{-1}\cdot)) = I(h)$.

The Riesz representation theorem gives a unique Radon measure representing I . It is nonzero and left invariant, so it is Haar. Functions in $C_c(G_S)$ extend by zero to G , since an open subgroup is also closed. Riesz uniqueness then gives $\mu|_{G_S} = \mu_S$, establishing the asserted restrictions and uniqueness.

For (4), f vanishes off G_S , and on that chart it is a finite product of local functions, independent of L_S . Each set $\{f_v \neq 0\}$ is σ -finite for μ_v : it is the union of the finite-measure sets $\{|f_v| > 1/j\}$. The rectangle and Tonelli theorems for Radon products therefore apply to the absolute values of these factors. They give

$$\int_G |f| d\mu = \prod_{v \in S} \int |f_v| d\mu_v < \infty.$$

Fubini now gives (4). This argument uses σ -finiteness of the nonzero sets of the particular functions, not of the ambient groups. $\square$

Thus the notation $\mu = \bigotimes_v \mu_v$ means the compatible Radon measure just constructed. Its definition does not require G to be second countable or σ -compact.

Characters see only finitely many integral tails

Now assume all G_v are abelian and write their law additively. Put $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. The dual $\widehat{H}$ of an LCA group consists of continuous homomorphisms $H \rightarrow \mathbb{T}$, with uniform convergence on compact sets. For a subgroup $K \leq H$, its annihilator is

$$K^\perp = \{\chi \in \widehat{H} : \chi(k) = 1 \text{ for every } k \in K\}.$$

The subgroup and quotient duality theorems imply that $K_v^\perp$ is compact and open when K_v is compact and open. These are the distinguished subgroups on the dual side.

We use a small fact about the circle. The arc $A = \{z \in \mathbb{T} : \text{Re } z > 0\}$ contains no nontrivial subgroup. Indeed, write a nonidentity element as $e^{2\pi it}$ with $0 < |t| \leq 1/2$. If $|t| \geq 1/4$, it is already outside A . Otherwise choose the least positive integer m with $m|t| \geq 1/4$. Then $1/4 \leq m|t| < 1/2$, and its m -th power is outside A .

Theorem 1.3. There is a canonical isomorphism of topological groups

$$\Phi : \prod'_v (\widehat{G}_v, K_v^\perp) \xrightarrow{\sim} \widehat{G}, \quad \Phi((\chi_v))(x) = \prod_v \chi_v(x_v). \quad (5)$$

The factors indexed by E remain unrestricted. Every continuous character of G has local restrictions trivial on K_v for almost all v .

Proof. For a tuple on the left and an element $x \in G$, both restrictions hold outside finite sets. Consequently all but finitely many factors in (5) equal one. On a chart containing the exceptional coordinates of the character tuple, (5) is a finite product of continuous local characters. It is therefore a continuous character on the open cover of G .

Conversely, let $\chi \in \widehat{G}$. Continuity provides a basic neighbourhood $B(S; U)$ of zero whose image lies in A . The compact subgroup

$$T_S = \{0\}^S \times \prod_{v \notin S} K_v$$

lies in this neighbourhood. Since $\chi(T_S)$ is a subgroup contained in A , it is trivial. Restricting χ to the embedded coordinate group gives $\chi_v \in \widehat{G}_v$, and $\chi_v \in K_v^\perp$ outside S . For any x , enlarge S to contain its nonintegral coordinates. Decompose x into a finite-coordinate part and an element of T_S . The latter contributes one, so $\chi(x) = \prod_v \chi_v(x_v)$. Uniqueness follows by evaluating on individual coordinates. This proves algebraic bijectivity.

We verify both topologies explicitly. Let $C \subseteq G$ be compact. Choose $S \supseteq E$ with $C \subseteq G_S$, and let C_v be its compact coordinate projections for $v \in S$. Restrict dual tuples to the open dual chart

$$H_S = \prod_{v \in S} \widehat{G}_v \times \prod_{v \notin S} K_v^\perp.$$

Their evaluations on C have only the factors indexed by S . Requiring each such factor to be uniformly within $\varepsilon / \max(1, |S|)$ of one on C_v gives an open neighbourhood in H_S . The inequality

$$\left| \prod_{j=1}^r z_j - 1 \right| \leq \sum_{j=1}^r |z_j - 1| \quad (z_j \in \mathbb{T})$$

shows that its image is uniformly within ε of one on C . Thus Φ is continuous.

For the inverse, take a basic identity neighbourhood with finite-coordinate conditions $\chi_v \in U_v$ for $v \in S$, and the condition $\chi_v \in K_v^\perp$ outside S . Restriction $\widehat{G} \rightarrow \widehat{G}_v$ is continuous, since a compact set in G_v embeds as a compact set in G . The remaining conditions say exactly that χ annihilates T_S . Its annihilator is open: the compact-open condition $\chi(T_S) \subseteq A$ is equivalent to triviality there. Intersecting this open annihilator with the finitely many inverse images of U_v gives the image of the chosen neighbourhood. This proves continuity of Φ^{-1} . $\square$

The theorem concerns a restricted product of **abelian** groups. Proposition 1.1 and the Haar construction require no commutativity.

Integral residues and finite adèles

The Chinese remainder theorem used below has a short algebraic proof. For pairwise coprime positive integers $m_1, \dots, m_r$, put $M = \prod_i m_i$. Bézout's identity gives b_i with $b_i(M/m_i) \equiv 1 \pmod{m_i}$. Given residues a_i , the integer $\sum_i a_i b_i (M/m_i)$ realizes all of them. An integer divisible by every m_i is divisible by M , proving uniqueness modulo M .

Define the **profinite integers** by

$$\widehat{\mathbb{Z}} = \varprojlim_{n \geq 1} \mathbb{Z}/n\mathbb{Z},$$

where indices are ordered by divisibility and the transition maps are reduction maps. A point is a compatible collection of residues. The topology is inherited from the product of the finite discrete rings. Compatibility is closed, so this is a compact Hausdorff ring.

For each prime p , we use $\mathbb{Z}_p = \varprojlim_k \mathbb{Z}/p^k\mathbb{Z}$ and its fraction field $\mathbb{Q}_p$. Its additive neighbourhoods are $p^k\mathbb{Z}_p$, for $k \in \mathbb{Z}$; $\mathbb{Z}_p$ is compact and open. Define

$$\mathbb{A}_f = \prod_p' (\mathbb{Q}_p, \mathbb{Z}_p).$$

Addition and multiplication are coordinatewise. This is a topological ring: on each open chart $\prod_{p \in S} \mathbb{Q}_p \times \prod_{p \notin S} \mathbb{Z}_p$ both operations are continuous and preserve the chart. Proposition 1.1 gives local compactness.

Proposition 1.4. There are canonical isomorphisms

$$\widehat{\mathbb{Z}} \simeq \prod_p \mathbb{Z}_p, \quad \widehat{\mathbb{Z}}^\times \simeq \prod_p \mathbb{Z}_p^\times, \quad (6)$$

of compact topological rings and of compact topological groups, respectively. Moreover,

$$\mathbb{A}_f \simeq \mathbb{Q} \otimes_{\mathbb{Z}} \widehat{\mathbb{Z}} = \bigcup_{n \geq 1} \frac{1}{n} \widehat{\mathbb{Z}} \quad (7)$$

as topological rings, when the tensor product is given the topology specified below. There are isomorphisms of discrete groups

$$\mathbb{A}_f / \widehat{\mathbb{Z}} \simeq \mathbb{Q} / \mathbb{Z} \simeq \bigoplus_p \mathbb{Q}_p / \mathbb{Z}_p, \quad \widehat{\widehat{\mathbb{Z}}} \simeq \mathbb{Q} / \mathbb{Z}. \quad (8)$$

In the last formula, the outer hat denotes the character group.

Proof. Restriction to the prime-power residues maps $\widehat{\mathbb{Z}}$ to $\prod_p \mathbb{Z}_p$. Conversely, given prime-power residues and $n = \prod_{p|n} p^{a_p}$, the Chinese remainder theorem supplies a unique residue modulo n . These residues are compatible as n varies. This proves (6) for rings. The map is continuous, and a continuous bijection from a compact space to a Hausdorff space is a homeomorphism. A tuple in a product ring is a unit precisely when every component is a unit; componentwise inversion is continuous on the compact product of the local unit groups. This proves the second assertion.

We will need the more precise residue calculation

$$\widehat{\mathbb{Z}} / n \widehat{\mathbb{Z}} \simeq \mathbb{Z} / n \mathbb{Z}. \quad (9)$$

In $\mathbb{Z}_p$, the kernel of reduction modulo p^a is $p^a \mathbb{Z}_p$: a compatible tuple in the kernel can be divided by p^a by using its residues modulo p^{a+k} . Multiplication by an integer prime to p is invertible, because its inverses modulo every p^k are compatible. The kernel of reduction modulo n in (6) is therefore exactly $n \widehat{\mathbb{Z}}$, and reduction is surjective by the Chinese remainder theorem. This proves (9). In particular, this kernel is compact and open and has index n .

Identify $\widehat{\mathbb{Z}}$ with the compact open subring $\prod_p \mathbb{Z}_p$ of $\mathbb{A}_f$. Given $x \in \mathbb{A}_f$, only finitely many coordinates have negative valuation. Choose

$$n = \prod_p p^{\max(0, -v_p(x_p))},$$

where zero coordinates contribute exponent zero. Then $nx \in \widehat{\mathbb{Z}}$. Thus the union in (7) is all of $\mathbb{A}_f$.

Every tensor can be put over a common denominator, hence written as $(1/n) \otimes z$. The map $q \otimes z \mapsto (qz_p)_p$ is surjective by the preceding paragraph. It is injective: if the image of $(1/n) \otimes z$ is zero, every component of z is zero. This identifies the tensor product algebraically with the localization of $\widehat{\mathbb{Z}}$ at the nonzero integers.

Give each stage $n^{-1} \widehat{\mathbb{Z}}$ the compact topology transported by $z \mapsto z/n$. Give their union the **stage topology**: a subset is open precisely when its intersection with each stage is open there. In $\mathbb{A}_f$ the stage is

$$\prod_{p|n} p^{-v_p(n)} \mathbb{Z}_p \times \prod_{p \nmid n} \mathbb{Z}_p,$$

so it is compact and open with exactly that transported topology. Since these open stages cover $\mathbb{A}_f$, their stage topology is its topology. This proves the topological assertion in (7); the algebraic tensor product alone does not specify a topology.

The diagonal map $\mathbb{Q} \rightarrow \mathbb{A}_f / \widehat{\mathbb{Z}}$ has kernel $\mathbb{Z}$: a rational number integral at every prime has no prime in its reduced denominator. To prove surjectivity, write $x = z/n$. By (9), choose $a \in \mathbb{Z}$ with $z - a \in n \widehat{\mathbb{Z}}$. Then $x - a/n \in \widehat{\mathbb{Z}}$. Hence the first isomorphism in (8) follows. The quotient is discrete because $\widehat{\mathbb{Z}}$ is open; give $\mathbb{Q} / \mathbb{Z}$ its discrete topology.

Coordinate residues give a surjection

$$\mathbb{A}_f \longrightarrow \bigoplus_p \mathbb{Q}_p / \mathbb{Z}_p$$

with kernel $\widehat{\mathbb{Z}}$: every finite collection of local cosets is represented by a tuple with zero in all other coordinates. This proves the middle identification, with the direct sum also discrete. In particular $\mathbb{Q}_p / \mathbb{Z}_p \simeq \mathbb{Z}[1/p] / \mathbb{Z}$. For

surjectivity of this local map, write $x = p^{-k}z$ and choose an integer with the same residue as z modulo p^k ; its kernel consists of the ordinary integers.

Finally, duality for surjective inverse systems of compact abelian groups gives

$$\widehat{\widehat{\mathbb{Z}}} \simeq \varinjlim \widehat{\mathbb{Z}/n\mathbb{Z}}.$$

This is the compact-limit statement in *Totally disconnected groups, the p -adic numbers and the adèles*, Proposition 4(b), also [Halter-Koch, Theorem 1.7.6.A]. A character of $\mathbb{Z}/n\mathbb{Z}$ is determined by an n -th root of unity, so its dual identifies with $n^{-1}\mathbb{Z}/\mathbb{Z}$. For $n \mid m$, pullback sends a/n to $(ma/n)/m$, the same rational class. The limit is $\mathbb{Q}/\mathbb{Z}$. Explicitly the pairing is

$$\langle z, a/n + \mathbb{Z} \rangle = \exp\left(2\pi i \frac{az_n}{n}\right), \quad (10)$$

where z_n is any integer representative of the residue of z modulo n . Compatibility makes it independent of all representatives. The dual of a compact group is discrete, so this is a topological isomorphism. $\square$

Worked example: one rational representative

Take $x_2 = 1/8$, $x_3 = 2/9$, and $x_p = 0$ at all other primes. Then $72x \in \widehat{\mathbb{Z}}$. Its residues modulo 8 and 9 are respectively 1 and 7. The integer 25 has both residues, so

$$x + \widehat{\mathbb{Z}} = 25/72 + \widehat{\mathbb{Z}}.$$

Indeed, the difference at 2 is $-2/9$, and the difference at 3 is $-1/8$, both integral in their respective fields. At every other prime the denominator 72 is a unit. This illustrates how independent local residues combine into one rational class.

The stages also have a useful countable form:

$$\mathbb{A}_f = \bigcup_{j \geq 1} \frac{1}{j!} \widehat{\mathbb{Z}}. \quad (11)$$

They increase because $j! \mid (j+1)!$, and every denominator divides some factorial. They are compact open **additive subgroups**. For $n > 1$, $n^{-1}\widehat{\mathbb{Z}}$ is not a subring: if $p \mid n$, the square of the diagonal element $1/n$ has valuation $-2v_p(n) < -v_p(n)$ at p , so it leaves that stage.

Finite idèles and the normalization table

An element $x \in \mathbb{A}_f$ is a ring unit exactly when every $x_p \neq 0$ and $x_p \in \mathbb{Z}_p^\times$ for almost all p . Necessity follows because both x and its inverse must be integral at almost all primes; sufficiency follows by taking the componentwise inverse. Thus, algebraically,

$$\mathbb{A}_f^\times = \prod_p' (\mathbb{Q}_p^\times, \mathbb{Z}_p^\times).$$

We give this **finite idèle group** the restricted product topology. Its compact open subgroup is $\widehat{\mathbb{Z}}^\times$. This differs from its additive subspace topology; Exercise 2 identifies the correct topology through the graph of inversion.

For example, the tuple $(p)_p$ is an adèle with no zero coordinate, yet is not a unit: its coordinatewise inverse fails integrality at every prime. In contrast, a nonzero rational number gives a finite idèle, because its numerator and denominator involve only finitely many primes. The valuation map identifies

$$\mathbb{A}_f^\times / \widehat{\mathbb{Z}}^\times \simeq \bigoplus_p \mathbb{Z}.$$

It is surjective by taking $x_p = p^{k_p}$ for a finitely supported integer tuple (k_p) , and its kernel is precisely the unit subgroup. The quotient is discrete because that subgroup is open.

All measures in this lesson use the following normalizations.

Group	Distinguished compact open subgroup	Its measure	Useful consequence
$(\mathbb{Q}_p, +)$	$\mathbb{Z}_p$	1	$\mu_p(p^k \mathbb{Z}_p) = p^{-k}$
$\mathbb{Q}_p^\times$	$\mathbb{Z}_p^\times$	1	$\mu_p^\times(p^k \mathbb{Z}_p^\times) = 1$
$(\mathbb{A}_f, +)$	$\widehat{\mathbb{Z}}$	1	$\mu(n^{-1} \widehat{\mathbb{Z}}) = n$
$\mathbb{A}_f^\times$	$\widehat{\mathbb{Z}}^\times$	1	multiplicative translates have measure 1

Here $k \in \mathbb{Z}$ and $n \geq 1$. The additive local formula follows by partitioning $\mathbb{Z}_p$ into its p^k residue cosets for $k \geq 0$; negative k follow by partitioning the larger ball into cosets of $\mathbb{Z}_p$. The multiplicative formula is Haar invariance. Equation (9) shows that $n^{-1} \widehat{\mathbb{Z}}$ has n additive cosets of $\widehat{\mathbb{Z}}$, proving its measure formula. Proposition 1.2 gives both global normalizations.

Worked example: a restricted box

The compact open set

$$B = 2^{-2} \mathbb{Z}_2 \times (1 + 9 \mathbb{Z}_3) \times \prod_{p \neq 2, 3} \mathbb{Z}_p$$

has additive measure $4/9$. The first factor contributes 4, the second contributes $1/9$, and every other factor contributes one. Translating the centre of a local ball leaves its additive measure unchanged. This example also explains why the tail factors in (2) must have measure one.

Test functions made from local pieces

On $\mathbb{Q}_p$, define a Bruhat–Schwartz function to be a locally constant, compactly supported complex function. Write this space as $\mathcal{S}(\mathbb{Q}_p)$. On $\mathbb{A}_f$ use the same definition. Then

$$\mathcal{S}(\mathbb{A}_f) = \text{span} \left\{ \bigotimes_p f_p : f_p \in \mathcal{S}(\mathbb{Q}_p), \quad f_p = 1_{\mathbb{Z}_p} \text{ for almost all } p \right\}. \quad (12)$$

Proof of (12). A tensor on the right is locally constant. Its support is a finite product of compact local supports with the compact integral tail, so it is compact. Finite sums retain both properties.

Conversely, let f be locally constant with compact support. At each point of its support, choose a compact open restricted rectangle on which f is constant. Such rectangles exist because the local balls $a + p^k \mathbb{Z}_p$ form a neighbourhood basis. The rectangles can lie inside $\{f \neq 0\}$: this set equals the support, since local constancy makes its complement open. Compactness gives a finite cover by these rectangles.

Include all their exceptional coordinates in one finite set S . They now have the same tail $\prod_{p \notin S} \mathbb{Z}_p$, and each local factor in S is compact and open. At each of these finitely many primes, partition the union of the local factors into the finitely many disjoint sets obtained by their intersections and differences. These sets are compact and open. The resulting finite rectangular partition makes f constant on each cell, and zero on cells outside its support. The indicator of every cell is a tensor of local compact open indicators with integral tail. Expressing f as its constant value times these indicators proves (12). $\square$

The same proof works for restricted products of groups whose local topologies have bases of compact open sets. More generally, given local test spaces $\mathcal{S}(G_v)$ containing 1_{K_v} outside E , define their restricted tensor space as the span of $\bigotimes_v f_v$ with $f_v \in \mathcal{S}(G_v)$ and $f_v = 1_{K_v}$ for almost all v . This definition specifies the local choices; it does not identify the resulting space with all of $C_c(G)$.

For arbitrary LCA factors, the intrinsic construction of local Bruhat–Schwartz spaces uses their structure theory and is beyond this lesson. Formula (12) proves the full identification needed for finite adèles. With a

Euclidean factor, the adelic test space is defined using finite sums of products of an ordinary Schwartz function on that factor and functions in (12).

For example,

$$f = 1_B - \frac{4}{9} 1_{\widehat{\mathbb{Z}}}$$

belongs to $\mathcal{S}(\mathbb{A}_f)$ and has integral zero. Both terms are factorizable, and their integrals are the measures already computed. Such finite local descriptions are what make adelic integration practical.

Exercises

Exercise 1 (easy). Prove that the restricted product topology equals the subspace topology from $\prod_v G_v$ if and only if $K_v = G_v$ for almost all $v \notin E$.

Solution. If equality holds outside a finite set, include that set in S . Then G is the full product, and (1) is its product topology. Conversely, suppose infinitely many K_v are proper. The subgroup G_E is open in the restricted topology. Any product neighbourhood of the identity constrains only a finite set F of coordinates. Choose $v \notin E \cup F$ with $K_v \neq G_v$, and an element $g_v \notin K_v$. The tuple with that single nonidentity coordinate belongs to G and to the chosen product neighbourhood but not to G_E . Thus G_E is not open in the subspace topology, and the two topologies differ.

Exercise 2 (medium). Show that inversion on $\mathbb{A}_f^\times$, with the subspace topology from $\mathbb{A}_f$, is discontinuous. Show also that

$$j : x \mapsto (x, x^{-1})$$

identifies the restricted product topology with the topology induced from $\mathbb{A}_f \times \mathbb{A}_f$.

Solution. Enumerate the primes increasingly as q_r . Let $u^{(r)}$ have coordinate q_r at q_r , and coordinate one elsewhere. Each tuple is an idèle. In an additive neighbourhood of one, only finitely many coordinates have extra local conditions; outside them, membership in $\mathbb{Z}_p$ suffices. Consequently $u^{(r)} \rightarrow 1$ in $\mathbb{A}_f$. But $(u^{(r)})^{-1} \notin \widehat{\mathbb{Z}}$ for every r . Since $\widehat{\mathbb{Z}}$ is an additive neighbourhood of one, these inverses do not converge to one. This proves discontinuity.

Take a basic product neighbourhood of (x, x^{-1}) in $\mathbb{A}_f \times \mathbb{A}_f$, and enlarge its two finite indexing sets to one S containing the exceptional coordinates of x and x^{-1} . Its inverse image under j has finite-coordinate conditions

$$y_p \in U_p \cap V_p^{-1} \quad (p \in S),$$

which are open in $\mathbb{Q}_p^\times$. Outside S , both y_p and y_p^{-1} must be integral, equivalently $y_p \in \mathbb{Z}_p^\times$. The inverse image is therefore open in the restricted topology.

Conversely, let a restricted basic neighbourhood of x prescribe $y_p \in W_p \subseteq \mathbb{Q}_p^\times$ for $p \in S$, and local units elsewhere. Each W_p is open in $\mathbb{Q}_p$, because $\mathbb{Q}_p^\times$ is open there. Use the additive basic sets with factors W_p and $\mathbb{Q}_p$, respectively, at $p \in S$, and integral tails for both. Their product pulls back under j to precisely that restricted neighbourhood. Hence both induced topologies agree, and injectivity of j proves the claimed identification.

Exercise 3 (medium). Compute the multiplicative measure and the additive measure of $\widehat{\mathbb{Z}}^\times$, with the normalizations in the table.

Solution. Its multiplicative measure is one, by Proposition 1.2. Additively, $\mathbb{Z}_p^\times = \mathbb{Z}_p \setminus p\mathbb{Z}_p$ has measure $1 - 1/p$. Let

$$C_N = \prod_{p \leq N} \mathbb{Z}_p^\times \times \prod_{p > N} \mathbb{Z}_p.$$

These compact sets decrease to $\widehat{\mathbb{Z}}^\times$, within the set $\widehat{\mathbb{Z}}$ of finite additive measure one. Continuity from above and the product formula give

$$\mu(\widehat{\mathbb{Z}}^\times) = \lim_{N \rightarrow \infty} \prod_{p \leq N} (1 - 1/p) = 0.$$

For completeness, the limit is zero without any asymptotic theorem about primes. Multiplying the finitely many geometric series gives

$$\prod_{p \leq N} (1 - 1/p)^{-1} = \sum_{\substack{m \geq 1 \\ \text{every prime divisor of } m \leq N}} \frac{1}{m} \geq \sum_{m=1}^{\lfloor N \rfloor} \frac{1}{m}.$$

The harmonic sums diverge: each block $2^{k-1} < m \leq 2^k$ contributes at least $1/2$. This proves the zero limit. The different answers arise from two different Haar measures on two different groups.

Exercise 4 (hard). Reconstruct the proof of Theorem 1.3 using an open annihilator to establish the topology. Treat the compact integral tail as a whole, rather than assuming a continuous character is a finite product in advance.

Solution. For a tuple (χ_v) in the restricted dual product, the formula $\chi(x) = \prod_v \chi_v(x_v)$ has only finitely many nontrivial factors. It is multiplicative and continuous on every chart containing the tuple's exceptional coordinates, hence on G .

For a continuous χ on G , choose $B(S; U)$ mapped into $\{\operatorname{Re} z > 0\}$. Its tail subgroup T_S has trivial image by the circle argument. The coordinate restrictions consequently annihilate K_v outside S . Splitting any element into its finitely many exceptional coordinates and a tail element recovers the product formula and proves bijectivity.

For continuity of the product map, put a given compact set C inside G_S . Within the dual chart H_S , the tail contributes one on C . Uniformly small evaluations on the finitely many compact projections C_v make the product uniformly small, using the product inequality in the theorem's proof.

For continuity in the other direction, the condition of belonging to H_S is exactly membership in $T_S^\perp$. The latter is open in $\widehat{G}$, because its defining triviality condition can be tested by uniform containment of the image of the compact subgroup T_S in the same circle arc. On that open subgroup, any finite-coordinate neighbourhood is obtained by the continuous restriction maps $\widehat{G} \rightarrow \widehat{G}_v$. Thus every restricted basic neighbourhood has open image. The two continuity statements prove the topological isomorphism, including for uncountable index sets and with unrestricted factors in E .

What this lesson does not prove

The following are the background results used here.

- The compact-product theorem and the fact that a continuous bijection from a compact space to a Hausdorff space is a homeomorphism, from point-set topology. The compact-product theorem is used in [Deitmar, §5.1]. Bézout's identity for coprime integers is assumed; the Chinese remainder argument was given above.
- Riesz representation for positive functionals on $C_c(X)$; existence and uniqueness of Haar measure on every locally compact Hausdorff group; finite Radon products and their integrable-function Fubini theorem. These are [Haar measure on locally compact groups, Theorem 2.2, Definition 4.3, Theorem 5.1, Theorems 8.3 and 9.2, and Proposition 12.1]. No global σ -finiteness is assumed.
- The dual of an LCA group is LCA; compact groups have discrete duals and discrete groups have compact duals. See [Halter-Koch, Definition and Theorem 1.6.3]. Pontryagin duality identifies H with $\widehat{\widehat{H}}$ by evaluation, as in *The Pontryagin duality theorem* and [Halter-Koch, Theorem 1.6.6].
- For closed $K \leq H$, pullback identifies $\widehat{H}/\widehat{K}$ with $K^\perp$, and restriction identifies $\widehat{H}/K^\perp$ with $\widehat{K}$, topologically. See *Subgroups, quotients and annihilators* and [Halter-Koch, Theorems 1.6.5 and 1.6.7]. Together with compact-discrete duality, these imply that the annihilator of a compact open subgroup is compact and open.
- For a surjective inverse system of compact abelian groups, every character of the limit factors through a coordinate group, and its discrete dual is the direct limit of the coordinate duals under pullback. See *Totally disconnected groups, the p -adic numbers and the adèles*, Proposition 4(b), and [Halter-Koch, Theorem 1.7.6.A]. We use this statement for finite cyclic groups, with explicit pullback maps.

- The construction of $\mathbb{Q}_p$, its valuation, and the compact open valuation ring $\mathbb{Z}_p$, whose topology has the balls used above. See [Halter-Koch, §5.1, items 7 and the valuation-ring discussion following it]. The elementary residue and unit calculations needed for (6)–(9) were proved here.

The self-duality of $\mathbb{Q}_p$ and of the adèles, the diagonal lattice in the full adèle ring, and Poisson summation are treated in *The adèle ring of a number field* and *Additive characters, self-dual measures and Poisson summation on the adèles*.

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The adèle ring of a number field

Written by GPT-6.1 Sol (OpenAI) in Codex, Ultra setting, October 2026. Self-checked by its author. Public domain (CC0).

A number field can be studied in every completion at once. The adèle ring makes this possible while retaining local compactness. Its topology expresses an arithmetic distinction: a number field is discrete when all places are present, but dense after any one place is removed. We will prove both assertions, construct fundamental domains, calculate their volume, and obtain simultaneous bounds on a nonzero field element.

We assume the results of [Restricted products and profinite completions](#), *Places of number fields in extensions and the product formula*, *Local fields: classification and Haar measure*, and *Lattices, Minkowski's theorem and the Minkowski embedding*. In particular, we use integral bases, the Chinese remainder theorem, the local decomposition of a finite field extension, and the Minkowski embedding. The exact facts used without proof are collected near the end. Basic references are [Getz], [Deitmar 2013], [Getz–Hahn draft], [Neukirch 1999], [Harari 2020], and [Halter–Koch 2022].

Local information with an integral tail

Let K be a number field of degree n , with r_1 real places and r_2 complex places, so $n = r_1 + 2r_2$. At each place v , let K_v be its completion. If v is finite, write $\mathcal{O}_v$ for its valuation ring, $\mathfrak{p}_v$ for the corresponding prime ideal of $\mathcal{O}_K$, and $q_v = \#(\mathcal{O}_K/\mathfrak{p}_v)$. Normalize ord_v to have value group $\mathbb{Z}$.

Our multiplicative local sizes are

$$|x|_v = \begin{cases} q_v^{-\text{ord}_v(x)}, & v \text{ finite,} \\ |x|, & K_v = \mathbf{R}, \\ |x|^2, & K_v = \mathbf{C}. \end{cases}$$

The last expression is the square of the ordinary complex modulus. It is useful for the product formula and measure scaling, but does not satisfy the ordinary triangle inequality. Whenever we use a geometric radius in $\mathbf{C}$, we mean the ordinary modulus. With these conventions, $\prod_v |x|_v = 1$ for $x \in K^\times$.

An **adèle** is a tuple $a = (a_v)_v$, with $a_v \in K_v$, such that $a_v \in \mathcal{O}_v$ at all but finitely many finite places. Thus

$$\mathbb{A}_K = \prod_v' (K_v, \mathcal{O}_v) = K_\infty \times \mathbb{A}_{K,f}, \quad K_\infty = \prod_{v|\infty} K_v \simeq \mathbf{R}^{r_1} \times \mathbf{C}^{r_2}.$$

There is no integral restriction at an infinite place. Addition and multiplication are coordinatewise. For a finite set S containing the infinite places, put

$$\mathbb{A}_{K,S} = \prod_{v \in S} K_v \times \prod_{v \notin S} \mathcal{O}_v.$$

These are open subrings, each with its product topology, and their union is $\mathbb{A}_K$. A basic open set is

$$\prod_{v \in S} U_v \times \prod_{v \notin S} \mathcal{O}_v, \quad (1)$$

where every $U_v \subset K_v$ is open. The requirement on the entire tail matters: this topology is finer than the topology inherited from the unrestricted product. An adèle may fail to be integral at finitely many places, but a neighborhood controls where new failures can occur.

Every element of K defines a diagonal adèle. Indeed, a nonzero field element has nonzero valuations at only finitely many finite places. We identify K with this diagonal subfield. For later use set

$$\widehat{\mathcal{O}}_K = \prod_{v < \infty} \mathcal{O}_v.$$

This is a compact open subring of $\mathbb{A}_{K,f}$. It is not open in the full adèle ring when embedded with zero infinite component.

Changing the number field

Proposition 2.1. The ring $\mathbb{A}_K$ is a Hausdorff locally compact topological ring. If L/K is finite, the canonical map

$$\Phi : L \otimes_K \mathbb{A}_K \longrightarrow \mathbb{A}_L, \quad \ell \otimes (a_v)_v \longmapsto (\ell a_v)_{w|v} \quad (2)$$

is an isomorphism of topological rings. On the tensor product, use the finite product topology from any K -basis of L . In particular,

$$\mathbb{A}_K \simeq K \otimes_{\mathbf{Q}} \mathbb{A}_{\mathbf{Q}}, \quad \mathbb{A}_K \simeq \mathbb{A}_{\mathbf{Q}}^n \quad (3)$$

as topological rings in the first assertion and as topological $\mathbb{A}_{\mathbf{Q}}$ -modules in the second.

Proof. Each stage $\mathbb{A}_{K,S}$ is a product of finitely many locally compact fields and a compact product of valuation rings. It is locally compact and Hausdorff. The stages are open and cover the ring, proving these assertions for the underlying space.

For continuity of addition or multiplication at (a, b) , enlarge S until both adèles belong to $\mathbb{A}_{K,S}$. Operations on that stage are continuous coordinatewise. Their values stay in the integral tail because each $\mathcal{O}_v$ is a subring. Continuity on these open neighborhoods proves continuity on the whole ring.

We use the local field-extension identity

$$L \otimes_K K_v \simeq \prod_{w|v} L_w. \quad (4)$$

Its map is the local version of (2), and is a homeomorphism of finite-dimensional K_v -vector spaces. To pass to restricted products we must check the integral lattices.

Here is the completion calculation. For a rational prime p , finite freeness over $\mathbf{Z}$, followed by the Chinese remainder theorem, gives

$$\begin{aligned}\mathcal{O}_K \otimes_{\mathbf{Z}} \mathbf{Z}_p &\simeq \varprojlim_m \mathcal{O}_K / p^m \mathcal{O}_K \\ &\simeq \prod_{v|p} \varprojlim_m \mathcal{O}_K / \mathfrak{p}_v^{me(v/p)} \simeq \prod_{v|p} \mathcal{O}_v.\end{aligned}\tag{5}$$

The powers $me(v/p)$ form a cofinal sequence of powers of $\mathfrak{p}_v$. Localizing before completing does not change these quotients: every denominator outside $\mathfrak{p}_v$ is invertible modulo $\mathfrak{p}_v^k$.

For the relative version, localize $\mathcal{O}_L$ at $\mathfrak{p}_v$ as a module over $(\mathcal{O}_K)_{\mathfrak{p}_v}$. It is finite and torsion-free over this discrete valuation ring, hence free. Tensoring this finite free module with the completed ring equals the inverse limit of its quotients by $\mathfrak{p}_v^m$. Decomposing $\mathfrak{p}_v \mathcal{O}_L$ into prime powers and applying the Chinese remainder theorem in this inverse limit gives

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_v \simeq \prod_{w|v} \mathcal{O}_w \tag{6}$$

at every finite place. Ramification causes no exception to this identity of completed integer rings.

Choose a K -basis $e_1, \dots, e_m$ of L consisting of algebraic integers. The lattice $M = \sum_j \mathcal{O}_K e_j$ lies in $\mathcal{O}_L$, and some nonzero $d \in \mathcal{O}_K$ satisfies $d\mathcal{O}_L \subset M$. Indeed, express a finite set of $\mathcal{O}_K$ -module generators of $\mathcal{O}_L$ in this basis and clear denominators. Thus $M \otimes \mathcal{O}_v = \mathcal{O}_L \otimes \mathcal{O}_v$ outside the finitely many primes dividing d .

In these coordinates the local map $K_v^m \rightarrow \prod_{w|v} L_w$ takes $\mathcal{O}_v^m$ onto $\prod_{w|v} \mathcal{O}_w$ at almost every finite place. The local maps and their inverses therefore take restricted tuples to restricted tuples. They are homeomorphisms on the open product stages after adjoining the exceptional places, and assemble to a global homeomorphism. Formula (2) respects multiplication, so it is a ring isomorphism.

Changes of K -basis act by invertible matrices over K on $\mathbb{A}_K^m$; the matrix and its inverse define continuous maps. The tensor-product topology is independent of the basis. Taking the base field to be $\mathbf{Q}$ proves (3). The module identification depends on a basis; the ring map does not. $\square$

Representatives for the additive quotient

Begin with $\mathbf{Q}$, and write $\hat{\mathbf{Z}} = \prod_p \mathbf{Z}_p$. A rational number belongs to every $\mathbf{Z}_p$ exactly when its reduced denominator has no prime divisor. Hence

$$\mathbf{Q} \cap (\mathbf{R} \times \hat{\mathbf{Z}}) = \mathbf{Z}. \tag{7}$$

Every finite rational adèle can be made integral by subtracting a rational number. Take its negative p -adic digits at each of its finitely many nonintegral places and add the resulting rational principal parts. A principal part at p has denominator a power of p , so is integral at every other prime. Thus

$$\mathbb{A}_{\mathbf{Q}} = \mathbf{Q} + (\mathbf{R} \times \hat{\mathbf{Z}}). \tag{8}$$

After this subtraction, subtract an integer to move the real coordinate into $[0, 1)$. The integer subtraction also changes every finite coordinate; it preserves their integrality.

Theorem 2.2. The diagonal subgroup K is discrete and closed in $\mathbb{A}_K$, and $\mathbb{A}_K/K$ is compact. Every coset in $\mathbb{A}_{\mathbf{Q}}/\mathbf{Q}$ has exactly one representative in

$$F_{\mathbf{Q}} = [0, 1) \times \hat{\mathbf{Z}}. \quad (9)$$

Proof. In the rational case, the open neighborhood

$$(-1/2, 1/2) \times \hat{\mathbf{Z}}$$

meets $\mathbf{Q}$ only at zero, by (7). This proves discreteness. A discrete subgroup of a Hausdorff topological group is closed: choose an open neighborhood V of zero whose difference set meets the subgroup only at zero. Any translate of V contains at most one subgroup element. A point in the closure must equal that element, since otherwise a smaller neighborhood excluding it contradicts the definition of closure.

Existence of representatives in (9) follows from (8). If two representatives differ by $q \in \mathbf{Q}$, their finite components imply $q \in \mathbf{Z}$. Their real components differ by a number strictly between -1 and 1 , so $q = 0$. This proves uniqueness.

The domain (9) is half-open and is not compact. Its compact closure $[0, 1] \times \hat{\mathbf{Z}}$ still maps onto the quotient. Thus the quotient is compact.

Choose a $\mathbf{Q}$ -basis of K . Under (3), the diagonal K becomes the subgroup $\mathbf{Q}^n \subset \mathbb{A}_{\mathbf{Q}}^n$. It is discrete and closed, and its quotient is the finite product of n compact rational quotients. This proves the assertions for K . $\square$

For volume calculations we want a domain adapted to the integer rings. Choose an integral basis $\omega_1, \dots, \omega_n$ of $\mathcal{O}_K$, let $j : K \rightarrow K_{\infty}$ be the Minkowski embedding, and put

$$P_K = \left\{ \sum_{i=1}^n t_i j(\omega_i) : 0 \leq t_i < 1 \right\}.$$

Equations (3) and (5), applied to this basis and the rational principal-part construction, give

$$\mathbb{A}_K = K + (K_{\infty} \times \hat{\mathcal{O}}_K), \quad K \cap (K_{\infty} \times \hat{\mathcal{O}}_K) = \mathcal{O}_K. \quad (10)$$

For the intersection, write an element of K in the integral basis. By (5), integrality at all finite places says its rational coordinates belong to every $\mathbf{Z}_p$, so they are integers. Reducing the infinite component modulo $j(\mathcal{O}_K)$ now shows that

$$F_K = P_K \times \hat{\mathcal{O}}_K \quad (11)$$

is a Borel fundamental domain with unique representatives. Its closure is compact. If $U \subset K_{\infty}$ is bounded, then

$$K \cap (U \times \hat{\mathcal{O}}_K)$$

is finite: it corresponds to the points of the full lattice $j(\mathcal{O}_K)$ in a bounded set.

Worked example: rational principal parts. Take an adèle with real component $13/10$, components $3/8$ at 2 and $7/25$ at 5 , and zero at all other primes. Subtract

$$q = 3/8 + 7/25 = 131/200.$$

At 2 the remaining value is $-7/25$, which is integral; at 5 it is $-3/8$, also integral. At every other prime the denominator of q is a unit. The remaining real component is $129/200 \in [0, 1)$, so this already gives the representative in (9).

Worked example: the solenoid quotient. Inclusion induces an isomorphism of topological groups

$$(\mathbf{R} \times \hat{\mathbf{Z}})/\mathbf{Z} \longrightarrow \mathbb{A}_{\mathbf{Q}}/\mathbf{Q}. \quad (12)$$

Here $\mathbf{Z}$ is embedded diagonally: $m \mapsto (m, (m)_p)$. Surjectivity and the kernel follow from (7)–(8). The source is compact, since $[0, 1] \times \hat{\mathbf{Z}}$ maps onto it. The target is Hausdorff by Theorem 2.2. Thus the continuous bijection is a homeomorphism. This quotient is called the rational solenoid; its further topological structure is studied in the lesson on solenoids. Formula (9) supplies measurable representatives, not a continuous product decomposition: crossing a real endpoint also changes the finite coordinate by a diagonal integer.

How the discriminant measures the quotient

Fix additive Haar measures dx_v by

$$dx_v = dx \quad (K_v = \mathbf{R}), \quad dx_v = 2 dx dy = |dz \wedge d\bar{z}| \quad (K_v = \mathbf{C}), \quad \mu_v(\mathcal{O}_v) = 1 \quad (v < \infty). \quad (13)$$

The restricted product measure $dx = \prod_v dx_v$ is characterized on product sets with integral tails by the product of local measures. Multiplication by $a \in K_v^\times$ scales dx_v by $|a|_v$. At a finite place this follows by decomposing $\mathcal{O}_v$ into q_v translates of its maximal ideal; at a complex place it follows from the real determinant $|a|^2$.

Let d_K be the field discriminant. We give the discrete additive subgroup K counting measure. The quotient measure is induced from dx with this choice; its total mass is the measure of a Borel fundamental domain such as (11).

Proposition 2.4. With (13),

$$\text{vol}(\mathbb{A}_K/K) = \sqrt{|d_K|}. \quad (14)$$

Proof. In ordinary real coordinates on K_∞ , form the matrix M whose columns are $j(\omega_i)$, recording real and imaginary parts at each complex place. The Euclidean volume of P_K is $|\det M|$.

Let $B = (\tau(\omega_i))_{\tau,i}$ be the matrix of all n complex embeddings, including both members of each conjugate pair. The trace matrix is $B^\top B$, so

$$d_K = \det(B^\top B) = (\det B)^2.$$

Replacing a real-coordinate pair (u, v) by $(u + iv, u - iv)$ has determinant $-2i$. Consequently

$$|\det B| = 2^{r_2} |\det M|, \quad |\det M| = 2^{-r_2} \sqrt{|d_K|}. \quad (15)$$

Our infinite measure multiplies Euclidean measure by 2^{r_2} . Thus $\mu_\infty(P_K) = \sqrt{|d_K|}$. The finite factor in (11) has measure 1, proving (14). $\square$

Using $dx dy$ at complex places would instead give $2^{-r_2} \sqrt{|d_K|}$. A change of integral basis changes M by a matrix in $\text{GL}_n(\mathbf{Z})$, whose determinant has absolute value 1, so does not change the answer.

A different finite normalization is used with Fourier transforms. For the standard trace character $\psi_v = \psi_{\mathbf{Q}_p} \circ \text{Tr}_{K_v/\mathbf{Q}_p}$, let $\mathfrak{D}_v^{-1}$ be the trace-dual lattice of $\mathcal{O}_v$. Its definition identifies it with the annihilator of $\mathcal{O}_v$. If a self-dual measure gives $\mathcal{O}_v$ mass c , Fourier inversion on its indicator gives

$$1 = c^2 N(\mathfrak{D}_v), \quad c = N(\mathfrak{D}_v)^{-1/2}.$$

Thus the self-dual finite measure is $N(\mathfrak{D}_v)^{-1/2}dx_v$. At unramified places the different is the unit ideal, so the measures agree [Neukirch 1999, Chapter III, §2, Theorem (2.6)]. The infinite factors in (13) are self-dual for the standard trace characters. Fourier inversion and its convention are used here from [Getz–Hahn draft, Appendix B, §B.2, equation (B.3)]; *Additive characters, self-dual measures and Poisson summation on the adèles* develops that theory.

A box large enough to contain a field element

An **idèle** is a tuple $\alpha = (\alpha_v)$ with $\alpha_v \in K_v^\times$ and $\alpha_v \in \mathcal{O}_v^\times$ at almost every finite place. These are precisely the units of $\mathbb{A}_K$: both a tuple and its coordinatewise inverse must have integral tails. Define

$$\|\alpha\|_{\mathbb{A}} = \prod_v |\alpha_v|_v.$$

Only finitely many factors differ from 1. We need the following bound before proving strong approximation.

Lemma 2.5 (adelic Minkowski lemma). One may take

$$C_K = \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|d_K|}. \quad (16)$$

For every idèle α with $\|\alpha\|_{\mathbb{A}} \geq C_K$, there exists $x \in K^\times$ such that

$$|x|_v \leq |\alpha_v|_v \quad \text{at every place } v. \quad (17)$$

Proof. First suppose $\|\alpha\|_{\mathbb{A}} > C_K$. Establish the counting principle behind Blichfeldt's argument. If a measurable set $E \subset \mathbb{A}_K$ satisfies $\mu(E) > \mu(F_K)$, two distinct points of E differ by a nonzero element of K . Indeed, the translates $F_K + k$, for $k \in K$, partition the ring. Translation invariance and Tonelli's theorem give

$$\int_{F_K} \sum_{k \in K} \mathbf{1}_E(f + k) df = \mu(E). \quad (18)$$

The group K is countable. If every coset met E at most once, the sum would be at most 1, contradicting the strict volume inequality. A coset meeting E twice gives the asserted difference.

Let R_v be the ordinary real or complex modulus of α_v at an infinite place. Choose

$$E = \prod_{v \text{ real}} [-R_v/2, R_v/2] \times \prod_{v \text{ complex}} \{z : |z| \leq R_v/2\} \times \prod_{v < \infty} \alpha_v \mathcal{O}_v. \quad (19)$$

This is a compact measurable subset of an open product stage. At real places its factor has measure $R_v = |\alpha_v|_v$. At complex places its measure is

$$2\pi(R_v/2)^2 = (\pi/2)|\alpha_v|_v.$$

At finite places its measure is $|\alpha_v|_v$. Hence

$$\mu(E) = (\pi/2)^{r_2} \|\alpha\|_{\mathbb{A}} > \sqrt{|d_K|} = \mu(F_K).$$

Choose distinct $y, z \in E$ with $x = y - z \in K^\times$. At a real place their difference has modulus at most R_v . At a complex place use the ordinary triangle inequality to obtain $|y_v - z_v| \leq R_v$, then square to obtain (17). At a finite place $\alpha_v \mathcal{O}_v$ is an additive subgroup, so the difference stays in it.

Now suppose $\|\alpha\|_{\mathbb{A}} = C_K$. Choose an infinite place w , and for $j \geq 1$ replace α_w by $(1 + 1/j)\alpha_w$, leaving every other component fixed. The resulting idèle has norm $(1 + 1/j)^{d_w} C_K > C_K$, where $d_w = 1$ at a real place and $d_w = 2$ at a complex place. The strict case gives a nonzero $x_j \in K$ with the corresponding bounds. All these elements lie in one fixed compact box: double only the ordinary radius at w , retain the other infinite radii, and retain every finite subgroup $\alpha_v \mathcal{O}_v$. By Theorem 2.2, this box meets K in a finite set. Some nonzero x therefore occurs for infinitely many j . Its bounds at $v \neq w$ already have the required size. Along that unbounded subsequence, its bound at w tends to $|\alpha_w|_w$. Thus (17) also holds at equality. $\square$

The half-radius choice is essential. Applying the counting principle directly to the full box would only bound the difference by twice its archimedean radii. Halving affects a complex factor by $1/4$, because it has two real dimensions. We need not halve the finite factors, which are additive subgroups.

Compactness and discreteness extend the strict volume argument to the closed boundary. The product formula explains why a size hypothesis is needed: (17) for nonzero x implies $1 \leq \|\alpha\|_{\mathbb{A}}$.

Removing a place frees approximation

For a fixed place v_0 , write

$$\mathbb{A}_K^{(v_0)} = \prod'_{v \neq v_0} (K_v, \mathcal{O}_v),$$

imposing integral restrictions only at finite places. Projection from the full ring forgets the v_0 -coordinate.

Theorem 2.3 (strong approximation). For every finite or infinite place v_0 , the diagonal image of K is dense in $\mathbb{A}_K^{(v_0)}$.

Equivalently, let S be a finite set disjoint from $\{v_0\}$, containing all infinite places other than v_0 . Given $a_v \in K_v$ and $\varepsilon_v > 0$ for $v \in S$, there exists $q \in K$ such that

$$|q - a_v|_v < \varepsilon_v \quad (v \in S), \quad q \in \mathcal{O}_v \quad (v < \infty, v \notin S \cup \{v_0\}). \quad (20)$$

Proof. Let $D = \overline{P_K} \times \widehat{\mathcal{O}_K}$. By (11), every adèle is $k + d$ for $k \in K$ and $d \in D$. For each infinite place set

$$B_v = \max \left(1, \sup_{d \in D} |d_v|_v \right),$$

and for finite places set $B_v = 1$. These are bounds on a fixed compact set, not norms on K_{∞} .

Choose an idèle β with

$$|\beta_v|_v B_v < \varepsilon_v \quad (v \in S), \quad |\beta_v|_v = 1 \quad (v < \infty, v \notin S \cup \{v_0\}).$$

At v_0 choose its component sufficiently large that $\|\beta\|_{\mathbb{A}} > C_K$. Every completion has elements of arbitrarily large size, so this is possible whether v_0 is finite or infinite. Lemma 2.5 supplies nonzero $c \in K$ with $|c|_v \leq |\beta_v|_v$ everywhere.

Let $t \in \mathbb{A}_K$ have components a_v for $v \in S$ and zero elsewhere, including at v_0 . Apply the representative property to $c^{-1}t$. There are $k \in K$ and $d \in D$ with

$$c^{-1}t = k + d.$$

Set $q = ck$. Then $t - q = cd$, so for $v \in S$,

$$|a_v - q|_v \leq |\beta_v|_v B_v < \varepsilon_v.$$

At a finite place outside $S \cup \{v_0\}$, both c and d_v are integral and $t_v = 0$, so $q = -cd_v$ is integral. This proves (20).

A neighborhood of a truncated adèle contains a set as in (20): include in S its finitely many nonintegral places, all retained infinite places, and the places with specified local neighborhoods. Choose smaller local balls inside those neighborhoods. Thus (20) proves density. $\square$

The omitted coordinate pays for the simultaneous small bounds on the other coordinates: it allows an arbitrarily large component before invoking Lemma 2.5. The proof uses neither a unit theorem nor multiplicative strong approximation.

Worked example: rational finite approximation. Suppose we want

$$q - 1/4 \in \mathbf{Z}_2, \quad q - 2/9 \in \mathbf{Z}_3,$$

and integrality at every other finite prime. Set $q = m/36$. The congruences become

$$m \equiv 1 \pmod{4}, \quad m \equiv 8 \pmod{9}.$$

Choose $m = 17$. Indeed, $17/36 - 1/4 = 2/9 \in \mathbf{Z}_2$, while $17/36 - 2/9 = 1/4 \in \mathbf{Z}_3$. Its denominator has no other prime factors. Higher precision gives higher prime-power congruences, solved by the same Chinese remainder theorem. In particular, $\mathbf{Q}$ is dense in $\mathbb{A}_{\mathbf{Q},f}$, the additive assertion used in [Bost–Connes 1995, §3, proof of Proposition 12].

This does not give density in the full ring. If a rational q is integral at every finite prime, it is an integer. No such q lies in $(1/3, 2/3)$. Thus $(1/3, 2/3) \times \hat{\mathbf{Z}}$ contains no rational number.

Exercises

Exercise 1 (easy). Prove (7) and (8) directly, without using Theorem 2.2. Explain why subtracting an arbitrary real number cannot replace rational subtraction in (8).

Exercise 2 (medium). Exhibit a Borel fundamental domain for $\mathbf{Q}(i)$ in $\mathbb{A}_{\mathbf{Q}(i)}$. Compute its volume with (13), including the discriminant.

Exercise 3 (medium). Prove strong approximation for $\mathbf{Q}$ directly. Treat both omission of infinity and omission of a finite prime. In the latter case include approximation at the real place.

Exercise 4 (hard). Prove Proposition 2.4 from an integral basis. Explain exactly where the factor 2^{r_2} enters, and why an arbitrary vector-space identification does not preserve the measure without a normalization check.

Solutions

Solution 1. Write $q = a/b$ with coprime integers a, b and $b > 0$. If $b > 1$, some prime $p \mid b$ has $\text{ord}_p(q) < 0$, so $q \notin \mathbf{Z}_p$. Conversely, every integer belongs to every $\mathbf{Z}_p$. This proves (7).

For $a = (a_\infty, (a_p))$, let T be its nonintegral finite places. Choose $D = \prod_{p \in T} p^{d_p}$ with $Da_p \in \mathbf{Z}_p$ for $p \in T$. The Chinese remainder theorem gives an integer m satisfying

$$m \equiv Da_p \pmod{p^{d_p}} \quad (p \in T).$$

Each right side denotes an integral residue class. Then $q = m/D$ satisfies $a_p - q \in \mathbf{Z}_p$ at primes in T . At other primes D is a unit and both a_p, q are integral. Thus $a - q \in \mathbf{R} \times \hat{\mathbf{Z}}$, proving (8). For empty T , take

$D = 1, m = 0$. A real number only supplies an infinite coordinate; it has no specified images in finite completions. Rational numbers supply the diagonal translation needed here.

Solution 2. First $\mathcal{O}_{\mathbf{Q}(i)} = \mathbf{Z}[i]$. If $a + bi$, with $a, b \in \mathbf{Q}$, is integral, traces of it and its product with i show $2a, 2b \in \mathbf{Z}$. Its norm $a^2 + b^2$ is integral. Writing $2a = m, 2b = n$, the condition $m^2 + n^2 \equiv 0 \pmod{4}$ forces both m, n even. Hence $a, b \in \mathbf{Z}$.

With basis $1, i$, formula (11) gives

$$\{x + iy : 0 \leq x < 1, 0 \leq y < 1\} \times \prod_{v < \infty} \mathcal{O}_v.$$

By (10), subtracting a field element makes the finite components integral. Subtracting a Gaussian integer then moves both real coordinates into the stated intervals. Two such representatives differ by a Gaussian integer whose real and imaginary parts lie strictly between -1 and 1 , so they agree. The complex square has ordinary area 1, hence measure 2; the finite product has measure 1. Its volume is 2. Finally,

$$d_{\mathbf{Q}(i)} = \det \begin{pmatrix} \text{Tr}(1) & \text{Tr}(i) \\ \text{Tr}(i) & \text{Tr}(i^2) \end{pmatrix} = \det \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = -4.$$

The volume is $\sqrt{|-4|}$, as required.

Solution 3. Finite local conditions may be written $q - a_p \in p^{r_p} \mathbf{Z}_p$, with $r_p \in \mathbf{Z}$. Let T be the finite set of constrained primes, including all nonintegral target components. Choose $d_p \geq 0$ so $Da_p \in \mathbf{Z}_p$ and $d_p + r_p \geq 0$, where $D = \prod_{p \in T} p^{d_p}$.

If the omitted place is infinity, choose

$$m \equiv Da_p \pmod{p^{d_p + r_p}} \quad (p \in T)$$

and put $q = m/D$. This solves all conditions and gives integrality at every other finite prime.

Now omit a finite prime ℓ , so $\ell \notin T$. For every $N \geq 0$, solve

$$m \equiv D\ell^N a_p \pmod{p^{d_p + r_p}} \quad (p \in T).$$

All solutions form $m_N + M\mathbf{Z}$, where $M = \prod_{p \in T} p^{d_p + r_p}$. The rationals

$$q = \frac{m_N + Mk}{D\ell^N} \quad (k \in \mathbf{Z})$$

satisfy the finite conditions. In $\mathbf{R}$ they form an arithmetic progression with mesh $M/(D\ell^N)$, tending to zero. Given a_∞ and $\varepsilon > 0$, choose N with mesh less than 2ε , then choose the nearest progression point. Its distance is at most half the mesh, hence less than ε . Its denominator is supported on $T \cup \{\ell\}$, so it is integral outside that set. This proves density for every omitted place. For empty T , use $D = M = 1$.

Solution 4. The Minkowski lattice theorem gives a parallelepiped for $j(\mathcal{O}_K)$ of Euclidean volume $2^{-r_2} \sqrt{|d_K|}$. To verify the convention, group the embeddings into real embeddings and complex conjugate pairs. Replacing real and imaginary coordinates by a conjugate pair multiplies the determinant's absolute value by 2 per pair. Thus the embedding determinant has absolute value 2^{r_2} times the real-coordinate determinant. Its square is the discriminant, giving the stated Euclidean volume.

By (10), the parallelepiped times $\widehat{\mathcal{O}}_K$ meets every additive coset exactly once. Counting measure on the diagonal subgroup identifies its measure with the total quotient mass. Each complex measure in (13) is twice ordinary area, so the infinite product measure multiplies Euclidean volume by 2^{r_2} . The finite factor has mass 1. The result is

$$2^{r_2} \cdot 2^{-r_2} \sqrt{|d_K|} \cdot 1 = \sqrt{|d_K|}.$$

An arbitrary $\mathbf{Q}$ -basis gives a topological module isomorphism, but its infinite determinant need not have absolute value 1, and its finite maps need not preserve the integer lattices. Haar measures transform by these local determinants. A topological isomorphism alone does not preserve their chosen numerical normalization.

What this lesson does not prove

We use the following prerequisite results with their stated conventions.

- A number field has an integral basis; the Minkowski image of $\mathcal{O}_K$ is a full lattice. For the canonical infinite measure, a nonzero integral ideal $\mathfrak{a}$ has lattice covolume $\sqrt{|d_K|}[\mathcal{O}_K : \mathfrak{a}]$. [Neukirch 1999, Chapter I, §5, Propositions (5.1)–(5.2)]. The determinant calculation needed for (14) is given above.
- Each finite completion is locally compact, its valuation ring is compact open, and that ring is the inverse limit of its quotients by powers of its maximal ideal. [Neukirch 1999, Chapter II, §5, Proposition (5.1) and its proof].
- For finite separable L/K , the canonical map $L \otimes_K K_v \rightarrow \prod_{w|v} L_w$ is an isomorphism. Finite extensions of number fields are separable. [Neukirch 1999, Chapter II, §8, Proposition (8.3)].
- With real modulus, squared complex modulus, and finite values $q_v^{-\text{ord}_v}$, the product formula holds for $K^\times$. [Neukirch 1999, Chapter III, §1, Proposition (1.3); Getz–Hahn draft, Proposition 2.1.1].
- The Chinese remainder theorem solves simultaneous conditions for distinct prime powers. Its rational approximation application is in [Deitmar 2013, §5.2, proof of Theorem 5.2.1(c)]. We also use the elementary ideal version of the Chinese remainder theorem in (5).
- Haar measures exist on locally compact additive groups. The restricted product measure has the product formula on integral-tail product sets. [Deitmar 2013, Theorem 5.2.2]. The same product construction applies to K .
- For a fixed nontrivial local character and perfect pairing, a unique self-dual Haar measure satisfies Fourier inversion. The standard characteristic-zero characters use local traces. [Getz–Hahn draft, Appendix B, §§B.1–B.2, equation (B.3)]. Only the normalization comparison after Proposition 2.4 uses this fact.
- An unramified finite local extension has unit different. [Neukirch 1999, Chapter III, §2, Theorem (2.6) and its reduction to completions].

The solenoid's classification and duality, multiplicative idèle class groups, and Poisson summation belong to subsequent lessons. The quotient proved compact here is the additive quotient $\mathbb{A}_K/K$.

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Idèles and the idèle class group

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An adèle can be integral at every finite place and still fail to have an adelic inverse. For example, the family whose component at each rational prime p is p , and whose real component is 1, lies in $\mathbb{A}_{\mathbb{Q}}$. Its coordinatewise inverse is not integral at any finite place, so it is not an adèle. The invertible adèles must have unit components almost everywhere. Their topology must also control the inverse.

The resulting group connects two kinds of information. Valuations record fractional ideals. Archimedean absolute values record the logarithms of units. After quotienting by multiplication by a field element, the part with total absolute value one is compact. We will prove this compactness from the adelic box lemma and then obtain both the finiteness of the ideal class group and Dirichlet's unit theorem. Neither of those two conclusions is assumed in the compactness proof.

Throughout, K is a number field, with r_1 real places and r_2 complex places. We use the normalized absolute values

$$|x|_v = \begin{cases} |x|, & v \text{ real,} \\ |x|^2, & v \text{ complex,} \\ q_v^{-\text{ord}_v(x)}, & v \text{ finite,} \end{cases} \quad q_v = \#(\mathcal{O}_v/\mathfrak{p}_v). \quad (1)$$

The product formula says $\prod_v |a|_v = 1$ for $a \in K^\times$. At complex places, geometric radii and triangle inequalities always use the ordinary modulus; the square in (1) is a multiplicative normalization.

We use three results from **The adèle ring of a number field**: K is discrete and closed in $\mathbb{A}_K$ (Theorem 2.2); finite-place integer rings are compact; and if an idèle β satisfies

$$\prod_v |\beta_v|_v > C_K, \quad C_K = (2/\pi)^{r_2} \sqrt{|d_K|}, \quad (2)$$

then there is a nonzero $a \in K$ with $|a|_v \leq |\beta_v|_v$ at every place (Lemma 2.5). That lemma was proved by additive quotient volume and a difference of two points in an adelic box. We also use the ordinary ideal theory of the Dedekind domain $\mathcal{O}_K$: fractional ideals have unique finite prime factorization, and the exponent in (a) is $\text{ord}_v(a)$. No class-number or unit-rank assertion enters these prerequisites.

A topology that controls multiplication and inversion

Define

$$J_K = \mathbb{A}_K^\times = \{(x_v) : x_v \in K_v^\times \text{ for all } v, x_v \in \mathcal{O}_v^\times \text{ for almost all finite } v\}. \quad (3)$$

The equality follows directly from the definition of the adèle ring. If both x and x^{-1} are adèles, then outside a finite set both components are integral, which forces $\text{ord}_v(x_v) = 0$. Conversely, the condition in (3) makes the coordinatewise inverse an adèle.

For a finite set S containing all archimedean places, put

$$J_{K,S} = \prod_{v \in S} K_v^\times \times \prod_{v \notin S} \mathcal{O}_v^\times. \quad (4)$$

Equip J_K with the restricted product topology: the subgroups (4) are open and have their product topologies. A basic neighbourhood of x , after S contains its nonunit components, has the form

$$\prod_{v \in S} W_v \times \prod_{v \notin S} \mathcal{O}_v^\times, \quad x_v \in W_v \subset K_v^\times \text{ open.} \quad (5)$$

Proposition 3.1. The restricted product topology makes J_K a locally compact Hausdorff abelian group. It is also the topology transported by

$$x \mapsto (x, x^{-1}) \quad (6)$$

from the subspace $\{(x, y) \in \mathbb{A}_K^2 : xy = 1\}$. The inclusion $J_K \rightarrow \mathbb{A}_K$ is continuous, but its image does not have the idèle topology as its additive subspace topology.

Proof. Each $\mathcal{O}_v^\times$ is compact and open in $K_v^\times$. Indeed, $\mathcal{O}_v^\times = \mathcal{O}_v \setminus \mathfrak{p}_v$, a closed subset of the compact ring $\mathcal{O}_v$, and it is open because valuations are locally constant away from zero. The restricted product theorem, Proposition 1.1 of [Restricted products and profinite completions](#), therefore gives all the asserted group and local compactness properties.

The coordinatewise inclusion into $\mathbb{A}_K$ is continuous on each open chart (4), hence is continuous globally. Inversion is continuous in the restricted product, so (6) is continuous. It is a bijection onto the displayed graph because every inverse in a ring is unique.

To check its inverse, take a basic neighbourhood (5). In the first adèle coordinate impose $x_v \in W_v$ for $v \in S$ and $x_v \in \mathcal{O}_v$ outside S . These are additive open conditions: $K_v^\times$ is open in K_v , so W_v is also open in K_v . In the second adèle coordinate impose $y_v \in \mathcal{O}_v$ outside S , with no restriction at S . On the graph $y = x^{-1}$, the two tail conditions are exactly $x_v \in \mathcal{O}_v^\times$. Their intersection pulls back to (5). Thus every restricted product neighbourhood is open in the graph topology. The graph is closed in $\mathbb{A}_K^2$, since multiplication is continuous and $\{1\}$ is closed.

For strictness of the topology comparison, choose pairwise distinct finite places v_n , and a uniformizer ϖ_n at v_n . Let $x^{(n)}$ have component ϖ_n at v_n and component 1 everywhere else. These are idèles. They tend to 1 in $\mathbb{A}_K$: any additive neighbourhood restricts only finitely many places and permits integral components at all remaining finite places. But none lies in the idèle neighbourhood

$$\prod_{v|\infty} K_v^\times \times \prod_{v \nmid \infty} \mathcal{O}_v^\times \quad (7)$$

of 1. Moreover, $(x^{(n)})^{-1}$ never belongs to the additive neighbourhood $K_\infty \times \prod_{v \nmid \infty} \mathcal{O}_v$, where $K_\infty = \prod_{v|\infty} K_v$. Inversion is therefore discontinuous at 1 for the additive subspace topology. $\square$

Equivalently, the idèle topology is the weakest topology containing the additive subspace topology for which inversion is continuous. Adding the inverse images of additive open sets gives precisely the graph topology; multiplication is then continuous by Proposition 3.1.

The norm-one constraint recovers the additive topology

Only finitely many factors differ from 1 in

$$|x| = \prod_v |x_v|_v, \quad J_K^1 = \{x \in J_K : |x| = 1\}. \quad (8)$$

This number is called the idèle norm, or content. It is not a vector-space norm.

Proposition 3.2. The map $|\cdot| : J_K \rightarrow \mathbb{R}_{>0}$ is a continuous surjective homomorphism. The diagonal subgroup $K^\times$ is discrete and closed in J_K and lies in J_K^1 . The subset J_K^1 is closed in $\mathbb{A}_K$, and its subspace topologies from J_K and $\mathbb{A}_K$ coincide.

Proof. On (4), the norm is the finite product $\prod_{v \in S} |x_v|_v$, hence is continuous. Each local absolute value is multiplicative. Choose an archimedean place w . Define a homomorphism $s : \mathbb{R}_{>0} \rightarrow J_K$ by setting all components except w equal to 1, and setting

$$s(t)_w = \begin{cases} t, & w \text{ real}, \\ \sqrt{t}, & w \text{ complex}. \end{cases} \quad (9)$$

It is continuous and $|s(t)| = t$, proving surjectivity. The product formula puts $K^\times$ in the kernel. Since K is discrete in $\mathbb{A}_K$, some additive open neighbourhood U of 1 meets K only at 1. Its inverse image in J_K isolates 1 in $K^\times$.

A discrete subgroup of a Hausdorff topological group is closed. Here is the argument we will use. Choose an identity neighbourhood V with VV^{-1} meeting the subgroup only at the identity. Each translate gV contains at most one subgroup element. If g lies in the closure of the subgroup, such a translate contains one element h . Were $g \neq h$, intersecting with a neighbourhood of g that avoids h would contradict closure. Thus every closure point belongs to the subgroup.

For the topology assertion, fix $x \in J_K^1$ and a basic idèle neighbourhood (5), with S containing all its nonunit components and all archimedean places. Shrink the finitely many W_v , if necessary, so that

$$\prod_{v \in S} |y_v|_v < 2 \quad \text{whenever } y_v \in W_v. \quad (10)$$

This is possible because the product at x equals 1. Permit $y_v \in \mathcal{O}_v$ outside S ; this gives an additive neighbourhood W of x . If $y \in W \cap J_K^1$ had a nonunit component at a finite place outside S , that component would have normalized absolute value at most $q_v^{-1} \leq 1/2$. All other tail factors are at most 1, so (10) would give $|y| < 1$, a contradiction. Consequently $W \cap J_K^1$ lies in (5). The opposite topology inclusion follows from the continuous inclusion $J_K \rightarrow \mathbb{A}_K$.

It remains to establish additive closedness. We give all the cases, since a closedness assertion in J_K alone would not justify intersecting J_K^1 with an additive compact box.

First take $z \in \mathbb{A}_K$ with a zero component. Choose a finite set S containing that place, every archimedean place and every finite place where z is nonintegral. The finite product $\prod_{v \in S} |z_v|_v$ is zero. By continuity it is less than 1 throughout some finite-coordinate neighbourhood. Together with integral tails, this gives an additive neighbourhood containing no norm-one idèle.

Next suppose every component is nonzero but z is not an idèle. Infinitely many integral finite components are nonunits. Starting with all archimedean and nonintegral places, add enough such nonunit places to make the finite product less than 1; each added factor is at most $1/2$. The same neighbourhood argument excludes J_K^1 .

Finally suppose z is an idèle with norm $c \neq 1$. If $c < 1$, take S containing all nonunit and archimedean places and again keep its finite product less than 1. If $c > 1$, enlarge this S to include all finite places with $q_v \leq 2c$. There are finitely many: such a place lies above a rational prime $p \leq 2c$, and a number field has finitely many places above each p . Choose finite-coordinate neighbourhoods so that

$$1 < \prod_{v \in S} |y_v|_v < 2c, \quad y_v \in \mathcal{O}_v \quad (v \notin S). \quad (11)$$

If a member of this additive neighbourhood were a norm-one idèle, either its tail would consist entirely of units, giving norm greater than 1, or some tail component would have absolute value at most $q_v^{-1} < (2c)^{-1}$, giving norm less than 1. Both are impossible. Every point outside J_K^1 therefore has an additive neighbourhood disjoint from it. $\square$

The mechanism in (10) is useful: a new tail valuation costs at least a factor of two, and a sufficiently small change at finitely many places cannot pay that cost while preserving norm one.

Compactness before class numbers and units

The idèle class group and its norm-one part are

$$C_K = J_K/K^\times, \quad C_K^1 = J_K^1/K^\times. \quad (12)$$

They have quotient topologies. Quotient maps of topological groups are open: the inverse image of the image of an open set is the union of its subgroup translates. Since $K^\times$ is closed, these quotients are Hausdorff and locally compact. The restriction of the quotient map to J_K^1 realizes C_K^1 as the kernel of the induced norm on C_K , with its subspace topology; this also follows explicitly from the splitting proved in Proposition 3.5 below.

Theorem 3.3 (norm-one compactness). The group C_K^1 is compact.

Proof. Choose an idèle η with $|\eta| > C_K$; the section (9) supplies one. The additive box

$$B_\eta = \{z \in \mathbb{A}_K : |z_v|_v \leq |\eta_v|_v \text{ at every } v\} \quad (13)$$

is compact. Each archimedean factor is a closed bounded interval or disk; each finite factor is the compact subgroup $\eta_v \mathcal{O}_v$; almost all of the latter equal $\mathcal{O}_v$. Thus the box is a product of compact sets in one additive restricted product chart.

By Proposition 3.2, $B_\eta \cap J_K^1$ is closed in this box and compact in the idèle topology. For any $x \in J_K^1$, apply the adelic box lemma (2) to ηx^{-1} . Its norm is $|\eta| > C_K$, so there is $a \in K^\times$ satisfying

$$|a|_v \leq |\eta_v x_v^{-1}|_v, \quad |ax_v|_v \leq |\eta_v|_v \quad \text{for every } v. \quad (14)$$

The product formula gives $|ax| = 1$. Thus $ax \in B_\eta \cap J_K^1$, and it represents the same class as x . The quotient is the continuous image of this compact set. $\square$

This proof uses no choice of ideal-class representatives and no fundamental domain for a unit lattice. Those two objects will follow from compactness.

Corollary 3.4. The ideal class group Cl_K is finite. The unit group has the form

$$\mathcal{O}_K^\times \simeq \mu_K \times \mathbb{Z}^{r_1+r_2-1}, \quad (15)$$

where μ_K is the finite cyclic group of roots of unity in K . More precisely, the logarithms of units form a full lattice in

$$H = \left\{ (t_v)_{v|\infty} \in \mathbb{R}^{r_1+r_2} : \sum_{v|\infty} t_v = 0 \right\}. \quad (16)$$

Proof. Let I_K be the group of nonzero fractional ideals, with the discrete topology. The map

$$\mathfrak{a} : J_K \longrightarrow I_K, \quad x \longmapsto \prod_{v|\infty} \mathfrak{p}_v^{\text{ord}_v(x_v)} \quad (17)$$

is a continuous surjective homomorphism. Its kernel is the open subgroup

$$J_{K,\infty} = K_\infty^\times \times U_f, \quad U_f = \prod_{v|\infty} \mathcal{O}_v^\times. \quad (18)$$

Surjectivity follows by placing uniformizer powers at the finitely many primes in an ideal factorization. Continuity follows because the kernel is open, or directly because valuation vectors are locally constant in idèle

charts. Multiplying a chosen lift by $s(|x|^{-1})$ changes no finite valuation and makes its norm one. Hence (17) is still surjective on J_K^1 . It takes a principal idèle a to (a) , and induces a continuous surjection

$$C_K^1 \longrightarrow \text{Cl}_K. \quad (19)$$

A compact discrete space is finite, proving class-number finiteness.

Set $J_{K,\infty}^1 = J_{K,\infty} \cap J_K^1$. The intersection $J_{K,\infty}^1 \cap K^\times$ is exactly $\mathcal{O}_K^\times$: zero finite valuations mean that both a and a^{-1} are algebraic integers. Moreover,

$$J_{K,\infty}^1 / \mathcal{O}_K^\times \simeq \ker(C_K^1 \rightarrow \text{Cl}_K) \quad (20)$$

as topological groups. To see surjectivity, if $\mathfrak{a}(x)$ is principal, divide x by a generator; its norm stays one and its finite valuations become zero. The kernel is the indicated intersection. For the topology, $J_{K,\infty}^1$ is open in J_K^1 , and the quotient map is open, so the induced bijection onto its image is open. The right side of (20) is closed in the compact group C_K^1 , hence the left side is compact.

On $J_{K,\infty}^1$, define

$$\ell(x) = (\log |x_v|_v)_{v|\infty} \in H. \quad (21)$$

This is a continuous surjection. For any $t \in H$, take component e^{t_v} at a real place and the positive real number $e^{t_v/2}$ at a complex place, and take all finite components equal to 1. These components give a lift with norm one. Its kernel is the compact group

$$\{\pm 1\}^{r_1} \times (S^1)^{r_2} \times U_f. \quad (22)$$

Let $\Gamma = \ell(\mathcal{O}_K^\times)$. We first prove that bounded sets meet Γ in finitely many points. For a fixed $M > 0$, all units whose logarithms satisfy $|\log |a|_v| \leq M$ lie in the compact idèle set consisting of finite units and archimedean annuli $e^{-M} \leq |x_v|_v \leq e^M$. Its intersection with the discrete closed subgroup $K^\times$ is finite. Consequently Γ is discrete and closed, and the kernel of ℓ on units is finite.

That finite kernel equals μ_K . Every element of a finite subgroup has finite order, while every root of unity is an algebraic unit with all normalized absolute values equal to one. It is cyclic: through any archimedean embedding it is a finite subgroup of $\mathbb{C}^\times$, contained in the cyclic group of N -th roots of unity for a common multiple N of its element orders.

The map (21) descends to a continuous surjection from the compact group in (20) onto H/Γ , so H/Γ is compact. We finish with the following elementary lattice argument rather than assuming the unit theorem.

If a subgroup $\Gamma \subset \mathbb{R}^d$ meets bounded sets in finitely many points, then it has a basis of at most d linearly independent vectors over $\mathbb{Z}$. For a nonzero group, choose $e \neq 0$ generating its intersection with the line $\mathbb{R}e$; one can choose a shortest nonzero vector on any line meeting the group, and subtract integer multiples to prove the cyclic assertion. Orthogonally project onto $e^\perp$. The projected group still meets bounded sets in finitely many points: lift points in a bounded set and subtract integer multiples of e so their e -coordinate lies in one fixed interval of length $\|e\|$. All resulting lifts lie in a bounded set, so there are finitely many. Induction on dimension gives a basis of the projected group; lifting that basis and adjoining e gives a basis of Γ . Independence follows by projecting a linear relation and then using the nonzero vector e .

If $\mathbb{R}^d/\Gamma$ is compact, these basis vectors must span $\mathbb{R}^d$. Otherwise a nonzero linear functional annihilating their span would descend to a continuous surjection $\mathbb{R}^d/\Gamma \rightarrow \mathbb{R}$, impossible for a compact source. Thus Γ has rank d . Applied to (16), this gives rank $r_1 + r_2 - 1$. Choose lifts $\varepsilon_1, \dots, \varepsilon_d$ of a lattice basis in the unit group. Every unit is uniquely a root of unity times $\varepsilon_1^{n_1} \dots \varepsilon_d^{n_d}$, proving (15). When $d = 0$, the lattice is zero and all units are roots of unity. $\square$

The same framework contains the usual theorem on S -units. Here are the details needed to keep track of the extra rank. Let $S = S_\infty \cup S_f$, with S_f a finite set of finite places, and let

$$E_{K,S} = \{a \in K^\times : \text{ord}_v(a) = 0 \text{ for finite } v \notin S\}.$$

The valuation map fits into

$$1 \longrightarrow \mathcal{O}_K^\times \longrightarrow E_{K,S} \longrightarrow \mathbb{Z}^{S_f}. \quad (23)$$

Its image has finite index. Indeed, if $h = \# \text{Cl}_K$, then $\mathfrak{p}_v^h = (a_v)$ for each $v \in S_f$, and $a_v \in E_{K,S}$ maps to h times the corresponding coordinate vector. Thus the image contains $h\mathbb{Z}^{S_f}$; the lattice argument above shows it is free of rank $\#S_f$. Lifting a basis splits (23) over its image and gives

$$E_{K,S} \simeq \mu_K \times \mathbb{Z}^{r_1+r_2-1+\#S_f}. \quad (24)$$

Its logarithms $(\log |a|_v)_{v \in S}$ form a full lattice in the hyperplane $\sum_{v \in S} t_v = 0$. For a direct check, the finite coordinates lie in the grid $\prod_{v \in S_f} \log q_v \mathbb{Z}$, and their image has finite index in that grid. After reducing these coordinates by the chosen lifted basis, the remaining logarithms lie in the archimedean unit lattice of (16). This proves discreteness and gives a bounded parallelepiped covering the hyperplane modulo the logarithms. The finite-coordinate grid spans $\mathbb{R}^{S_f}$, while the archimedean unit lattice spans its kernel, so the full span has dimension $\#S - 1$. The logarithmic kernel is again μ_K .

Separating the norm from the compact group

Proposition 3.5. A choice of section (9) gives topological group isomorphisms

$$J_K \simeq J_K^1 \times \mathbb{R}_{>0}, \quad C_K \simeq C_K^1 \times \mathbb{R}_{>0}. \quad (25)$$

Proof. The maps

$$(y, t) \mapsto y s(t), \quad x \mapsto (x s(|x|)^{-1}, |x|) \quad (26)$$

are continuous inverse homomorphisms. The second map sends $K^\times$ onto $K^\times \times \{1\}$. Taking quotients yields (25), since the product of a quotient map with the identity is open and surjective, hence a quotient map. Explicitly the class inverse is $[x] \mapsto ([x s(|x|)^{-1}], |x|)$, so its continuity also follows from the defining quotient topology. $\square$

The norm homomorphism is canonical; its splitting is a choice. For a general number field the connected component of C_K can include connected groups inside C_K^1 . Therefore (25) does not identify the connected component with its $\mathbb{R}_{>0}$ factor in general.

Rational representatives

For $K = \mathbb{Q}$, write

$$\widehat{\mathbb{Z}}^\times = \prod_p \mathbb{Z}_p^\times.$$

We embed this group in $J_{\mathbb{Q}}$ with real component 1, embed $\mathbb{R}_{>0}$ with every finite component 1, and embed $\mathbb{Q}^\times$ diagonally.

Proposition 3.6. These embeddings give internal direct products of topological groups

$$J_{\mathbb{Q}} = \mathbb{Q}^{\times} \times \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}, \quad \mathbb{A}_f^{\times} = \mathbb{Q}_{>0}^{\times} \times \widehat{\mathbb{Z}}^{\times}. \quad (27)$$

Here the rational factors carry the discrete topology, and the finite idèles carry their restricted product topology. Consequently

$$C_{\mathbb{Q}} \simeq \mathbb{R}_{>0} \times \widehat{\mathbb{Z}}^{\times}, \quad C_{\mathbb{Q}}^1 \simeq \widehat{\mathbb{Z}}^{\times}. \quad (28)$$

The identity component of $C_{\mathbb{Q}}$ is $\mathbb{R}_{>0} \times \{1\}$, and its quotient by that component is $\widehat{\mathbb{Z}}^{\times}$.

Proof. For $x \in J_{\mathbb{Q}}$, set

$$q_0 = \prod_p p^{\text{ord}_p(x_p)} > 0, \quad q = \text{sgn}(x_{\infty})q_0, \quad t = \frac{x_{\infty}}{q} > 0, \quad u_p = \frac{x_p}{q}. \quad (29)$$

The product for q_0 is finite, and $\text{ord}_p(q) = \text{ord}_p(x_p)$, so every u_p is a unit. Thus $x = qtu$. Finite valuations determine q_0 , and the real sign determines q ; the other two factors then follow. This proves existence and uniqueness.

The multiplication map from the product in (27) is continuous. Its inverse is continuous too: near a fixed idèle, the finitely many relevant finite valuations and the real sign can be kept constant in a basic neighbourhood (5), so q is locally constant. Then $t = x_{\infty}/q$ and $u = x/q$ with the real component removed vary continuously. Also

$$|x| = |x_{\infty}| \prod_p p^{-\text{ord}_p(x_p)} = |x_{\infty}|/q_0 = t. \quad (30)$$

The same argument without the real sign gives the finite decomposition, with positive rational factor q_0 . Its discreteness can also be seen from $\mathbb{Q}_{>0}^{\times} \cap \widehat{\mathbb{Z}}^{\times} = \{1\}$. Quotienting the first decomposition by its diagonal rational factor proves (28), and (30) identifies the norm-one factor.

Finally $\widehat{\mathbb{Z}}^{\times}$ is totally disconnected. Reduction modulo p^n is continuous into a finite discrete unit group, so a connected subset has constant image in every such quotient. These reductions separate points in $\mathbb{Z}_p^{\times}$, and the coordinate projections separate points in the product. Thus every connected subset is a singleton. In contrast $\mathbb{R}_{>0}$ is connected. Projecting a connected subset of (28) to the second factor proves the component assertion. $\square$

For an explicit representative, take an idèle with real component $-7/10$, component 8 at 2, component $1/25$ at 5, and component 1 at all other primes. Formula (29) gives

$$q_0 = 8/25, \quad q = -8/25, \quad t = 35/16,$$

and

$$u_2 = -25, \quad u_5 = -1/8, \quad u_p = -25/8 \ (p \neq 2, 5).$$

Every displayed u_p is a unit in its own local ring, and the idèle class is represented by $(35/16, u)$. Its norm is $35/16$, even though its real component has absolute value $7/10$.

For $K = \mathbb{Q}(i)$, whose integer ring $\mathbb{Z}[i]$ was identified in Exercise 2 of *The adèle ring of a number field*, the corresponding ideal-class quotient is trivial:

$$C_K / \text{image}(\mathbb{C}^{\times} \times U_f) \simeq \text{Cl}_K = 1. \quad (31)$$

Indeed, $\mathbb{Z}[i]$ is Euclidean for $N(a+bi) = a^2 + b^2$. Given a complex quotient, round each real coordinate to the nearest integer. The error has squared modulus at most $1/2 < 1$, so division by a nonzero Gaussian integer has a remainder of smaller norm. A nonzero ideal contains an element of least positive norm; division by that element shows it generates the ideal. All fractional ideals are principal. Formula (17) then proves (31). In this case every norm-one class has a representative with all finite components units; its complex component must have ordinary modulus one. Hence

$$C_{\mathbb{Q}(i)}^1 \simeq (S^1 \times U_f)/\mu_4, \quad \mu_4 = \{1, -1, i, -i\}, \quad (32)$$

where μ_4 is embedded diagonally in all components. The unit calculation follows either by solving $a^2 + b^2 = 1$, or from the zero-rank case of (15). Unlike the rational compact factor, (32) visibly has a connected circle coming from S^1 .

There is a different quotient, $\mathbb{A}_K/K^\times$, where multiplication acts on all adèles, including zero. It is not the idèle class group and is not Hausdorff. For example, the orbit of the adèle with component 1 at a chosen archimedean place and zero at every other place is not closed: multiplication by positive rational numbers tending to zero makes it converge to the zero adèle, which is not in that orbit. If the quotient were Hausdorff, the inverse image of every singleton would be closed. This observation explains why the topology assertions above concern invertible adèles.

Exercises

1. **Easy.** Starting with the finite valuations and real sign of a rational idèle, construct its factors q, t, u in (27). Show that t is its idèle norm and that all three factors are unique. Carry out the construction when $x_\infty = 9, x_3 = 1/9, x_5 = 5$, and every other finite component is 1.
2. **Medium.** For an arbitrary number field, construct idèles tending additively to 1 whose inverses do not tend additively to 1. Explain separately why the same idèles fail to converge in the idèle topology.
3. **Medium.** Prove directly from the product formula, without additive discreteness as a black box, that $K^\times$ is discrete and closed in J_K . Include complex places with the normalization (1).
4. **Hard.** Recover Dirichlet's unit theorem from compactness of C_K^1 . Identify the compact quotient used in the proof, show the logarithmic image is a lattice of full rank, and explain why the finite kernel is precisely μ_K . Then show how each finite place allowed in an S -unit group adds one to the rank.

Solutions

Solution 1. Finite valuations determine $q_0 = \prod p^{\text{ord}_p(x_p)}$. Choose q to have magnitude q_0 and the sign of x_∞ , put $t = x_\infty/q$, and divide each finite component by q . Its valuation becomes zero at every prime, giving u . The normalized finite product is q_0^{-1} , so $t = |x_\infty|/q_0 = |x|$. Any proposed decomposition has these finite valuations and real sign, so yields the same q , then the same t, u . In the example, $q = q_0 = 5/9$, $t = 81/5, u_3 = 1/5, u_5 = 9$, and $u_p = 9/5$ at every other prime. These are local units. Directly, the norm is $9 \cdot 9 \cdot (1/5) = 81/5$.

Solution 2. Enumerate pairwise distinct finite places v_n . They exist because there are infinitely many rational primes and each has at least one place above it. Take a uniformizer at v_n , set that component of $x^{(n)}$ equal to it, and all others equal to 1. An additive basic neighbourhood of 1 restricts finitely many coordinates and has integral tails, so eventually contains $x^{(n)}$. Each inverse has a nonintegral component at v_n , and therefore none belongs to the additive open set $K_\infty \times \prod_{v \nmid \infty} \mathcal{O}_v$ containing 1. Also none of the original idèles lies in (7), which is an idèle neighbourhood. This proves both claims. The example does not contradict Proposition 3.2: these idèles have norm $q_{v_n}^{-1}$, not norm one.

Solution 3. At every real place require $|x_v - 1| < 1$, at every complex place require ordinary $|x_v - 1| < 1$, and at every finite place require $x_v \in \mathcal{O}_v^\times$. This defines an idèle neighbourhood U of 1. Suppose a principal idèle $a \in U$ has $a \neq 1$. At each finite place, the ultrametric inequality gives $|a - 1|_v \leq 1$. At each archimedean place, $|a - 1|_v < 1$; squaring the ordinary complex inequality preserves this strict inequality. There is at

least one archimedean place, so $\prod_v |a - 1|_v < 1$, contradicting the product formula for $a - 1 \neq 0$. Thus $U \cap K^\times = \{1\}$, proving discreteness. Choose V with $VV^{-1} \subset U$. Each translate gV contains at most one principal idèle; a point in the closure must equal that element, because otherwise a smaller neighbourhood could avoid it. Hence the subgroup is closed.

Solution 4. Use $J_{K,\infty}^1$, whose diagonal intersection is $\mathcal{O}_K^\times$. The ideal map identifies its quotient by units with the kernel of $C_K^1 \rightarrow \text{Cl}_K$; the identification is open, and the kernel is closed, so the quotient is compact. The logarithm map onto the hyperplane (16) has compact kernel (22). On units, bounded logarithms put the units in a compact idèle set. Its intersection with discrete closed $K^\times$ is finite. Thus the logarithmic subgroup Γ is discrete and closed, with finite kernel. The kernel is a finite group of units, hence consists of roots of unity; every root of unity lies in it. A complex embedding proves its cyclicity.

The compact unit quotient maps onto H/Γ . Apply the lattice induction in Corollary 3.4: project away from a one-dimensional cyclic subgroup, reduce lifts into a bounded strip to preserve discreteness, and lift the resulting basis. This gives at most $\dim H$ independent generators. A proper real span would allow a nonzero real linear functional on H/Γ with unbounded image, contradicting compactness. Thus the rank is exactly $\dim H = r_1 + r_2 - 1$. Lifting this basis splits units as $\mu_K \times \mathbb{Z}^{r_1+r_2-1}$.

For S -units, use the finite valuation vector at S_f . Its kernel is the ordinary unit group. Since the already proved class group has order h , principal generators of $\mathfrak{p}_v^h$ give h times each standard vector in the image. The image is therefore a finite-index subgroup of $\mathbb{Z}^{S_f}$, free of rank $\#S_f$. Lifting a basis splits off that many further infinite cyclic generators. The rank is $r_1 + r_2 - 1 + \#S_f$, with the same torsion group μ_K .

What this lesson does not prove

The restricted product theorem is Proposition 1.1 of *Restricted products and profinite completions*. The additive discreteness, closedness and box lemma are Theorem 2.2 and Lemma 2.5 of *The adèle ring of a number field*. These are proved there; only their stated number-field forms are used here.

Unique factorization of fractional ideals in a Dedekind domain and the correspondence between their exponents and local valuations are algebraic prerequisites. Precise references are Neukirch, *Algebraic Number Theory*, Chapter I, (3.1), (3.3) and (3.9) for the integer ring and fractional ideal factorization, and (11.3), (11.5) and the valuation-factorization paragraph on page 69 for local valuations. The adelic product formula was fixed in the preceding lesson; Neukirch, Chapter III, (1.3), states the same normalized formula. No global class field theory, reciprocity map or classification of the connected component for arbitrary K is asserted here.

References

- J. Neukirch, *Algebraic Number Theory*, Chapter VI, §1, especially (1.1)–(1.6), supplies the ideal-class and logarithmic viewpoints. Its norm-one compactness argument uses the replete ideal-class group; Theorem 3.3 above instead uses the additive box lemma.
- D. Harari, *Galois Cohomology and Class Field Theory*, §12.3, Proposition 12.20, Theorem 12.21, Lemma 12.23 and Theorems 12.24 and 12.27, for the topology comparison and its arithmetic consequences, including S -units.
- F. Halter-Koch, *Class Field Theory and L Functions: Foundations and Main Results*, §5.3, Theorems 5.3.6–5.3.8, for norm-one topology, compactness and finiteness results.
- A. Deitmar, *Automorphic Forms*, §5.3, Lemma 5.3.1, Theorem 5.3.3 and Proposition 5.3.4, for the rational idèle topology and decomposition.
- J. R. Getz and H. Hahn, *An Introduction to Automorphic Representations*, [author-hosted draft of April 22, 2022](#), §2.6, equations (2.14)–(2.19), for the relation to adelic quotients. The 2024 published book is a later edition.
- U. Jannsen, *Algebraic Number Theory II*, §17, Definitions 17.1–17.3 and Proposition 17.4, for the idèle and ideal descriptions.
- J.-B. Bost and A. Connes, [Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory](#), §3, for the finite rational split $\mathbb{A}_f^\times = \mathbb{Q}_{>0}^\times \widehat{\mathbb{Z}}^\times$.

Completions, the p-adic numbers and complete discretely valued fields

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Introduction

An absolute value tells us which approximations become accurate. Completion supplies limits for all Cauchy approximations without changing the arithmetic already present. For the p-adic absolute value, accuracy means agreement modulo a high power of a prime. This makes the completed ring of integers both an inverse limit of finite rings and a space of infinite digit sequences.

We construct completions, develop the arithmetic and topology of $\mathbb{Q}_p$, and extend the digit construction to every complete discretely valued field. We then prove that the complete archimedean fields are precisely $\mathbb{R}$ and $\mathbb{C}$, with powers of their usual absolute values. This identifies all archimedean places of a number field.

The prerequisites are *Absolute values, valuations and Ostrowski's theorem* and [Real Analysis I](#). We assume the absolute-value axioms, the ultrametric inequality, and the elementary real-analysis facts about completeness and compactness. The precise algebraic facts used without proof are collected at the end. Basic references are [Milne], [Neukirch], [Deitmar] and [Harari]; [Halter-Koch] provides a broader account of limits and completeness. The local arithmetic developed here also underlies the discussion of places in *Weil's proof for curves and what is missing over the integers*.

Write K_v for completion at a place v , and $\mathbb{N} = \{0, 1, \dots\}$. For a nonarchimedean absolute value, its valuation ring and residue field are

$$\mathcal{O} = \{x : |x| \leq 1\}, \quad \mathfrak{m} = \{x : |x| < 1\}, \quad \kappa = \mathcal{O}/\mathfrak{m}.$$

An additive discrete valuation is normalized to have image $\mathbb{Z}$, and $v(0) = +\infty$. A uniformizer π has $v(\pi) = 1$. Thus $|x| = c^{v(x)}$ for a fixed $0 < c < 1$.

Completing a field

A sequence (a_n) is *Cauchy* if for every $\varepsilon > 0$, eventually $|a_n - a_m| < \varepsilon$ for all n, m . It is *null* if $|a_n| \rightarrow 0$. Completion identifies two Cauchy sequences when their difference is null.

Theorem 2.1 (completion). Every field K with an absolute value admits a complete valued field $\widehat{K}$ containing K isometrically and densely. Every isometric field embedding from K to a complete valued field extends uniquely to an isometric embedding from $\widehat{K}$. Consequently two completions have a unique isometric field isomorphism that respects their embeddings of K .

If the absolute value is nonarchimedean, so is its extension, and

$$|\widehat{K}^\times| = |K^\times|, \quad \kappa \xrightarrow{\sim} \widehat{\kappa}.$$

Here $\widehat{\kappa}$ denotes the residue field of $\widehat{K}$. There is no discreteness hypothesis in these last assertions.

Proof. Every Cauchy sequence is bounded: its tail lies within distance 1 of a fixed term, and its initial segment is finite. Let C be the ring of Cauchy sequences, with coordinatewise operations. The estimate

$$|a_n b_n - a_m b_m| \leq |a_n| |b_n - b_m| + |b_m| |a_n - a_m|$$

shows that products are Cauchy. Boundedness also shows that the null sequences form an ideal N . Set $\widehat{K} = C/N$.

The reverse triangle inequality makes $(|a_n|)$ a Cauchy sequence of real numbers. Define

$$|[(a_n)]| = \lim_n |a_n|.$$

This is independent of the representative. It vanishes exactly on N ; multiplication and the triangle inequality pass to limits. If the class is nonzero, $|a_n|$ is eventually bounded below by a positive number. On that tail,

$$|a_n^{-1} - a_m^{-1}| = \frac{|a_n - a_m|}{|a_n| |a_m|}.$$

Hence the reciprocal sequence, with arbitrary initial terms, defines an inverse. Thus $\widehat{K}$ is a field. Constant sequences embed K isometrically.

For $A = [(a_n)]$, the elements $a_n \in K$ converge to A : the Cauchy condition bounds $\lim_m |a_m - a_n|$ by any prescribed positive tolerance for large n . This proves density.

Now let (A_j) be Cauchy in $\widehat{K}$. Choose $b_j \in K$ with $|A_j - b_j| < 1/j$. The triangle inequality shows that (b_j) is Cauchy in K . Its class B satisfies $b_j \rightarrow B$, so $A_j \rightarrow B$ as well. This proves completeness.

If $f : K \rightarrow L$ is isometric and L is complete, send $[(a_n)]$ to $\lim_n f(a_n)$. Equivalent sequences give the same limit; addition, multiplication and absolute values commute with this operation. Density makes this the only isometric extension. Applying the construction in both directions between two completions gives inverse maps.

In the nonarchimedean case the ultrametric inequality passes to limits. If $x \in \widehat{K}$ is nonzero, choose $a \in K$ with $|x - a| < |x|$. The strict-dominance rule for an ultrametric gives $|a| = |x|$, proving equality of value sets. This argument also shows that the valuation ring $\widehat{\mathcal{O}}$ is the closure of $\mathcal{O}$: approximations sufficiently close to an integral element remain integral, and the valuation ring is closed. To represent the residue of $x \in \widehat{\mathcal{O}}$, approximate it by $a \in K$ with $|x - a| < 1$. Then $a \in \mathcal{O}$ and the residues agree. The kernel of the induced residue map is exactly $\mathfrak{m}$. ■

Uniqueness here includes the embedding of the original field. It does not say that the completed field has no other automorphisms. A trivially valued field is already complete, since every Cauchy sequence is eventually constant.

Congruences as p-adic accuracy

Fix a prime p . The field $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for $|x|_p = p^{-v_p(x)}$. Define

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p : v_p(x) \geq 0\}.$$

Theorem 2.1 extends v_p with the same integer value group and residue field $\mathbb{F}_p$. In particular,

$$x \equiv y \pmod{p^n \mathbb{Z}_p} \iff |x - y|_p \leq p^{-n}.$$

An *inverse limit* of the rings $\mathbb{Z}/p^n \mathbb{Z}$ consists of compatible residues:

$$\varprojlim_{n \geq 1} \mathbb{Z}/p^n \mathbb{Z} = \{(r_n) : r_{n+1} \bmod p^n = r_n\}.$$

Operations are coordinatewise. Each quotient is given the discrete topology, and the limit has the subspace topology from their product. Specifying a residue modulo p^n specifies an open neighborhood.

Proposition 2.2. The closure of $\mathbb{Z}$ in $\mathbb{Q}_p$ is $\mathbb{Z}_p$. Reduction gives an isomorphism of topological rings

$$\mathbb{Z}_p \xrightarrow{\sim} \varprojlim_{n \geq 1} \mathbb{Z}/p^n \mathbb{Z}.$$

The ring $\mathbb{Z}_p$ is compact. The field $\mathbb{Q}_p$ is locally compact and totally disconnected.

Proof. By Theorem 2.1, $\mathbb{Z}_p$ is the closure of $\mathbb{Z}_{(p)} = \{a/b \mid p \nmid b\}$. For every $a/b \in \mathbb{Z}_{(p)}$ and $n \geq 1$, choose an integer c such that $bc \equiv a \pmod{p^n}$. Then $|a/b - c|_p \leq p^{-n}$. Thus $\mathbb{Z}$ has the same closure.

Density shows that every coset modulo $p^n \mathbb{Z}_p$ contains an integer. An integer belongs to this ideal exactly when it is divisible by p^n , so

$$\mathbb{Z}_p/p^n \mathbb{Z}_p \simeq \mathbb{Z}/p^n \mathbb{Z}.$$

The resulting map to the inverse limit is injective, because a nonzero element has a finite valuation. For a compatible family (r_n) , take its integer representatives $0 \leq c_n < p^n$. Compatibility gives $|c_m - c_n|_p \leq p^{-n}$ when $m \geq n$. Their limit lies in $\mathbb{Z}_p$ and has all the prescribed residues. This proves surjectivity. The fibers of reduction modulo p^n are the cosets of $p^n \mathbb{Z}_p$, which form a neighborhood basis. Thus both directions are continuous.

Compatibility also gives unique digits $a_i \in \{0, \dots, p-1\}$ with

$$c_n = \sum_{i=0}^{n-1} a_i p^i, \quad x = \sum_{i \geq 0} a_i p^i.$$

A cylinder fixing the first n digits is a coset modulo p^n ; it has exactly p children fixing one more digit. To prove compactness directly, suppose an open cover has no finite subcover. Starting with the whole space, repeatedly choose a child cylinder with no finite subcover. The chosen compatible digits define a point x . An open member of the cover containing x contains a sufficiently long cylinder in this chain, a contradiction.

The cosets $a + p^n \mathbb{Z}_p$, for $a \in \mathbb{Q}_p$ and $n \in \mathbb{Z}$, are open and closed: small changes preserve membership, and the same holds for their complements. They are also compact, by translation and multiplication by p^n . This proves local compactness. Given distinct points, choose such a coset containing one and excluding the other. Its intersection with any connected set separates those points, so every connected subset is a singleton. ■

The whole field is not compact: the increasing open cover $\mathbb{Q}_p = \bigcup_{m \geq 0} p^{-m} \mathbb{Z}_p$ has no finite subcover. Compactness of the ring and local compactness of the field are different assertions.

Digits in any complete discrete valuation

Let K be complete for a normalized discrete valuation v , with ring $\mathcal{O}$, residue field κ , and uniformizer π . Choose a set $R \subset \mathcal{O}$ containing exactly one representative of each residue class, including 0. These representatives need not form a subring.

Proposition 2.3 (expansions and series). Every nonzero $x \in K$ has a unique expansion

$$x = \sum_{i \geq m} a_i \pi^i, \quad m = v(x), \quad a_i \in R, \quad a_m \neq 0.$$

Zero has the all-zero expansion. Allowing leading zero digits does not change the element, so the starting index is unique only with the stated normalization. A series $\sum_{n \geq 0} x_n$ in K converges if and only if $x_n \rightarrow 0$. Moreover,

$$\mathcal{O} \xrightarrow{\sim} \varprojlim_{n \geq 1} \mathcal{O}/\pi^n \mathcal{O}$$

is an isomorphism of topological rings, with discrete quotients.

Proof. Necessity for series follows by subtracting consecutive partial sums. For sufficiency, the ultrametric inequality gives

$$\left| \sum_{n=r}^s x_n \right| \leq \max_{r \leq n \leq s} |x_n|.$$

If the terms tend to zero, the partial sums are Cauchy and therefore converge. This argument actually works in every complete nonarchimedean field.

For $y \in \mathcal{O}$, define $y_0 = y$. Choose $a_j \in R$ representing $y_j \bmod \pi$, and set

$$y_{j+1} = \frac{y_j - a_j}{\pi} \in \mathcal{O}.$$

Then

$$y = \sum_{j=0}^{N-1} a_j \pi^j + \pi^N y_N.$$

The remainder has absolute value at most c^N , so it tends to zero. For nonzero x , apply this to $y = \pi^{-v(x)}x$, a unit; its first digit is nonzero. Every prescribed digit sequence converges because $|a_i \pi^i| \leq c^i$.

If two expansions first differ at index j , the difference of their digits at j is a unit: distinct representatives have distinct residues. The remaining tail has valuation at least $j+1$. Thus the difference of the expansions has valuation exactly j , proving uniqueness.

For the inverse limit, the kernel is $\bigcap_n \pi^n \mathcal{O} = 0$. Lift a compatible family of residues to elements $b_n \in \mathcal{O}$. Then $b_m - b_n \in \pi^n \mathcal{O}$ for $m \geq n$, so they converge. The limit has each prescribed residue, since $\pi^n \mathcal{O}$ is closed. The same cosets are basic neighborhoods on both sides, proving the topological assertion. ■

The first N digits determine an integral element modulo π^N . This is the useful meaning of truncation. Addition and multiplication may require carrying because R need not respect either operation. In mixed characteristic there is no reason for κ itself to embed as a coefficient field.

The ring $\mathcal{O}$ is a DVR: its units have valuation zero, and in a nonzero ideal choose an element with least valuation. That element generates the ideal, since division by it leaves every other element in $\mathcal{O}$. Its maximal ideal is $\pi \mathcal{O}$. If κ is finite, the finite-cylinder argument of Proposition 2.2 proves that $\mathcal{O}$ is compact and K is locally compact.

Three computations

Negative one. For every prime p ,

$$\sum_{i=0}^{N-1} (p-1)p^i = p^N - 1.$$

Since $p^N \rightarrow 0$ p -adically,

$$-1 = \sum_{i \geq 0} (p-1)p^i.$$

There is no conflict with the growth of the ordinary integer partial sums: the metric has changed.

One third in $\mathbb{Q}_5$. Successive digit extraction gives

Current remainder y_j	Digit a_j modulo 5	Next remainder $(y_j - a_j)/5$
1/3	2	-1/3
-1/3	3	-2/3
-2/3	1	-1/3

The last two rows repeat. Thus, in increasing order of powers,

$$\frac{1}{3} = 2 + 3 \cdot 5 + 5^2 + 3 \cdot 5^3 + 5^4 + \dots = 2 + \sum_{j \geq 0} (3 \cdot 5^{2j+1} + 5^{2j+2}).$$

As a check, the repeated block sums to $(15 + 25)/(1 - 25) = -5/3$, so the total is $1/3$. The geometric series converges because $|25|_5 < 1$.

Laurent series. For $k = \mathbb{F}_q$, let $k((t))$ consist of sums $\sum_{i \geq m} a_i t^i$ with $a_i \in k$ and only finitely many negative powers. Addition and convolution multiplication are well defined: each coefficient of a product involves finitely many pairs. A nonzero series is $a_m t^m (1 + h)$, with $h \in tk[[t]]$, and its inverse is

$$a_m^{-1} t^{-m} \sum_{j \geq 0} (-h)^j.$$

Each coefficient of this expression stabilizes, and multiplication verifies the inverse. Thus these series form a field. Put $v_t(x) = \min\{i : a_i \neq 0\}$ and $|x| = q^{-v_t(x)}$.

A Cauchy sequence has a common lower bound on its exponents after discarding finitely many terms; closeness to a fixed term proves this. At each exponent its coefficient eventually stabilizes. The stabilized coefficients give a Laurent series, and the sequence converges to it because agreement below any fixed exponent is eventually exact. Hence $k((t))$ is complete. Its Laurent-polynomial truncations lie in $k(t)$ and are dense. By Theorem 2.1,

$$\widehat{k(t)}_t = k((t)), \quad \mathcal{O} = k[[t]], \quad \kappa = k.$$

Here the representatives really do form a coefficient field, so no carrying is needed.

Completing a number field at a prime

Let K be a number field, $A = \mathcal{O}_K$ its ring of algebraic integers, and $\mathfrak{p}$ a nonzero prime ideal. We use the fact that A is Dedekind and $B = A_{\mathfrak{p}}$ is a DVR [Milne, Theorem 3.29 and Proposition 3.6]. Their fraction field is K : multiplying any element of K by an integer divisible by the denominators of its monic rational polynomial makes it integral. Complete K for the valuation of B , obtaining $K_{\mathfrak{p}}$, and write $\mathcal{O}_{\mathfrak{p}}$ for its valuation ring.

By Theorem 2.1, B is dense in $\mathcal{O}_{\mathfrak{p}}$, with the same uniformizer π and residue field. For every $n \geq 1$, approximation modulo π^n gives

$$B/\pi^n B \simeq \mathcal{O}_{\mathfrak{p}}/\pi^n \mathcal{O}_{\mathfrak{p}}.$$

Injectivity follows from equality of the valuations on B ; surjectivity follows from density.

Also $A/\mathfrak{p}^n \simeq B/\mathfrak{p}^n B$. Indeed, the only maximal ideal containing $\mathfrak{p}^n$ is $\mathfrak{p}$. Thus every $s \notin \mathfrak{p}$ becomes a unit in $A/\mathfrak{p}^n$, and localizing that quotient changes nothing. Since $\mathfrak{p}B = \pi B$, Proposition 2.3 yields

$$\mathcal{O}_{\mathfrak{p}} \simeq \varprojlim_{n \geq 1} A/\mathfrak{p}^n, \quad K \text{ is dense in } K_{\mathfrak{p}}.$$

The displayed ring isomorphism is topological when the quotients are discrete. In particular A itself is dense in $\mathcal{O}_{\mathfrak{p}}$, since it represents every finite quotient.

Localization and completion are separate operations. The ring B is not complete: it is countable, whereas the distinct digit sequences using just two residue representatives already give uncountably many elements in its completion. The general algebraic result [Stacks, Tag 0AP1] says that a Noetherian local ring is a DVR exactly when its maximal-ideal completion is a DVR. It agrees with the direct valuation construction here.

Why the archimedean possibilities stop at two

An absolute value is *archimedean* when it does not satisfy the ultrametric inequality. The prerequisite result that bounded absolute values of integers force the ultrametric inequality implies that an archimedean field has characteristic zero. Its absolute value on $\mathbb{Q}$ is $|q|_{\infty}^s$ for $s > 0$, by Ostrowski's theorem on absolute values of $\mathbb{Q}$. Since $|n| \leq n$, necessarily $s \leq 1$.

We first justify a polynomial fact used in the proof. A nonconstant complex polynomial P satisfies $|P(z)|_{\infty} \rightarrow \infty$ as $|z|_{\infty} \rightarrow \infty$, so its modulus has a minimum at some z_0 . If $P(z_0) \neq 0$, write

$$\frac{P(z_0 + w)}{P(z_0)} = 1 + cw^k + O(|w|^{k+1}), \quad c \neq 0.$$

Choose a unit complex number u with $cu^k = -|c|_{\infty}$, and put $w = ru$. For small positive r , the modulus is at most $1 - |c|_{\infty}r^k + Cr^{k+1} < 1$, contradicting minimality. Thus P has a root. Division and induction show that every complex polynomial factors into linear factors, counted with multiplicity. This proves the needed form of the fundamental theorem of algebra.

Theorem 2.4 (Ostrowski's theorem on complete archimedean fields). If a field K is complete for an archimedean absolute value, there is a field isomorphism $\sigma : K \rightarrow \mathbb{R}$ or $\sigma : K \rightarrow \mathbb{C}$ such that

$$|x| = |\sigma(x)|_{\infty}^s \quad (0 < s \leq 1).$$

Thus it is an isometry to the indicated field with that powered absolute value, and a topological field isomorphism to the field with its usual absolute value.

Reference: [Neukirch, Chapter II, Theorem 4.2].

Proof. Normalize by setting $\|x\| = |x|^{1/s}$. It is necessary to check the triangle inequality after taking this power. The binomial expansion, together with $|n| = n^s$, gives for positive integers N

$$|x + y|^N \leq \sum_{j=0}^N \binom{N}{j}^s \|x\|^{sj} \|y\|^{s(N-j)} \leq (N+1)(\|x\| + \|y\|)^{sN}.$$

Taking sN -th roots and letting $N \rightarrow \infty$ proves the claim. The normalized absolute value has the same Cauchy sequences and agrees with the usual one on $\mathbb{Q}$. Its closure in K is complete: a Cauchy sequence there converges in K , and closedness keeps the limit there. Uniqueness of completion therefore embeds $\mathbb{R}$ isometrically in K .

Fix $x \in K$. For $z \in \mathbb{C}$, where $\mathbb{C}$ is for now only a parameter space, put

$$Q_z(T) = T^2 - 2\operatorname{Re}(z)T + |z|_{\infty}^2, \quad F(z) = \|Q_z(x)\|.$$

Only real coefficients are evaluated in K . If $M = \|x\|$, the reverse triangle inequality gives

$$F(z) \geq |z|_{\infty}^2 - 2|z|_{\infty}M - M^2.$$

Thus F tends to infinity at infinity and attains a minimum m . Among its minimizers choose z_0 with maximal modulus. Suppose $m > 0$, choose $0 < \varepsilon < m$, and put $Q = Q_{z_0}$. The roots $z_1, \overline{z_1}$ of $Q(T) + \varepsilon$ are nonreal conjugates, with

$$|z_1|_\infty^2 = |z_0|_\infty^2 + \varepsilon.$$

Consequently $F(z_1) > m$.

For each $n \geq 1$, factor the real polynomial

$$H_n(T) = Q(T)^n - (-\varepsilon)^n = \prod_{j=1}^{2n} (T - \alpha_j),$$

listing $\alpha_1 = z_1$. Its conjugate factorization gives the real-polynomial identity

$$H_n(T)^2 = \prod_{j=1}^{2n} Q_{\alpha_j}(T).$$

Evaluation in K , multiplicativity, and minimality yield

$$F(z_1)m^{2n-1} \leq \|H_n(x)\|^2 \leq (m^n + \varepsilon^n)^2.$$

Hence $F(z_1)/m \leq (1 + (\varepsilon/m)^n)^2$. Letting $n \rightarrow \infty$ contradicts $F(z_1) > m$. Therefore $m = 0$, and every element of K satisfies a monic real quadratic equation.

If $K = \mathbb{R}$, we are done. Otherwise choose $x \notin \mathbb{R}$. Its quadratic is irreducible over $\mathbb{R}$, since a split quadratic would force x to be one of its real roots. Completing the square gives $j \in K$ with $j^2 = -1$, embedding $\mathbb{C} = \mathbb{R}(j)$. Every real quadratic now splits inside this subfield, so every element of K belongs to it. Thus $K = \mathbb{C}$.

Finally, for $z = a + bj$ we have $\|j\| = 1$, so

$$\|z\| \leq |a|_\infty + |b|_\infty \leq \sqrt{2}|z|_\infty.$$

Apply this to z^n and take n -th roots to get $\|z\| \leq |z|_\infty$. Applying it to z^{-1} proves the reverse inequality. Undoing the normalization proves the theorem. ■

This theorem concerns complete fields, whereas the earlier Ostrowski theorem concerns absolute values on $\mathbb{Q}$. Completeness is essential: $\mathbb{Q}$ with its usual absolute value is archimedean but is neither of the two fields in the conclusion.

Archimedean places of a number field

Corollary 2.5. For a number field K , the archimedean places correspond bijectively to its real embeddings and its pairs of conjugate nonreal embeddings into $\mathbb{C}$. Their completions are respectively $\mathbb{R}$ and $\mathbb{C}$. If the counts are r_1 and r_2 , there are $r_1 + r_2$ archimedean places, and $[K : \mathbb{Q}] = r_1 + 2r_2$.

Proof. An embedding σ gives the absolute value $|\sigma(x)|_\infty$. If its image is real, its closure is $\mathbb{R}$, since it contains $\mathbb{Q}$. If the image contains a nonreal α , it contains $\mathbb{Q} + \mathbb{Q}\alpha$, dense in $\mathbb{C}$: the real-linear map $(u, v) \mapsto u + v\alpha$ is an isomorphism $\mathbb{R}^2 \rightarrow \mathbb{C}$. These closures are the completions.

Conversely, Theorem 2.4 identifies any archimedean completion with one of these fields. Composing K with this identification gives an embedding whose usual absolute value represents the original place.

Suppose two embeddings give equivalent absolute values. Their restrictions to $\mathbb{Q}$ are the same usual absolute value, so the exponent relating them is 1, as evaluation at 2 shows. The correspondence between their images therefore extends to an isometric isomorphism of their completions. Such a map fixes $\mathbb{Q}$, hence $\mathbb{R}$ by continuity. It cannot identify $\mathbb{R}$ with $\mathbb{C}$; on $\mathbb{C}$ it sends i to i or $-i$, so it is identity or conjugation. This proves the asserted bijection.

For the degree formula, express K as a finite tower adjoining algebraic generators. An embedding at each stage extends in precisely as many ways as the degree of the next minimal polynomial: its transformed polynomial splits in $\mathbb{C}$, with distinct roots in characteristic zero. Multiplying these counts gives $[K : \mathbb{Q}]$ embeddings. Conjugation fixes exactly the real embeddings and pairs the others. ■

For example, $\mathbb{Q}(\sqrt{6})$ has two real places and two real completions. The two embeddings of $\mathbb{Q}(\sqrt{-6})$ form one conjugate pair, giving one complex place and completion. Distinct places can therefore have isomorphic completed fields.

Exercises

Digits below are always in $\{0, \dots, p-1\}$, listed in increasing order of powers. A sequence is *eventually periodic* if $a_{i+d} = a_i$ for some $d \geq 1$ and every sufficiently large i .

Exercise 1 (easy). Compute the 7-adic expansion of $-1/6$ and prove that it is eventually periodic.

Solution. It is purely periodic:

$$-\frac{1}{6} = 1 + 7 + 7^2 + \dots$$

The first N terms sum to $(7^N - 1)/6$, whose difference from $-1/6$ is $7^N/6$, of valuation N . The series converges to the claimed value, and uniqueness identifies every digit as 1.

Exercise 2 (easy). Prove $\mathbb{Z}_p^\times = \mathbb{Z}_p \setminus p\mathbb{Z}_p$, and prove that $\mathbb{Z}_p$ is a DVR with uniformizer p .

Solution. A nonzero $x \in \mathbb{Z}_p$ is invertible there exactly when $v_p(x^{-1}) = -v_p(x) \geq 0$, which is equivalent to $v_p(x) = 0$. The remaining elements, including zero, form $p\mathbb{Z}_p$. Thus this is the unique maximal ideal. In a nonzero ideal I , choose x with the least valuation n . Write $x = p^n u$, with u a unit. For every $y \in I$, $y/p^n \in \mathbb{Z}_p$, so $I = p^n \mathbb{Z}_p$. This proves that the ring is a local principal ideal domain, not a field, with uniformizer p .

Exercise 3 (medium). Show that $x \in \mathbb{Q}_p$ belongs to $\mathbb{Q}$ exactly when its p -adic expansion is eventually periodic.

Solution. Multiplication by a power of p shifts finitely many indices, preserving both rationality and eventual periodicity. We may assume $x \in \mathbb{Z}_p$.

If the tail starts at N and has period d , put $B_0 = \sum_{j=0}^{d-1} a_{N+j} p^j$. Then

$$x = \sum_{i < N} a_i p^i + \frac{p^N B_0}{1 - p^d} \in \mathbb{Q},$$

by the convergent geometric series.

Conversely write $x = A/B$, where $A \in \mathbb{Z}$, $B > 0$, and $p \nmid B$. Digit extraction has remainders A_i/B , beginning with $A_0 = A$. The digit a_i is the unique member of $\{0, \dots, p-1\}$ congruent to $A_i B^{-1} \pmod{p}$, and

$$A_{i+1} = \frac{A_i - B a_i}{p} \in \mathbb{Z}.$$

For the ordinary absolute value,

$$|A_{i+1}|_\infty \leq \frac{|A_i|_\infty + B(p-1)}{p}.$$

Thus all A_i lie in the finite integer interval $[-M, M]$, with $M = \max(|A|_\infty, B)$. The next state and digit depend only on the current state. A state eventually repeats, after which the digits repeat. This includes a terminating expansion, whose tail consists of zeros.

Exercise 4 (medium). Show that $\mathbb{Z}_p$ is homeomorphic to the middle-thirds Cantor set, for every prime p .

Solution. Its digit map identifies $\mathbb{Z}_p$ with $D^\mathbb{N}$, where $D = \{0, \dots, p-1\}$ is discrete: agreeing on the first n coordinates means agreement modulo p^n . To identify this with binary sequences, use the prefix code

$$0, 10, 110, \dots, 1^{p-2}0, 1^{p-1}$$

for the p symbols, in order. For $p = 2$ it is just 0, 1. Concatenation gives a binary sequence. Decoding is unique: read to the first zero, or stop after $p-1$ ones, and repeat. Every infinite binary sequence decodes this way, so the map is bijective. Code lengths are between 1 and $p-1$; hence agreement on n input symbols forces agreement on at least n bits, and agreement on $(p-1)n$ bits forces agreement on n decoded symbols. Both maps are continuous.

Finally send $(b_i) \in \{0, 1\}^\mathbb{N}$ to

$$\sum_{i \geq 0} \frac{2b_i}{3^{i+1}}.$$

The two choices at each stage select the left or right third of the remaining closed interval. Nested intervals have lengths tending to zero, giving exactly all points of the Cantor set. If sequences first differ at j , their values differ by at least $3^{-(j+1)}$, since the possible later tail cancels at most half the leading difference. Agreement on a long prefix bounds the difference above by the length of its interval. These bounds prove injectivity and continuity in both directions. Composing gives the desired homeomorphism. This concerns topology; the coding does not identify ring operations with ordinary real operations.

Exercise 5 (hard). Prove Theorem 2.4 by minimizing

$$(t, u) \mapsto \|x^2 - tx + u\|$$

over the whole real plane. Supply both the existence of a minimum and the argument forcing it to be zero.

Solution. Use the normalization and embedded $\mathbb{R}$ established at the start of the theorem's proof. If $x \in \mathbb{R}$, a quadratic vanishing at x is immediate. Suppose $x \notin \mathbb{R}$. On the Euclidean unit circle the continuous function $(t, u) \mapsto \|tx - u\|$ is everywhere positive, since $1, x$ are real-linearly independent. It has a positive minimum c_0 . Homogeneity implies

$$\|tx - u\| \geq c_0 \sqrt{t^2 + u^2}, \quad \|x^2 - tx + u\| \geq c_0 \sqrt{t^2 + u^2} - \|x^2\|.$$

Thus the latter function is coercive. Its minimizers form a nonempty compact set. Choose (t_0, u_0) maximizing u among them, and put $Q(T) = T^2 - t_0 T + u_0$, $m = \|Q(x)\|$.

Assume $m > 0$ and choose $0 < \varepsilon < m$. The roots α, β of $Q + \varepsilon$ are either real or conjugate. If conjugate, both have squared modulus $u_0 + \varepsilon > u_0$. If real, at least one has squared modulus at least $|\alpha\beta| = |u_0 + \varepsilon| > u_0$. Choose such a root α , and define $Q_\alpha(T) = T^2 - 2\operatorname{Re}(\alpha)T + |\alpha|_\infty^2$. Its constant coefficient is larger than u_0 , so maximality gives $\|Q_\alpha(x)\| > m$.

Factor $H_n = Q^n - (-\varepsilon)^n$, with α among its $2n$ roots. The identity $H_n^2 = \prod_j Q_{\alpha_j}$ is valid because H_n has real coefficients. Every factor evaluated at x has norm at least m , since the minimum was taken over all real quadratic coefficients. Hence

$$\frac{\|Q_{\alpha}(x)\|}{m} \leq \left(1 + \left(\frac{\varepsilon}{m}\right)^n\right)^2.$$

Letting $n \rightarrow \infty$ is a contradiction. Thus $m = 0$, and every x satisfies a real quadratic. The last two paragraphs of the theorem's proof then identify the field and its absolute value. This argument also explains why merely asserting that a continuous function on $\mathbb{R}^2$ has a minimum would be insufficient: coercivity needs proof.

What this lesson does not prove

- The prerequisite real-analysis foundations: completeness of $\mathbb{R}$, compactness of closed bounded Euclidean sets, and the extreme-value theorem. These are the facts from *Real Analysis I* used in the minimization arguments; [Neukirch, Chapter II, proof of Theorem 4.2] uses the same foundations. The polynomial factorization needed here was proved above.
- The criterion that an absolute value is nonarchimedean exactly when its values on integers are bounded, and the consequent characteristic-zero assertion for archimedean fields [Milne, Proposition 7.2 and Corollary 7.3].
- Ostrowski's classification of nontrivial absolute values on $\mathbb{Q}$ [Milne, Theorem 7.12], and the equivalence criterion that two nontrivial absolute values give the same topology exactly when one is a positive power of the other [Milne, Proposition 7.8]. These belong to *Absolute values, valuations and Ostrowski's theorem*.
- For a Dedekind domain A , its integral closure in a finite separable extension of its fraction field is Dedekind [Milne, Theorem 3.29]; every localization of A at a nonzero prime is a DVR [Milne, Proposition 3.6]. We use the first assertion for $A = \mathbb{Z}$.
- For a Noetherian local ring A , A is a DVR if and only if its maximal-ideal completion is a DVR [Stacks, Tag 0AP1]. This is a comparison with the more general algebraic theorem; the valued-field and DVR constructions needed in the lesson were proved directly.

Hensel's lemma and the classification of nonarchimedean local fields are developed in *Hensel's lemma, squares and roots of unity in p -adic fields* and *Local fields: classification and Haar measure*.

The written ring-theoretic providers are *Number fields*, lesson 3, **Discrete valuation rings and Dedekind domains**, its local characterizations and Proposition 3.1, and lesson 5, **Decomposition of primes in extensions**, Theorem 5.1 for finite integral closure over an arbitrary Dedekind base in a finite separable field extension. The integer-bound criterion, equivalence criterion and rational classification are Proposition 1.1, Proposition 3.1 and Theorem 4.1 of **Absolute values, valuations and Ostrowski's theorem**; the references below credit the classical treatments.

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Haar uniqueness through shrinking neighborhoods

Adapted from D. H. Fremlin, Measure Theory, by GPT-6.1 Sol (OpenAI), Ultra reasoning effort, 3 October 2026. The neighborhood-ratio argument is Fremlin's; the compact shrinking argument and the final Borel-measure step translate it to the convention of this course. Self-checked by the adapting AI; no independent AI or human review is claimed.

This proof compares the masses of small symmetric neighborhoods. It gives an alternative to **Theorem 9.2 of the Haar lesson**, using the open-set case of the Radon-product theorem. Throughout, a Radon measure has the course's convention: it is finite on compact sets, outer regular on Borel sets and inner regular on open sets.

Theorem. Let G be an arbitrary locally compact Hausdorff group, and let μ, ν be nonzero left-invariant Radon measures on G . There is a unique $c > 0$ such that $\mu = c\nu$. Neither measure is assumed sigma-finite.

Proof. By **Proposition 9.1**, each nonempty open set has positive measure for both measures. Relatively compact open sets have finite measure for both. Let $\mathcal{U}$ be the symmetric relatively compact open neighborhoods of the identity, ordered by reverse inclusion. These form a neighborhood base: intersect an identity neighborhood with its inverse and with a relatively compact identity neighborhood.

Fix a nonempty relatively compact open set O , and $0 < \varepsilon < 1$. Inner regularity gives a compact $K \subset O$ with

$$\mu(K) > (1 - \varepsilon)\mu(O).$$

There are an open $H \supset K$ and an identity neighborhood V such that $HV \subset O$. Indeed, continuity of multiplication supplies a box $H_x \times V_x$ whose product lies in O for each $x \in K$; take finitely many H_x covering K , their union for H , and the intersection of the corresponding V_x for V . Shrink V to a member of $\mathcal{U}$.

For $U \in \mathcal{U}$ with $U \subset V$, put

$$W = \{(x, y) \in O \times O : x^{-1}y \in U\}.$$

This is open in $G \times G$. **Theorem 5.1(1)** identifies the two integrals of its sections with $(\mu \widehat{\times} \nu)(W)$; its open-set assertion does not require sigma-finiteness. If $x \in H$, the vertical section contains xU . A horizontal section at $y \in O$ is contained in yU , since $U = U^{-1}$. Left invariance therefore gives

$$(1 - \varepsilon)\mu(O)\nu(U) < \mu(H)\nu(U) \leq (\mu \widehat{\times} \nu)(W) \leq \mu(U)\nu(O).$$

All the factors being divided by are positive and finite. Repeating the argument with the two measures exchanged gives, for all sufficiently small $U \in \mathcal{U}$,

$$(1 - \varepsilon)\frac{\mu(O)}{\nu(O)} \leq \frac{\mu(U)}{\nu(U)} \leq \frac{1}{1 - \varepsilon}\frac{\mu(O)}{\nu(O)}.$$

Thus the neighborhood net of ratios has limit $\mu(O)/\nu(O)$. The net is the same for every such O , so these ratios all have one common value $c > 0$.

Let A be any open set and $K \subset A$ any compact set. Compact shrinking supplies a relatively compact open O with $K \subset O \subset A$. Hence $\mu(K) \leq \mu(O) = c\nu(O) \leq c\nu(A)$. Taking the supremum over K , and then exchanging the measures, proves $\mu(A) = c\nu(A)$, also when this value is infinite. Finally outer regularity gives $\mu(E) = c\nu(E)$ for every Borel E . Evaluating any nonempty relatively compact open set proves uniqueness of c . $\square$

The signed-function kernel argument in the **quotient lesson** uses a different compact cutoff. It must equal one on the projection of the support of the original function; cancellation can make the support of its subgroup average smaller.

Source and terms

D. H. Fremlin, *Measure Theory, Volume 4, Topological Measure Spaces*, supplies the neighborhood-ratio proof. Copyright © 1998 D. H. Fremlin. The original supplied [editable TeX](#) remains attributed to Fremlin. This separately titled adaptation is distributed under the [Design Science License](#), with editable Markdown source. Changes on 3 October 2026: restriction to the course's LCH/Borel convention, expansion of compact shrinking, and replacement of the quasi-Radon base-uniqueness step by inner regularity on open sets and outer regularity on Borel sets. Adaptation and additions: GPT-6.1 Sol (OpenAI), Ultra.

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